

KE-430HX KE-430HS BE-438HX BE-438HS

SERVICE MANUAL

Please read this manual before making any adjustments.

ELECTRONIC DIRECT DRIVE LOCKSTITCH BAR TACKER
ELECTRONIC DIRECT DRIVE LOCKSTITCH BUTTON SEWER



brother

This service manual is intended for KE-430HX/KE-430HS and BE-438HX/BE-438HS; be sure to read the KE-430HX/KE-438HS and BE-438HX/BE-438HS instruction manual before this manual.

Carefully read the "SAFETY INSTRUCTIONS" below and the whole of this manual to understand this product before you start maintenance.

As a result of research and improvements regarding this product, some details of this manual may not be the same as those for the product you purchased.

If you have any questions regarding this product, please contact a Brother dealer.

SAFETY INSTRUCTIONS

[1] Safety indications and their meanings

This service manual and the indications and symbols that are used on the machine itself are provided in order to ensure safe operation of this machine and to prevent accidents and injury to yourself or other people.

The meanings of these indications and symbols are given below.

Indications

 DANGER	The instructions which follow this term indicate situations where failure to follow the instructions will result in death or serious injury.
 WARNING	The instructions which follow this term indicate situations where failure to follow the instructions could result in death or serious injury.
 CAUTION	The instructions which follow this term indicate situations where failure to follow the instructions may result in minor or moderate injury.

Symbols



.....

This symbol () indicates something that you should be careful of. The picture inside the triangle indicates the nature of the caution that must be taken.
(For example, the symbol at left means “beware of injury”.)



.....

This symbol () indicates something that you must not do.



.....

This symbol () indicates something that you must do. The picture inside the circle indicates the nature of the thing that must be done.
(For example, the symbol at left means “you must make the ground connection”.)

[2] Notes on safety

DANGER



Wait at least 5 minutes after turning off the power switch and disconnecting the power cord from the wall outlet before opening the control box cover. Touching areas where high voltages are present will result in serious injury from electric shocks.

WARNING



Do not allow any liquids to get onto this sewing machine, otherwise fire, electric shocks or operating problems may occur.



If any liquid gets inside the sewing machine (machine head or control box), immediately turn off the power and disconnect the power plug from the electrical outlet, and then contact the place of purchase or a qualified technician.

CAUTION

Environmental requirements



Use the sewing machine in an area which is free from sources of strong electrical noise such as electrical line noise or static electric noise. Sources of strong electrical noise may cause problems with correct operation.



Any fluctuations in the power supply voltage should be within $\pm 10\%$ of the rated voltage for the machine. Voltage fluctuations which are greater than this may cause problems with correct operation.



The power supply capacity should be greater than the requirements for the sewing machine's power consumption. Insufficient power supply capacity may cause problems with correct operation.



The ambient temperature should be within the range of 5°C to 35°C during use. Temperatures which are lower or higher than this may cause problems with correct operation.



The relative humidity should be within the range of 45% to 85% during use, and no dew formation should occur in any devices. Excessively dry or humid environments and dew formation may cause problems with correct operation.



In the event of an electrical storm, turn off the power and disconnect the power cord from the wall outlet. Lightning may cause problems with correct operation.



Do not connect anything to the USB port other than the USB memory. If this is not observed, problems with operation may result.

Installation



Machine installation should only be carried out by a qualified technician.



Contact your Brother dealer or a qualified electrician for any electrical work that may need to be done.



The sewing machine weighs approximately 60 kg (132lb). The installation should be carried out by two or more people.



Do not connect the power cord until installation is complete. The machine may operate if the foot switch is depressed by mistake, which may result in injury.



Turn off the power switch before inserting or removing the plug, otherwise damage to the control box could result.



Be sure to connect the ground. If the ground connection is not secure, you run a high risk of receiving a serious electric shock, and problems with correct operation may also occur.



When securing the cords, do not bend the cords excessively or fasten them too hard with staples, otherwise there is the danger that fire or electric shocks could occur.



If using a work table which has casters, the casters should be secured in such a way so that they cannot move.



Secure the table so that it will not move when tilting back the machine head. If the table moves, it may crush your feet or cause other injuries.



Use both hands to hold the machine head when tilting it back or returning it to its original position. If only one hand is used, the weight of the machine head may cause your hand to slip, and your hand may get caught.



Be sure to wear protective goggles and gloves when handling the lubricating oil and grease, so that they do not get into your eyes or onto your skin, otherwise inflammation can result. Furthermore, do not drink the oil or eat the grease under any circumstances, as they can cause vomiting and diarrhea. Keep the oil out of the reach of children.

CAUTION

Lubrication

-  Turn off the power switch before carrying out work.
The machine may operate if the foot switch is depressed by mistake, which may result in injury.
-  Be sure to wear protective goggles and gloves when handling the lubricating oil, so that it does not get into your eyes or onto your skin.
If care is not taken, inflammation can result. Furthermore, do not drink the lubricating oil. Diarrhea or vomiting may result.
Keep the oil out of the reach of children.
-  When cutting the nozzle of the oil tank, hold the base of the nozzle securely.
If you hold the end of the nozzle, injury from the scissors may result.

Sewing

-  To prevent problems, do not use objects with sharp points to operate the operation panel.
-  This sewing machine should only be used by operators who have received the necessary training in safe use beforehand.
-  The sewing machine should not be used for any applications other than sewing.
-  Be sure to wear protective goggles when using the machine.
If goggles are not worn, there is the danger that if a needle breaks, parts of the broken needle may enter your eyes and injury may result.
-  Turn off the power switch at the following times.
The machine may operate if the foot switch is depressed by mistake, which may result in injury.
 - When replacing the bobbin and needle
 - When not using the machine and when leaving the machine unattended
-  Use threading mode or turn off the power first in order to carry out threading.
-  If using a work table which has casters, the casters should be secured in such a way so that they cannot move.
-  Attach all safety devices before using the sewing machine. If the machine is used without these devices attached, injury may result.
-  Do not touch any of the moving parts or press any objects against the machine while sewing, as this may result in personal injury or damage to the machine.
-  Secure the table so that it will not move when tilting back the machine head. If the table moves, it may crush your feet or cause other injuries.
-  Use both hands to hold the machine head when tilting it back or returning it to its original position. If only one hand is used, the weight of the machine head may cause your hand to slip, and your hand may get caught.
-  If an error occurs in machine operation, or if abnormal noises or smells are noticed, immediately turn off the power switch. Then contact your nearest Brother dealer or a qualified technician.
-  If the machine develops a problem, contact your nearest Brother dealer or a qualified technician.

Cleaning

-  Turn off the power switch before carrying out cleaning. The machine may operate if the foot switch is depressed by mistake, which may result in injury.
-  Secure the table so that it will not move when tilting back the machine head. If the table moves, it may crush your feet or cause other injuries.
-  Use both hands to hold the machine head when tilting it back or returning it to its original position. If only one hand is used, the weight of the machine head may cause your hand to slip, and your hand may get caught.
-  Be sure to wear protective goggles and gloves when handling the lubricating oil and grease, so that they do not get into your eyes or onto your skin, otherwise inflammation can result.
Furthermore, do not drink the oil or eat the grease under any circumstances, as they can cause vomiting and diarrhea.
Keep the oil out of the reach of children.

CAUTION

Maintenance and inspection



Maintenance and inspection of the sewing machine should only be carried out by a qualified technician.



Ask your Brother dealer or a qualified electrician to carry out any maintenance and inspection of the electrical system.



Turn off the power switch and disconnect the power cord from the wall outlet at the following times, otherwise the machine may operate if the foot switch is depressed by mistake, which may result in injury.

- When carrying out inspection, adjustment and maintenance
- When replacing consumable parts such as the hook and knife



Always be sure to turn off the power switch and then wait one minute before opening the rear cover. If you touch the surface of the motor, it may cause burns.



If the power switch needs to be left on when carrying out some adjustment, be extremely careful to observe all safety precautions.



Secure the table so that it will not move when tilting back the machine head. If the table moves, it may crush your feet or cause other injuries.



Use both hands to hold the machine head when tilting it back or returning it to its original position.

If only one hand is used, the weight of the machine head may cause your hand to slip, and your hand may get caught.



When replacing parts and installing optional accessories, be sure to use only genuine Brother parts.

Brother will not be held responsible for any accidents or problems resulting from the use of non-genuine parts.



If any safety devices have been removed, be absolutely sure to re-install them to their original positions and check that they operate correctly before using the machine.



To prevent accidents and problems, do not modify the machine yourself.

Brother will not be held responsible for any accidents or problems resulting from modifications made to the machine.

[3] Warning labels

The following warning labels appear on the sewing machine.

Please follow the instructions on the labels at all times when using the machine. If the labels have been removed or are difficult to read, please contact your nearest Brother dealer.

1		⚠ 危険 高電圧部分にふれて、大けがをすることがある。 電源を切り、5分たってからカバーをはずすこと。		⚠ 危险 触摸高电压部分，会导致受伤。 在切断电源5分钟后，再开启盖罩。	
		⚠ DANGER Hazardous voltage will cause injury. Turn off main switch and wait 5 minutes before opening this cover.	⚠ GEFAHR Hochspannung verletzungsgefahr! Bitte schalten sie den hauptschalter aus und warten sie 5 minuten, bevor sie diese abdeckung öffnen.	⚠ DANGER Un voltage non adapte provoque des blessures. Eteindre l'interrupteur et attendre 5 minutes avant d'ouvrir le capot.	⚠ PERIGRO Un voltaje inadecuado puede provocar las heridas. Apagar el interruptor principal y esperar 5 minutos antes de abrir esta cubierta.

- 2 Touching areas where high voltages are present will result in severe injury. Turn off the power before removing the cover.

- 3
- CAUTION**
Moving parts may cause injury.
- Operate with safety devices* installed.
- Turn off the power before carrying out operations such as threading, changing the needle, bobbin, knives or hook, cleaning and adjusting.

* Safety devices
Eye guard, Finger guard, DT solenoid cover, Thread take-up cover, Side cover, Rear cover, etc.

- 4 Be careful to avoid injury from the moving thread take-up.

- 5 Be careful not to get your hands caught when returning the machine head to its original position after it has been tilted.

- 6 Be sure to connect the ground. If the ground connection is not secure, you run a high risk of receiving a serious electric shock, and problems with correct operation may also occur.

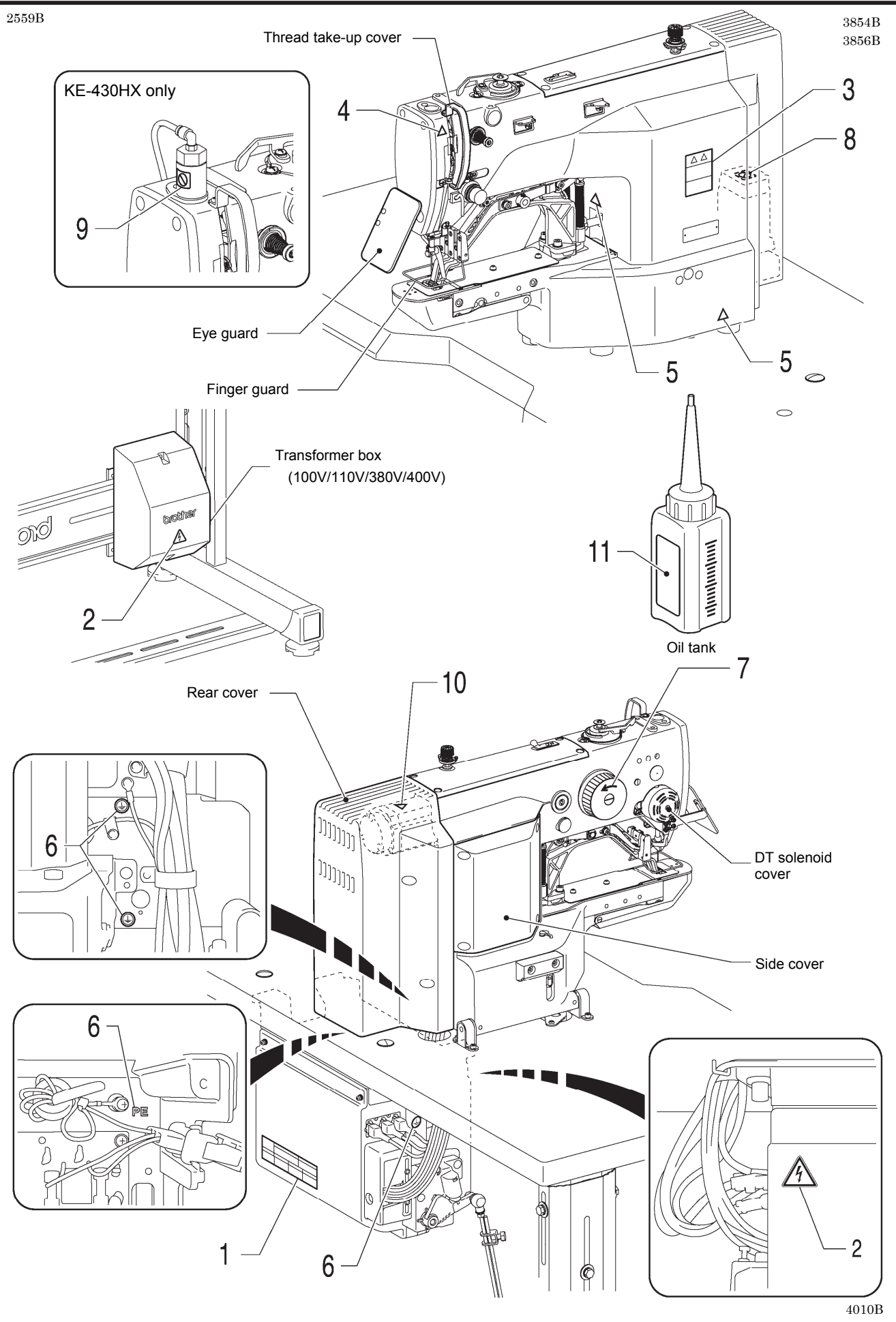
- 7 Direction of operation

- 8 Do not hold, otherwise problems with operation or injury may occur.

- 9 (KE-430HX only)
Do not hold this part when tilting the machine head back, otherwise problems with operation or injury may occur.

- 10 High temperature warning display

- 11
- brother**
Lubricating Oil for Machining
- ⚠ CAUTION**
Lubricating oil may cause inflammation to eye and skin.
Wear protective glasses and gloves.
- Swallowing oil can cause diarrhoea and vomiting.
Do not swallow.
- Keep away from children.**
FIRST AID
Eye contact :
-Rinse with plenty of cold water.
-Seek medical help.
Skin contact :
-Wash with soap and water.
If swallowed :
-Seek medical help immediately.
-Do not induce vomiting.



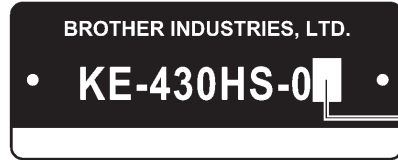
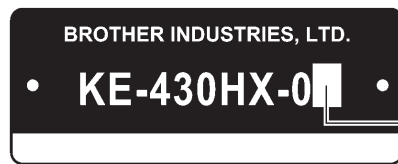
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1. SPECIFICATIONS



1	Light-weight materials
3	Medium-weight materials
5	Heavy-weight materials
K	Knitted materials

3	Medium-weight materials
5	Heavy-weight materials



3956B

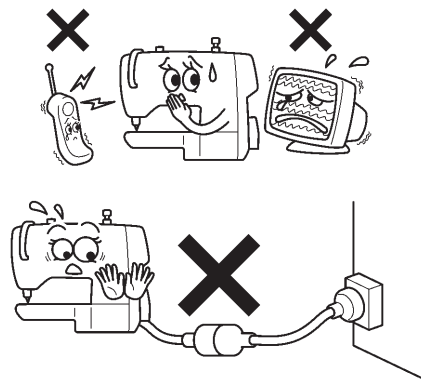
	KE-430HX	KE-430HS	BE-438HX	BE-438HS
	Electronic direct drive lockstitch bar tacker		Electronic direct drive lockstitch button sewer	
Stitch formation	Single needle lock stitch			
Maximum sewing speed	3,300 sti/min (*1)		2,800 sti/min	
Pattern size (X x Y)	50 x 40 mm max. (*2)		6.4 x 6.4 mm max.	
Dimensions of buttons that can be sewn			Outer diameter of button 8 - 30 mm (*3)	
Feed mechanism	Y-θ intermittent feed mechanism (pulse-motor driven mechanism)			
Stitch length	0.05 - 12.7 mm		0.05 – 6.4 mm	
Number of stitches	Refer to “List of sewing patterns” for the number of stitches in sewing patterns which have already been entered.			
Maximum stitch number	Approx. 5,000 stitches (1 pattern)			
Work clamp lifter	Pulse-motor driven mechanism			
Work clamp height Button clamp height	17 mm max.		13 mm max.	
Hook	Shuttle hook (double shuttle hook, optional)		Shuttle hook	
Lubrication method	Semi dry	Minimum lubrication	Semi dry	Minimum lubrication
Digital tension set	Standard equipment		Standard equipment	
Thread wiper device	Standard equipment			
Thread trimmer device	Standard equipment			
Thread nipper device	-03, -0K, -01 specifications: Standard equipment; -05 specifications: Option	-03, -05 specifications: option	Option	
Illumination LED	Standard equipment			
Data storage method	Flash memory (Any sewing pattern can be added using USB memory)			
Data recording media	USB memory (*4)			
Number of cycle programs	Up to 30 can be registered (up to 50 steps each)			
Number of stored data	89 sewing patterns are set already		64 sewing patterns are set already	
	(Up to a maximum of 500 additional types of sewing patterns. Total number of stitches of stored data which can be added is within 500,000.) (*5)			
Motor	AC servo motor 550 W			
Weights	Machine head: approx. 60 kg, Operation panel: approx. 0.4 kg Control box: approx. 7 kg			
Power source	100V, 110V, 200V - 230V, 380V - 400V * The voltages which can be used and the available regions may vary depending on the specifications.			
Power consumption	450 VA			

- *1 Reduce the maximum sewing speed in accordance with the sewing conditions.
The maximum sewing speed for additional sewing patterns with 66 stitches or more is 2,800 sti/min.
When using the double shuttle hook, use a sewing speed of 2,700 sti/min or less.
- *2 Use the optional 50 X 40 sewing area set (other than for Europe: SB7777-001, for Europe: SB7777-0E1) for a sewing area of 50 x 40 mm.
The normal sewing area size is 40x30.
- *3 Use the optional button clamp L (SA9303-201) for diameters of 20 mm or greater.
- *4 No guarantees of operation can be given for any media.
- *5 The number of sewing patterns and the number of stitches that can be recorded will vary depending on the number of stitches in each sewing pattern.
- *6 If using a shank button, use the shank device (other than for Europe: SB7900-001, for Europe: SB7900-0E1).

2. NOTES ON HANDLING

About the machine set-up location

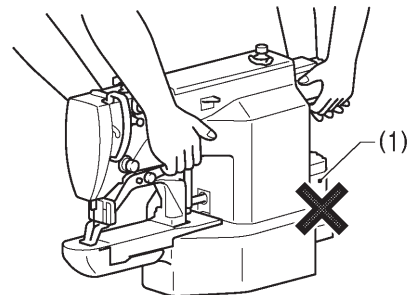
- Do not set up this sewing machine near other equipment such as televisions, radios or cordless telephones, otherwise such equipment may be affected by electronic interference from the sewing machine.
- The sewing machine should be plugged directly into an AC wall outlet. Operation problems may result if extension cords are used.



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Carrying the machine

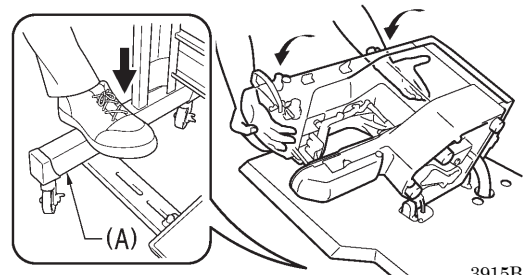
- The machine should be carried by the arm by two people as shown in the illustration.
- When holding the machine head, do not hold it by the pulse motor (1), otherwise it may damage the pulse motor.



3914B

Tilting back the machine head

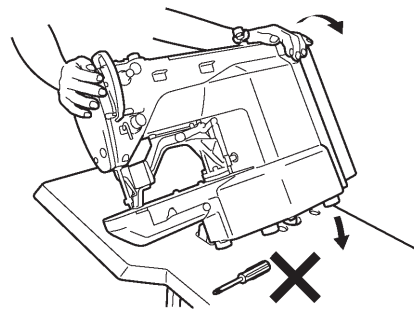
1. Pack away any tools which are near the table.
 2. Secure the foot (A) so that the table will not move, and then pull the arm with both hands to tilt back the machine head.
- * While supporting the arm with both hands, gently lower it.



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Returning the machine head to the upright position

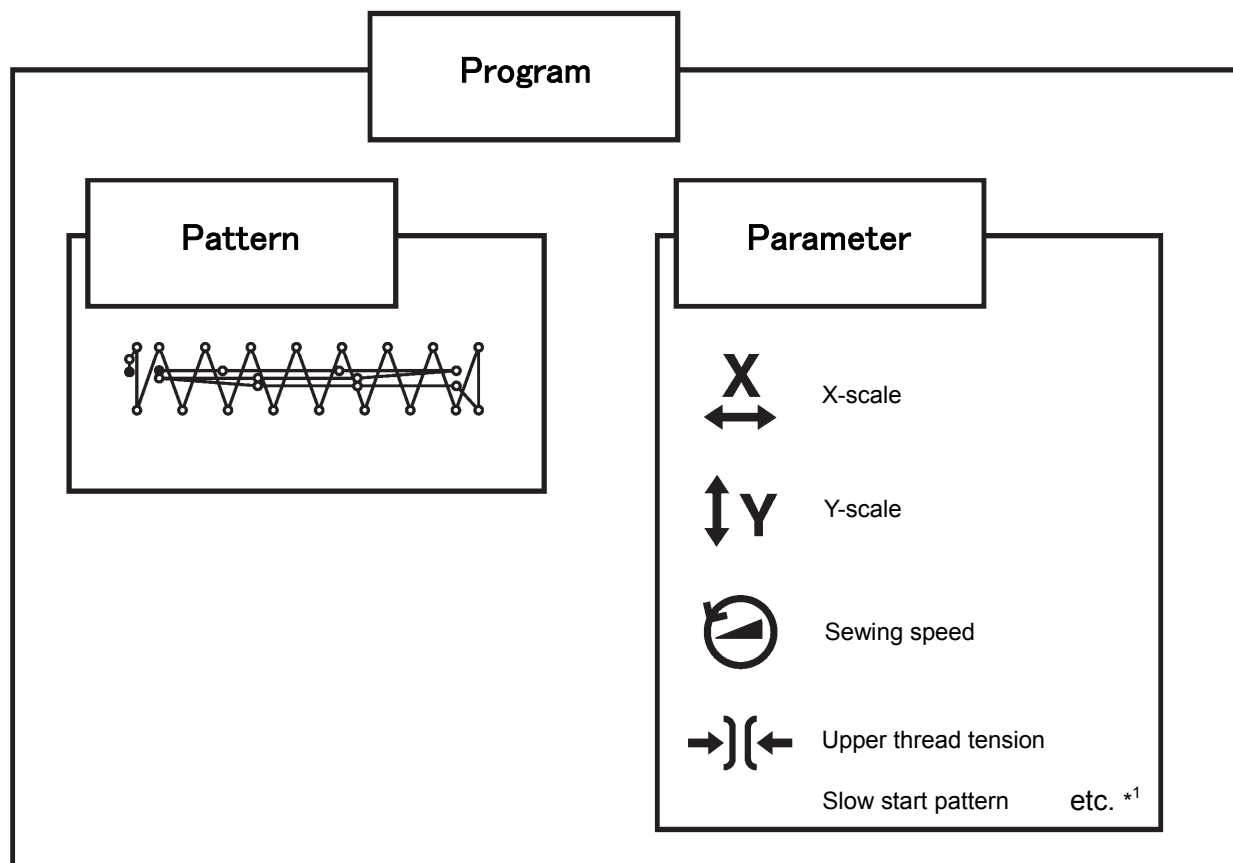
1. Pack away any tools which are near the table.
2. While supporting the arm with both hands, gently return the machine head to its original position.



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3. RELATIONSHIP BETWEEN PROGRAMS AND PATTERNS

Programs specify the way that patterns (stitches) are to be sewn (parameters).



4050B

<Programming method>

1. Use a standard pattern which has been registered in the actual sewing machine (430HX/HS: S001 to S089, 438HX/HS: S001 to S064) ^{*2}, and save the pattern that you want to sew into that program number.
2. Set the parameters (such as X-scale, Y-scale, sewing speed and upper thread tension). This completes the program.

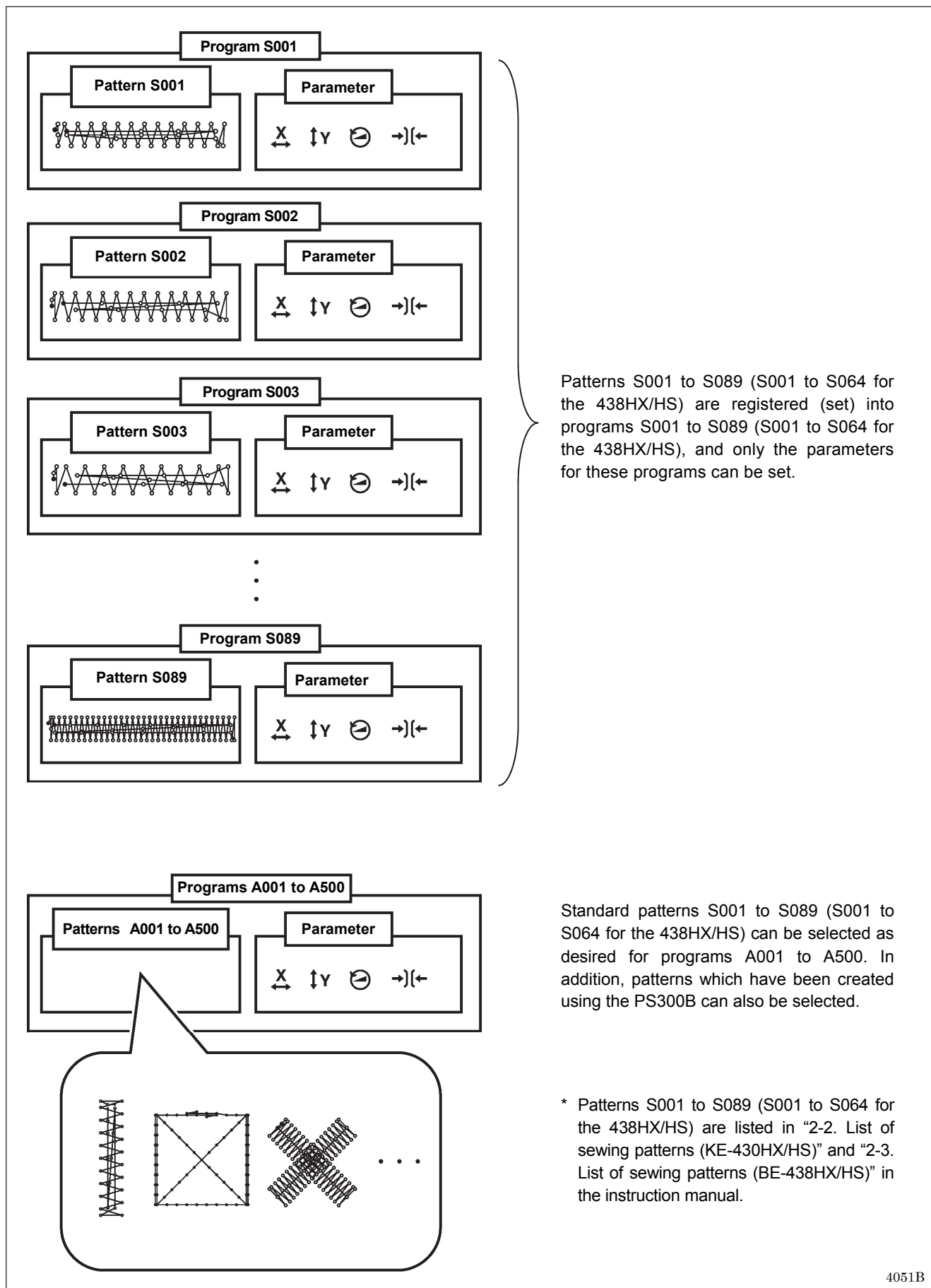
* When actually sewing, select the program number and then start sewing.

*1 Depending on the memory switch settings, the height of the work clamp/button clamp can also be set by means of parameters.

*2 The PS300B can be used to create patterns and load them into actual sewing machines.

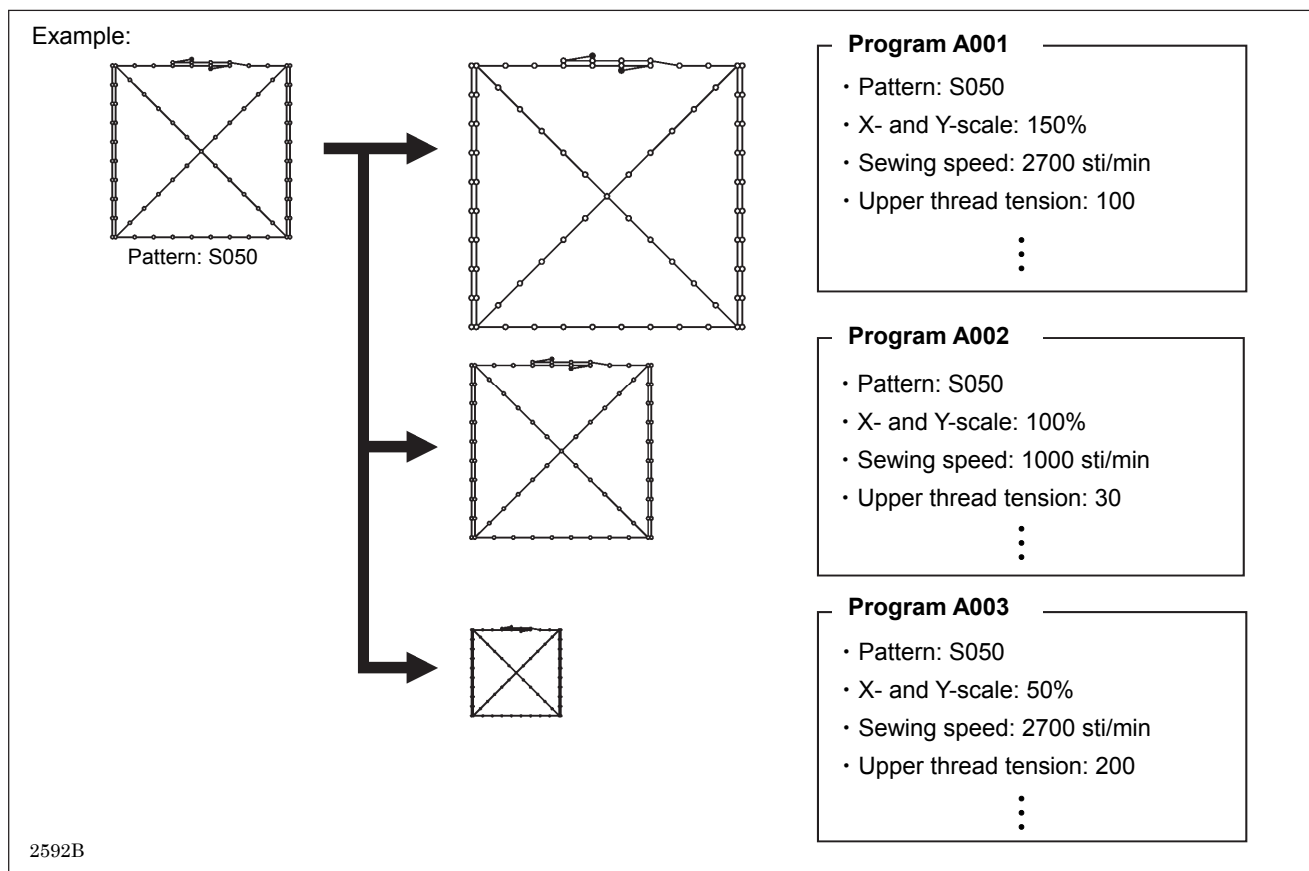
3. RELATIONSHIP BETWEEN PROGRAMS AND PATTERNS

- This sewing machine contains programs S001 to S089 (S001 to S064 for the 438HX/HS) which contain standard patterns, and can also be programmed with additional programs A001 to A500.



3. RELATIONSHIP BETWEEN PROGRAMS AND PATTERNS

- After a program has been finished, you can revise the program by changing the setting details.
- You can organize the use of program numbers so that, for instance, programs with the same pattern but different sizes or upper thread tensions to suit the material being sewn are preset. (Refer to “5-8. User program setting method”.)



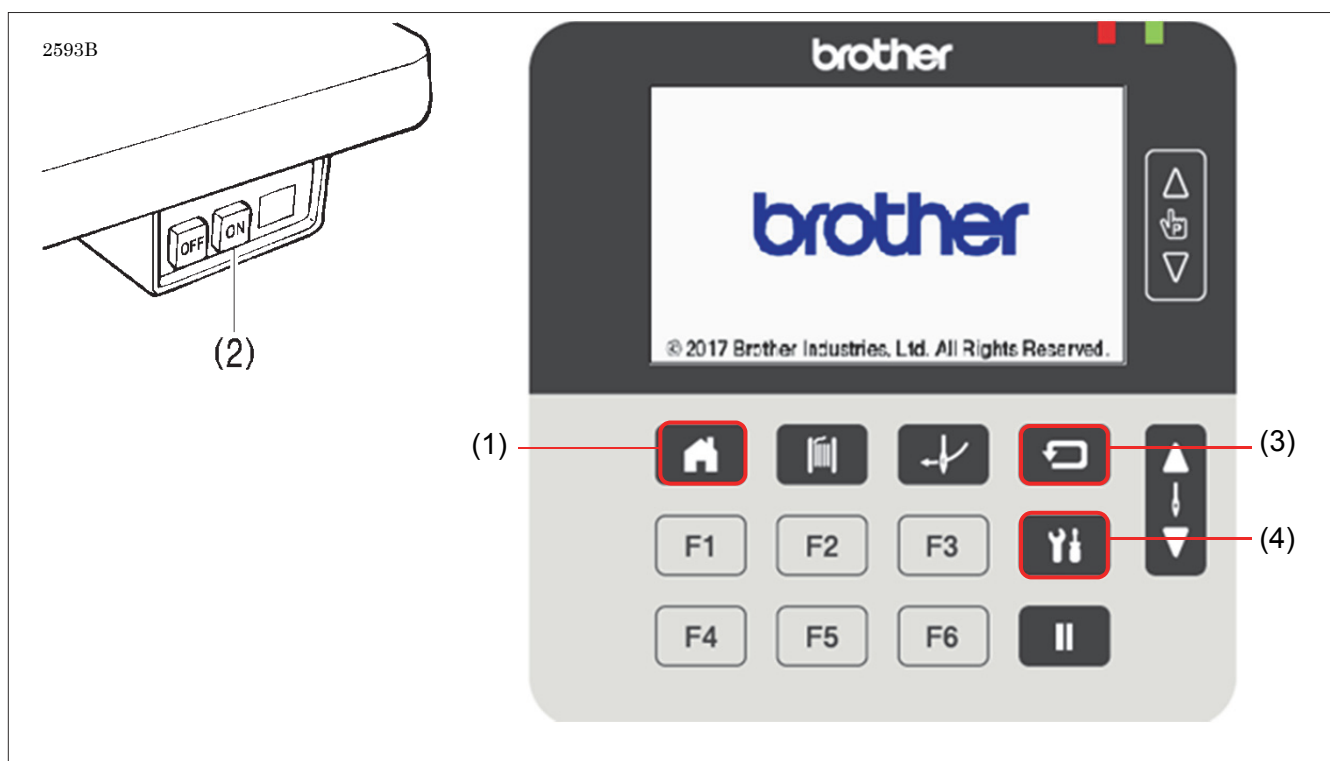
The following table shows restrictions for sewing program numbers, pattern numbers and parameters.

		Standard programs		Additional available patterns	Initial value	
		Preset		Set by user		
Model		430H	438H	430H / 438H	430H	438H
Sewing program No.		S001 – S089	S001 – S064	A001 to A500 (500 programs)		
Pattern No.		S001 – S089	S001 – S064	A001 - A500		
		Fixed at same number as sewing program		Number set when program is created		
Parameter	X-scale (%)	20 - 200			100	
	Y-scale (%)	20 - 200			100	
	Sewing speed (sti/min)	430H: 200 - 3300 / 438H: 200 - 2800			2700	2300
	Slow start pattern	1 - 9			8	7
	X parallel movement amount (mm) When memory switch No. 465 is set to 3	-9.9 - 9.9			0.0	
	Y parallel movement amount (mm) When memory switch No. 465 is set to 3	-9.9 - 9.9			0.0	
	Work clamp/button clamp height (high) (mm) When memory switch No. 470 is set to ON.	430H: 10 - 17 / 438H: 6 - 13			14	10
	Work clamp/button clamp height (medium) (mm) When memory switch No. 470 is set to ON.	430H: 1 - 17 / 438H: 1 - 13			1	1
	Upper thread tension setting value	0 - 300			75	

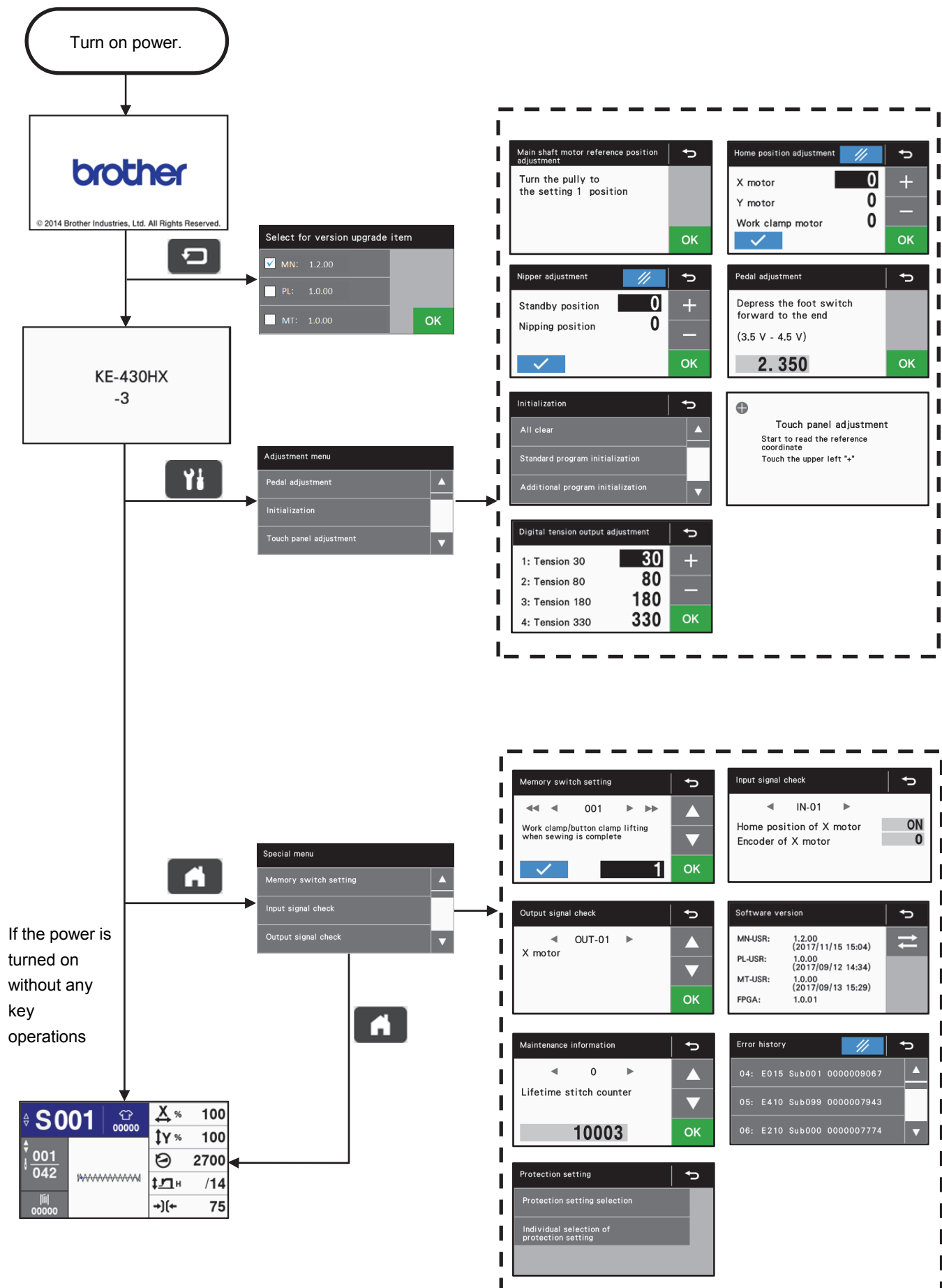
4. FUNCTION SETTINGS

4-1. List of special functions when power is turned on

If you turn on the power switch (2) while pressing a panel key, you can load special functions.



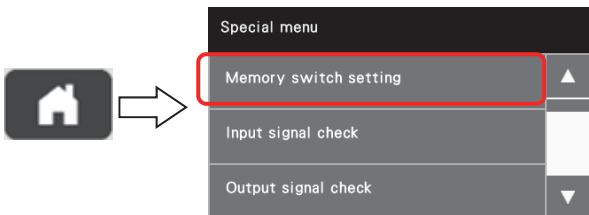
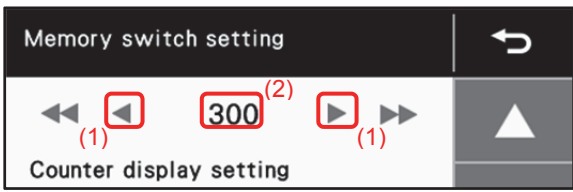
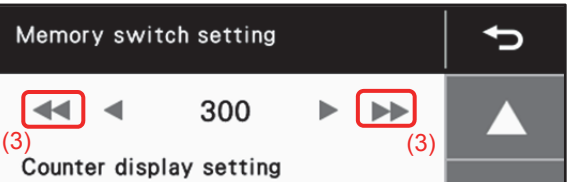
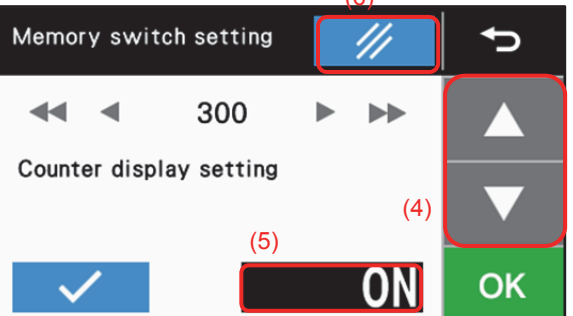
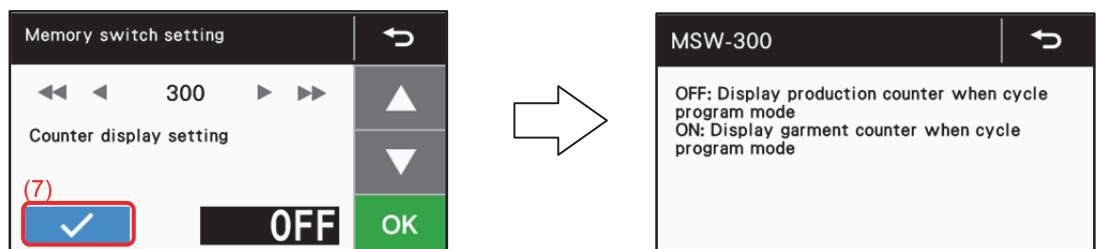
Key operation when power is turned on	Target function
+	Special menu • “4-2. Memory switch setting method (for technicians)” • “4-4. Checking input” • “4-5. Checking output” • “4-6. Checking the software version” • “4-7. Maintenance information list” • “4-8. Checking the error history” • “4-9. Making protection settings”
+	Adjustment menu • “9-18. Adjusting the main shaft motor reference position” • “9-19. Adjusting the X- and Y-feed motor home position” • “9-20. Adjusting the work clamp motor home position” • “9-21. Adjusting the thread nipper motor position” • “9-22. Setting method for standard depression strokes (Foot switch)” • “9-23. Data initialization method” • “9-24. Adjusting the touch panel key position” • “9-25. Correcting the digital tension output (KE-430HX, BE-438HX)”
+	• “4-10. Updating the software version”

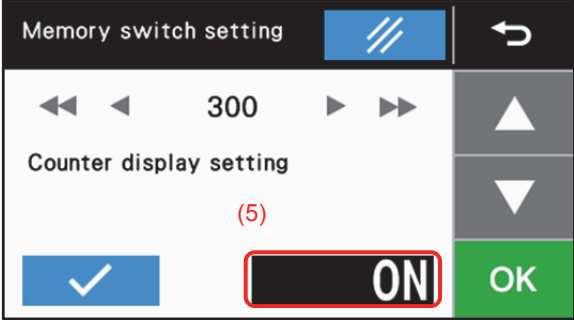


4-2. Memory switch setting method (for technicians)

The settings for the memory switches are enabled for all programs.

Refer to "4-3. List of memory switches (for technicians)" for details on memory switch numbers and settings.

1	<Switching to memory switch setting mode (for technicians)>  1. While pressing the home key, turn on the power to switch to the special menu screen. 2. At the special menu screen, select "Memory switch setting" to switch to memory switch setting mode.
2	<Memory switch settings>  1. Press the [◀] or [▶] key (1) to change the memory switch number (2). If you would like to display only the numbers of memory switches that have been changed from default settings  2. When the [◀◀] or [▶▶] key (3) is pressed, the memory switches with current values that are different from their initial values are displayed in order.
3	Change the memory switch parameters.  1. Press the [▲] or [▼] key (4) to change the memory switch number (5). If the value that is displayed is different from the value that is currently set, the value in the display will flash. 2. If you would like to return the setting to the initial value, press the RESET key (6). * The RESET key is only displayed when the setting value (the value being displayed) is different from the initial value. When checking detailed explanations of memory switches You can touch the check key (7) to check the detailed explanation of the memory switch which is currently selected. 

4	Apply the changed parameter setting.  Touch the OK key . • The setting value (5) will stop blinking and illuminate, and this means that the parameter setting is confirmed. • If you change the memory switch number or touch the back key  to switch the screen without touching the OK key, you can cancel the changes.
5	<Exiting the memory switch setting screen> • At the special menu screen, press the home key  to close the special menu and switch to the home screen.

4-3. List of memory switches (for technicians)

NOTE:

In standard memory switch setting mode which lets you check from the setting screen (refer to "5-3. Memory switch setting method (for operators)"), the bottom two digits only display memory switch Nos. 00 to 49. The bottom two digits do not display memory switch Nos. 50 to 99.

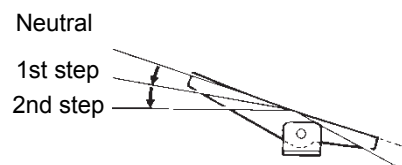
Work clamp/button clamp settings (001 – 099)

No.	Setting range	Initial value	Setting units	Setting details
001	OFF, 1 – 2	1	1	Work clamp/button clamp lift timing when sewing is complete OFF: Does not lift automatically. 1: Lifts at the final stitch position. 2: Lifts after moving to the home position.
051	ON/OFF	OFF	-	Work clamp/button clamp operation before moving to the sewing start position ON: The work clamp/button clamp can operate vertically even before moving to the sewing start position. OFF: The work clamp/button clamp cannot operate vertically before moving to the sewing start position.
052	ON/OFF	ON	-	Work clamp/button clamp operation during pause (split program) OFF: The work clamp/button clamp cannot rise automatically during a pause (split program). ON: The work clamp/button clamp can rise automatically during a pause (split program).
060	0 – 3000 (ms)	0	10	Time after the work clamp/button clamp drops until the upper shaft starts rotating
070	1 – 2	1	1	Type of foot switch used 1: Foot switch 2: 2-pedal foot switch (option)
071	1 – 2	1	1	Foot switch operating mode * Enabled when No. 070 is set to "1". 1: (1st step) Maximum drop → (2nd step) Start 2: (1st step) Drop to intermediate height → (2nd step) Maximum drop and start
072	1 – 5	1	1	2-pedal foot switch operating mode * Enabled when No. 070 is set to "2". 1: (Work clamp switch 1st step) Maximum drop, (Start switch) Start 2: (Work clamp switch 1st step) Drop to intermediate height → (Work clamp switch 2nd step) Maximum drop, (Start switch) Start 3: (While work clamp switch depressed to 1st step) Work clamp/button clamp rises, (Start switch) Start 4: (Work clamp switch 1st step) Maximum drop → (Work clamp switch 2nd step) Start 5: (Work clamp switch 1st step) Drop to intermediate height → (Work clamp switch 2nd step) Maximum drop and start

<Foot switch operating mode settings using memory switches>

- When using a foot switch

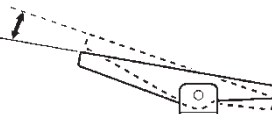
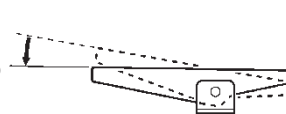
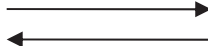
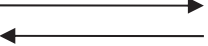
The following table describes the operation of the work clamp/button clamp during sewing standby mode when using the foot switch.



4441Q

1. Set memory switch No. 070 to "1".
2. Use memory switch No. 071 to select the desired operation mode from the following.

* In the following table, the term "work clamp" refers to the work clamp/button clamp.

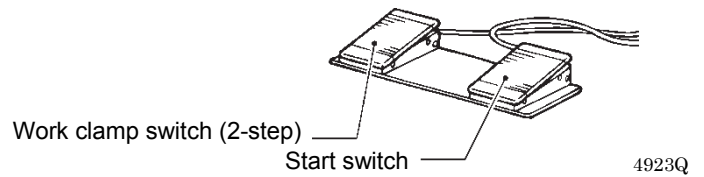
Setting value No. 071	Foot switch				
	Neutral 1st step 		1st step 2nd step 		
	Neutral	→ Depress ← Release	1st step	→ Depressed further	2nd step
1	Work clamp raised		Work clamp maximum drop		Sewing machine start
2	Work clamp raised		Work clamp drop to intermediate height		Work clamp maximum drop + Sewing machine start

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4. FUNCTION SETTINGS

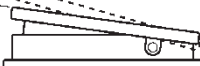
- When using a 2-pedal foot switch (option)

The following table describes the operation of the work clamp/button clamp during sewing standby mode when using a 2-pedal foot switch.



- Set memory switch No. 070 to "2".
- Use memory switch No. 072 to select the desired operation mode from the following.

* In the following table, the term "work clamp" refers to the work clamp/button clamp.

Setting value No. 072	Work clamp switch (left side)					Start switch (right side)
	Neutral 1st step  1st step 2nd step 					
	Neutral	→ Depress ← Release	1st step	→ Depressed further ← Release slightly	2nd step	
1	Work clamp raised Work clamp maximum drop *1 Work clamp raised	→ ← → ←	Work clamp maximum drop Work clamp raised			Sewing machine start *1
2	Work clamp raised Work clamp maximum drop *1 Work clamp raised	→ ← → ← → ←	Work clamp drop to intermediate height Work clamp drop to intermediate height Work clamp maximum drop Work clamp raised			
3	Work clamp maximum drop *1	→ ←	Work clamp raised			Sewing machine start *1
4	Work clamp raised	→ ←	Work clamp maximum drop	→	Sewing machine start	
5	Work clamp raised	→ ←	Work clamp drop to intermediate height	→	Work clamp maximum drop + Sewing machine start	

*1: Enabled when the work clamp switch is at neutral and the work clamp/button clamp is fully lowered.

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Main shaft motor settings (100–199)

No.	Setting range	Initial value	Setting units	Setting details
100	ON/OFF	ON	-	Slow start pattern setting method ON: Follows the slow start patterns for each program. (Refer to "5-4. Slow start pattern setting method".) OFF: The sewing speed for the first 1 to 5 stitches is set by memory switch numbers 151 to 155.
150	ON/OFF	OFF	-	Needle lift by reverse operation ON: When the upper shaft stops, the motor operates in reverse to return the needle bar close to its highest position. NOTE: When the motor operates in reverse to raise the needle, the thread take-up will stop at a position which is lower than its normal stopping position. As a result, the thread take-up will rise slightly at the sewing start. This may result in the thread pulling out under certain conditions. OFF: Disabled
151	430H: 200 – 3300 (sti/min) 438H: 200 – 2800 (sti/min)	*1	100	1st stitch sewing speed at the sewing start * Enabled when No. 100 is set to "OFF".
152	430H: 200 – 3300 (sti/min) 438H: 200 – 2800 (sti/min)	*1	100	2nd stitch sewing speed at the sewing start * Enabled when No. 100 is set to "OFF".
153	430H: 200 – 3300 (sti/min) 438H: 200 – 2800 (sti/min)	*1	100	3rd stitch sewing speed at the sewing start * Enabled when No. 100 is set to "OFF".
154	430H: 200 – 3300 (sti/min) 438H: 200 – 2800 (sti/min)	*1	100	4th stitch sewing speed at the sewing start * Enabled when No. 100 is set to "OFF".
155	430H: 200 – 3300 (sti/min) 438H: 200 – 2800 (sti/min)	*1	100	5th stitch sewing speed at the sewing start * Enabled when No. 100 is set to "OFF".
156	430H: 400 – 3300 (sti/min) 438H: 400 – 2800 (sti/min)	*1	100	Sewing speed for 5th stitch before the sewing end
157	430H: 400 – 3300 (sti/min) 438H: 400 – 2800 (sti/min)	*1	100	Sewing speed for 4th stitch before the sewing end
158	430H: 400 – 3300 (sti/min) 438H: 400 – 2800 (sti/min)	*1	100	Sewing speed for 3rd stitch before the sewing end
159	430H: 400 – 3300 (sti/min) 438H: 400 – 2800 (sti/min)	*1	100	Sewing speed for 2nd stitch before the sewing end

*1: The initial value will vary depending on the specifications. (Refer to "Initial values for each specification" on the following page.)

4. FUNCTION SETTINGS

Initial values for each specification

No.	KE-430HX -01	KE-430HX -03 KE-430HS -03	KE-430HX -05 KE-430HS -05	KE-430HX -0K	BE-438HX BE-438HS
151	800	800	1500	400	400
152	1200	1200	2000	800	400
153	2500	2500	2500	1200	600
154	3300	3300	3300	2500	900
155	3300	3300	3300	3300	2800
156	3300	3300	3300	3300	2800
157	3300	3300	3300	3300	2800
158	3300	3300	3300	3300	2800
159	2700	2700	3000	2700	2800

No.	Setting range	Initial value	Setting units	Setting details
161	ON/OFF	OFF	-	Effective hammer operation when needle does not pierce ON: Enabled OFF: Disabled
162	ON/OFF	OFF	-	Regulation of sewing speed changes due to sewing pitch changes ON: Restricted (Sewing speed is fixed at the minimum speed for the maximum pitch of the sewing data.) * Set to ON if there may be a problem with sewing speed changes as a result of pitch changes. OFF: Not restricted (Sewing speed varies depending on sewing pitch of the sewing data.)
163	430H: 1200 – 3300 (sti/min) 438H: 1200 – 2800 (sti/min)	3300 2800	100	Sewing speed limit value * When setting the sewing speed, the speed will be limited by this value.
164	ON/OFF	OFF	-	Thread trimming operation disallowed ON: All thread trimming operations are disabled. OFF: Thread trimming is carried out in accordance with the sewing data.
165	-20 – 20 (degrees)	0	1	Stop position for needle lift by reverse operation –: Becomes higher +: Becomes lower
166	10 – 500 (ms)	150	10	Delay time until reverse operation starts during reverse operation needle position stop
167	ON/OFF	OFF	-	Servo lock operation ON: Enabled OFF: Disabled
168	OFF, 1 – 120 (seconds)	OFF	1	Servo lock timer setting * Enabled when No. 167 is set to “ON”. OFF: No timer operation 1 to 120: Timer operation (1 to 120 secs.)
169	30 – 89 (degrees)	45	1	Servo lock release rotation angle * Enabled when No. 167 is set to “ON”.
170	430H: 0 – 2800 (sti/min) 438H: 0 – 2300 (sti/min)	0	100	Lowers the allowable speed for the sewing pitch by the amount set. However, the minimum allowable speed value is 400 sti/min.
171	ON/OFF	ON	-	Automatic needle lifter operation ON: If the needle bar is not at the needle up stop position during work clamp/button clamp home position detection, it moves automatically to the needle up stop position. OFF: An error is displayed without the needle bar moving automatically. (No. 655 can also be used to ignore this sensor.)

Feed settings (200 – 299)

No.	Setting range	Initial value	Setting units	Setting details
200	ON/OFF	430H: OFF 438H: ON	-	Single stitch test feed method ON: During test feeding, the mechanism moves one stitch at a time when the foot switch is depressed to the 2nd step (for a 2-pedal foot switch, when the start switch is depressed). In addition, during test mode, test feeding will move forward one stitch at a time when the machine pulley is turned by hand. OFF: During test feeding, sewing starts and automatically continues to the last stitch when the foot switch is depressed to the 2nd step (for a 2-pedal foot switch, when the start switch is depressed).
250	ON/OFF	OFF	-	Home position detection at the sewing end ON: At the sewing end, the work clamp/button clamp will detect the home position and then move to the sewing start position. OFF: At the sewing end, the work clamp/button clamp will move to the sewing start position.
251	1 – 5	3	1	Feeding speed 1: 100 mm/s (Slow) 2: 200 mm/s 3: 300 mm/s 4: 400 mm/s 5: 500 mm/s (Fast)
252	ON/OFF	OFF	-	High-speed test feed method * Enabled when No. 200 is set to "OFF". ON: High-speed feeding starts at the same time test feeding starts. OFF: Normally slow, but becomes faster when the foot switch is depressed to the 1st step (for a 2-pedal foot switch, when the work clamp switch is depressed).
253 *1	0 – 2	0	1	Moving method to the sewing start position 0: Depress the foot switch to the 2nd step (for a 2-pedal foot switch, depress the start switch) while the program number is flashing. 1: Press the Pause key while the program number is flashing. 2: Press the special external input switch (option input (IN3)) while the program number is flashing.
260	430H: -80 – 80 (degrees) 438H: -100 – 100 (degrees)	*2	1	Changes the overall feed timing –: Feed timing becomes earlier. +: Feed timing becomes later.
261	430H: -80 – 80 (degrees) 438H: -100 – 100 (degrees)	*2	1	Changes the feed timing for the 1st stitch at the sewing start –: Feed timing becomes earlier. +: Feed timing becomes later.
262	430H: -80 – 80 (degrees) 438H: -100 – 100 (degrees)	*2	1	Changes the feed timing for the 2nd stitch at the sewing start –: Feed timing becomes earlier. +: Feed timing becomes later.

4. FUNCTION SETTINGS

No.	Setting range	Initial value	Setting units	Setting details
263	430H: -80 – 80 (degrees)	*2	1	Changes the feed timing for the 3rd stitch at the sewing start –: Feed timing becomes earlier. +: Feed timing becomes later.
	438H: -100 – 100 (degrees)			
264	430H: -80 – 80 (degrees)	*2	1	Changes the feed timing for the 3rd stitch before the sewing end –: Feed timing becomes earlier. +: Feed timing becomes later.
	438H: -100 – 100 (degrees)			
265	430H: -80 – 80 (degrees)	*2	1	Changes the feed timing for the 2nd stitch before the sewing end –: Feed timing becomes earlier. +: Feed timing becomes later.
	438H: -100 – 100 (degrees)			
266	430H: -80 – 80 (degrees)	*2	1	Changes the feed timing for the 1st stitch before the sewing end –: Feed timing becomes earlier. +: Feed timing becomes later.
	438H: -100 – 100 (degrees)			

*1: This is not initialized during initialization mode.

*2: The initial value will vary depending on the specifications. (Refer to “Initial values for each specification” on the following page.)

Initial values for each specification

No.	KE-430HX -01	KE-430HX -03 KE-430HS -03	KE-430HX -05 KE-430HS -05	KE-430HX -0K	BE-438HX BE-438HS
260	-23	-23	-23	-23	2
261	37	37	42	37	-88
262	37	37	37	37	-58
263	37	37	37	37	-58
264	0	0	25	0	0
265	35	35	35	35	15
266	35	35	0	35	15

No.	Setting range	Initial value	Setting units	Setting details
267	OFF, 1 – 99 (stitches)	OFF	1	If the overall feed timing (setting No. 260) has been changed from the initial value, this specifies the effective number of stitches. OFF: No limit 1 – 99: When the specified number of stitches from the sewing start is exceeded, the feed timing returns to the standard timing.
268	0 – 2	1	1	Changes the overall feed timing reference. 0: [Feed start reference] Makes the timing uniform at the start of feed. 1: [Needle up reference] Changes the timing at the start of feed so that the feed moves half the feed width when the needle bar is at its highest position. 2: [Feed end reference] Makes the timing uniform at the end of feed.
269	0 – 2	430H: 0 438H: 1	1	Changes the feed timing reference for the first three stitches at the sewing start. 0: [Feed start reference] Makes the timing uniform at the start of feed. 1: [Needle up reference] Changes the timing at the start of feed so that the feed moves half the feed width when the needle bar is at its highest position. 2: [Feed end reference] Makes the timing uniform at the end of feed.
270	0/1	1	1	Moving method to the sewing start point after the program is changed * However, operation for the "0" setting is always carried out when No. 273 is set to ON. 0: Moves to sewing start position and then stops. 1: Home position detection is carried out, then moves to the sewing start position and stops.
271	ON/OFF	OFF	-	Work clamp/button clamp standby position ON: For the 430H, the center of the sewing area is the standby position (escape point). For the 438H, the front is the standby position (escape point). OFF: The sewing start position becomes the standby position.
272	ON/OFF	OFF	-	Reduced feed noise ON: Enabled (Operating speed is limited to 2500 sti/min.) OFF: Disabled
273	ON/OFF	OFF	-	Start switch operation after program is changed ON: Moves to sewing start position and sewing starts (program No. illuminates) OFF: Only moves to sewing start position (program No. flashes)

4. FUNCTION SETTINGS

Operation panel settings (300 – 399)

No.	Setting range	Initial value	Setting units	Setting details
300	ON/OFF	OFF	-	Counter display when using cycle programs ON: Garment counter display OFF: Production counter display
353	ON/OFF	OFF	-	Lower thread counter counting timing ON: Counted at the sewing start OFF: Counted at the sewing end

Program settings (400 – 499)

No.	Setting range	Initial value	Setting units	Setting details
402	ON/OFF	OFF	-	X-scale/Y-scale mm display * The mm display may vary from actual sewing dimensions. ON: Displayed as mm. OFF: Displayed as %.
406	ON/OFF	OFF	-	Assignment of program numbers when switching using F keys OFF: No assignment ON: When the F1 to F4 keys are pressed, operation switches to the program number which has been pre-assigned to the respective key. Assignment to F1 key: Memory switch No. 407 Assignment to F2 key: Memory switch No. 408 Assignment to F3 key: Memory switch No. 409 Assignment to F4 key: Memory switch No. 410
407	OFF, 1-89, 1001 – 1500, 2001 – 2100, 3001 – 3030	OFF	-	Program No. assigned to the F1 key * Enabled when No. 406 is set to "ON". 1 – 89: S001 – S089 1001 – 1500: A001 – A500 2001 – 2100: U001 – U100 3001 – 3030: C001 – C030
408	OFF, 1 – 89, 1001 – 1500, 2001 – 2100, 3001 – 3030	OFF	-	Program No. assigned to the F2 key * Enabled when No. 406 is set to "ON". 1 – 89: S001 – S089 1001 – 1500: A001 – A500 2001 – 2100: U001 – U100 3001 – 3030: C001 – C030
409	OFF, 1 – 89, 1001 – 1500, 2001 – 2100, 3001 – 3030	OFF	-	Program No. assigned to the F3 key * Enabled when No. 406 is set to "ON". 1 – 89: S001 – S089 1001 – 1500: A001 – A500 2001 – 2100: U001 – U100 3001 – 3030: C001 – C030
410	OFF, 1 – 89, 1001 – 1500, 2001 – 2100, 3001 – 3030	OFF	-	Program No. assigned to the F4 key * Enabled when No. 406 is set to "ON". 1 – 89: S001 – S089 1001 – 1500: A001 – A500 2001 – 2100: U001 – U100 3001 – 3030: C001 – C030
411	OFF, 1	OFF	-	Function assigned to F5 key OFF: No assignment 1: Home position movement
451	ON/OFF	OFF	-	Split mode selection ON: Independent split mode OFF: Continuous split mode
452	-5.0 – 5.0 (mm)	0.0	0.1	Final stitch offset in X direction
453	-5.0 – 5.0 (mm)	0.0	0.1	Final stitch offset in Y direction
460	0.0 – 40.0 (mm)	430H: 40.0 438H: 6.4	0.1	Sewing area limit in X direction * Setting range is 0.0 to 50.0 (mm) when No.473 is set to "1".

No.	Setting range	Initial value	Setting units	Setting details
461	0.0 – 30.0 (mm)	430H: 30.0 438H: 6.4	0.1	Sewing area limit in Y direction * Setting range is 0.0 to 40.0 (mm) when No.473 is set to “1”.
462	0 – 8	0	1	Zoom reference point 0: Center of sewing frame 1: Sewing start point 2: Center of pattern 3: Front of pattern center 4: Back of pattern center 5: Back-left corner of sewing area 6: Front-left corner of sewing area 7: Back-right corner of sewing area 8: Front-right corner of sewing area
463	ON/OFF	OFF	-	Zoom for pitch of less than 1 mm ON: Bar tacking stitch lengths (pitch less than approx. 1 mm) are also zoomed. OFF: Bar tacking stitch lengths (pitch less than approx. 1 mm) are not zoomed.
465	1 – 3	1	1	Setting offset amount for sewing pattern 1: The offset amount is initialized when the power is turned off and when the program number, X-scale or Y-scale is changed. 2: The offset amount is retained when the power is turned off, but is initialized when the program number, X-scale or Y-scale is changed. 3: The offset amount can be set for each program using parameters. The offset amount is retained when the power is turned off and when the program number, X-scale or Y-scale is changed.
469	ON/OFF	OFF	-	Tension value setting ON: Operation is according to individual tension values. * Setting is only possible for additional programs * For standard programs, the common tension value is used. * For user programs and cycle programs, this setting is enabled when additional programs are used, and when standard programs are used, the common tension value is used. OFF: Operation is according to the common tension value. * Switching to pattern editing mode is not possible. * If a tension value is set using a programmer, etc., the setting is ignored.
470	ON/OFF	OFF	-	Work clamp/button clamp height setting for each program ON: Work clamp/button clamp height is set separately for each program. OFF: A common work clamp/button clamp height is set for all programs. * The work clamp/button clamp height is set by Nos. 471 and 472.
471	430H: 10 – 17 (mm) 438H: 6 – 13 (mm)	14 10	1	Work clamp/button clamp raise position setting * Enabled when No. 470 is set to “OFF”.
472	430H: 1 – 17 (mm) 438H: 1 – 13 (mm)	1	1	Work clamp/button clamp intermediate drop position setting * Enabled when No. 470 is set to “OFF”.
473	0/1	430H: 0	0	Sewing area (430H only) 0: 40×30mm 1: 50×40mm

4. FUNCTION SETTINGS

Device settings (500 – 599)

No.	Setting range	Initial value	Setting units	Setting details
500	ON/OFF	OFF	-	Thread nipper operation ON: Enabled * However, all of the following conditions must be satisfied. Memory switch No. 100 = ON Memory switch No. 566 = ON Slow start speed is 8 or 9 Sewing speed is 1500 sti/min. or more OFF: Disabled
550	OFF, 100 – 2000 (ms)	OFF	100	Output continue time for needle cooler device when sewing machine is stopped OFF: Needle cooler device is not used 100 to 2000: Output is continued for the specified time.
551	OFF, 1 – 3 (stitches)	430H-05: 1	1	Upper thread tension release at the sewing start * Only enabled when the thread nipper is not operating. OFF: Not released 1 to 3: Released during the specified number of stitches.
		OFF		
554	ON/OFF	OFF	-	Use of the thread breakage sensor (available by special order) ON: Used OFF: Not used
555	ON/OFF	OFF	-	Sewing start sensing sensitivity for thread breakage sensor (available by special order) ON: 10 stitches at sewing start OFF: 5 stitches at sewing start
565 *1	0 – 2	0	1	Tension release solenoid power setting 0: Standard 1: Greater force 2: Maximum
566 *2	ON/OFF	430H-01: ON	-	Thread nipper device installation ON: Installed * The thread nipper device only operates when all of the following conditions are satisfied. Memory switch No. 100 = ON Memory switch No. 500 = ON Slow start speed is 8 or 9 Sewing speed is 1500 sti/min. or more OFF: Not installed NOTE: If the thread nipper device is installed, do not set this to "OFF". If this is not observed, damage may occur.
		430H-03: ON		
		430H-05: OFF		
		430H-0K: ON		
		438H: OFF		
567	ON/OFF	OFF	-	Upper thread tension during feeding while upper shaft is stopped ON: Not released OFF: Released
580	-80 – 115 (degrees)	10	5	Nip timing offset for thread nipper device
581	-160 – 110 (degrees)	-110	5	Release timing offset for thread nipper device

*1: This is not displayed for models with digital tension.

*2: This is not initialized during initialization mode.

No.	Setting range	Initial value	Setting units	Setting details
582	ON/OFF	ON	-	Upper thread tension tightening at the sewing start ON: Enabled OFF: Disabled
583	-150 – 150 (degrees)	0	5	Upper thread tension change (tightening) angle offset at the sewing start
584	-110 – 40 (degrees)	40	5	Upper thread tension release angle offset at the sewing end
585	0 – 500	470	1	Upper thread tension value (tightening) at the sewing start
587	0 – 300	0	1	Upper thread tension value (release) at the sewing start
588	0 – 50 (ms)	35	1	Thread trimmer motor ON timing
589	0 – 50 (ms)	20	1	Thread trimmer motor ON time
590	5 – 10	5	1	Thread trimmer motor holding force
591	0 – 580 (degrees)	430H: 280 438H: 270	5	Specified final speed output angle offset
592	200 – 700 (sti/min)	600	100	Last stitch sewing speed for upper shaft motor
593	0 – 15 (ms)	8	1	Speed change delay until thread nipper operation is complete * If the thread nipper device is not to operate, set to "0".

Error processing settings (600 – 699)

No.	Setting range	Initial value	Setting units	Setting details
650 *1	0 – 2	0	1	Error release method when operation has stopped 0: Press the RESET key on the operation panel. 1: Press the RESET key on the operation panel or the STOP switch (available by special order). 2: Press the RESET key on the operation panel or the external switch (EXIN2).
651	ON/OFF	OFF	-	Needle stop position when sewing is interrupted by the STOP switch (available by special order) ON: Needle stops in the up position. OFF: Needle stops in the down position.
652	ON/OFF	ON	-	Thread trimming operation when sewing is paused ON: Thread trimming is carried out when the pause is canceled. OFF: Thread trimming is not carried out when the pause is canceled.
654	ON/OFF	OFF	-	Use of STOP switch (available by special order) ON: Used OFF: Not used
655	ON/OFF	ON	-	Needle up stop position monitoring function ON: Enabled OFF: Disabled
657	OFF, 2 – 30 (seconds)	OFF	2	Time from error occurring to buzzer stopping OFF: Buzzer does not stop. 2 – 30: Buzzer stops after the specified time.

*1: This is not initialized during initialization mode.

4. FUNCTION SETTINGS

Maintenance settings (700 – 799)

No.	Setting range	Initial value	Setting units	Setting details
750	ON/OFF	OFF	-	Continuous operation mode (for technicians) ON: Continuous operation is repeated while the foot switch is being depressed to the 2nd step (for a 2-pedal foot switch, while the start switch is being depressed). NOTE: This setting is for use by an administrator in test running mode. It is dangerous to set it to "ON" for normal sewing, so it should always be set back to "OFF" after use. OFF: Disabled
751	0 – 3000 (ms)	200	10	Continuous operation cycle time adjustment timer (for technicians) * Enabled when No. 750 is set to "ON".
752	0 – 99	0	1	Sewing machine ID code (for specifying folder in external media)
756	ON/OFF	ON	-	Thread winding speed limit ON: Upper limit is limited to 2000 sti/min. * If the speed specified at the operation panel exceeds 2000 sti/min, thread winding will be limited to 2000 sti/min. OFF: Upper limit is not limited to 2000 sti/min. * Thread winding is carried out at the speed specified at the operation panel.

Specification and destination settings (800 – 899)

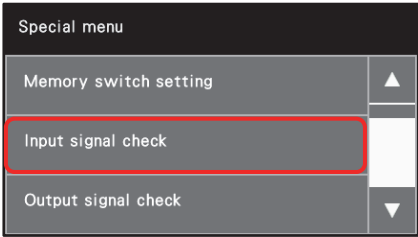
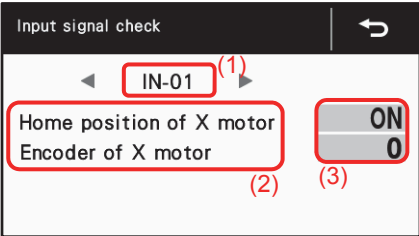
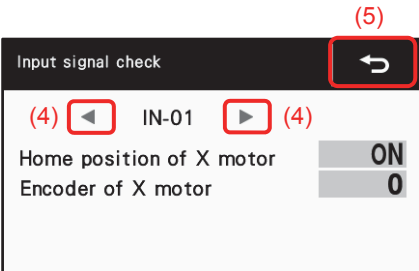
No.	Setting range	Initial value	Setting units	Setting details
850 *1	430H-01: 1,3,5,K	4	1	Specification code setting * The 438H has only -03 specifications, so this will not be displayed. 1: -03 specifications (medium-weight material) 2: -05 specifications (heavy-weight material) 3: -0K specifications (knitted wear) 4: -01 specifications (light-weight material)
	430H-03: 1,3,5,K	1		
	430H-05: 1,3,5,K	2		
	430H-0K: 1,3,5,K	3		
	438H-03: 3	1		
851	0 – 4	0	1	Language 0: English 1: Japanese 2: Chinese 3: Vietnamese 4: Indonesian

*1: This is not initialized during initialization mode.

Unique settings for each model (900 – 999)

No.	Setting range	Initial value	Setting units	Setting details
951	OFF, 1 – 2	OFF	1	Eyelet bar tack closing device (old 432 series) OFF: Disabled 1: Eyelet bar tack closing device operates after the work clamp has dropped. 2: Eyelet bar tack closing device operates at the sewing start.

4-4. Checking input

1	<Switching to input check mode and input check method> 1. While pressing the home key, turn on the power to switch to the special menu screen.  +  2. At the special menu screen, select "Input signal check".  3. The display will switch to the input signal check screen, and the item number (1), item name (2) and input status (3) will be displayed.  4. Press the [◀] or [▶] key (4) to select the item to be checked.  Press the back key  (5) to return to the special menu screen.
2	<To exit from the special menu> • At the special menu screen, press the home key  to close the special menu and switch to the home screen.

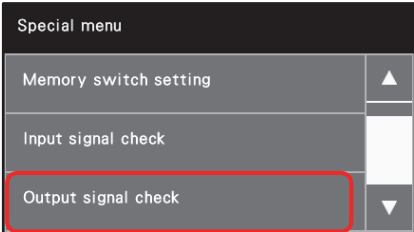
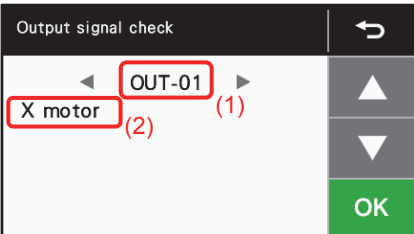
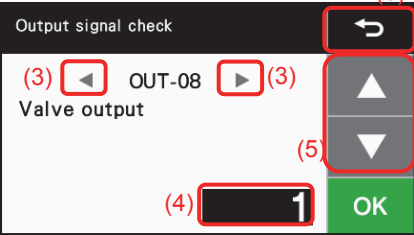
4. FUNCTION SETTINGS

List of check items

Item No.	Check details	Panel display		Checking method
		Input status (minimum value - maximum value)	Judgment	
IN-01	Home position signal of X motor	ON / OFF	ON when at right side, OFF when at left side	Move the work clamp arm assembly by hand.
	Encoder of X motor	-9999 – 9999	---	Move the work clamp arm assembly by hand.
IN-02	Home position signal of Y motor	ON / OFF	ON when at back, OFF when at front	Move the work clamp arm assembly by hand.
	Encoder of Y motor	-9999 – 9999	---	Move the work clamp arm assembly by hand.
IN-03	Home position signal of P motor	ON / OFF	ON when raised, OFF when lowered	Remove the side cover, and then operate the cam shaft by hand.
	Encoder of P motor	-9999 – 9999	---	Remove the side cover, and then operate the cam shaft by hand.
IN-04	Home position signal of T motor	ON / OFF	ON when at thread nipping side, OFF when at thread trimming side	Remove the cover, and then operate the thread trimmer cam by hand.
	Encoder of T motor	-9999 – 9999	---	Remove the cover, and then operate the thread trimmer cam by hand.
IN-05	Needle up signal	ON / OFF	ON in the needle up region, OFF in any other region.	Turn the pulley by hand.
	Main shaft angle	0 – 359°	The forward turning direction (the direction of the arrow) = up.	Turn the pulley by hand.
IN-06	Sewing machine safety SW	ON / OFF	ON when pressed	Tilt back the machine head and then return it.
	Power supply voltage	0 – 300V	---	---
IN-07	Foot switch position	1,3,6,7	Depressed backward : 1 Neutral : 3 Work clamp : 6 Start : 7	Depress the foot switch and then return it.
	Foot switch analog value	0 – 255	The value increases when the foot switch is depressed.	Depress the foot switch and then return it.
IN-08	Key input	1 – 16	When pressed, a number is displayed. ProNo change Δ : 1 ProNo change ∇ : 2 Home: 3 Bobbin winding: 4 Threading: 5 Test: 6 Needle movement change $\blacktriangle$: 7 F1: 8 F2: 9 F3: 10 Menu: 11 Needle movement change $\blacktriangledown$: 12 F4: 13 F5: 14 F6: 15 Stop: 16	Press the respective panel keys.

Panel display				Checking method
Item No.	Check details	Input status (minimum value - maximum value)	Judgment	
IN-09	Work clamp switch 1	ON / OFF	ON when pressed	Depress the work clamp switch of the 2-step foot switch (option) to the 1st step
	Work clamp switch 2	ON / OFF	ON when pressed	Depress the work clamp switch of the 2-step foot switch (option) to the 2nd step
	Start switch	ON / OFF	ON when pressed	Depress the start switch of the 2-step foot switch (option)
IN-10	STOP switch	ON / OFF / NO	ON when detected, OFF when not detected, NO when not connected	Connect the STOP key (option) and then press it.
	Thread breakage detector	ON / OFF	ON when detected, OFF when not detected, NO when not connected * ON is only displayed at the instant the thread breakage sensor detects a breakage.	Pass a thread under the thread breakage sensor (option).
IN-11	Home position sensor of X motor	ON / OFF	ON when detected, OFF when not detected	---
	Home position sensor of Y motor	ON / OFF	ON when detected, OFF when not detected	---
	Home position sensor of P motor	ON / OFF	ON when detected, OFF when not detected	---
IN-12	Option sensor	ON / OFF	ON when detected, OFF when not detected	---
	Air switch	ON / OFF	ON when detected, OFF when not detected	---
IN-13	External input 1	ON / OFF	ON when detected, OFF when not detected	---
	External input 2	ON / OFF	ON when detected, OFF when not detected	---
	External input 3	ON / OFF	ON when detected, OFF when not detected	---

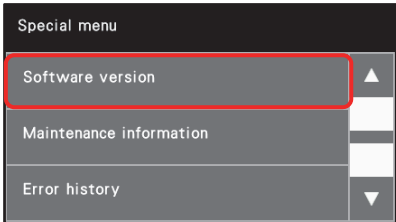
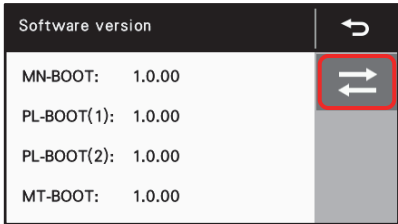
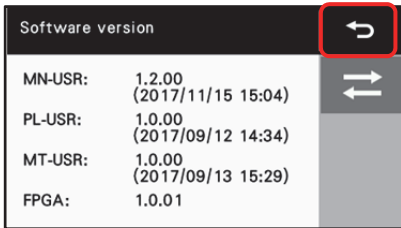
4-5. Checking output

1	<Switching to output check mode and output check method> 1. While pressing the home key, turn on the power to switch to the special menu screen.  2. At the special menu screen, select "Output signal check".  3. The item number (1) and item name (2) will be displayed.  4. Press the [◀] or [▶] key (3) to select the item to be checked.  * If you select items [OUT-08] or [OUT-09], the output number (4) will also be displayed. The output number can be changed by touching the [▲] or [▼] key (5). Press the back key  (6) to return to the special menu screen.
2	<To exit from the special menu> At the special menu screen, press the home key  to close the special menu and switch to the home screen.

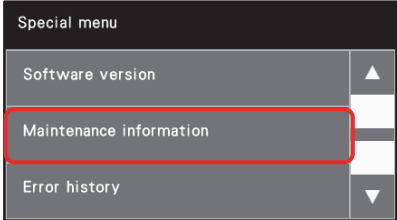
<List of output check items>

Panel display		Check method
Item no.	Check details	
OUT-01	X motor	Press [▲] to move to the left, and press [▼] to move to the right.
OUT-02	Y motor	Press [▲] to move forward, and press [▼] to move backward.
OUT-03	Work clamp motor	Press [▲] to move down, and press [▼] to move up.
OUT-04	Thread trimming motor	Press [▲] to move to the thread trimming side, and press [▼] to move to the thread nipping side.
OUT-05	Tension solenoid	ON when OK key  is pressed, and OFF automatically after the specified time has elapsed
OUT-06	Illumination LED	ON while OK key  is being pressed, and OFF while it is released
OUT-07	Caution LED	ON while OK key  is being pressed, and OFF while it is released
OUT-08	Valve output	[Press [▲] or [▼] to select the valve number. ON while OK key  is being pressed, and OFF while it is released 1: Port output 1 2: Port output 2 3: Port output 3 4: Needle cooling 5: Eyelet closing 6: Optional valve output 1 7: Optional valve output 2 8: Optional valve output 3
OUT-09	External output	Press [▲] or [▼] to select the valve number. ON while OK key  is being pressed, and OFF while it is released 1: External output 1 2: External output 2 3: External output 3
OUT-10	Optional solenoid	ON when OK key  is pressed, and OFF automatically after the specified time has elapsed

4-6. Checking the software version

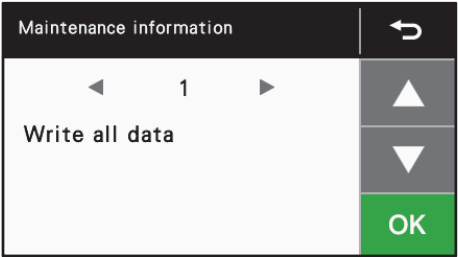
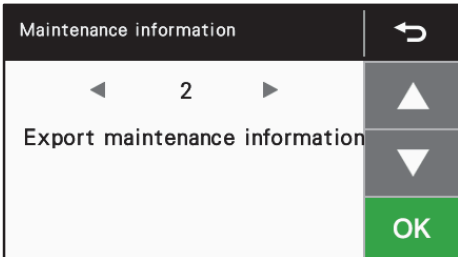
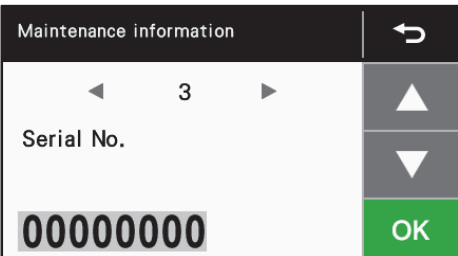
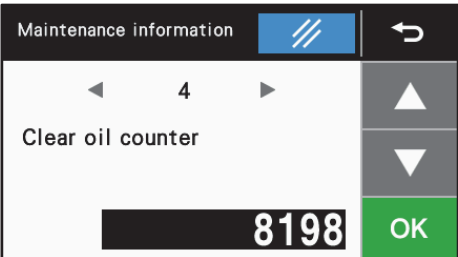
1	<Switching to software version checking mode>  +   1. While pressing the home key, turn on the power to switch to the special menu screen. 2. At the special menu screen, select "Software version" to switch to the software version checking screen.
2	<Software version checking mode>    The display switches between the application version screen and the boot program version screen each time you press the change key . - At the application version screen, the main control program version, the panel control program version, the motor control program version and the FPGA version are displayed. - At the boot program version screen, the main boot program version, the panel boot program (1) version, the panel boot program (2) version and the motor boot program version are displayed.
3	<Exiting software version checking mode>  Press the back key  to return to the special menu screen.
4	<Exiting the special menu> At the special menu screen, press the home key  to close the special menu and switch to the home screen.

4-7. Maintenance information list

1	<Switching to maintenance information mode and maintenance information checking method> 1. While pressing the home key, turn on the power to switch to the special menu screen.  +  2. When "Maintenance information" is selected in the special menu screen, the display switches to the maintenance information screen.  3. The maintenance screen number (2), maintenance information (3) and the corresponding value (4) will be displayed.  4. Press the [◀] or [▶] key (5) to select the screen (0 to 4) that you would like to check.  Press the back key  (6) to return to the special menu screen.
2	<To exit from the special menu> • At the special menu screen, press the home key  to close the special menu and switch to the home screen.

4. FUNCTION SETTINGS

<List of maintenance information>

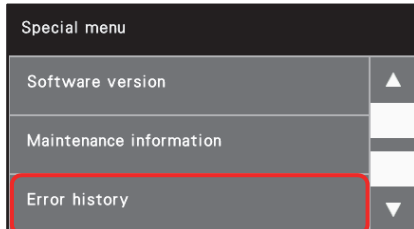
Screen No.	Panel display	Display details
0		• Lifetime stitch counter This shows the total number of stitches sewn since the sewing machine was shipped from the factory.
1		• Write all data When the OK key  is touched, the following machine information is written to the storage media. • Sewing data for additional programs • Sewing parameters for all sewing programs • Data for all memory switches
2		• Export maintenance information When the OK key  is touched, the following machine information is written to the storage media. • Data for all memory switches • Error log
3		• Serial No. The unique 8-digit number which is assigned to the sewing machine is displayed.
4		• Clear oil counter The total number of stitches sewn since the lubricating oil was replaced and the actual display value can be cleared. * The oil counter value can be cleared by touching the RESET key  and then touching the OK key .

4.8. Checking the error history

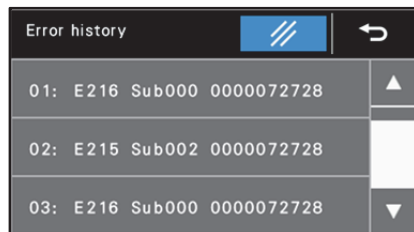
1

<Switching to error history mode and error history checking method>

1. While pressing the home key, turn on the power to switch to the special menu screen.



2. When "Error history" is selected in the special menu screen, the display switches to the error history screen.

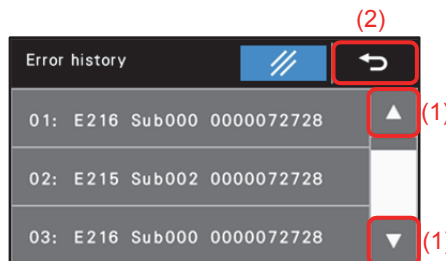


3. Error items are displayed on the screen three at a time. The error items show the history number, error code, sub-code and the lifetime stitch counter (10 digits) at the time the error occurred, in order starting from the left.

NOTE:

The following errors which can be reset (without needing to turn off the power) are not entered into the error log.

E010, E011, E012, E020, E025, E035, E110, E420, E422, E424, E425, E428, E474, E500, E502, E510, E511, E512, E600

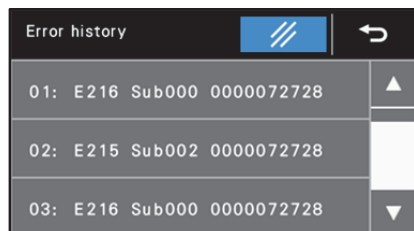


4. Press the [▲] or [▼] key (1) to change the screen one page at a time. Up to 99 error items (01 - 99) are stored in order starting from the newest. The error with history number "01" is the newest in the history.

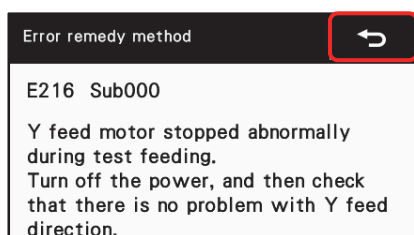
Press the back key (2) to return to the special menu screen.

2

<When checking the error remedy>



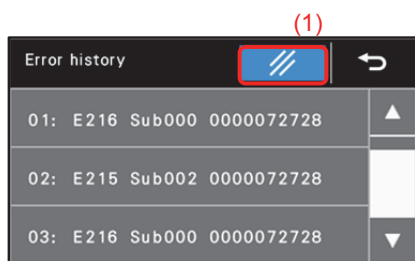
1. If you press an error item, the display will switch to the error handling screen for the selected item.



2. The remedy for the error code is displayed.

Press the back key (2) to return to the error history screen.

3 <Clearing the error codes>



1. Touch the RESET key  (1) to switch to the confirmation screen for clearing the error history.



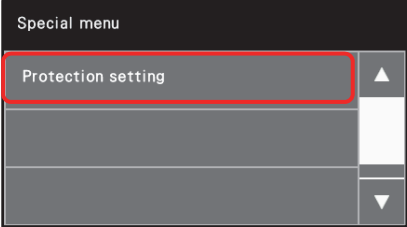
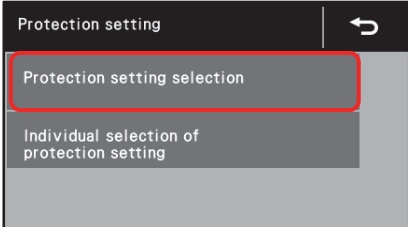
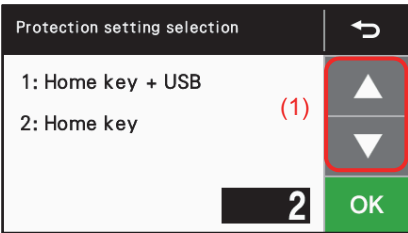
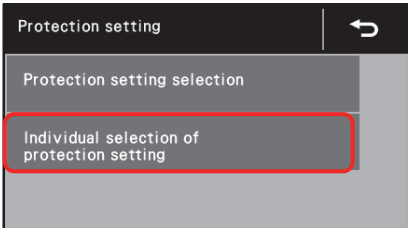
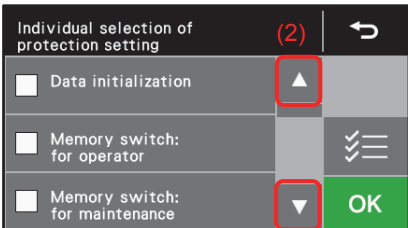
2. Press the OK key  (2) to clear the whole of the error history. Once clearing is finished, the buzzer will sound for 1 second, and all error items will be displayed as "[E---] [-----] [---]". Press the back key (3) to return to the error history screen.

If you press the back key  (3) without pressing the OK key, the error history will not be cleared.

4 <To exit from the special menu>

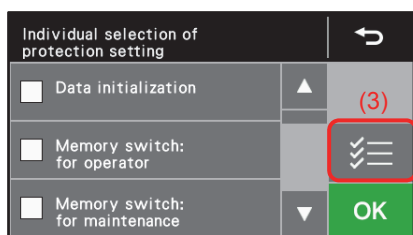
- At the special menu screen, press the home key  to close the special menu and switch to the home screen.

4.9. Making protection settings

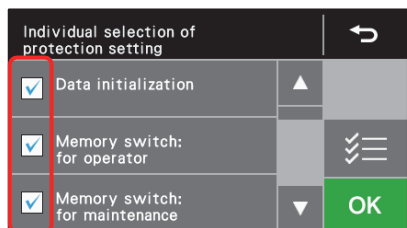
1	<Protection settings>  1. While pressing the home key, turn on the power to switch to the special menu screen.  2. At the special menu screen, select "Protection setting". * If startup method 1 has been set, this option cannot be selected unless USB memory has been inserted. (The initial setting is "2".)
2	<Protection setting selection> You can change the protection setting startup method. 1. At the protection setting screen, select "Protection setting selection".   2. Press the [▲] or [▼] key (1) to select the desired setting. 1: Protection settings cannot be carried out unless USB memory is inserted. 2: Protection settings can be carried out at any time. 3. Press the OK key  to confirm the setting.
3	<Individual selection of protection setting> Protection settings can be carried out individually for each function*. * For details, refer to "List of functions that can be protected". 1. At the protection setting screen, select "Individual selection of protection setting".   2. Press the [▲] or [▼] key (2) to select the items to be protected. 3. Touch an item, and then select whether to enable or disable protection for that item.  : Protection setting is enabled (changing settings is disallowed)  : Protection setting is disabled (changing settings is allowed) 4. Press the OK key  to confirm the protection setting.

4 <Selecting all items to disallow changing them>

You can turn protection on for all functions by following the operation below.



1. At the Individual selection of protection setting screen, select the ALL key  (3).



2. All items will have a check mark  alongside them.

* You can clear all of the check marks by pressing the ALL key  again.

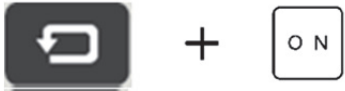
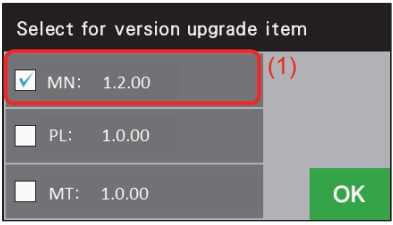
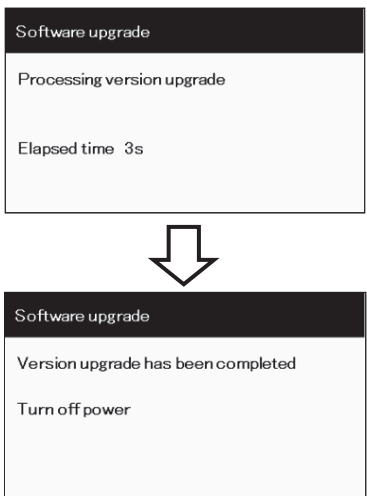
3. Press the OK key  to confirm the setting.

<List of functions that can be protected>

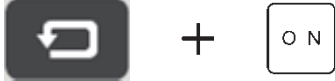
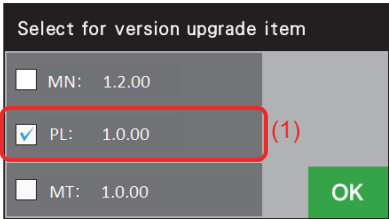
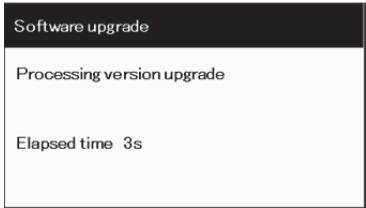
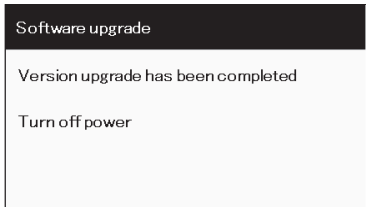
Item name	Operation when protection setting is enabled
Data initialization	Data initialization operations in the adjustment menu are disallowed.
Memory switches : for operator	Memory switch operations in the setting menu are disallowed.
Memory switches : for technicians	Memory switch operations in the special menu are disallowed.
Reference/Home position adjustment	Main shaft motor standard position adjustment, home position adjustment and thread nipper position adjustment operations in the adjustment menu are disallowed.
Change sewing program No.	Sewing program number change operations in the home screen are disallowed.
Change X / Y scale	X direction and Y direction zoom ratio change operations in the home screen are disallowed.
Change sewing speed	Sewing speed change operations in the home screen are disallowed.
Change work clamp height	Work clamp height change operations in the home screen are disallowed.
Change tension	Tension change operations in the home screen are disallowed.
Change section	Pattern editing operations in the setting menu are disallowed.
Slow start setting	Slow start setting operations in the setting menu are disallowed.
Parallel movement setting	Offset setting operations in the setting menu are disallowed.
Lower thread counter setting	Lower thread counter settings in the setting menu and home screen are disallowed.
Production counter setting	Production counter settings in the setting menu and home screen are disallowed.
Garment counter setting	Garment counter settings in the setting menu and home screen are disallowed.
User program setting	Creating and editing user programs in the setting menu are disallowed.
Cycle program setting	Creating and editing cycle programs in the setting menu are disallowed.
Copy program	Program copy operations in the setting menu are disallowed.
Read/Write data	Data reading and writing operations in the setting menu and maintenance data writing operations in the special menu are disallowed.
Restart sewing	Restarting sewing from an interrupted position is disallowed.

4-10. Updating the software version

When updating the main control program version

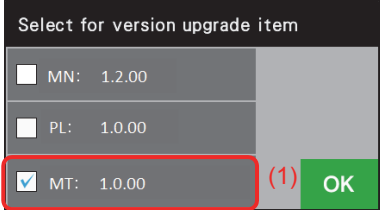
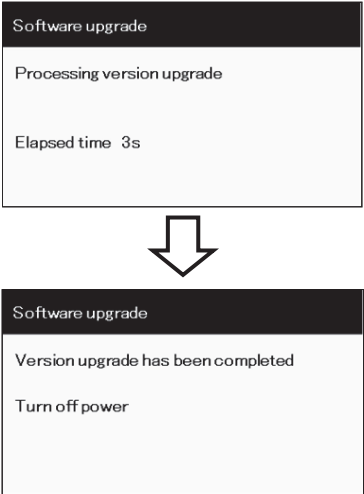
1	<Preparation for updating the program version> 1. The main control program (ISM22MN.BVP) can be found in the following folder in the USB memory. \BROTHER\ISM\ISMSYS\ 2. Connect the USB memory to the USB port.
2	<Starting software version update mode> 1. While pressing the TEST key, turn on the power to switch to the version update selection screen. 
3	<Version update selection mode> 1. Check that the version number (1) for the main control program that you are going to update is correct. 2. Touch the main control program item (1) in the list, and then touch the OK key  to start the version update. 
4	<While version updates are running> 1. While version updating is in progress, the progress status will be displayed. * Do not turn off the power while version updating is in progress. 2. Once version updating is finished, turn off the power. 

When updating the panel control program version

1	<Preparation for updating the program version> 1. The panel control program (ISM22PL.upf) can be found in the following folder in the USB memory. \BROTHER\ISM\ISMSYS\ 2. Connect the USB memory to the USB port.
2	<Starting software version update mode> 1. While pressing the TEST key, turn on the power to switch to the version update selection screen. 
3	<Version update selection mode> 1. Check that the version number (1) for the panel control program that you are going to update is correct. 2. Touch the panel control program item (1) in the list, and then touch the OK key  to start the version update. 
4	<While version updates are running> 1. While version updating is in progress, the progress status will be displayed. * Do not turn off the power while version updating is in progress. 2. Once version updating is finished, turn off the power.   

4. FUNCTION SETTINGS

When updating the motor control program version

1	<Preparation for updating the program version> 1. The motor control program (ISM22MT.BVP) can be found in the following folder in the USB memory. \BROTHER\ISM\ISMSYS\ 2. Connect the USB memory to the USB port.
2	<Starting software version update mode>  1. While pressing the TEST key, turn on the power to switch to the version update selection screen.
3	<Version update selection mode>  1. Check that the version number (1) for the motor control program that you are going to update is correct. 2. Touch the motor control program item (1) in the list, and then touch the OK key  to start the version update.
4	<While version updates are running>  1. While version updating is in progress, the progress status will be displayed. * Do not turn off the power while version updating is in progress. 2. Once version updating is finished, turn off the power.

5. SETTING FUNCTION

5-1. Edit pattern

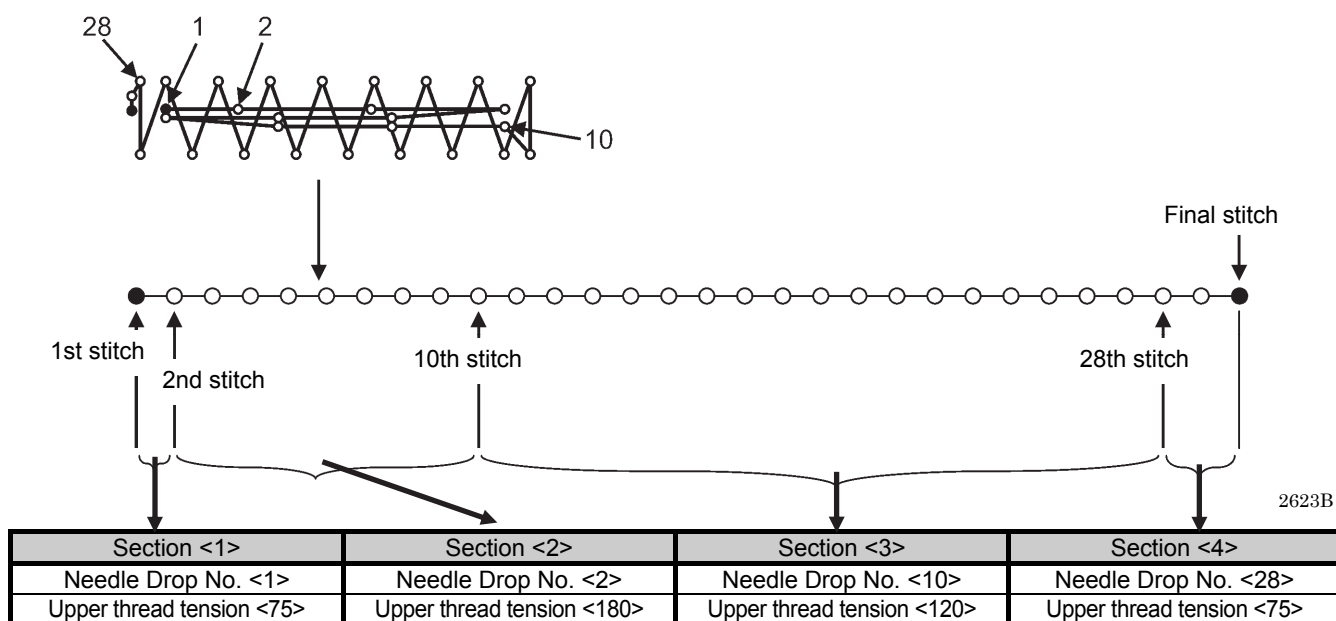
Tension codes are recorded in the pattern data to provide tension information.

The tension codes indicate the type of tension.

In addition, the units which tension codes are divided into are called sections, and sequential numbers are assigned to them starting from the first stitch. Up to a maximum of 99 sections are possible.

* The tension values for the sections are set in program mode for each program.

Example: Additional program No. A002 (stitch count 30)



Example: Operation when pattern editing for additional program No. A002 has been set as follows for sewing

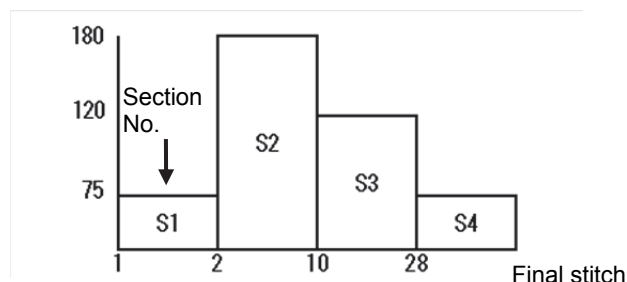
Example: Pattern editing settings

Pattern No. 200		
Needle Drop No.	Section No.	Tension value
1st stitch	1	75
2nd stitch (to 9th stitch)	2	180
10th stitch (to 27th stitch)	3	120
28th stitch (to final stitch)	4	75

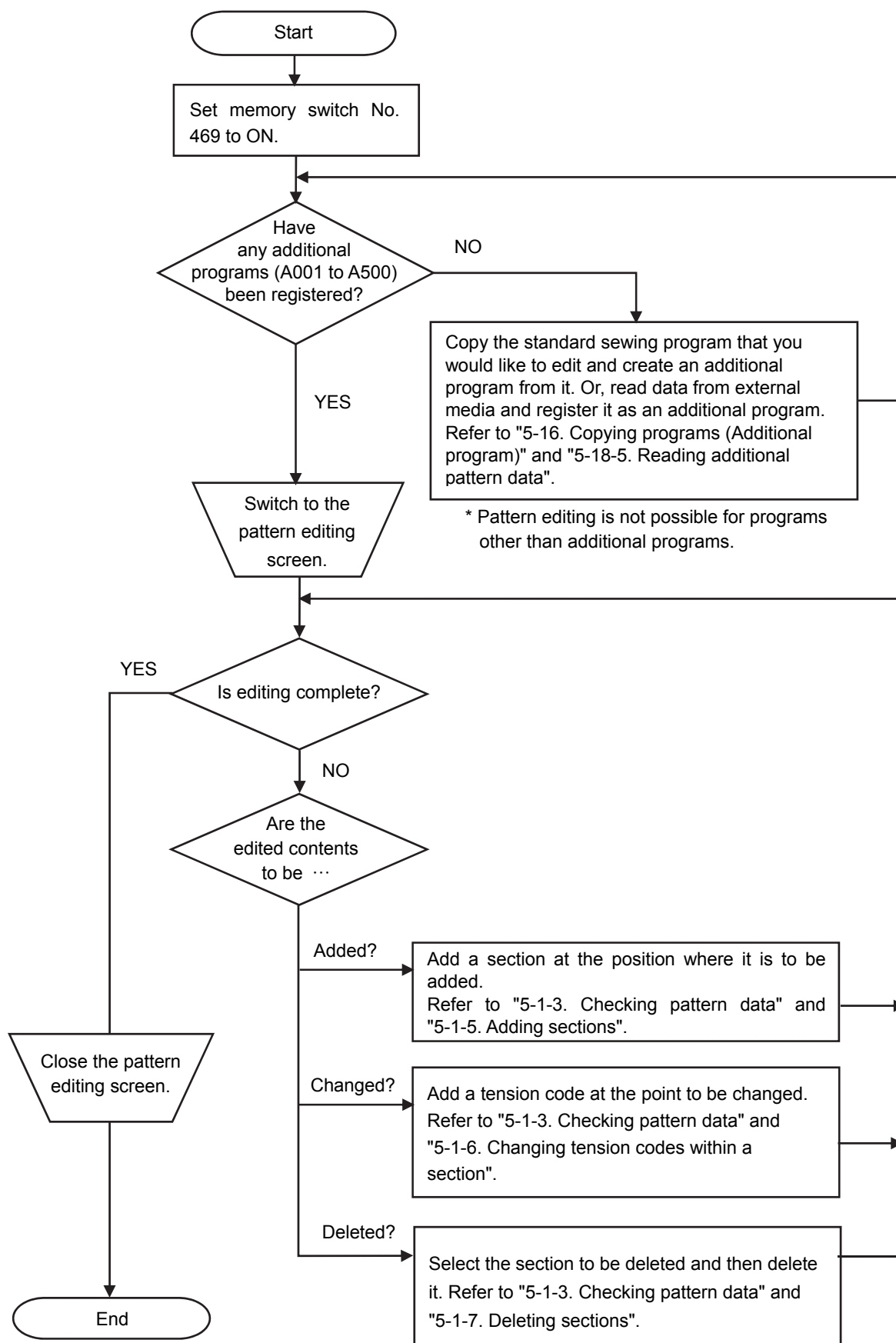
Tension code diagram

Vertical axis: Tension value

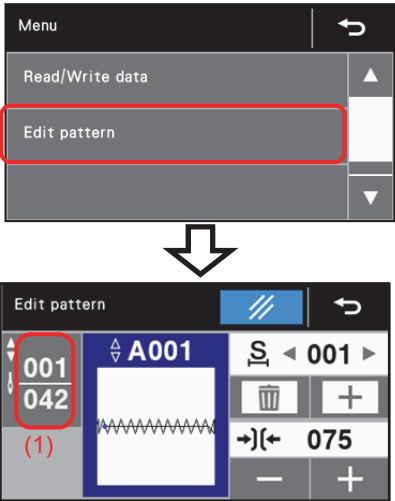
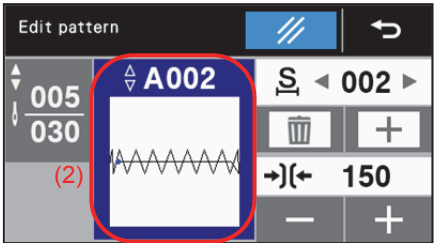
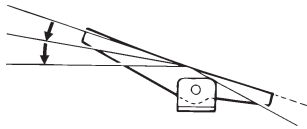
Horizontal axis: Number of stitches



5-1-1. Work flowchart



5-1-2. Switching to pattern editing mode

1		Press the menu key  to switch to the setting menu. Press "Edit pattern" in the setting menu to switch to the edit pattern screen. • The number of stitches until the final stitch (1), the section number and the tension value for the current section are displayed on the screen. * In order to switch to the the edit pattern screen, it is necessary to first create at least one additional program, and to set memory switch No. 469 to "ON". * Switching to the edit pattern screen is only possible with the 430HX/438HX.
2		Touch the program No. (the pattern image can also be touched) (2) to switch to the program selection screen, or press the program number change key  or  to select the additional program number to be added. * Only the additional program numbers which are currently registered will be displayed.
3	2nd step  4441Q	Depress the foot switch to the 2nd step. • Home position is detected, and then the sewing machine moves to the sewing start position. From this point on, the condition is referred to as pattern editing mode.

5-1-3. Checking pattern data

You can check the tension information (section numbers and tension codes) in pattern data.

1	Carry out steps 1 to 3 in "5-1-2. Switching to pattern editing mode" to switch to pattern editing mode. Press the needle movement change key [▲] or [▼] to move the work clamp/button clamp by one stitch. The section number, tension code and number of stitches are displayed at the current position. You can press the section change key [◀] or [▶] (1) on the section number side to move the work clamp/button clamp automatically to the start of the section. ◀ : Moves in the forward direction ▶ : Moves in the reverse direction
2	Exit pattern editing mode Touch the back key  or the menu key . The display will return to the setting menu screen.

5-1-4. Copying sewing data

Editing (adding, changing or deleting) of the tension codes within pattern data is limited to additional programs.

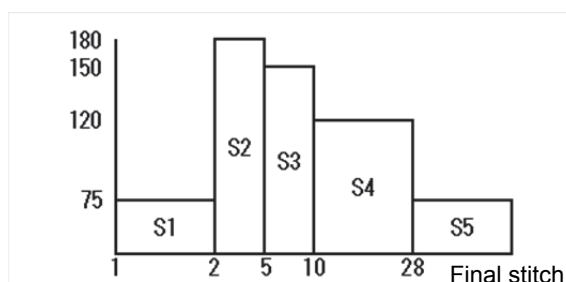
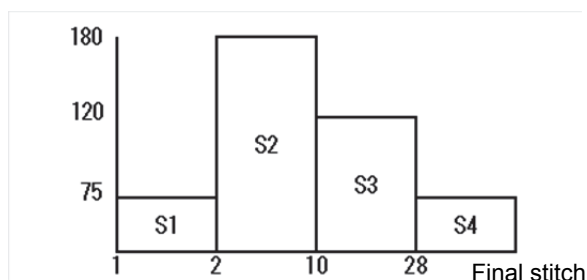
Because of this, if you would like to edit a standard program which has already been registered, you will need to copy the program to an additional program.

For details on the program copying method, refer to "5-16. Copying programs (additional programs)".

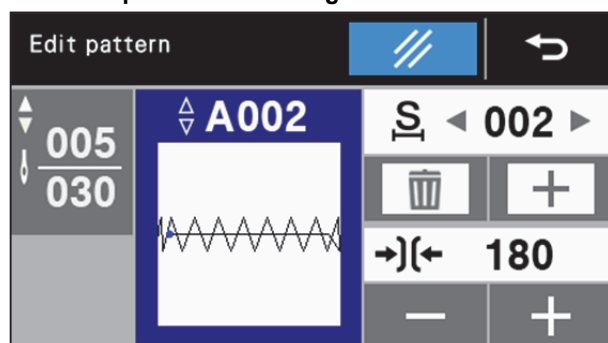
- * You can use the initialization or additional program deletion functions to delete any sewing program data which has been added to that program by copying. Refer to "9-23. Data initialization method" and "5-7. Deleting additional programs".

5-1-5. Adding sections

<Example: If adding a new section at the 5th stitch in section 2>



1 Select the position for adding the section.

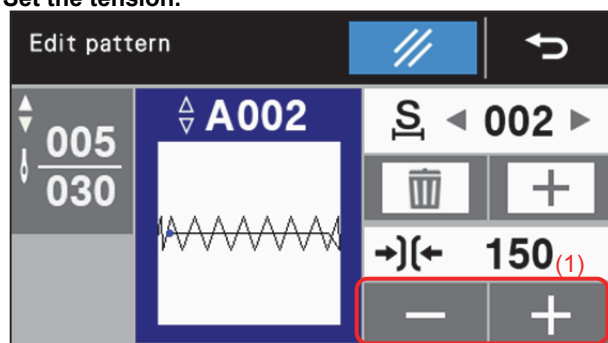


Carry out steps 1 to 3 in "5-1-2. Switching to pattern editing mode" to switch to pattern editing mode.

Press the needle movement change key or to move the work clamp/button clamp to the desired position.

* To add a section, move to anywhere other than the start of the section.

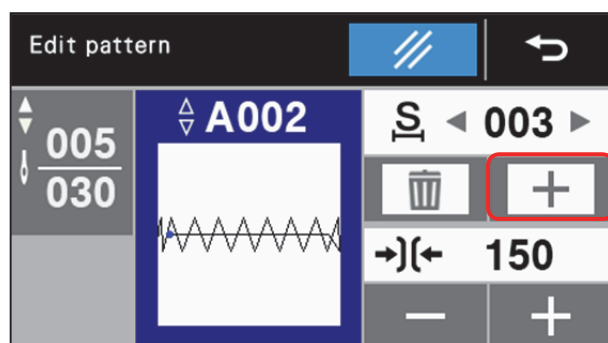
2 Set the tension.



Example: Set the tension to "150".

Press the tension change keys (1) to select the tension (0 to 300) to be used in the newly-added section.

3 Add the section.



To confirm the addition, press the add section key (2).

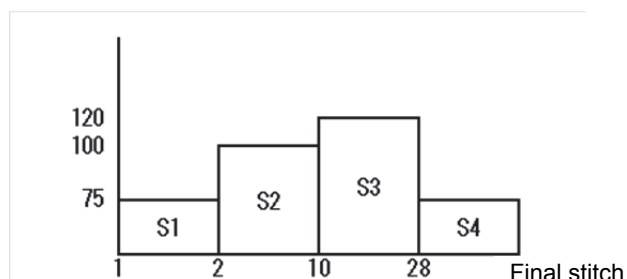
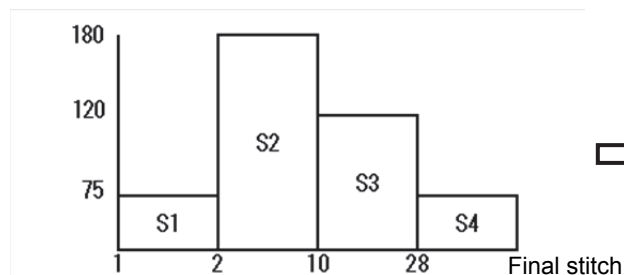
• A new section with the selected tension will be added from the current needle position. The section numbers for all subsequent sections will be incremented by 1 as a result.

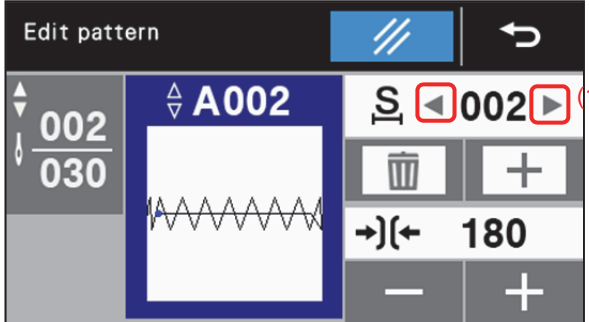
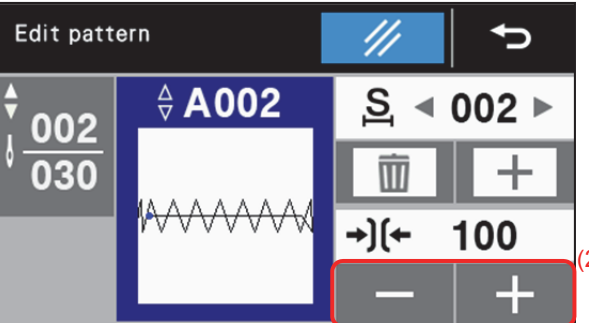
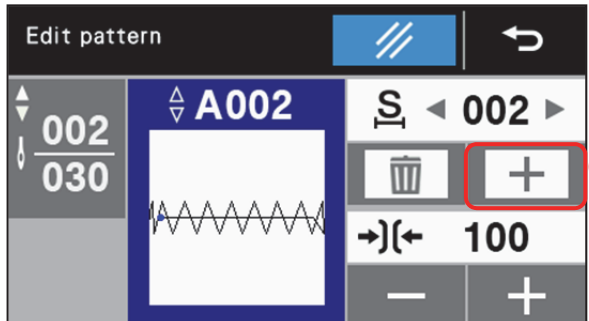
If you press the needle movement change key or before pressing the add section key, the adding of the new section will be canceled, and the displayed tension value will return to the value which was displayed before it was changed.

If you would like to end pattern editing without adding any sections, press the back key , the menu key or the home key .

5-1-6. Changing tension codes within a section

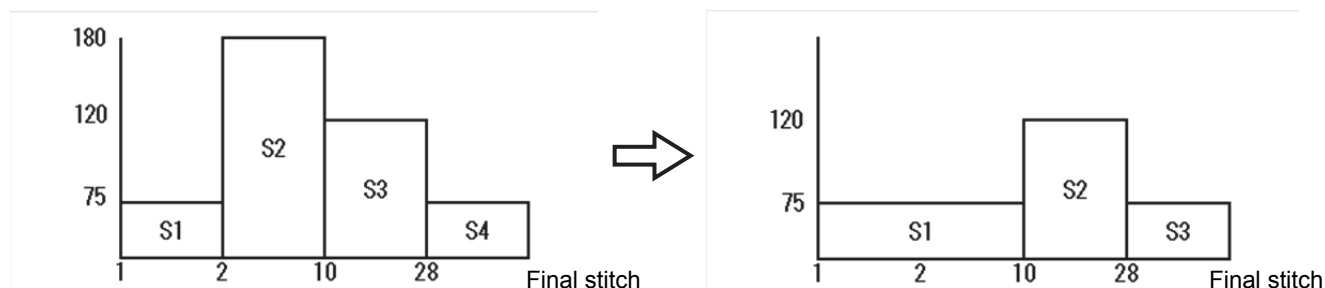
<Example: To change the tension of section 2>



1	Select the section to change the tension for.  Carry out steps 1 to 3 in "5-1-2. Switching to pattern editing mode" to switch to pattern editing mode. You can press the section change key [◀] or [▶] (1) on the section number side to move the work clamp/button clamp automatically to the start of the section. Move the work clamp/button clamp to the start of the section you would like to change.
2	Change the tension.  Press the tension change keys [−] [−] (2) to select the desired tension (0 to 300). Example: Change the tension code to "100" in section 2
3	Confirm the change.  To confirm the change, press the key [✓] (3). The change will be applied, and the tension for the currently-selected section will change to the newly-set tension value. If you press the needle movement change key [▲] or [▼] before pressing the [✓] key, the tension change will be canceled. If you would like to end pattern editing without changing any sections, press the back key [↩], the menu key [≡] or the home key [🏠].

5-1-7. Deleting sections

< Example: Deleting section 2 >

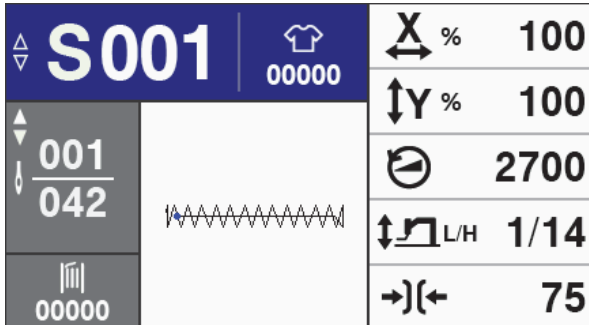
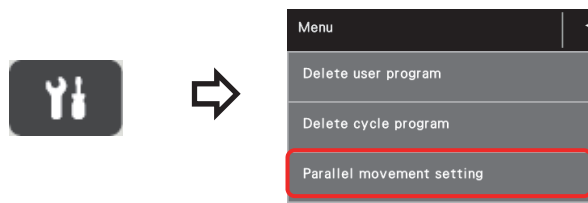
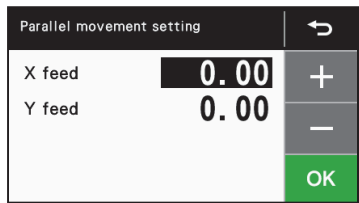
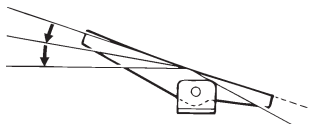
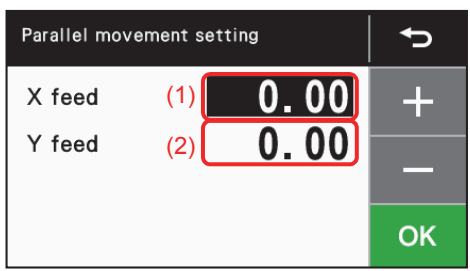


1	Select the section to be deleted.	Carry out steps 1 to 3 in “5-1-2. Switching to pattern editing mode” to switch to pattern editing mode. You can press the section change key [◀] or [▶] (1) on the section number side to move the work clamp/button clamp automatically to the start of the section. Move the work clamp/button clamp to the start of the section you would like to delete.
2	Delete the section.	To delete the section, press the delete section key (2). - The selected section will be deleted. As well as the current section being deleted, the section numbers for all subsequent sections will be decremented by 1 as a result. - The start section cannot be deleted. - All sections after the deleted section will move forward by one section, and the tension values will become the same.
3	To delete all sections	You can delete all sections by pressing the reset key (3). - Only the settings for section 1 will remain, and all section settings will be deleted. - Whether deleting sections individually or all at once, the tension setting for section 1 at the beginning of the program alone cannot be deleted. If you would like to end pattern editing without deleting any sections, making any changes, press the back key , the menu key or the home key .

5-2. X and Y parallel movement of sewing patterns

The work clamp/button clamp can be moved to any desired position so that sewing patterns which have already been programmed can be moved in parallel directions forward, back or to the left or right.

Use this when you would like to correct the sewing position with respect to the work clamp/button clamp.

1	 For example, program number 1	Touch the program number change key  or a program number to switch to the program change screen, and then select the program number in that screen. * If a cycle program is selected, the following operation cannot be carried out. Select a sewing program other than a cycle program.
2	The display will switch to the parallel movement setting screen. 	 1. Press the menu key to switch from the home screen to the setting menu. 2. Touch "Parallel movement setting" in the menu to switch to the parallel movement setting screen. - The amount of parallel movement in which has currently been set the X and Y directions will be displayed in the parallel movement setting screen in mm units.
3	2nd step  4441Q	Depress the foot switch to the 2nd step. - Home position detection will be carried out. - The settings can be changed.
4	Switch to X or Y direction. 	You can touch the feed amount setting value for the direction you would like to set to switch to the respective setting. * You can switch between the "X feed" (1) and "Y feed" (2) setting values as often as desired.
5	Set the feed in the X direction  For example, set to 3.00.	Each time the  or  key is pressed, the work clamp/button clamp will move by 0.05 mm. * In X-scale parallel movement mode, the work clamp/button clamp will move to the left when the  key is pressed, and to the right when the  key is pressed. * If the sewing pattern goes outside the sewing area as a result of parallel movement, sewing will not be possible.

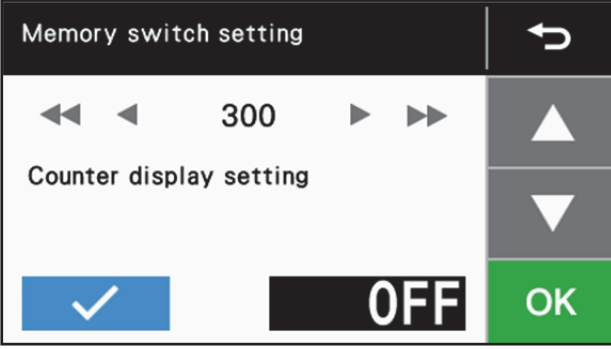
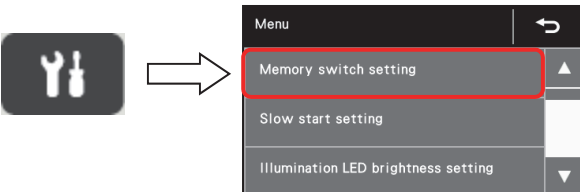
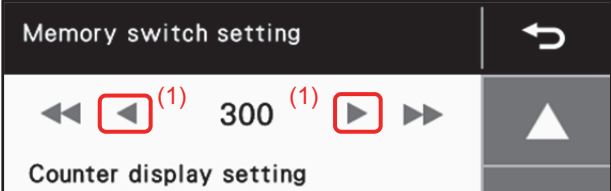
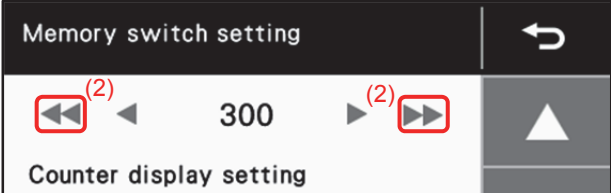
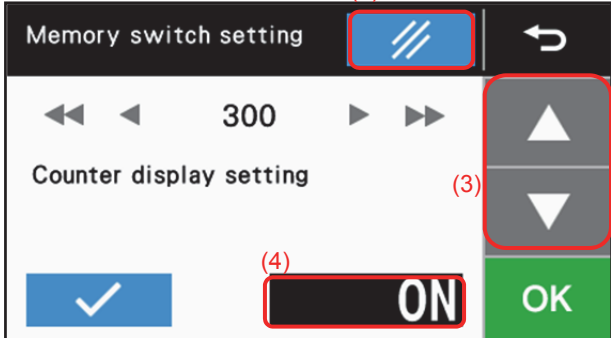
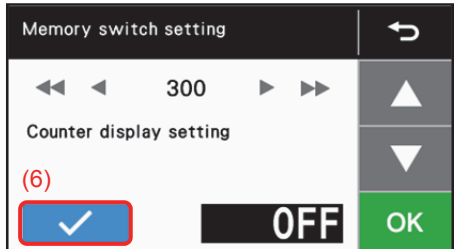
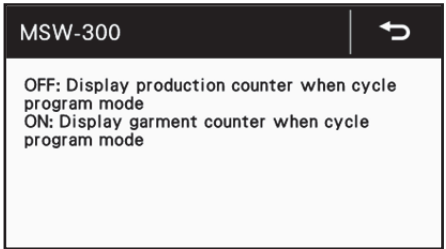
6	Set the feed in the Y direction For example, set to -3.00. Each time the or key is pressed, the work clamp/button clamp will move by 0.05 mm. * In Y-scale parallel movement mode, the work clamp/button clamp will move forward when the key is pressed, and backward when the key is pressed. * If the sewing pattern goes outside the sewing area as a result of parallel movement, sewing will not be possible.
7	Confirm the feed amount and exit the parallel movement settings. Touch the OK key . • The parallel movement amounts in the X direction and Y direction will be confirmed, and the display will return to the setting menu screen. • If you touch the back key , the menu key or the home key without touching the OK key, you can cancel the changes without saving them.

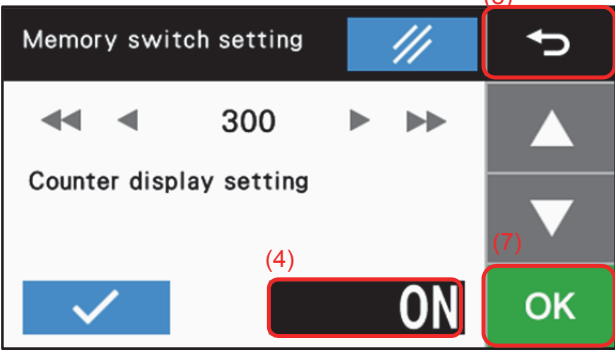
NOTE:

- If you change the program number, the X-scale or the Y-scale, the movement amount which has been stored will be reset.

When the power switch is turned off, the movement amount which has been stored will be reset. However, if memory switch No. 465 is set to "2", the movement amount will remain stored and will not be reset.

5-3. Memory switch setting method (for operators)

1	Change the mode to the memory switch setting screen. 	Touch the menu key to switch to the setting menu screen. Touch "Memory switch setting" in the menu.  The display will switch to the memory switch setting screen.
2	Select the memory switch that you would like to change the setting for.  If you would like to display only the numbers of memory switches that have been changed from default settings 	Touch the [◀] or [▶] key (1) to select the memory switch number. Touch the [◀◀] or [▶▶] key (2). The numbers of memory switches that have been changed from default settings will appear in order.
3	Change the memory switch parameters.  When checking detailed explanations of memory switches You can touch the check key (6) to check the detailed explanation of the memory switch which is currently selected. 	Touch the key ▲ or the key ▼ (3) to change the setting value (4). - If the setting value has not been confirmed, the display will blink. - You can make the initial setting appear in the display by touching the reset key (5). 

4	Apply the changed parameter setting.  Touch the OK key  (7). • The setting value (4) will stop blinking and illuminate, and this means that the parameter setting is confirmed. • If you change the memory switch number or touch the menu key  or the back key  (8) to switch the screen without touching the OK key, you can cancel the changes.
5	Repeat steps 2 to 4 above to set each memory switch.
6	Exit setting mode.  Touch the back key  or the menu key . • The display will return to the setting menu screen.

<List of memory switches (for operators)>

In the memory switch setting mode which can be checked from the adjustment menu (refer to "4-2. Memory switch setting method (for technicians)"), the memory switches with the bottom two digits from 50 to 99 will also be displayed.

No.	Setting value	Setting detail	Initial value
001	Work clamp / button clamp lift timing after sewing is completed		1
	OFF	Does not lift automatically.	
	1	Lifts at the final stitch position.	
	2	Lifts after moving to the home position.	
100	Sewing speed at the sewing start		ON
	OFF	The sewing speed for the first 1 to 5 stitches is set by memory switch Nos. 151 to 155. (Refer to "4-3. List of memory switches (for technicians)" for details on memory switch Nos. 151 to 155.)	
	ON	Slow start patterns can be selected from the slow start settings in the setting menu screen.	
200	Single-stitch test feed		430H: OFF
	OFF	When the foot switch is depressed to the 2nd step (start switch), test feed moves automatically to the final stitch, and when the foot switch is depressed to the 1st step (work clamp switch), quick feeding is carried out.	438H: ON
	ON	The test feed moves 1 stitch at a time when the foot switch is depressed to the 1st step (work clamp switch). In addition, test feed advances one stitch at a time when the machine pulley is turned by hand.	
300	Production counter display		OFF
	OFF	The production counter is displayed when using cycle programs.	
	ON	The garment counter is displayed when using cycle programs.	
402	X scale/Y scale unit (mm/%) (* The mm display may differ slightly from the actual sewing size.)		OFF
	OFF	Displayed as %.	
	ON	Displayed as mm.	
406	Switching program numbers by F keys		OFF
	OFF	The program number switching function is disabled.	
	ON	When the F1 to F4 keys are pressed, operation switches to the program number which has been pre-assigned to the respective key. Assignment to F1 key: memory switch No. 407 Assignment to F2 key: memory switch No. 408 Assignment to F3 key: memory switch No. 409 Assignment to F4 key: memory switch No. 410	

5. SETTING FUNCTION

No.	Setting value	Setting detail	Initial value
407	Assignment to F1 key		OFF
	OFF 1-100: S001-S089 1001-1500: A001-A500 2001-2100: U001-U100 3001-3030: C001-C030	If memory switch No. 406 is set to "ON", operation switches to the program number which has been set.	
408	Assignment to F2 key		OFF
	OFF 1-100: S001-S089 1001-1500: A001-A500 2001-2100: U001-U100 3001-3030: C001-C030	If memory switch No. 406 is set to "ON", operation switches to the program number which has been set.	
409	Assignment to F3 key		OFF
	OFF 1-100: S001-S089 1001-1500: A001-A500 2001-2100: U001-U100 3001-3030: C001-C030	If memory switch No. 406 is set to "ON", operation switches to the program number which has been set.	
410	Assignment to F4 key		OFF
	OFF 1-100: S001-S089 1001-1500: A001-A500 2001-2100: U001-U100 3001-3030: C001-C030	If memory switch No. 406 is set to "ON", operation switches to the program number which has been set.	
411	Assignment to F5 key		OFF
	OFF	No assignment	
	1	Moves to the home position	
500	Thread nipper device		OFF
	OFF	Disabled	
	ON	Thread nipper device can be used. (*1)	

*1 The setting is enabled when memory switch No. 100 is set to "ON", the slow start pattern is set to 8 or 9 and when the sewing speed value is set to 1,500 sti/min or more.

5-4. Slow start pattern setting method

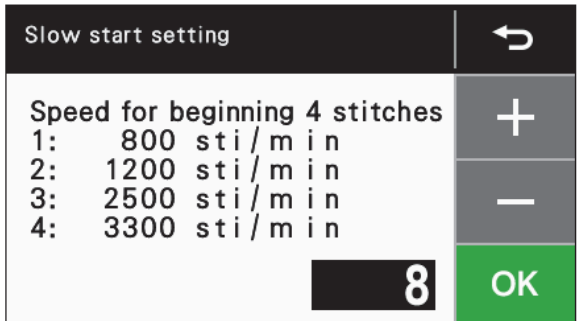
The starting-up speed at the sewing start can be adjusted by setting a slow start pattern.

* When memory switch No. 100 is OFF, slow start cannot be set.

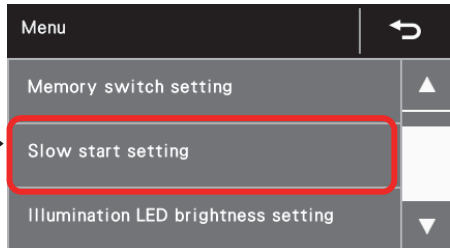
* The smaller the number, the slower the start.

* This is used to stop the thread from pulling out at the sewing start, and at times when skipped stitches might easily occur.

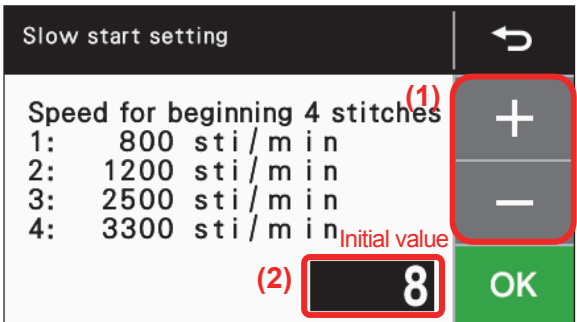
1 Switch to the slow start setting screen.



Touch the menu key to switch to the setting menu screen. Touch "Slow start setting" in the menu to switch to the slow start setting screen.



2



You can touch the or key (1) to change the slow start pattern (2).
(Initial value: 430HX/HS: 8, 438HX/HS: 7)

When the setting is complete, touch the OK key on the screen.

- Touch the OK key to switch to the setting menu screen, and the slow start setting value which is set will be applied from the next time sewing is carried out.

Slow start pattern

1. Slow start is set as shown in the following table.
2. You can change the starting-up speed by changing the slow start pattern.

430HX/HS: Medium-weight materials (-03), knitted wear (-0K), light-weight materials (-01) (sti/min)

	1	2	3	4	5	6	7	8	9
Sewing speed for the 1st stitch	200	200	300	400	400	400	400	800	1,500
Sewing speed for the 2nd stitch	200	300	400	400	500	600	800	1,200	3,000
Sewing speed for the 3rd stitch	300	400	500	600	800	800	1,200	2,500	3,300
Sewing speed for the 4th stitch	500	600	700	900	1,200	1,200	2,500	3,300	3,300

430HX/HS: Heavy-weight materials (-05) (sti/min)

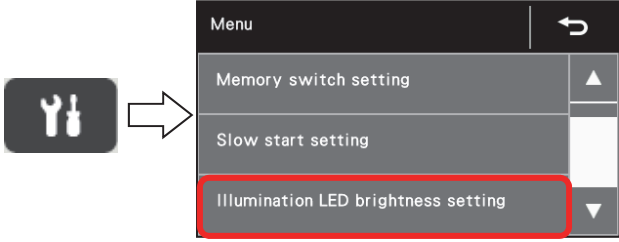
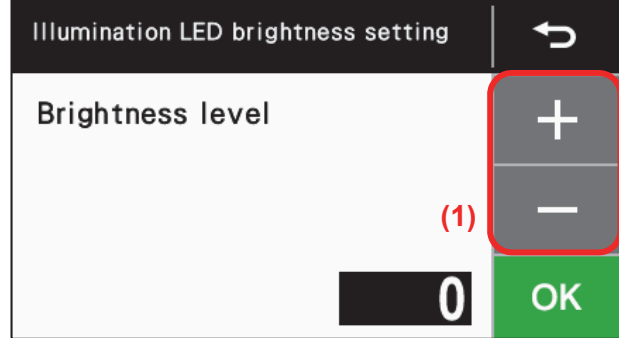
	1	2	3	4	5	6	7	8	9
Sewing speed for the 1st stitch	200	200	300	400	400	400	800	1,500	1,500
Sewing speed for the 2nd stitch	200	300	400	400	600	800	1,200	2,000	3,000
Sewing speed for the 3rd stitch	300	400	500	600	800	1,200	2,500	2,500	3,300
Sewing speed for the 4th stitch	500	600	700	900	1,200	2,500	3,300	3,300	3,300

438HX/HS (sti/min)

	1	2	3	4	5	6	7	8	9
Sewing speed for the 1st stitch	200	200	300	300	300	400	400	400	400
Sewing speed for the 2nd stitch	200	200	300	300	400	400	400	600	900
Sewing speed for the 3rd stitch	300	300	300	400	400	400	600	900	1,500
Sewing speed for the 4th stitch	300	400	400	400	400	400	900	2,000	2,000

* The speed will not be faster than the sewing speed which has been set.
* The thread nipper device will not operate for settings other than 8 and 9.

5-5. Illumination LED brightness setting method

1	Switch to the illumination LED brightness setting screen. 	Touch the menu key to switch to the setting menu screen. Touch "Illumination LED brightness setting" in the menu to switch to the illumination LED brightness setting screen.
2		You can touch the + or - key (1) to change the illumination LED brightness (2). When the setting is complete, touch the OK key OK on the screen. • When you touch the OK key to switch to the setting menu screen, the LED will illuminate at the set brightness. • If you would like to cancel the setting, press the back key , the menu key  or the home key  to switch the screen without applying the changes. • The brightness level can be set to a value from "0" to "9". • If the brightness level is set to "0", the LED will switch off, and if it is set to a number from 1 to 9, the LED brightness will be greater if a larger number is selected as the setting.

5-6. Setting additional programs

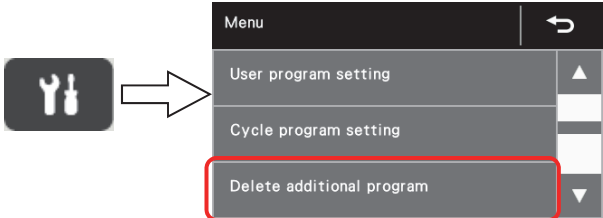
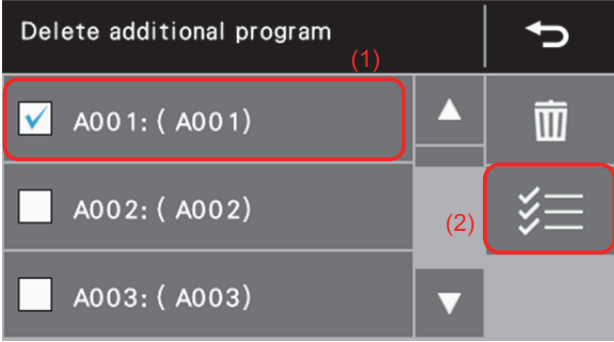
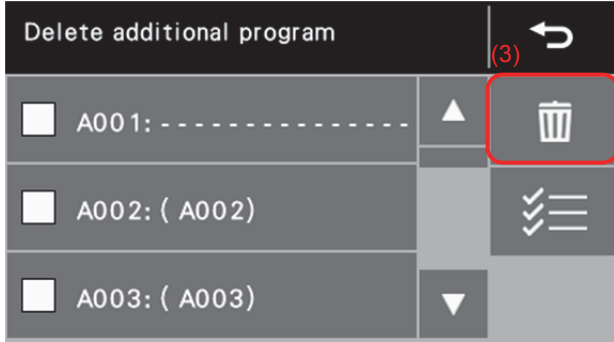
Sewing data which has been created externally can be loaded from USB memory and registered as additional programs. In addition, standard programs can also be registered as additional programs using the program copy function.

With additional programs, you can edit the tension codes (adding, changing and deleting), and also register them as user programs.

- * For details on loading additional programs from USB memory, refer to "5-18-5. Reading additional pattern data".
- * For details on the program copying method, refer to "5-16. Copying programs (additional programs)".
- * For details on editing tension codes, refer to "5-1. Editing patterns".

5-7. Deleting additional programs

Additional programs which are not needed can be deleted.

1	Switch to the delete additional program setting screen. 	Press the menu key to switch to the setting menu screen. Touch "Delete additional program" in the setting menu screen to switch to the select delete additional program screen.
2	Select the program number for the program that you would like to delete. Example: program number A001 	Set the program number (1) for the cycle program which you would like to delete from the list on the screen. • Select the number by which the program has been registered. • When you touch a program number, a check mark  will appear against the program number in the list. (Multiple program numbers can be selected.) • You can select all cycle programs in the list (A001 to A500) by pressing the ALL key  (2).
3	Delete the additional programs. 	Once all the additional programs to be deleted have been selected, you can press the delete key  (3) to delete the registered sewing patterns which have been assigned to the additional program numbers which are currently selected. • "----" will be displayed after the program numbers of the additional programs which have been deleted.
4	<To exit from the additional program deleting screen> • At the delete additional program screen, you can press the back key  or the menu key  to return to the setting menu. • In addition, when the home key  is pressed, the display switches to the home screen.	

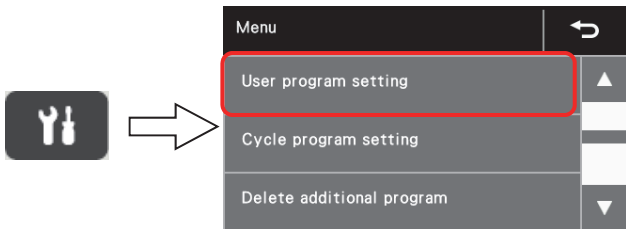
5-8. User program setting method

Patterns, X scale, Y scale, sewing speed and work clamp height can be registered into programs.

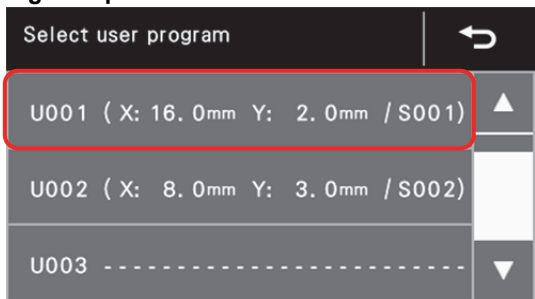
When a program number is selected, the pattern which has been set for that number can then be sewn.

Standard program numbers S001 to S089 (S001 to S064 for the 438HX/HS) have patterns preprogrammed into the program numbers with the same numbers as the pattern numbers, and these pattern numbers cannot be changed. All items in user program numbers U001 to U100 and additional program numbers A001 to A500 can be set as desired by the user.

1 Switch to the user program setting screen.



2 Select a program number that you would like to change the parameter for.



For example, program number U001

Restrictions due to program type selected

	Standard program	User and additional program
Program No.	S001 to S089 (430HX/HS), S001 to S064 (438HX/HS)	U001 to U100 A001 to A500
Pattern selection operation	Not possible	Can be registered as desired
Patterns that can be registered	Patterns with same number as program number already registered	All patterns registered in the sewing machine

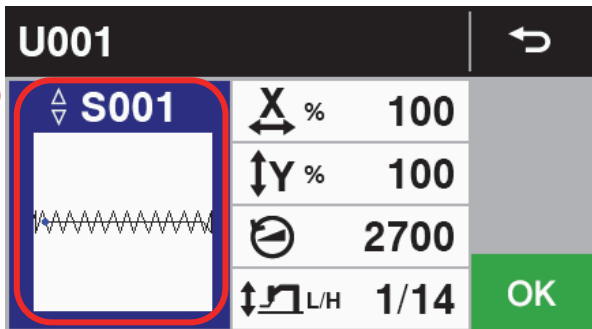
Press the menu key to switch to the setting menu screen. Touch "User program setting" in the menu to switch to the select user program screen.

- If the pattern has been registered in a program, the "Program No." (X: "X scale", Y: "Y scale" / "Base program No." will be displayed in the list; if it has not been registered in a program, "----" will be displayed below the program numbers.

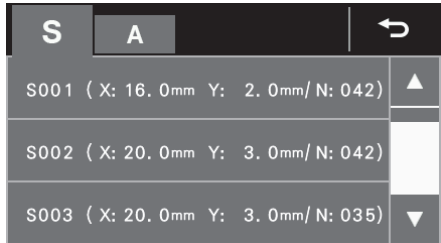
Set the program number that you would like to register from the on-screen list.

- Touch the back key  or the menu key  to return to the setting menu screen.

3 (If a program number from U001 to U100 is selected) Register the pattern.



Press the  or  key to select the program, or touch the sewing program number or the pattern image (1) to switch to the select program screen, and select the program.

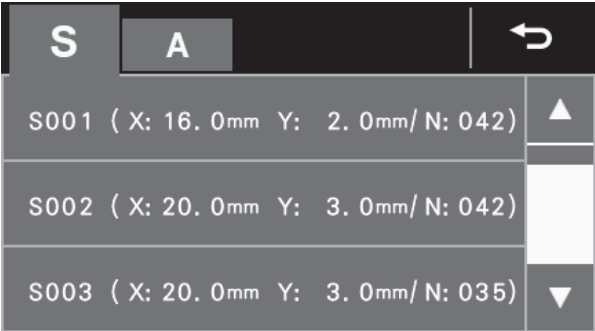
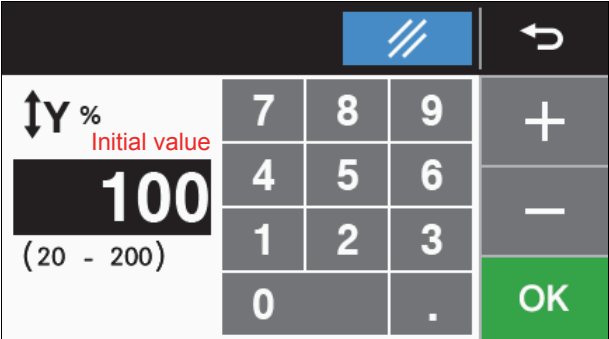
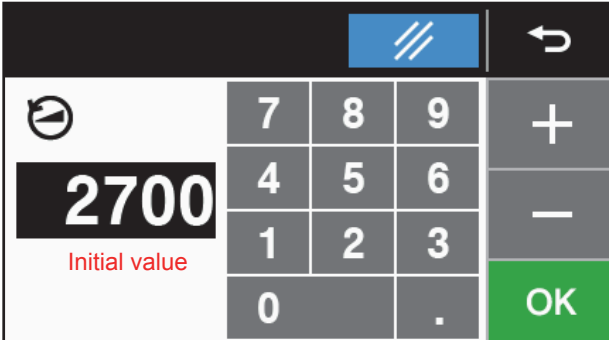

or


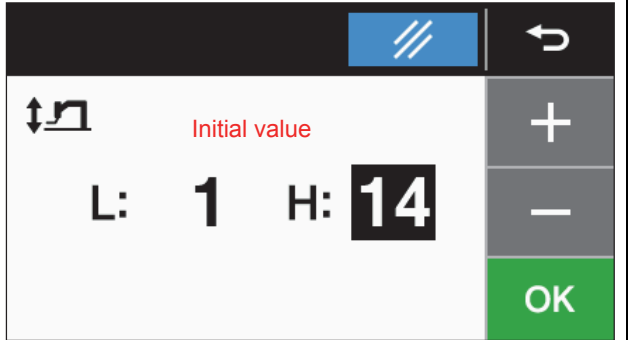
- The setting ranges for other items will vary in accordance with the pattern which is registered, so register the pattern first.

4	Select an item to be changed.		Touch the item to be changed.
5	Change the setting for the parameter. (Refer to "List of items" on the next page for details on parameter changes.) Example: X scale setting (4)		Touch the or key (2), or touch the numeric keypad keys (3) to change the parameter setting.
6	Apply the changed parameter setting.		Touch the OK key . - The display will return to the user program editing screen, and the edited parameters will be applied. - If you press the back key (5) without pressing the OK key, you can cancel the parameter changes.
7	Repeat steps 4 to 6 to register the settings for each item.		
8	Register the settings for the user program.		Once all the items have been set, touch the OK key in the user program editing screen. - The display will return to the select user program screen, and the program which was set will be registered. - If you press the back key without pressing the OK key, you can return to the select user program screen and cancel the changes to parameter settings.
9	If you would like to continue another program, repeat steps 2 to 8 above.		

5. SETTING FUNCTION

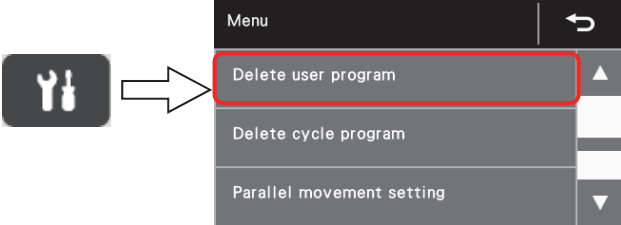
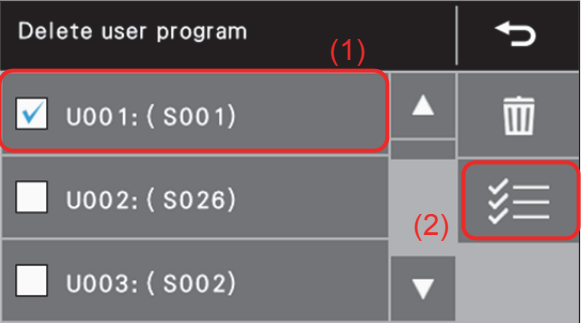
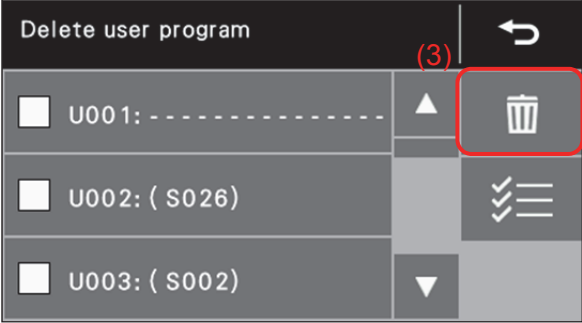
<List of items>

Item	Setting range and initial value	Display
Pattern	Standard programs and additional programs which have already been registered can be selected. Touch the tabs at the top of the screen to switch between the standard program and additional program lists. - Standard program [Program No. S001 to S089 (S001 to S064 for the 438HX/HS)] - Additional program [Program numbers A001 to A500] (These are programs which have been added to memory from an external USB source or by using the program copy function.) * If there are no additional programs which have been registered, there will be no response when this tab is touched.	
X scale	20% to 200% (Limited by available sewing area.) (Initial value is 100%.) * The setting can be displayed in "mm" units by setting memory switch No. 402 to "ON".	
Y scale	20% to 200% (Limited by available sewing area.) (Initial value is 100%.) * The setting can be displayed in "mm" units by setting memory switch No. 402 to "ON".	
Sewing speed	430HX/HS: 200 sti/min to 3,300 sti/min 430HX/HS: 200 sti/min to 2,800 sti/min The setting is in units of 100 sti/min. (Initial value: 2,700 sti/min for the 430HX/HS 2,300 sti/min for the 438HX/HS)	

Work clamp height	430HX/HS: Intermediate work clamp height (L): 1 (mm) to 17 (mm) (Initial value is 1.) Maximum work clamp height (H): 10 (mm) to 17 (mm) (Initial value is 14.) 438HX/HS: Intermediate work clamp height (L): 1 (mm) to 13 (mm) (Initial value is 1.) Maximum work clamp height (H): 6 (mm) to 13 (mm) (Initial value is 10.) * It is possible to change the value of intermediate work clamp height (L) only when memory switch No. 071 is set to "2" or memory switch No. 072 is set to "2" or "5".	
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5-9. Deleting user programs

User programs which are not needed can be deleted.

1	Switch to the delete user program setting screen. 	Press the menu key to switch to the setting menu screen. Touch "Delete user program" in the setting menu screen to switch to the select delete user program screen. If a pattern has been registered in a program, the "Program No." (X: "X scale", Y: "Y scale" / "Base program No.") will be displayed in the list; if it has not been registered in a program, "---" will be displayed after the program numbers.
2	Select the program number for the program that you would like to delete. Example: program number U001 	Set the program number (1) for the user program which you would like to delete from the list on the screen. - Select the number by which the program has been registered. - When you touch a program number, a check mark  will appear against the program number in the list. (Multiple program numbers can be selected.) - You can select all user programs in the list (U001 to U100) by pressing the ALL key  (2).
3	Delete the user programs. 	Once all the user programs to be deleted have been selected, you can press the delete key  (3) to delete the registered sewing patterns which have been assigned to the user program numbers which are currently selected. "---" will be displayed after the program numbers of the user programs which have been deleted.
4	<To exit from the user program deleting screen> - At the delete user program screen, you can press the back key  or the menu key  to return to the setting menu. - In addition, when the home key  is pressed, the display switches to the home screen.	

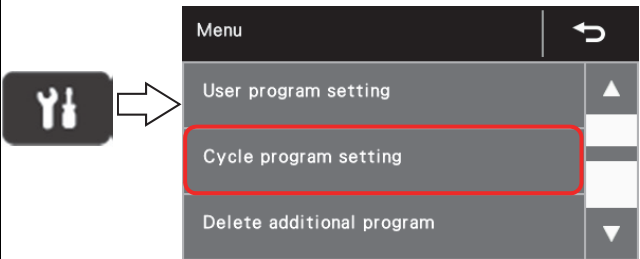
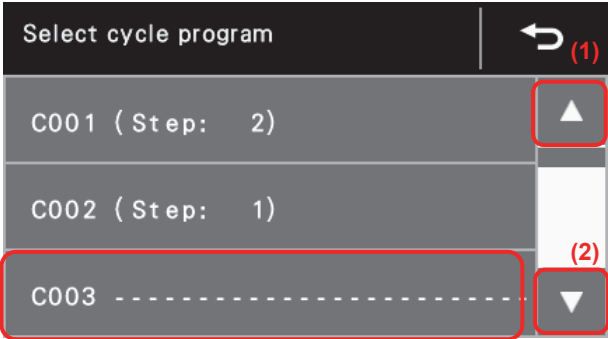
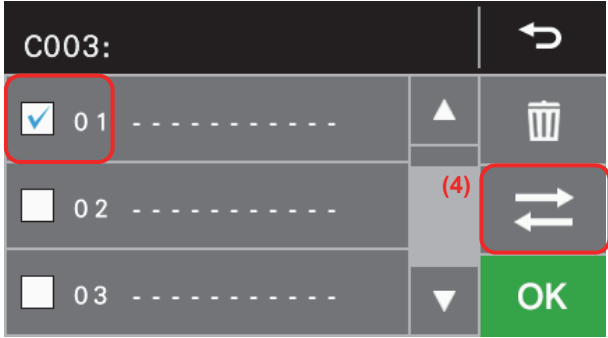
5-10. Cycle program setting

Sewing patterns that have been registered in user programs can be registered in up to 30 cycle programs (C001 to C030).

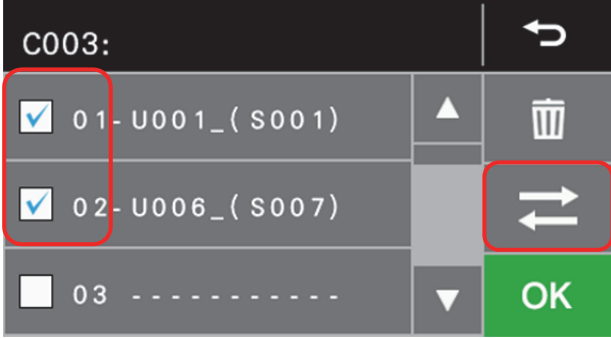
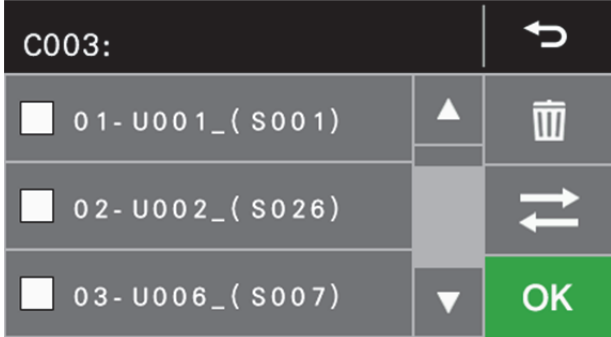
One cycle program can contain user programs up to a maximum of 50 steps.

When sewing the sewing patterns in a certain order, it can be useful to register them in a cycle program beforehand.

<Recording method>

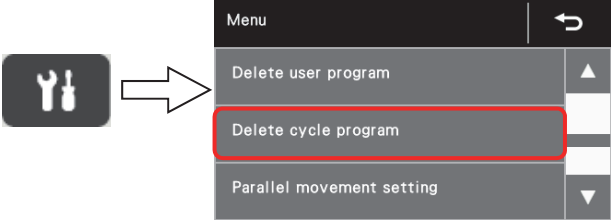
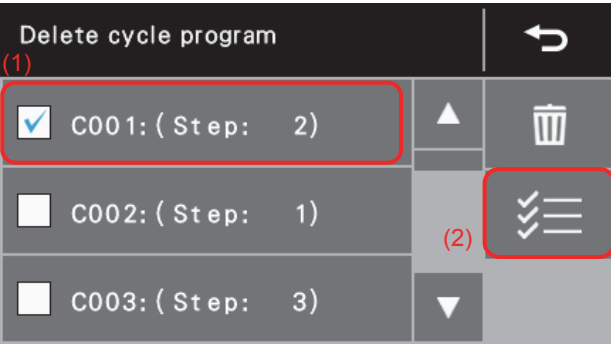
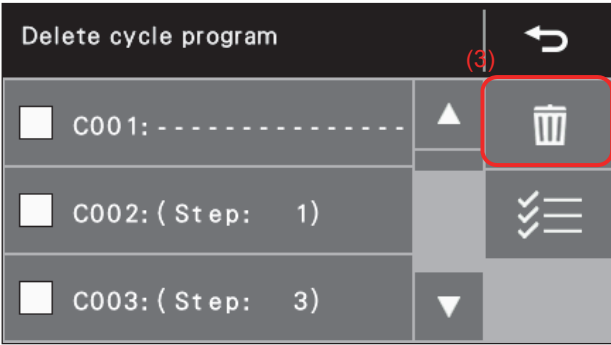
1	Switch to the cycle program setting screen.  Touch the menu key to switch to the setting menu screen, and touch "Cycle program setting" in the menu. The display will switch to the select cycle program screen.
2	Select a cycle program number.  Touch the  key (1) or the  key (2) to select the cycle program number. - If no cycle program has been registered, "----" will be displayed after the program number. Select the program number for the program you would like to edit, and touch the list. - The cycle program edit screen will be displayed.
3	Select the program number to be registered in step 1.  Select the step No.01 on the top of the list. - A check mark will appear in the check column of the list which was touched. - If you would like to delete the registration, you can do it by touching a step to select it, and then touching the clear key  (3).
4	Switch to the select program screen.  Once a step has been selected and a check mark appears, touch the step select key  (4). - The display will switch to the select user program screen. - The user programs which have been registered will be set in the selected step.

5. SETTING FUNCTION

5	Select a program.  Touch the user program which has been registered as a step to select it from the list. The display will switch to the cycle program edit screen and the selected program will be registered in the selected step.
6	Repeat steps 3 to 5 above to register step 2 and subsequent steps in the same way. - If you would like to insert a new step in between two existing steps  If two or more steps are registered and you would like to insert a new step in between two already-registered steps, add check marks to the two existing steps to select them, and then touch the step select key . - The selected user program will be registered in between the two selected steps at this time.
7	End cycle program setting mode.  Touch the OK key . The display will return to the select cycle program screen. This completes the registering of cycle programs.
8	If you would like to continue setting another cycle program, repeat steps 2 to 7 above.
9	<To exit from the cycle program setting screen> - At the delete user program screen, you can press the back key  or the menu key  to return to the setting menu. - In addition, when the home key  is pressed, the display switches to the home screen.

5-11. Deleting cycle programs

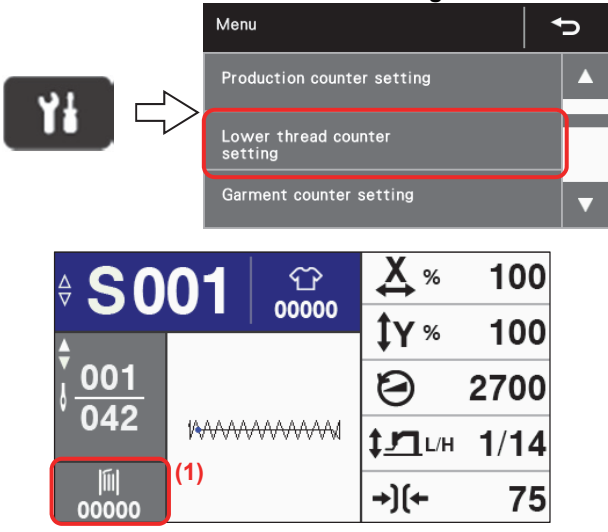
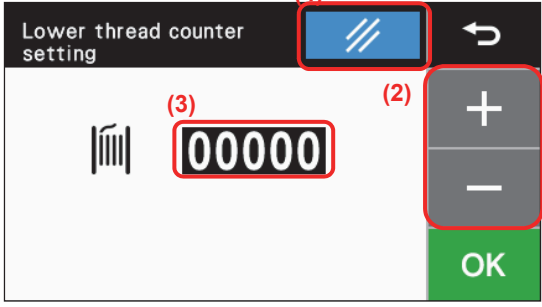
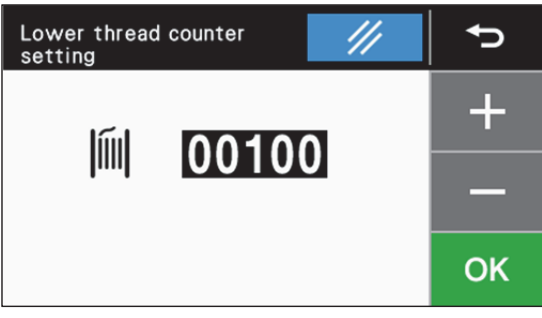
Cycle programs which are not needed can be deleted.

1	Switch to the delete cycle program setting screen. 	Press the menu key to switch to the setting menu screen. Touch "Delete cycle program" in the setting menu screen to switch to the select delete cycle program screen. If any cycle programs have been registered, they will be displayed in the list in the format of "Program No." (Step: "Cycle program step number"; if no cycle programs have been registered, "---" will be displayed after the program numbers.
2	Select the program number for the program that you would like to delete. Example: program number C001 	Set the program number (1) for the cycle program which you would like to delete from the list on the screen. - Select the number by which the program has been registered. - When you touch a program number, a check mark  will appear against the program number in the list. (Multiple program numbers can be selected.) - You can select all cycle programs in the list (C001 to C030) by pressing the ALL key  (2).
3	Delete the cycle programs. 	Once all the cycle programs to be deleted have been selected, you can press the delete key  (3) to delete the cycle programs which are currently selected. "---" will be displayed after the program numbers of the cycle programs which have been deleted.
4	<To exit from the cycle program deleting screen> - At the delete cycle program screen, you can press the back key  or the menu key  to return to the setting menu. - In addition, when the home key  is pressed, the display switches to the home screen.	

5-12. Using the lower thread counter

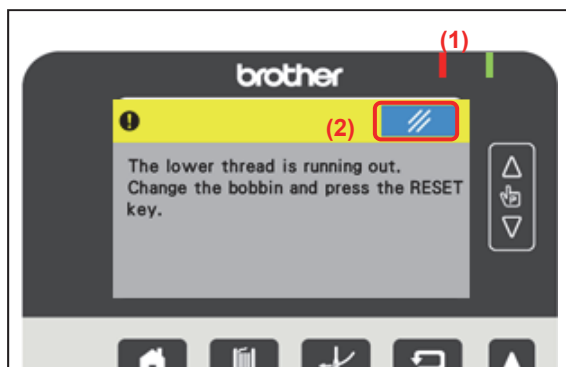
If you use the lower thread counter to set the number of articles which can be sewn with the amount of lower thread available, you can stop the lower thread running out in the middle of sewing a pattern.

<Initial value setting>

1	Switch to lower thread counter setting mode. 	From the setting menu screen Press the menu key to switch to the setting menu screen, and then touch "Lower thread counter setting" in the menu. From the home screen At the home screen, you can also touch the lower thread counter icon  (1) to switch the screen. The 5 digits of the current counter value will be displayed.
2	Change the initial value. 	Touch the  key or the  key (2) to change the counter value (3). - The initial value can be set from 1 "00001" to 9999 "09999". - If the initial value is set to "00000", the lower thread counter will not operate. - If you touch the reset key  (4), the setting will become "00000".
3	Apply the changed parameter setting. 	Touch the OK key . - The value will be applied and the display will return to the menu (*1). - If you touch the back key  without touching the OK key, you can cancel the changes. *1: If you have switched from the home screen, the display will return to the home screen.

<Lower thread counter operation>

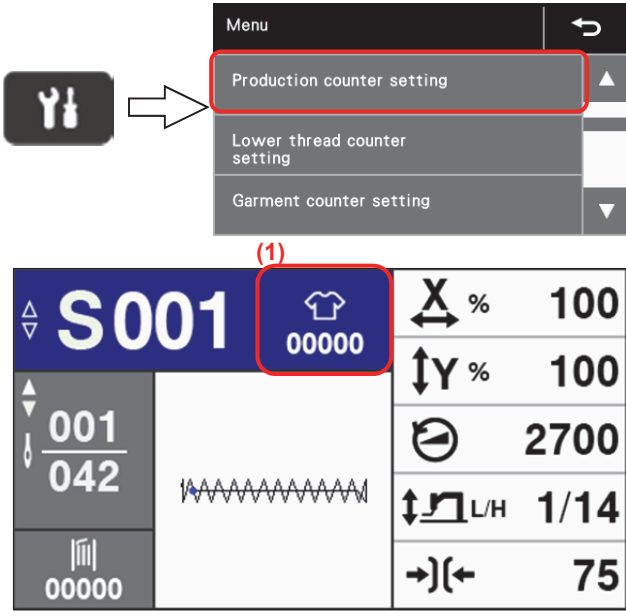
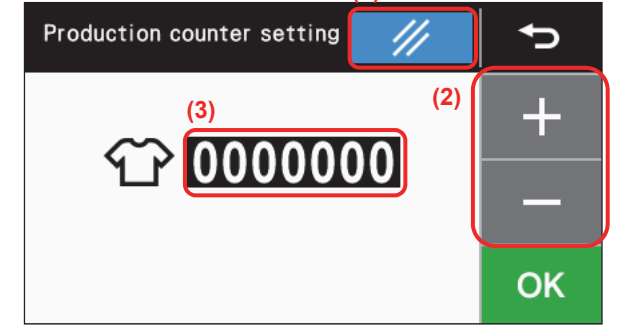
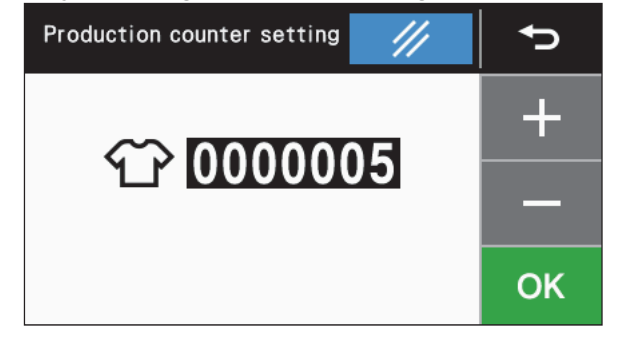
If the initial value is set to other than "0", the lower thread counter will operate.



- Each time the sewing of a single article is completed, the value displayed on the home screen decreases by 1.
- When the lower thread counter becomes "00000", the lower thread counter warning screen will be displayed, the caution LED (1) will blink, and the buzzer will sound. The sewing machine will not operate during this time, even if the foot switch is depressed.
- If you touch the reset key  (2), the display will switch to the home screen, the lower thread counter value will return to the setting value, and sewing will be possible.

5-13. Using the production counter

<Setting the counter value>

1	Switch to production counter setting mode. 	From the setting menu screen Press the menu key to switch to the setting menu screen, and then touch "Production counter setting" in the menu. From the home screen At the home screen, you can also touch the production counter icon (1) to switch the screen. The 7 digits of current counter value will be displayed.
2	Change the counter value. 	Touch the  key or the  key (2) to change the counter value (3). - The counter value can be set from "0000000" to "9999999". - If you press the reset key  (4) during setting mode, the value will become "0000000".
3	Apply the changed parameter setting. 	Touch the OK key . - The value will be applied and the display will return to the menu (*1). - If you touch the back key  without touching the OK key, you can cancel the changes. *1: If you have switched from the home screen, the display will return to the home screen.

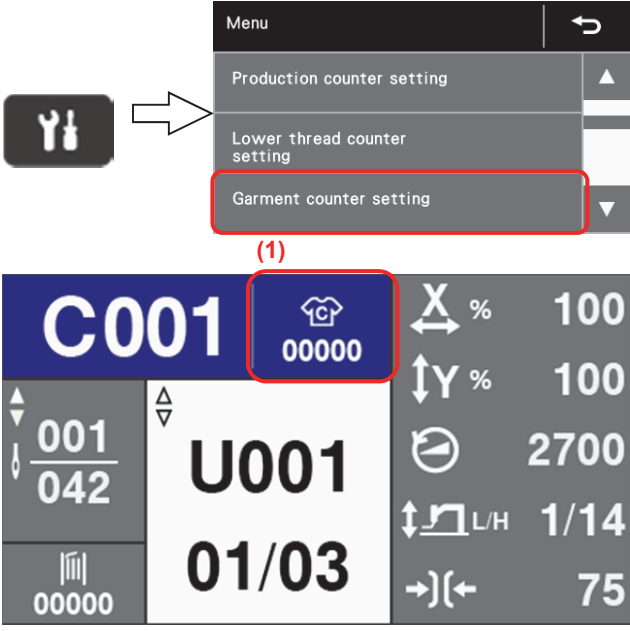
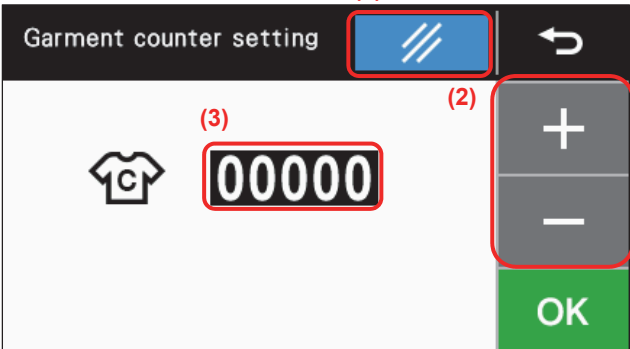
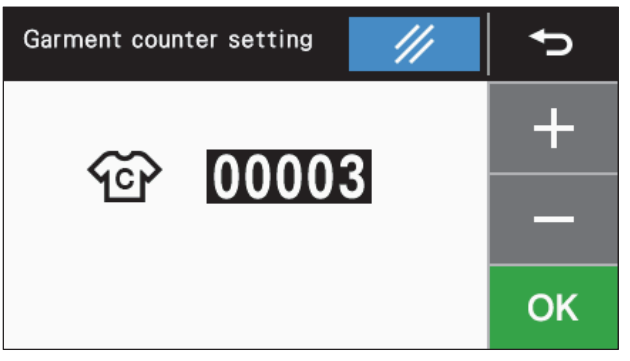
<Production counter operation>

The production counter is always operating.

- The counter value displayed in the home screen is incremented by 1 each time one sewing operation is completed.
- * The lower 5 digits are displayed in the home screen. If you would like all digits to be displayed, check using the production counter setting screen.

5-14. Using the garment counter

<Setting the counter value>

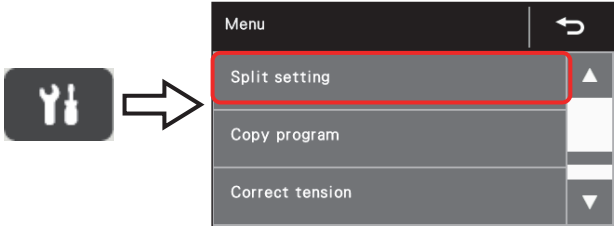
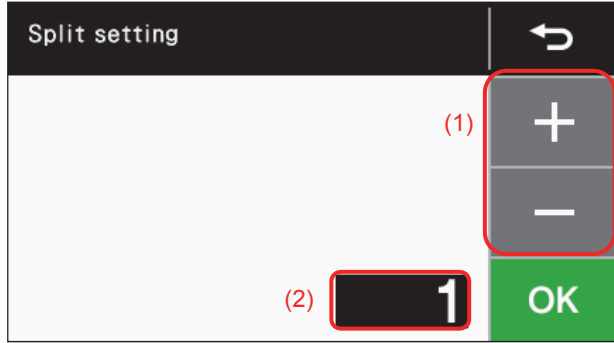
1	Switch to garment counter setting mode. 	From the setting menu screen Touch the menu key to switch to the setting menu screen, and touch "Garment counter setting" in the menu. From the home screen When a cycle program is selected or if memory switch No. 300 is set to "ON", you can also touch the garment counter icon (1) at the home screen to switch the screen. The 5 digits of current counter value will be displayed.
2	Change the counter value. (4) 	Touch the  key or the  key (2) to change the counter value (3). - The counter value can be set from "00000" to "09999". - If you press the reset key  (4) during setting mode, the value will become "00000".
3	Apply the changed parameter setting. 	Touch the OK key . - The value will be applied and the display will return to the setting menu screen (*1). - If you touch the back key  without touching the OK key, you can cancel the changes. *1: If you have switched from the home screen, the display will return to the home screen.

<Garment counter operation>

The garment counter always operates when sewing cycle programs.

- If you would like to display the garment counter on the home screen, set memory switch No. 300 to "ON".
- The garment counter will not be displayed on the home screen when sewing programs other than cycle programs.
- The counter value which is displayed on the home screen will be incremented by 1 each time all of the steps in the selected cycle program have been sewn once.

5-15. Split setting

1	Display the sewing programs which use split codes. Select the sewing programs which use split codes can be selected by pressing the program number change key  or in the program select screen, so that they are displayed in the home screen. * The 430H does not have any sewing programs which use split codes registered as standard programs, so it is necessary to read sewing programs which have had split codes inserted when they were created from USB memory and register them as additional programs. (For details on reading data, refer to "5-18-5. Reading additional pattern data".)
2	Switch to the split setting screen. Press the menu key to switch to the setting menu screen. Touch the split setting in the setting menu screen to switch to the split setting screen. 
3	Select the split number. From the split setting screen, touch the  or  key (1) to set the number (2) of the sewing split section. The setting values are displayed in the order in which the split codes appear, and "1" indicates the sewing start position. 
4	Confirm the split number. Once the split section to be sewn has been selected, you can press the OK key  to confirm the value which is currently selected. The next sewing operation will start from the selected split section. If you touch the back key  without touching the OK key, you can cancel the changes. 

* You can set whether to use single split mode or continuous split mode by using memory switch No. 451.

• ON=Single split mode

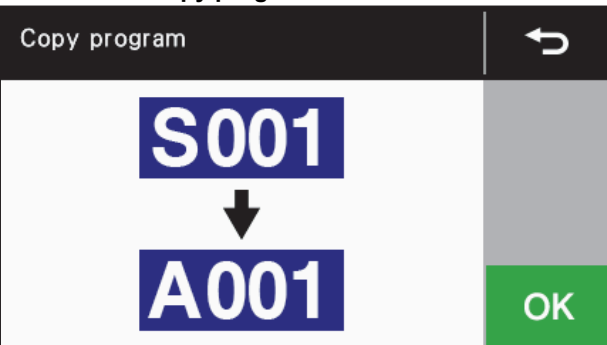
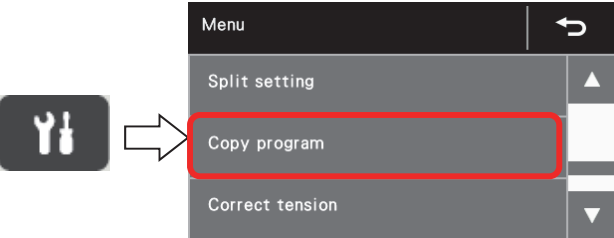
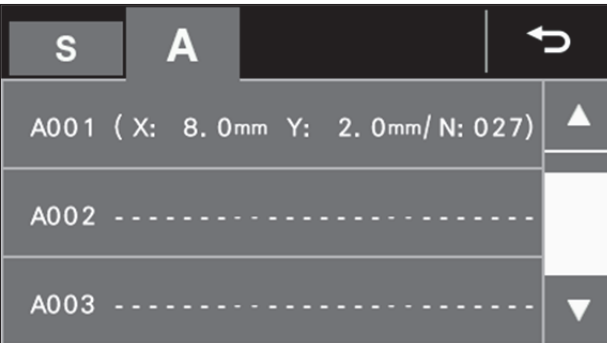
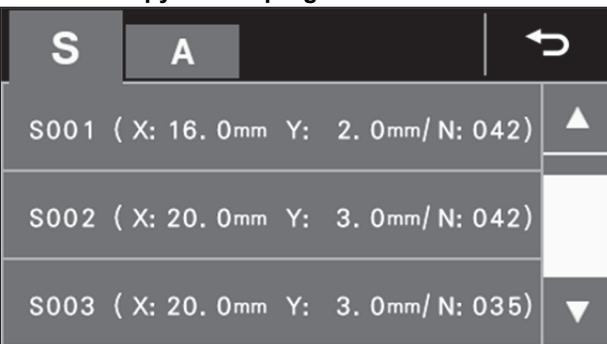
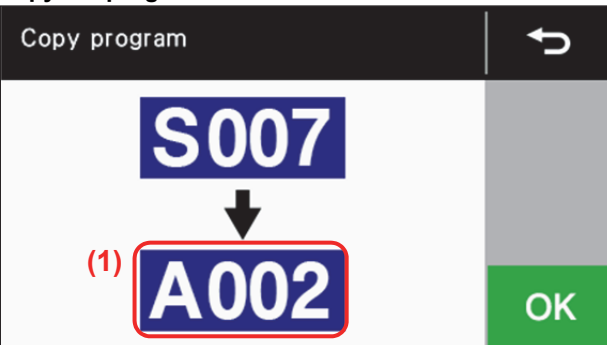
Only the split section which is selected in the split settings can be sewn.

• OFF=Continuous split mode

Sewing can be carried out from the split section which is selected in the split settings to the sewing end.

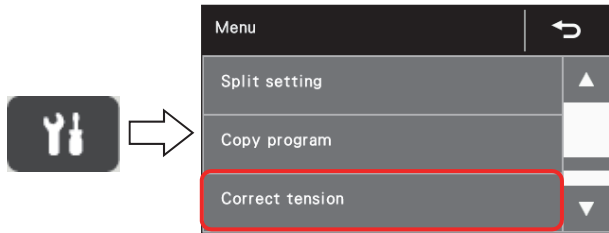
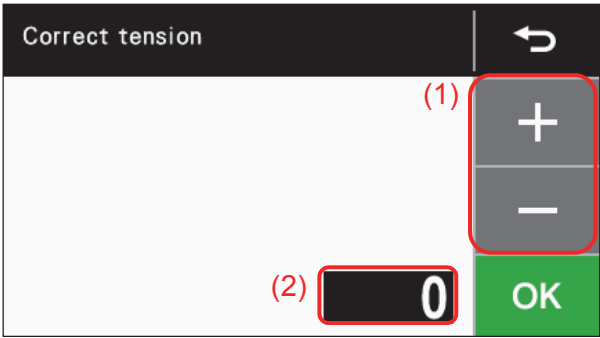
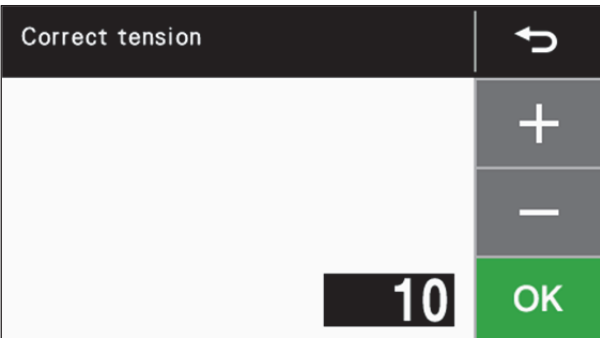
5-16. Copying programs (Additional program)

If you would like to reuse an existing program which has almost exactly the same parameters, but change only the necessary parts, you can copy the existing program and register the copy as a new additional program.

1	Switch to the copy program screen. 	Touch the menu key to switch to the setting menu screen. Touch "Copy program" in the menu to switch to the copy program screen. 
2	Select the copy destination program. 	When you touch the program number at the bottom of the copy program screen, the screen switches to the list for selecting the copy destination program number. Select from the list to change the copy destination program number. • Select from additional program numbers A001 to A500. • When a program number is selected, the display will switch immediately to the copy program screen. • The destination program number for copying will blink.
3	Select the copy source program. 	Select from the list to change the number of the standard program or additional program to be copied. • When a program number is selected, the display will switch immediately to the copy program screen.
4	Copy the program. 	Touch the OK key.  • All items in the program will be copied, and the copy destination program number (1) will stop blinking. • If you touch the back key  without touching the OK key, you can cancel the copy.
5	Change the necessary items.	Carry out the steps in "5-4. Setting the X scale and Y scale", "5-5. Setting the sewing speed" and "5-6. Setting the work clamp height" in the instruction manual, and change the necessary items.

5-17. Uniform tension correction

This function is useful if you would like to change the tension values for all sewing programs to correct the tension all at once.

1	The display will switch to the correct tension screen. 	Touch the menu key to switch to the setting menu. Touch "Correct tension" in the menu to switch to the correct tension screen.
2	Select the correction value. 	Press the + or - key (1) to change the uniform correction value (2) in the tension value display.
3	Confirm the correction value. 	Touch the OK key OK. The correction value will be confirmed, and it will be applied to the upper thread tension value the next time sewing is carried out.

5-18. Using USB memory

Programs and memory switch setting details can be copied between sewing machines by means of USB memory.

CAUTION



Do not connect anything to the USB port other than the USB memory. If this is not observed, problems with operation may result.

Notes on handling USB memory (commercially available)

Do not disassemble or alter the USB memory.

Do not bend, drop, scratch or place heavy objects on top of the USB memory.

Do not allow the USB memory to become wet, such as with water, oil, solvents, drinks or any other liquids.

Do not use or store the USB memory in locations exposed to strong static electricity or electrical interference.

Do not use or store the USB memory in locations exposed to vibrations or impacts, direct sunlight, extreme dust (or lint), high temperatures, high humidity, severe temperature fluctuations, or strong magnetic forces (such as from speakers).

Do not subject the USB memory to vibration or shocks or remove it from the sewing machine while data reading or writing is in progress.

Data on the USB memory may be lost or damaged due to some malfunction or accident. We recommend backing up important data.

The USB memory that you purchased is already formatted. We recommend that the USB memory not be reformatted.

The recommended USB memory is those sold by SanDisk and Panasonic. USB memory from other manufacturers may use different formatting methods and may not work correctly as a result.

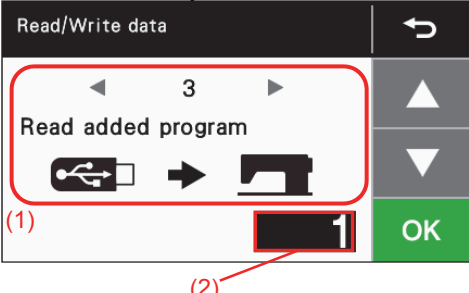
For more information, refer to the documentation provided with the purchased USB memory.

- * This product is compatible with USB memory that has been formatted using the FAT16/32 method. USB memory that has been formatted using other formatting methods cannot be used.
- * All other company and product names mentioned in this instruction manual are trademarks or registered trademarks of their respective companies. However, the explanations for markings such as TM are not clearly described within the text.

5-18-1. Structure of a USB memory

Data type	Folder name	File name
Control program	\BROTHER\ISM\ISMSYS\	ISM22MN.BVP (Main control program) ISM22MT.BVP (Motor control program) ISM22PL.upf (Panel control program)
Additional sewing programs	\BROTHER\ISM\ISMDA**\ * " ***" represents the value for memory switch No. 752. If storing additional pattern data from other sewing machines in the same USB memory, change the name of the folder.	ISMS0***.sew * "****" represents the pattern data number (001 to 500).
Memory switch settings	Same as above	ISMMSW.SEW
Sewing parameters of programs	Same as above	ISMUPG.SEW
Cycle program	Same as above	ISMCYC.SEW
Error log	Same as above	Stores the files which relate to error logs. (The data can only be written from the maintenance information screen.)

5-18-2. Preparation for reading/writing data

1	First connect the USB memory to the sewing machine. NOTE: • Make sure the USB memory is facing the correct way. • Always be sure to keep the cover closed except when inserting and removing USB devices. If this is not done, dust may get inside and cause problems with operation.
2	Turn on the power switch.
3	Switching to read/write data screen  Touch the menu key  to switch to the setting menu screen. Touch "Read/Write data" in the menu to switch to the read/write data screen. * Some multi card readers may not be recognized. * If there is no USB memory connected at this time, the message "USB memory is not connected or inserted." will be displayed.
4	For example, item No. 3  1. The currently-selected read/write item and illustration (1) are displayed, so press the ◀ or ▶ key to change the read/write item. 2. Press the OK key  to run the selected read/write item. The read/write in progress screen will be displayed. * To read additional programs from or write them to read/write items "3" and "4", use the ▲ or ▼ keys to select the program number (2) to read from or write to. • Touch the back key  to return to the setting menu screen.

[List of Read/Write data screen items]

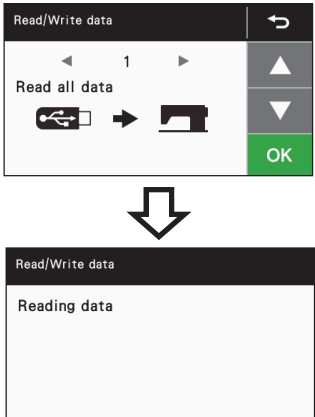
Item no.	Menu display	Setting item
1	Read all data	The contents of items 3 to 8 are read from the USB memory.
2	Write all data	The contents of items 3 to 8 are written to the USB memory.
3	Read added program	Additional program data is read from the USB memory.(*1)
4	Write added program	Additional program data is written to the USB memory.
5	Read program	Data for all sewing programs (additional, user and cycle) and sewing parameters (standard, additional, user, cycle) is read from the USB memory.
6	Write program	Data for all sewing programs (additional, user and cycle) and sewing parameters (standard, additional, user, cycle) is written to the USB memory.
7	Read memory switch	Memory switch settings are read from the USB memory.
8	Write memory switch	Memory switch settings are written to the USB memory.

[List of maintenance information screen items]

1	Write all data	Same as for "Export all data" in the "Read/Write data" list.
2	Export maintenance information	Memory switch settings and error logs are written to the USB memory.

The additional sewing data that can be used with this sewing machine is data that has been created for KE-430H/BE-438H models.

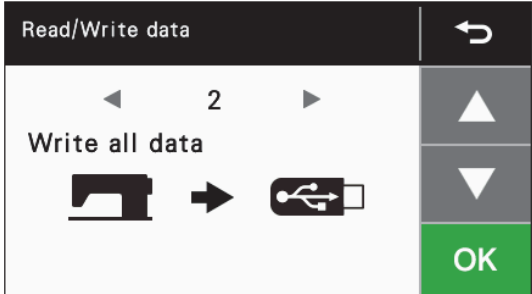
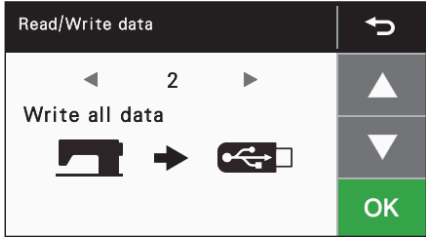
5-18-3. Reading all data

1	 Select item No. "1" in the Read/Write data screen. (Refer to "5-18-2. Preparation for reading/writing data".)
2	 Press the OK key . • The buzzer will sound and the sewing program data, cycle program data, memory switch settings and additional pattern data will be read from the USB memory and stored in the sewing machine's memory. • If any one of additional program data, memory switch data, sewing pattern data or cycle program data does not exist in the USB memory, an error buzzer will sound.
3	 When the display returns to the status in step 1, writing of the sewing machine data is complete. • If you would like to use any other read/write modes, press the ◀ or ▶ key to select the mode and then run it.
4	Exit read/write mode • At the Read/Write data screen, you can press the back key  or the menu key  to return to the setting menu. • In addition, when the home key  is pressed, the display switches to the home screen.

5-18-4. Writing all data to USB memory

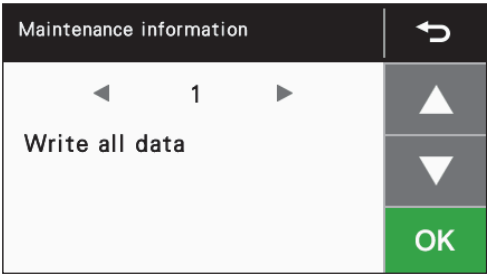
Writing all data can be carried out using "Read/Write data" in the setting menu and "Maintenance information" in the special menu.

<When using the Read/Write data screen>

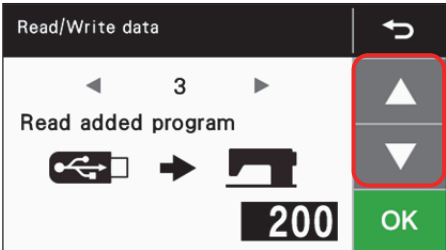
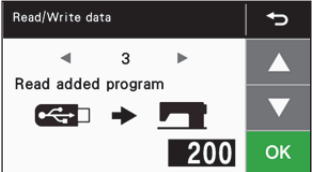
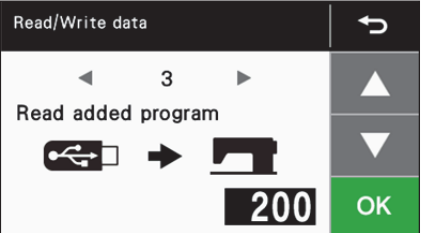
1	 Select item No. "2" in the Read/Write data screen. (Refer to "5-18-2. Preparation for reading/writing data".)
2	 Press the OK key . The buzzer will sound and the sewing program data, cycle program data, memory switch settings and additional pattern data will be copied to the USB memory.
3	 When the screen display returns to the status before writing started, writing of the sewing machine data is complete. • If you would like to use any other read/write modes, press the ◀ or ▶ key to select the mode and then run it.
4	Exit read/write mode • At the Read/Write data screen, you can press the back key  or the menu key  to return to the setting menu. • In addition, when the home key  is pressed, the display switches to the home screen.

5. SETTING FUNCTION

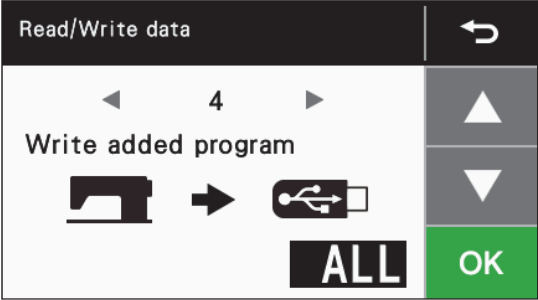
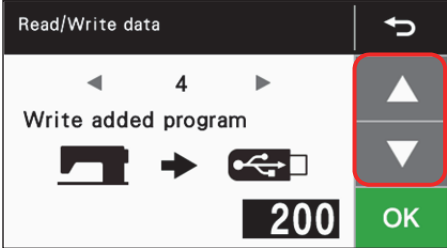
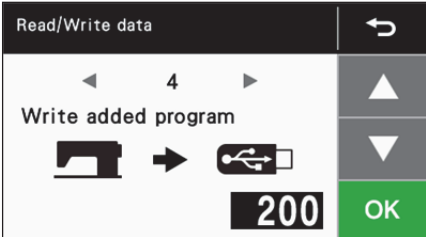
<When using the Maintenance information screen>

1	Switch to the Maintenance information screen. 1. While referring to "4-7. Maintenance information list", switch from the special menu to the Maintenance information screen. 2. In the Maintenance information screen, select "Write all data" (item "1"). (Refer to "4-7. Maintenance information list".)
2	 Press the OK key . • The buzzer will sound and the sewing program data, cycle program data, memory switch settings and additional pattern data will be copied to the USB memory.
3	 When the screen display returns to the status before writing started, writing of the sewing machine data is complete. • If you would like to use any other read/write modes, press the ◀ or ▶ key to select the mode and then run it.
4	Exit maintenance information list mode. You can return to the special menu by pressing the back key . <To exit from the special menu> • At the special menu screen, press the home key  to close the special menu and switch to the home screen.

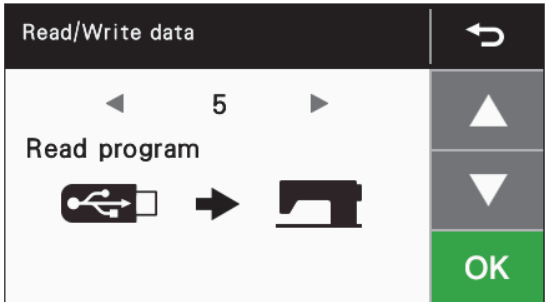
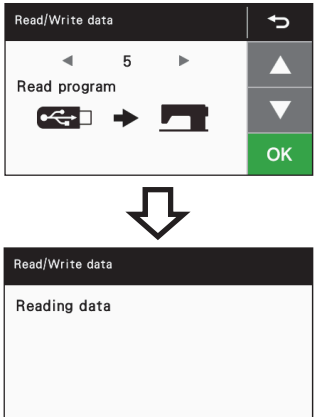
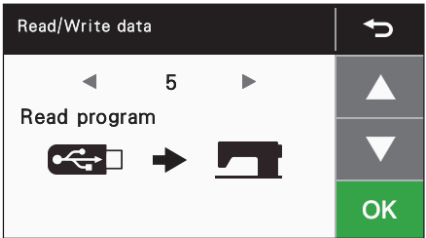
5-18-5. Reading additional pattern data

1	 Select item No. "3" in the Read/Write data screen. (Refer to "5-18-2. Preparation for reading/writing data".) The numbers for the additional sewing program data contained in the USB memory folder are displayed on the screen. (The initial value displayed is "ALL".)
2	 Press the  or  key to select the number for the additional pattern that you would like to be read (ALL, 1 to 500). - When "ALL" is displayed, all of the additional pattern data will be read at once. - The additional pattern numbers displayed will only be those that are in the USB memory folders. (If there is no data in the memory, "ALL" will be the only thing displayed.) For example, to read pattern No. 200
3	 Press the OK key . "Reading data" will be displayed on the screen until reading is complete. - Immediately after the key is pressed, the buzzer will sound and the selected pattern data will be read from the USB memory and copied into the sewing machine's memory. - If no additional program data exists in the USB memory, an error will be generated. 
4	 When the screen display returns to the status before reading started, reading of the additional pattern data is complete. If you would like to use any other read/write modes, press the  or  key to select the mode and then run it.
5	Exit read/write mode - At the Read/Write data screen, you can press the back key  or the menu key  to return to the setting menu. - In addition, when the home key  is pressed, the display switches to the home screen.

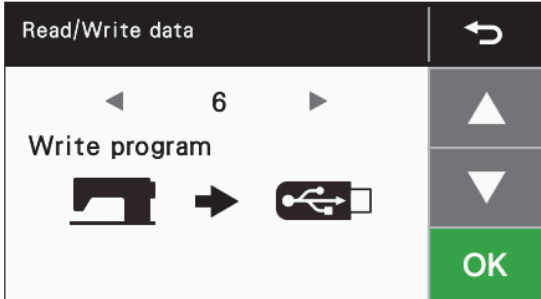
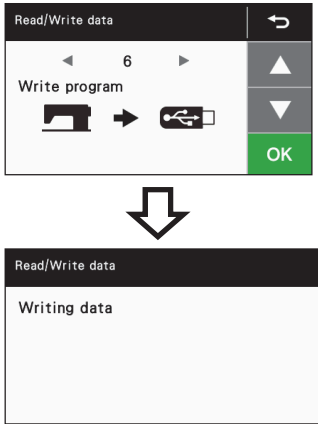
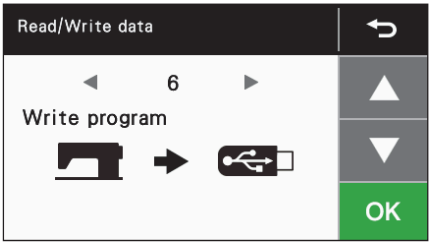
5-18-6. Writing additional pattern data to an USB memory

1	 Select item No. "4" in the Read/Write data screen. (Refer to "5-18-2. Preparation for reading/writing data".) The numbers for the additional sewing program data contained in the sewing machine's memory are displayed on the screen. (The initial value displayed is "ALL".)
2	 Press the  or  key to select the number for the additional pattern that you would like to write (ALL, 1 to 500). - When "ALL" is displayed, all of the additional pattern data will be read at once. - The additional pattern numbers displayed will only be those that are in the sewing machine. (If there is no data in the machine, "ALL" will be the only thing displayed.) For example, to read pattern No. 200
3	 Press the OK key . "Writing data" will be displayed on the screen until writing is complete. - Immediately after the key is pressed, the buzzer will sound and the selected pattern data will be read from the USB memory and copied into the sewing machine's memory.
4	 When the screen display returns to the status before writing started, writing of the additional pattern data is complete. If you would like to use any other read/write modes, press the  or  key to select the mode and then run it.
5	Exit read/write mode - At the Read/Write data screen, you can press the back key  or the menu key  to return to the setting menu. - In addition, when the home key  is pressed, the display switches to the home screen.

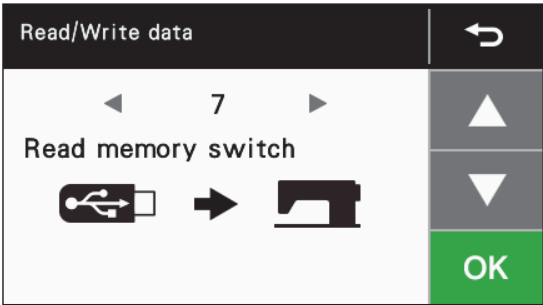
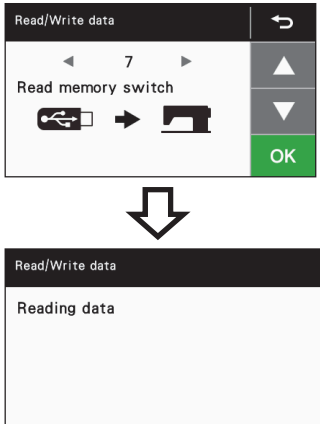
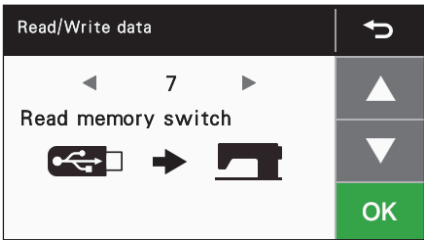
5-18-7. Reading program data

1	 Select item No. "5" in the Read/Write data screen. (Refer to "5-18-2. Preparation for reading/writing data".)
2	 Press the OK key . • The buzzer will sound and the sewing program data and cycle program data will be read from the USB memory and stored in the sewing machine's memory, and the sewing programs and cycle programs will then be set. • If no sewing program data or cycle program data exists, an error buzzer will sound.
3	 When the screen display returns to the status before reading started, reading of the program data is complete. • If you would like to use any other read/write modes, press the ◀ or ▶ key to select the mode and then run it.
4	Exit read/write mode • At the Read/Write data screen, you can press the back key  or the menu key  to return to the setting menu. • In addition, when the home key  is pressed, the display switches to the home screen.

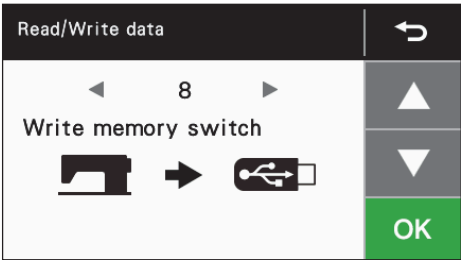
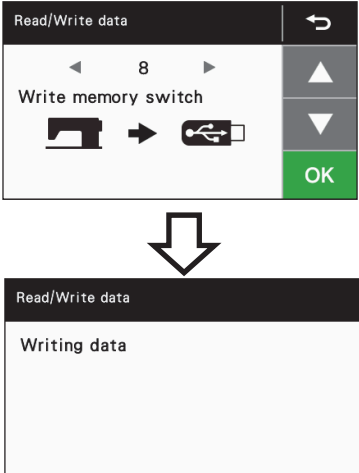
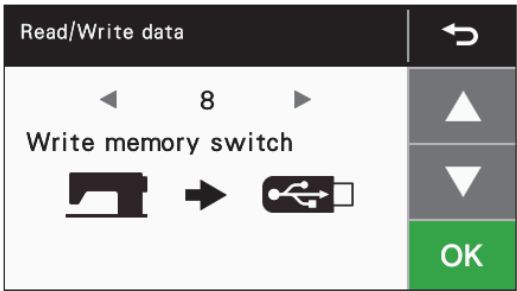
5-18-8. Writing program data to an USB memory

1	 Select item No. "6" in the Read/Write data screen. (Refer to "5-18-2. Preparation for reading/writing data".)
2	 Press the OK key . The buzzer will sound, and the sewing program data and cycle program data will be copied to the USB memory.
3	 When the screen display returns to the status before writing started, writing of the program data is complete. If you would like to use any other read/write modes, press the ◀ or ▶ key to select the mode and then run it.
4	Exit read/write mode - At the Read/Write data screen, you can press the back key  or the menu key  to return to the setting menu. - In addition, when the home key  is pressed, the display switches to the home screen.

5-18-9. Reading memory switch data

1	 Select item No. "7" in the Read/Write data screen. (Refer to "5-18-2. Preparation for reading/writing data".)
2	 Press the OK key . • The buzzer will sound, all memory switch setting data will be read from the USB memory and written into the sewing machine's memory, and the memory switch settings will then be applied. • If no memory switch data exists, an error will be generated.
3	 When the screen display returns to the status before reading started, reading of the memory switch data is complete. • If you would like to use any other read/write modes, press the ◀ or ▶ key to select the mode and then run it.
4	Exit read/write mode • At the Read/Write data screen, you can press the back key  or the menu key  to return to the setting menu. • In addition, when the home key  is pressed, the display switches to the home screen.

5-18-10. Writing memory switch data to USB memory

1		Select item No. "8" in the Read/Write data screen. (Refer to "5-18-2. Preparation for reading/writing data".)
2		Press the OK key . The buzzer will sound, and all memory switch setting data will be copied to the USB memory.
3		When the screen display returns to the status before writing started, writing of the memory switch data is complete. If you would like to use any other read/write modes, press the ◀ or ▶ key to select the mode and then run it.
4	Exit read/write mode - At the Read/Write data screen, you can press the back key  or the menu key  to return to the setting menu. - In addition, when the home key  is pressed, the display switches to the home screen.	

5-18-11. Writing error log data and memory switch data to USB memory

Error logs are handled as maintenance information, and so they can only be written to the USB memory from the Maintenance information screen.

1	Switch to the Maintenance information screen. - While referring to "4-7. Maintenance information list", switch from the special menu to the Maintenance information screen. - In the Maintenance information screen, select "Export maintenance information" (item "2"). (Refer to "4-7. Maintenance information list".)
2	Press the OK key . • The buzzer will sound and the error log data and memory switch log data will be copied to the USB memory.
3	When the screen display returns to the status before writing started, writing of the log data is complete. • If you would like to use any other read/write modes, press the ◀ or ▶ key to select the mode and then run it.
4	Exit maintenance information list mode. You can return to the special menu by pressing the back key . <To exit from the special menu> • At the special menu screen, press the home key  to close the special menu and switch to the home screen.

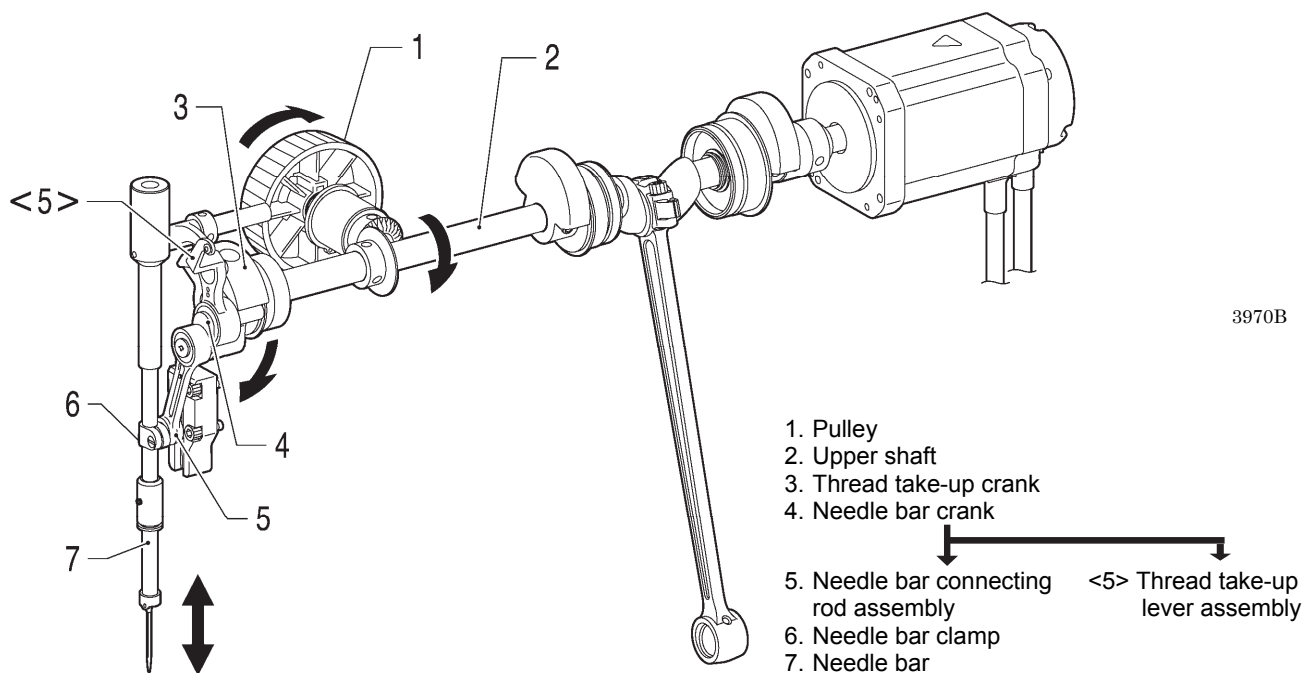
6. MECHANICAL DESCRIPTIONS

The mechanisms operate in the order of the numbers given in the illustrations.

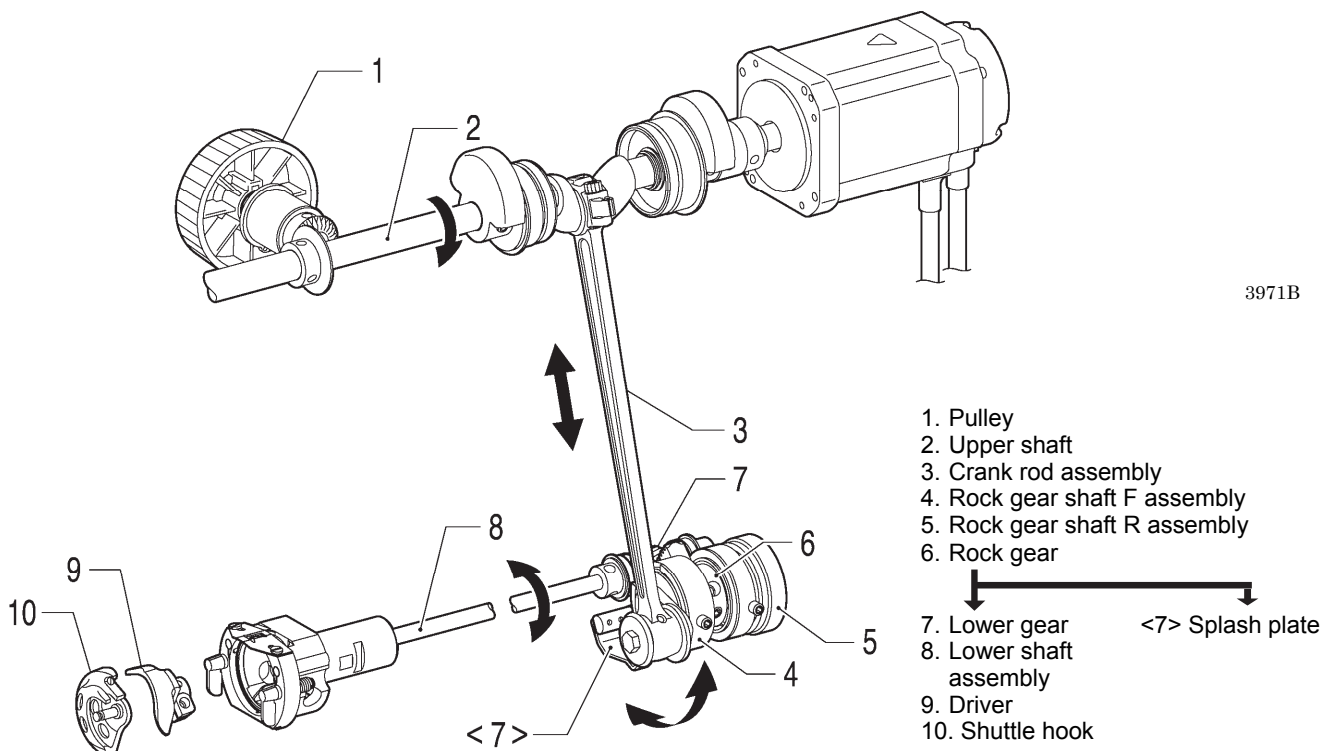
* <number> and [number] indicates the flow of each operations given.

* (number) indicates part names only. (They do not represent the flow of operations.)

6-1. Needle bar and thread take-up mechanisms

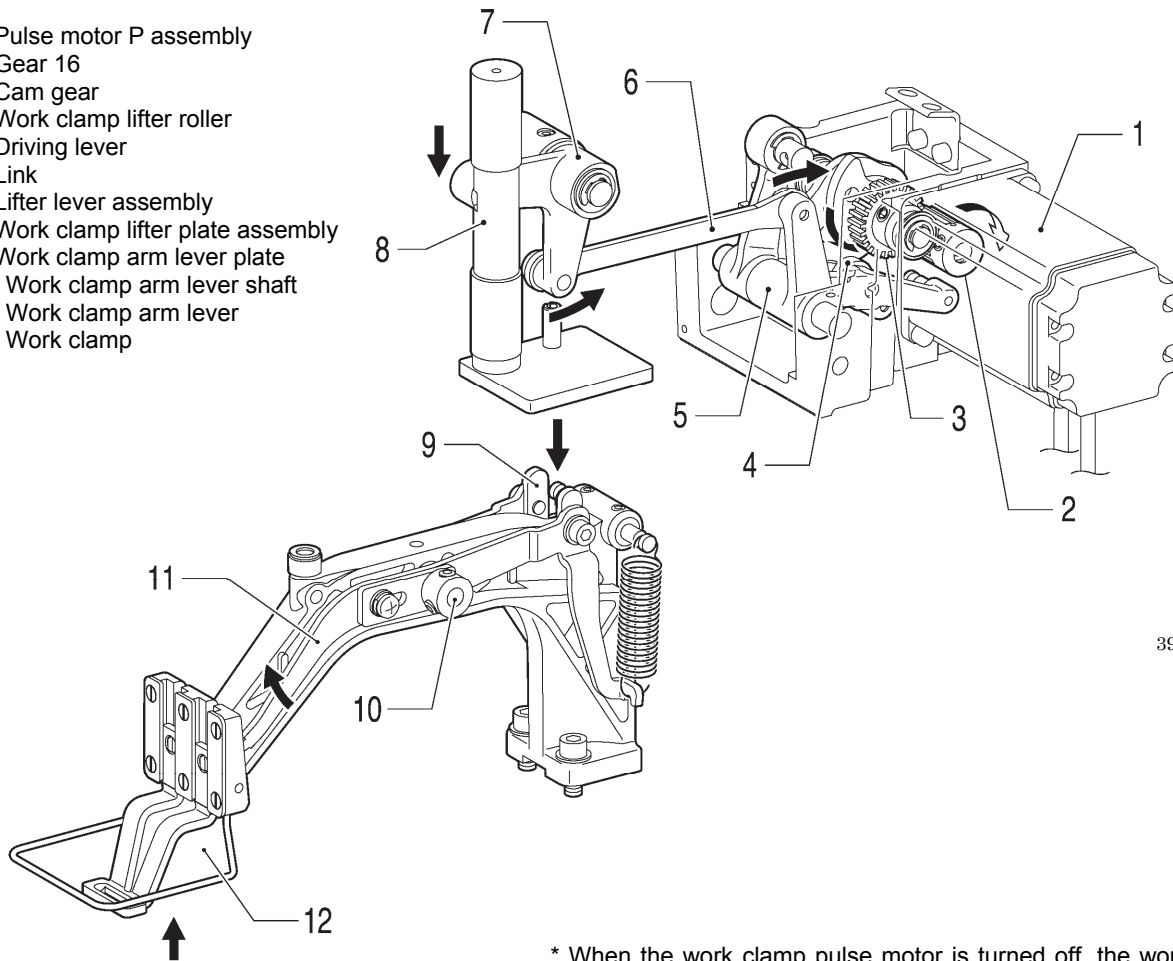


6-2. Lower shaft and hook mechanisms



6-3. Work clamp lifter mechanism

1. Pulse motor P assembly
2. Gear 16
3. Cam gear
4. Work clamp lifter roller
5. Driving lever
6. Link
7. Lifter lever assembly
8. Work clamp lifter plate assembly
9. Work clamp arm lever plate
10. Work clamp arm lever shaft
11. Work clamp arm lever
12. Work clamp

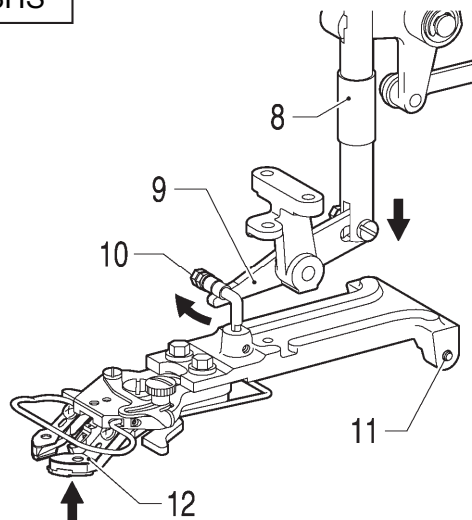


3972B

* When the work clamp pulse motor is turned off, the work clamp lifter plate rises, and work clamps lower.

BE-438HX/BE-438HS

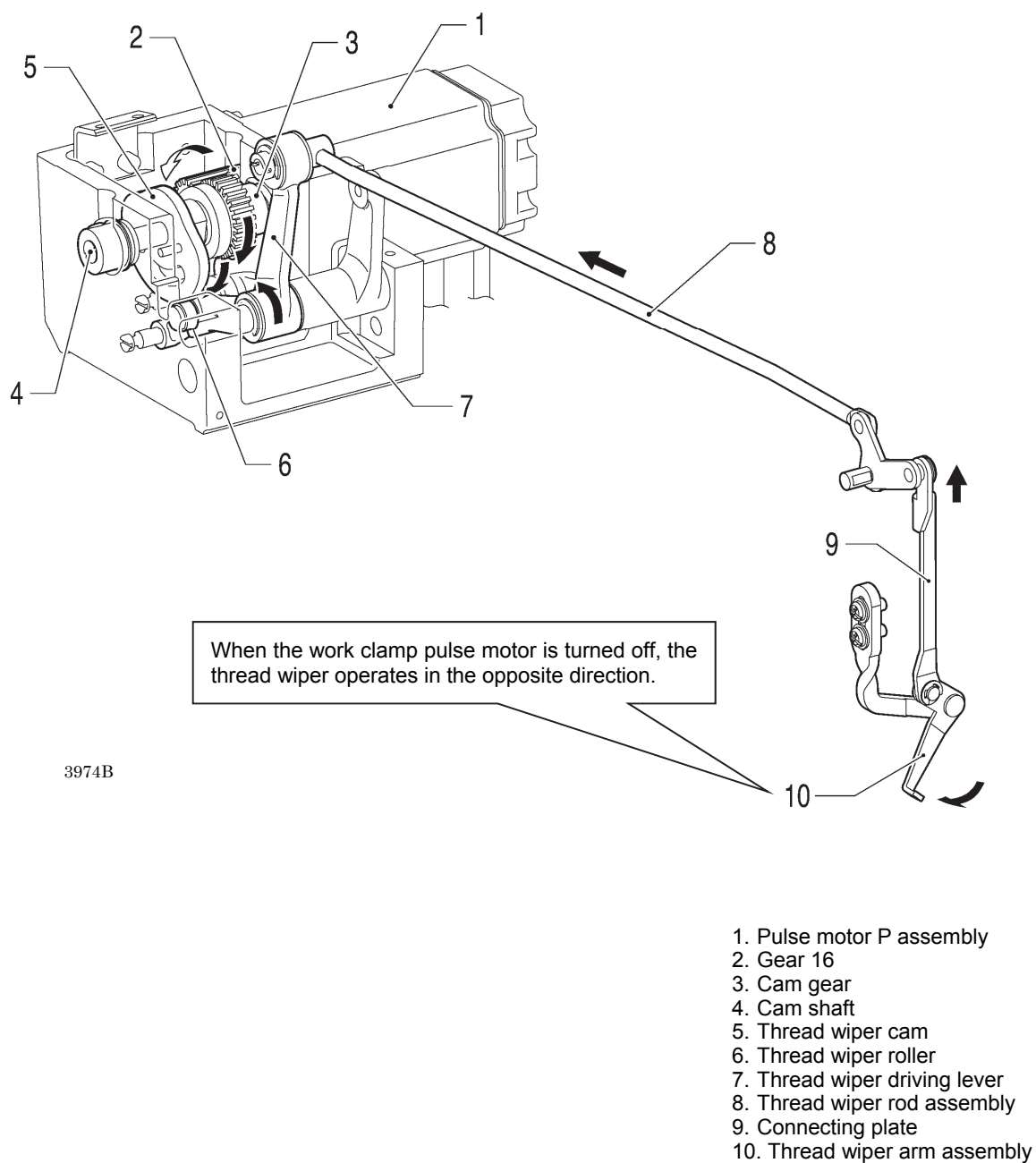
3973B



* When the work clamp pulse motor is turned off, the button clamp rod rises, and button clamps lower.

8. Button clamp rod
9. Button clamp lever
10. Button clamp holder hook assembly
11. Button clamp holder shaft
12. Button clamp

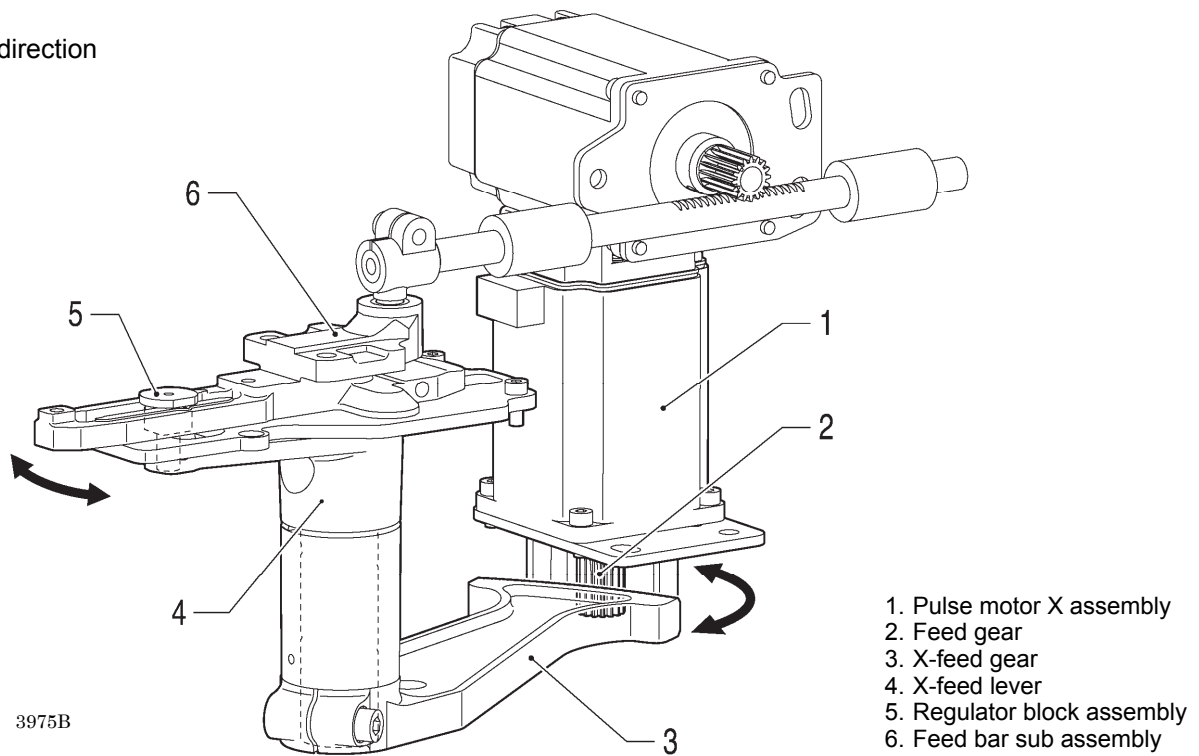
6-4. Thread wiper mechanism



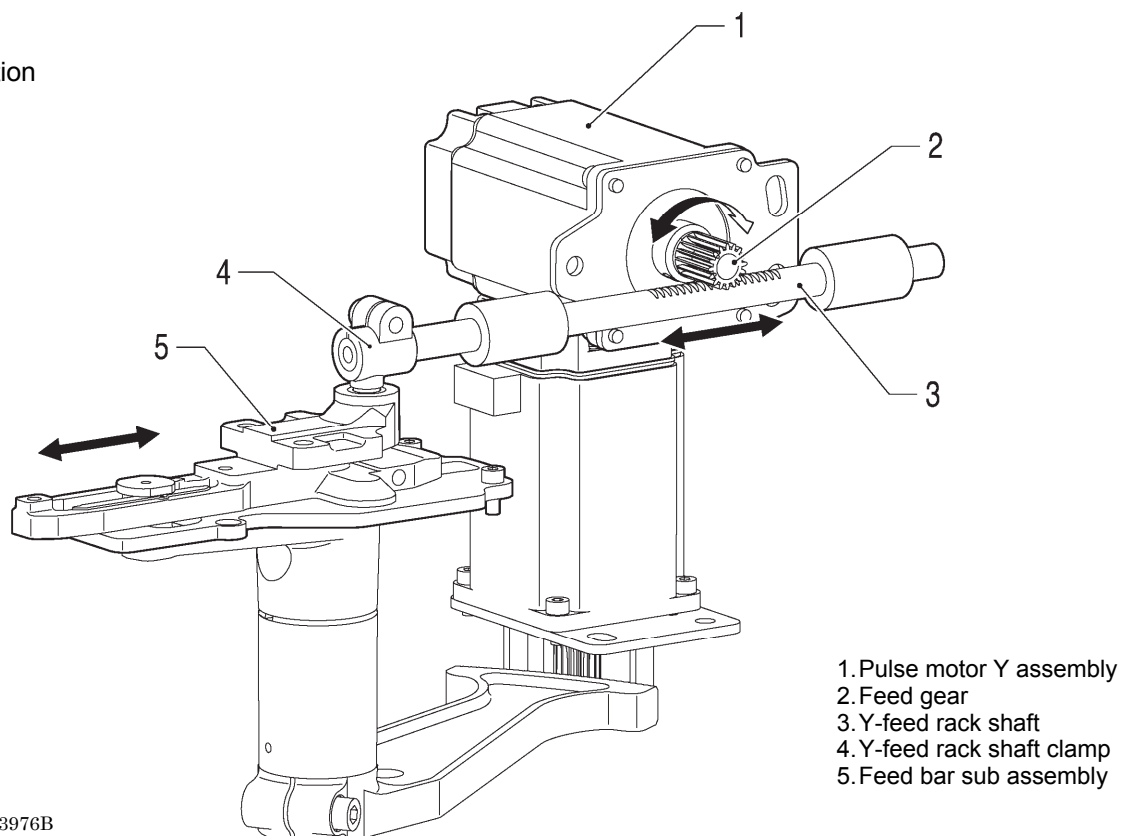
6-5. Feed mechanism

Sewing patterns are created through combinations of X and Y movements.

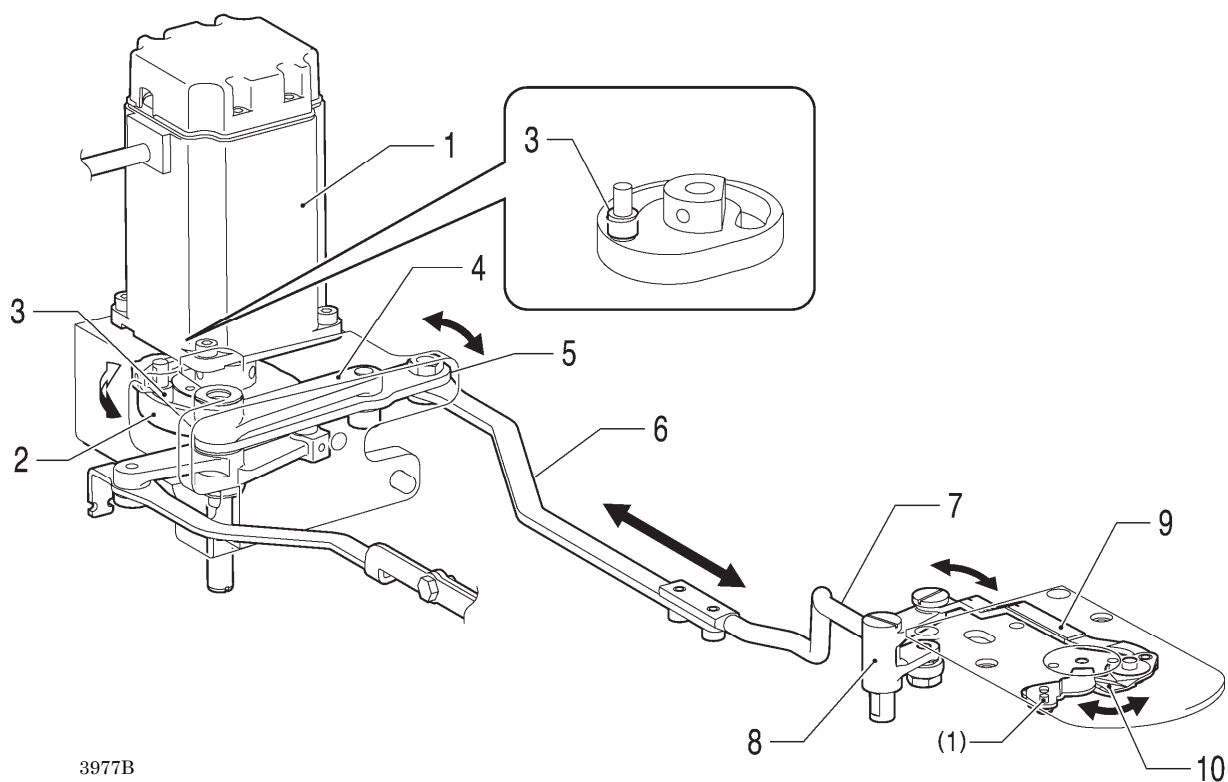
X direction



Y direction



6-6. Thread trimmer mechanism

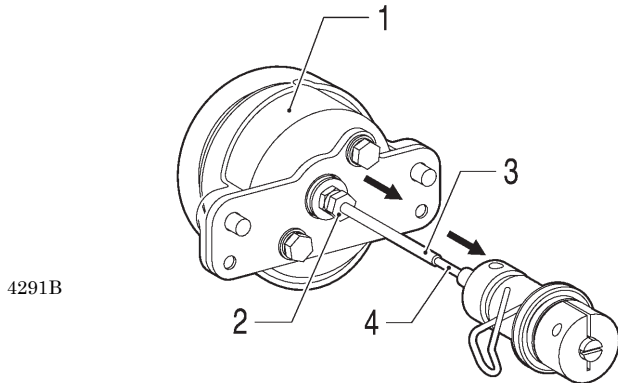


- 1. Pulse motor Y assembly
- 2. Thread trimmer cam
- 3. Thread trimmer collar H
- 4. Thread trimmer cam lever
- 5. Thread trimmer lever
- 6. Thread trimmer link
- 7. Thread trimmer rod
- 8. Movable knife lever
- 9. Movable knife connecting plate
- 10. Movable knife

(1) Fixed knife

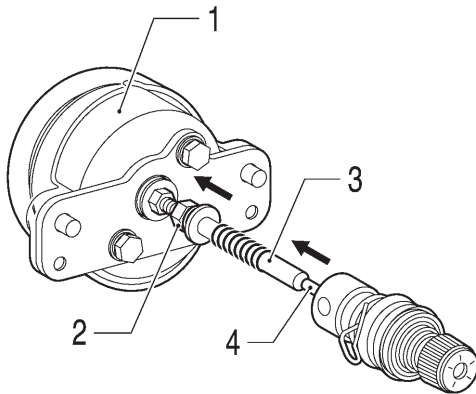
6-7. Tension release mechanism

KE-430HX, BE-438HX



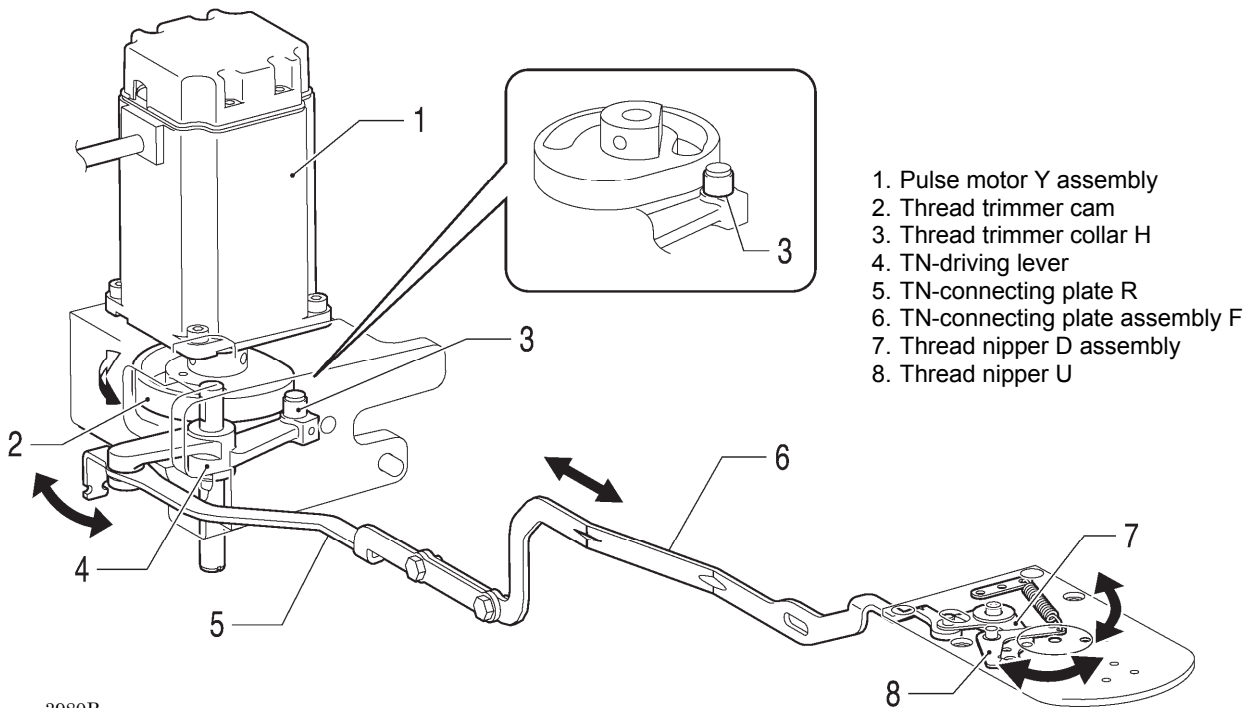
1. Tension solenoid
2. Bolt
3. Tension release bar
4. Tension release pin

KE-430HS, BE-438HS



1. Tension release solenoid
2. Bolt
3. Tension release bar
4. Tension release pin

6-8. Thread nipper mechanism (KE-430HX -03, -0K, -01 specifications only)



1. Pulse motor Y assembly
2. Thread trimmer cam
3. Thread trimmer collar H
4. TN-driving lever
5. TN-connecting plate R
6. TN-connecting plate assembly F
7. Thread nipper D assembly
8. Thread nipper U

7. DISASSEMBLY

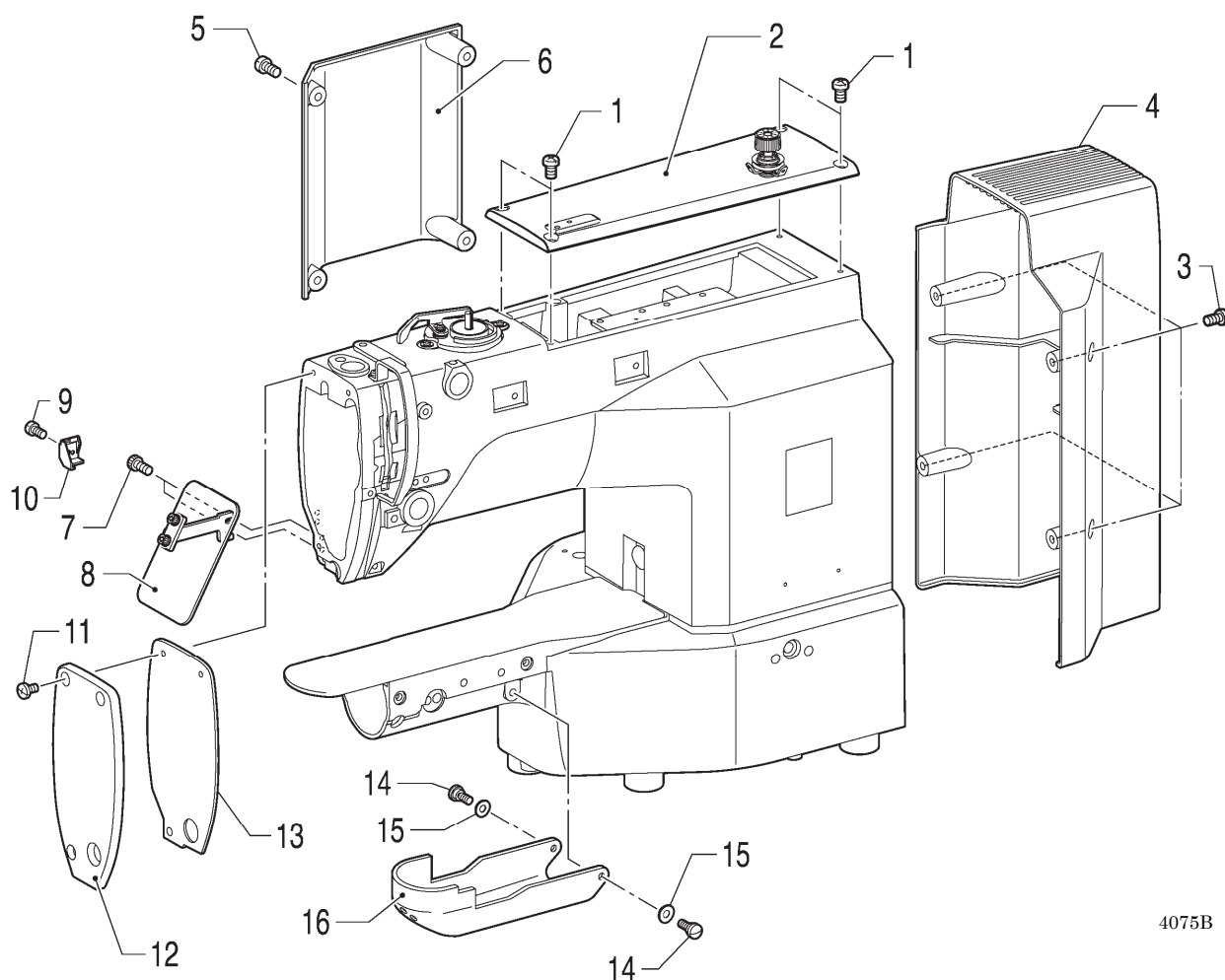
CAUTION

- | | |
|---|--|
| <p> Disassembly should only be carried out by a qualified technician.</p> <p> Turn off the power switch before carrying out disassembly, otherwise the machine may operate if the foot switch is depressed by mistake, which may result in injury.</p> <p> Be sure to wear protective goggles and gloves when handling the lubricating oil and grease, so that they do not get into your eyes or onto your skin, otherwise inflammation can result.
Furthermore, do not drink the oil or eat the grease under any circumstances, as they can cause vomiting and diarrhea.
Keep the oil out of the reach of children.</p> | <p> When replacing parts and installing optional accessories, be sure to use only genuine Brother parts.
Brother will not be held responsible for any accidents or problems resulting from the use of non-genuine parts.</p> <p> If any safety devices have been removed, be absolutely sure to re-install them to their original positions and check that they operate correctly before using the machine.</p> <p> To prevent accidents and problems, do not modify the machine yourself.
Brother will not be held responsible for any accidents or problems resulting from modifications made to the machine.</p> |
|---|--|

Disassemble each part in order of the numbers.

* (number) indicates part names only. (They do not represent the flow of operations.)

7-1. Covers

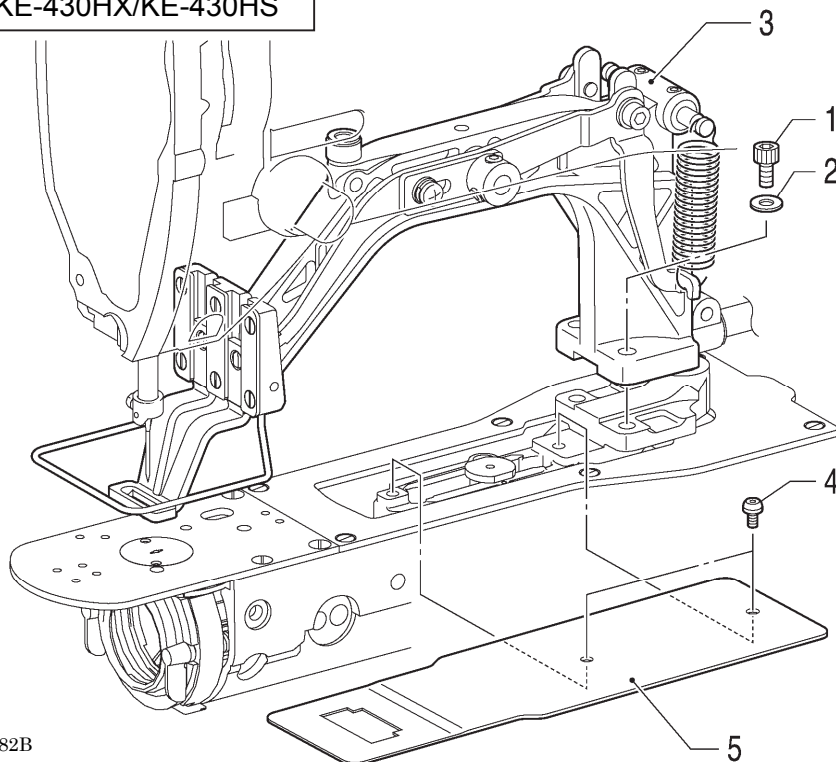


4075B

- | | |
|-----------------------|----------------------------------|
| 1. Screws [4 pcs] | 9. Screw |
| 2. Top cover | 10. LED cover |
| 3. Screws [4 pcs] | 11. Screws [3 pcs] |
| 4. Rear cover | 12. Face plate |
| 5. Screws [4 pcs] | 13. Face plate packing |
| 6. Side cover | 14. Shoulder screws 8-2 [2 pcs] |
| 7. Bolts [2 pcs] | 15. Washers, wave spring [2 pcs] |
| 8. Eye guard assembly | 16. Shuttle race cover assembly |

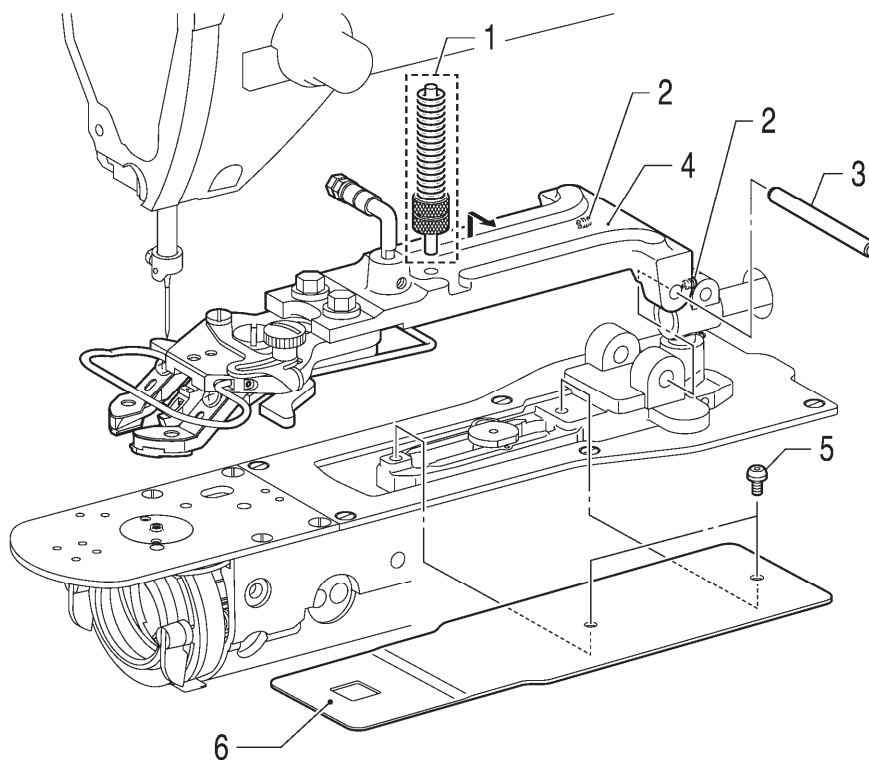
7-2. Work clamp arm mechanism

KE-430HX/KE-430HS



- 1. Bolts [2 pcs]
- 2. Plain washers [2 pcs]
- 3. Work clamp arm assembly
- 4. Screws [2 pcs]
- 5. Feed plate

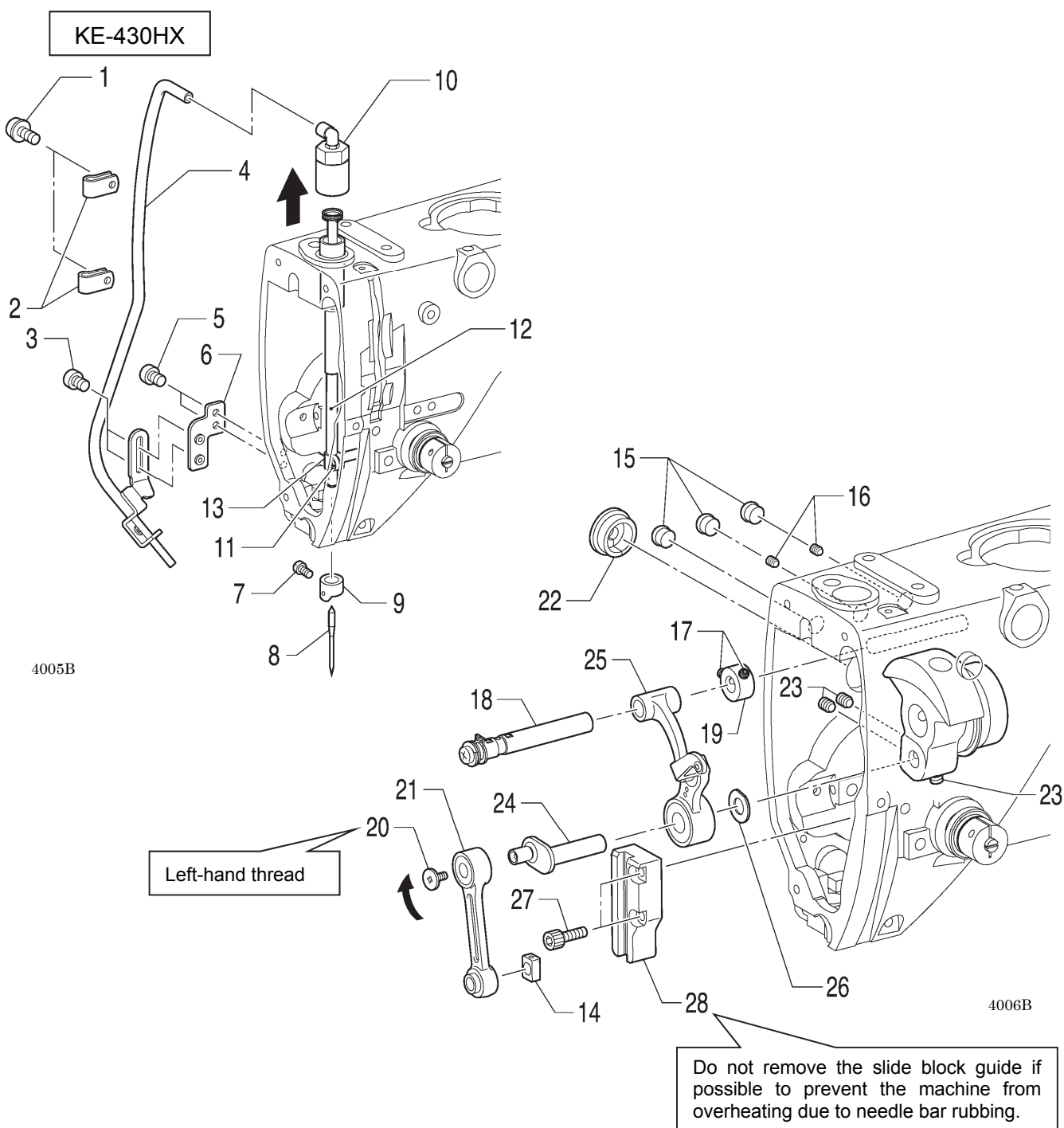
BE-438HX/BE-438HS



- 1. Tension nuts [2 pcs]
Adjusting nut
Compression spring [Push upward and remove horizontally]
- 2. Set screws [2 pcs: Loosen]
- 3. Button clamp holder shaft
- 4. Button clamp holder
- 5. Screws [2 pcs]
- 6. Feed plate

3983B

7-3. Needle bar mechanism



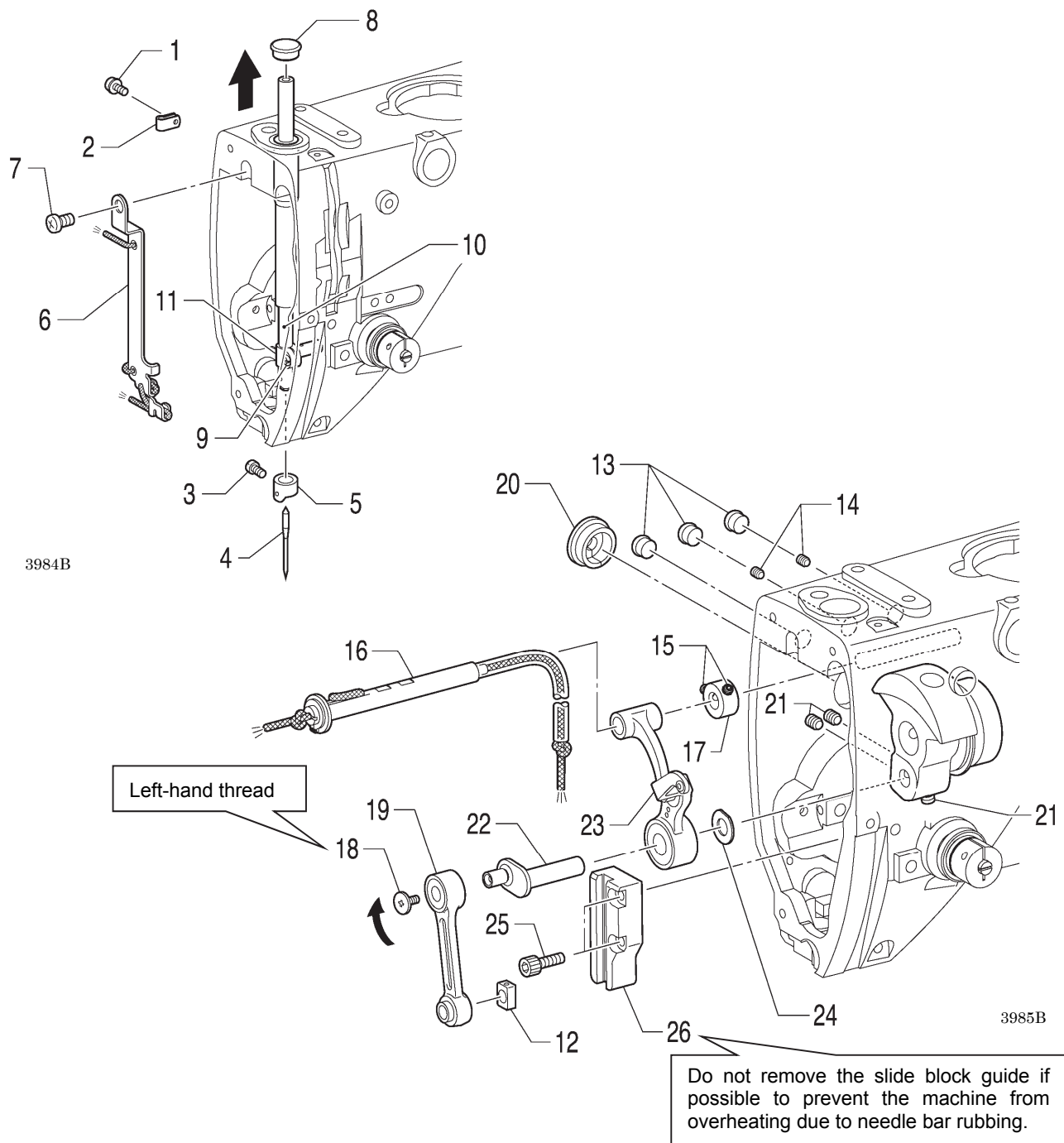
1. Screws [2 pcs]
2. Cord holders [2 pcs]
3. Screws [2 pcs]
4. Air nozzle assembly
5. Screws [2 pcs]
6. Middle plate
7. Set screw
8. Needle
9. Needle bar thread guide

10. Air cylinder cap assembly
11. Screw
12. Needle bar D assembly
[Pull out]
13. Needle bar clamp
14. Slide block
15. Rubber caps [3 pcs]
16. Set screws [2 pcs: Loosen]
17. Set screws [2 pcs: Loosen]
18. Thread take-up spring shaft assembly
19. Set screw collar

20. Screw [Loosen]
21. Needle bar connecting rod assembly
22. Rubber cap
23. Set screws [3 pcs: Loosen]
24. Needle bar crank
25. Thread take-up lever assembly
26. Washer
27. Bolts [2 pcs]
28. Slide block guide

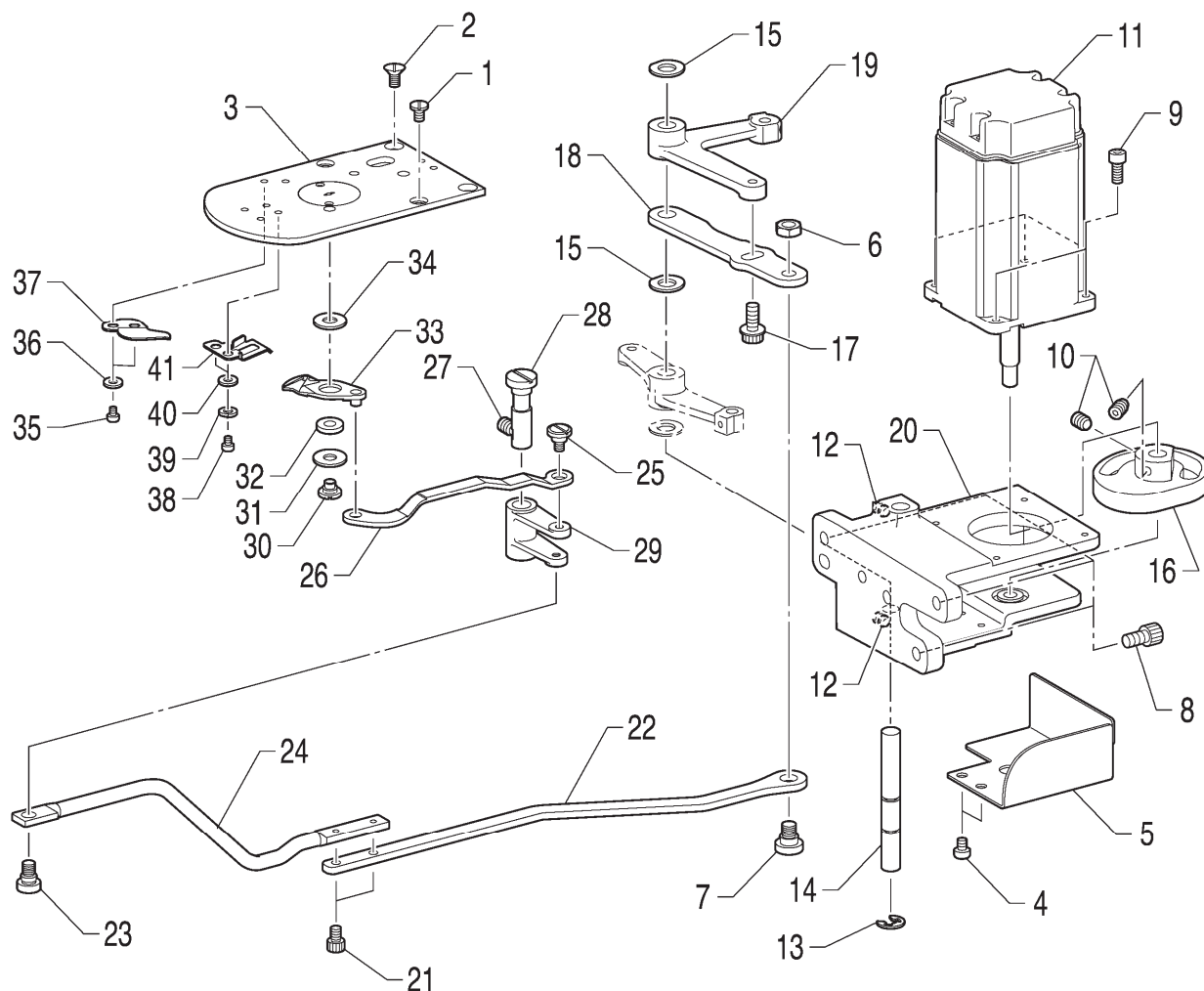
(Continued on next page.)

KE-430HS, BE-438HX/BE-438HS



- | | | |
|----------------------------|--|--|
| 1. Screws | 10. Needle bar [Pull out] | 19. Needle bar connecting rod assembly |
| 2. Cord holder | 11. Needle bar clamp | 20. Rubber cap |
| 3. Set screw | 12. Slide block | 21. Set screws [3 pcs: Loosen] |
| 4. Needle | 13. Rubber caps [3 pcs] | 22. Needle bar crank |
| 5. Needle bar thread guide | 14. Set screws [2 pcs: Loosen] | 23. Thread take-up lever assembly |
| 6. Screw | 15. Set screws [2 pcs: Loosen] | 24. Washer |
| 7. Wick holder assembly | 16. Thread take-up spring shaft assembly | 25. Bolts [2 pcs] |
| 8. Rubber cap | 17. Set screw collar | 26. Slide block guide |
| 9. Screw | 18. Screw [Loosen] | |

7-4. Thread trimmer mechanism

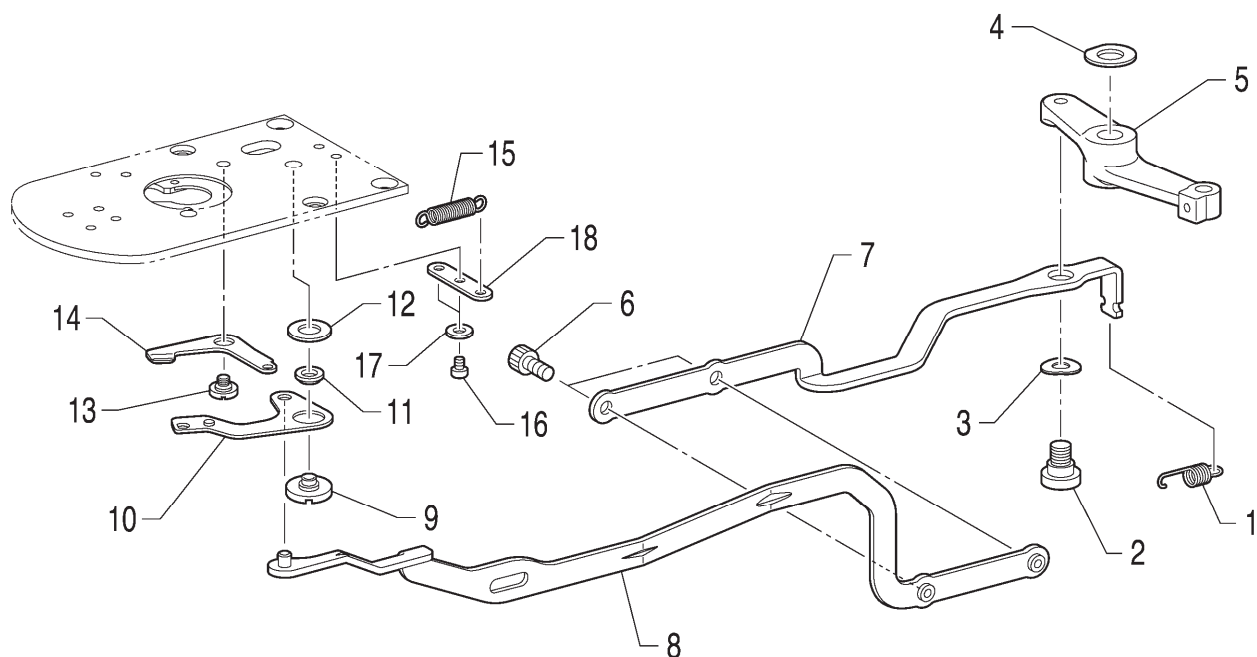


4003B

- | | | |
|------------------------------------|------------------------------------|----------------------------------|
| 1. Screws [2 pcs] | 19. Thread trimmer cam lever | 32. Movable knife collar |
| 2. Flat screws [2 pcs] | 20. Motor bracket rear assembly | 33. Movable knife assembly |
| 3. Needle plate | 21. Bolts [2 pcs] | 34. Movable knife spacer |
| 4. Screws [2 pcs] | 22. Thread trimmer link | 35. Screws [2 pcs] |
| 5. Motor bracket rear cover | 23. Shoulder screw 6.5-3 | 36. Plain washers [2 pcs] |
| 6. Nut | 24. Thread trimmer rod | 37. Fixed knife |
| 7. Link shoulder screw | 25. Shoulder screw 6-1.8 | * 38. Screws [2 pcs] |
| 8. Bolts [3 pcs] | 26. Movable knife connecting plate | * 39. Spring washers [2 pcs] |
| 9. Bolts [4 pcs] | 27. Set screw [Loosen] | * 40. Plain washers [2 pcs] |
| 10. Set screws [2 pcs: Loosen] | 28. Shaft movable knife lever | * 41. Bobbin thread clamp spring |
| 11. Pulse motor Y assembly | 29. Movable knife lever | |
| 12. Set screws [2 pcs: Loosen] | 30. Movable knife shoulder screw | |
| 13. Retaining ring E | 31. Thrust washer | |
| 14. Thread trimmer cam lever shaft | | |
| 15. Washers [2 pcs] | | |
| 16. Thread trimmer cam | | |
| 17. Socket bolt (S/P washer) | | |
| 18. Thread trimmer lever | | |

For the KE-430HX/HS (-05 specifications) and BE-438HX/HS, parts marked with * are supplied as accessories. Install them if necessary.

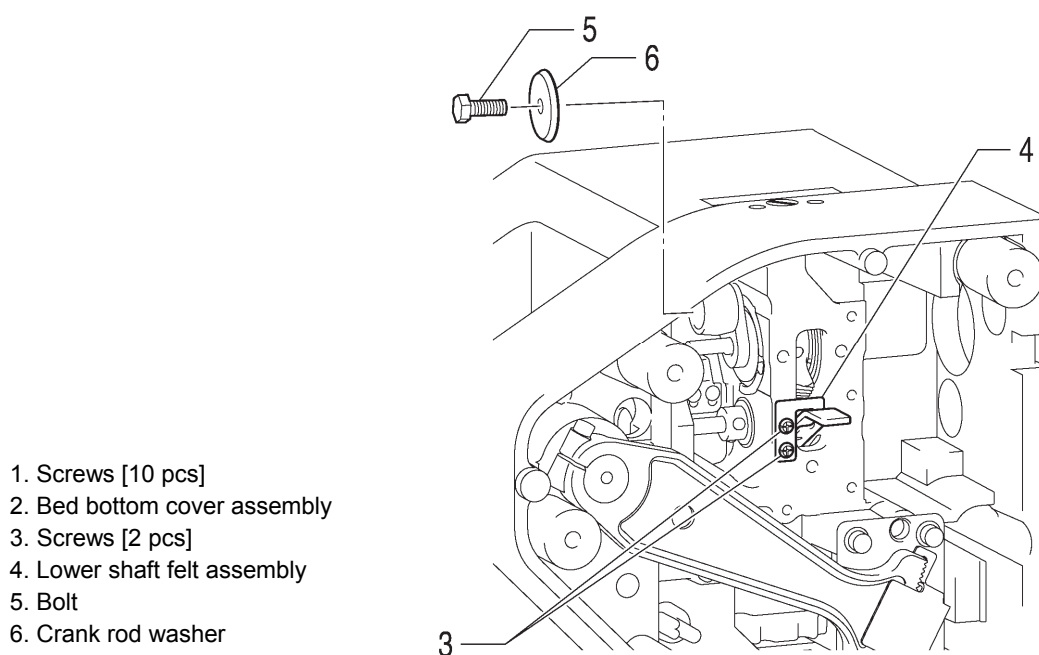
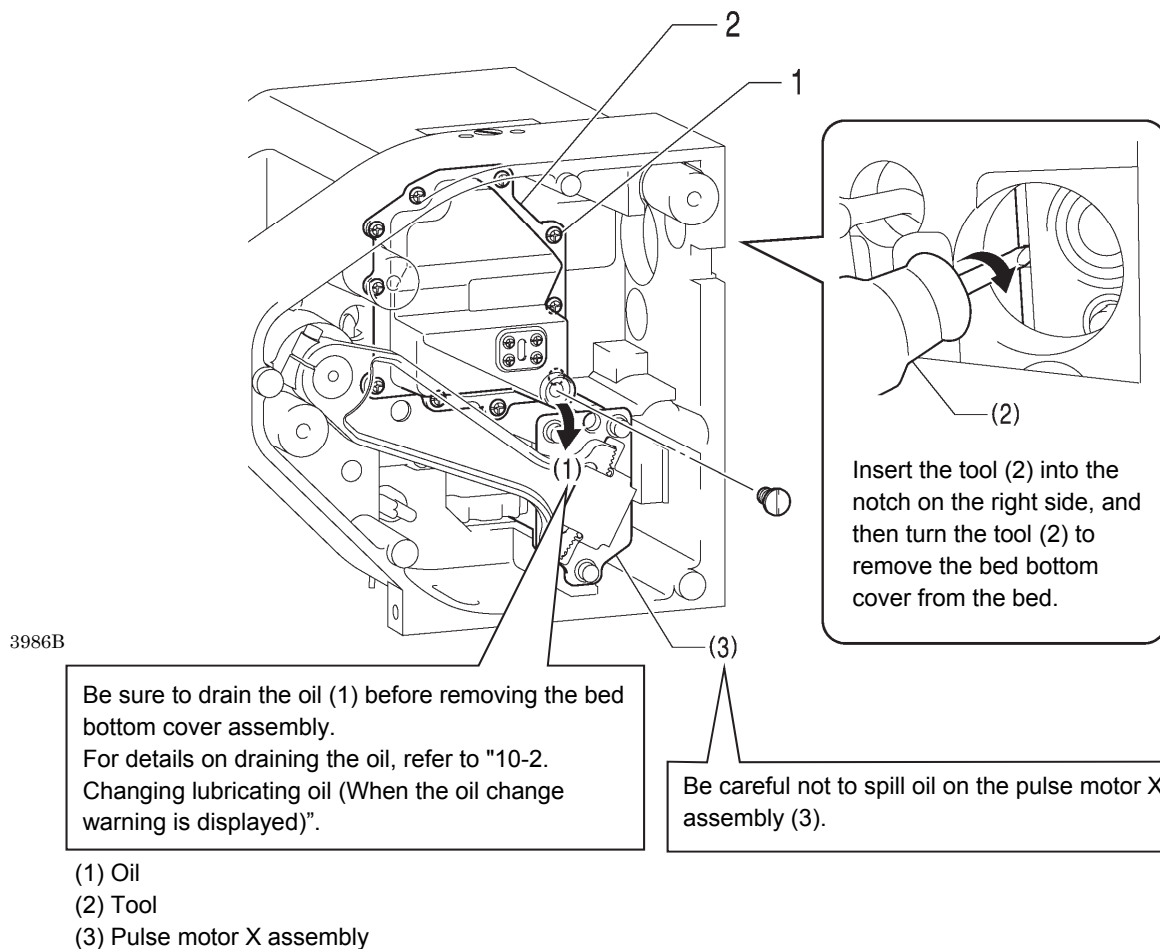
7-5. Thread nipper mechanism (KE-430HX-03, -0K, -01 specifications only)



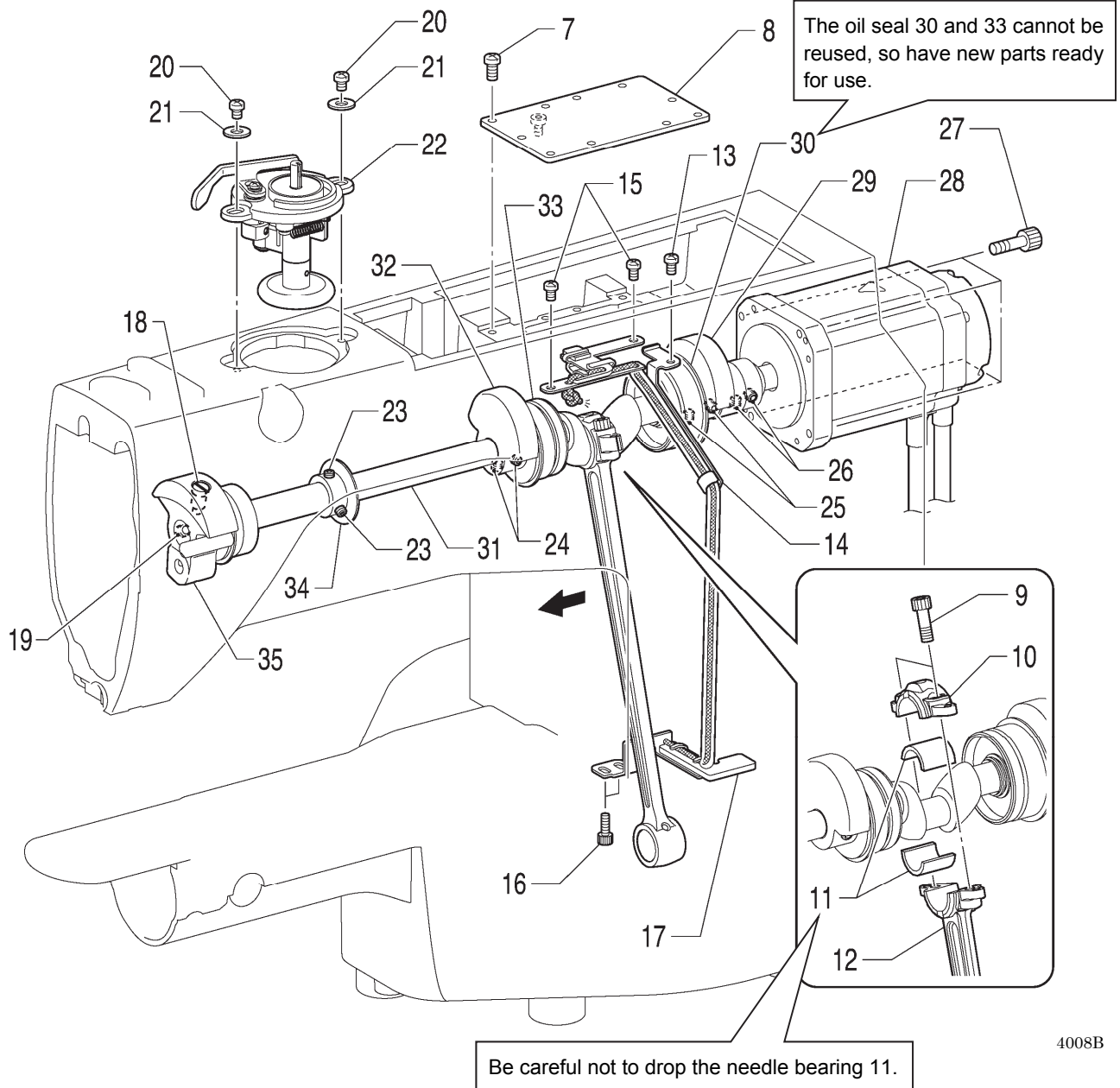
4002B

- | | | |
|---------------------------------------|--|---------------------------|
| 1. Thread nipper driving lever spring | 7. Thread nipper connecting plate R | 13. Shoulder screw 6-0.9 |
| 2. Link shoulder screw | 8. Thread nipper connecting plate F assembly | 14. Thread nipper U |
| 3. Washer, wave spring | 9. Thread nipper shoulder screw | 15. Spring extension |
| 4. Washer | 10. Thread nipper D assembly | 16. Screws [2 pcs] |
| 5. Thread nipper lever | 11. Thread nipper collar | 17. Plain washers [2 pcs] |
| 6. Bolts [2 pcs] | 12. Thread nipper spacer | 18. Spring hook plate |

7-6. Upper shaft mechanism



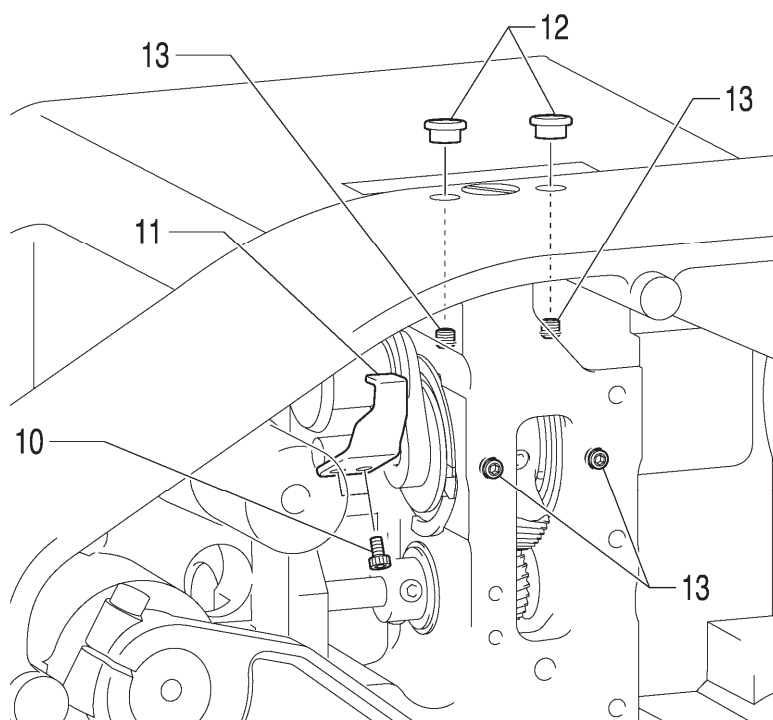
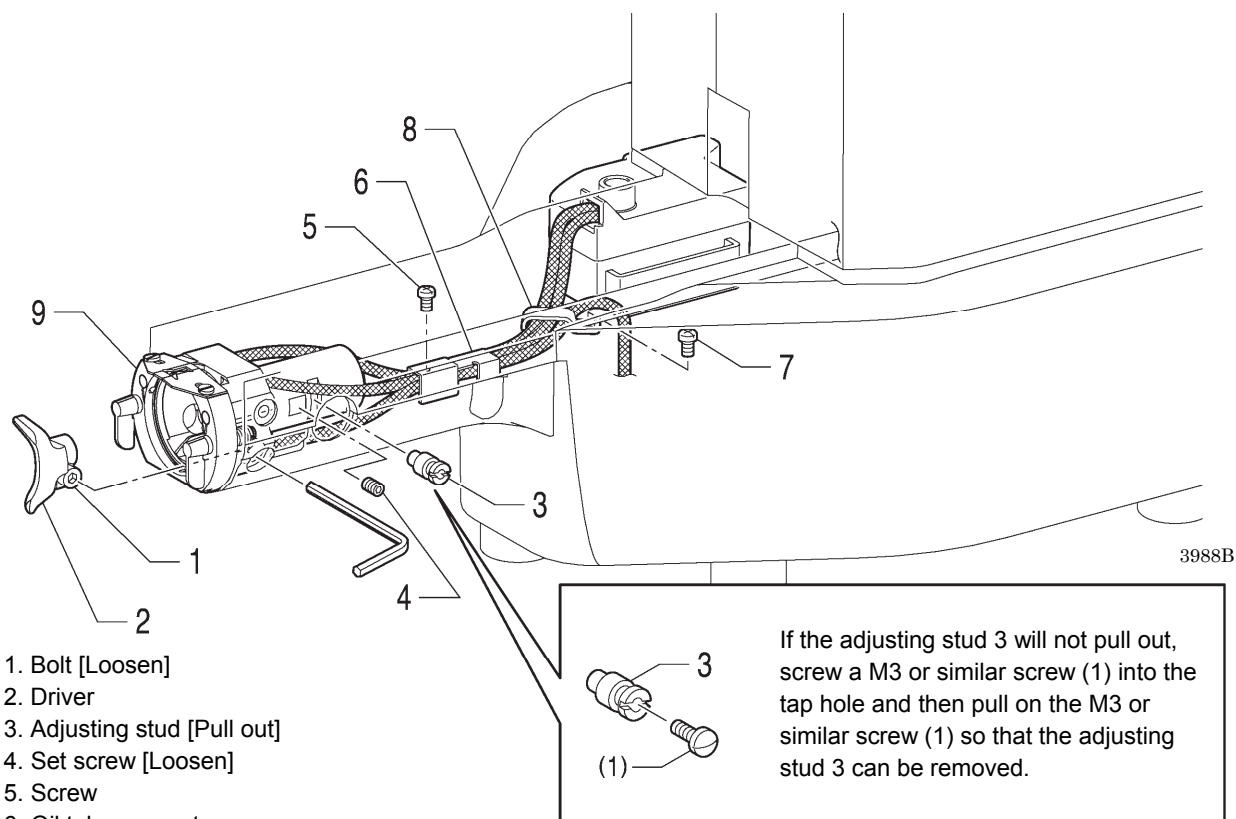
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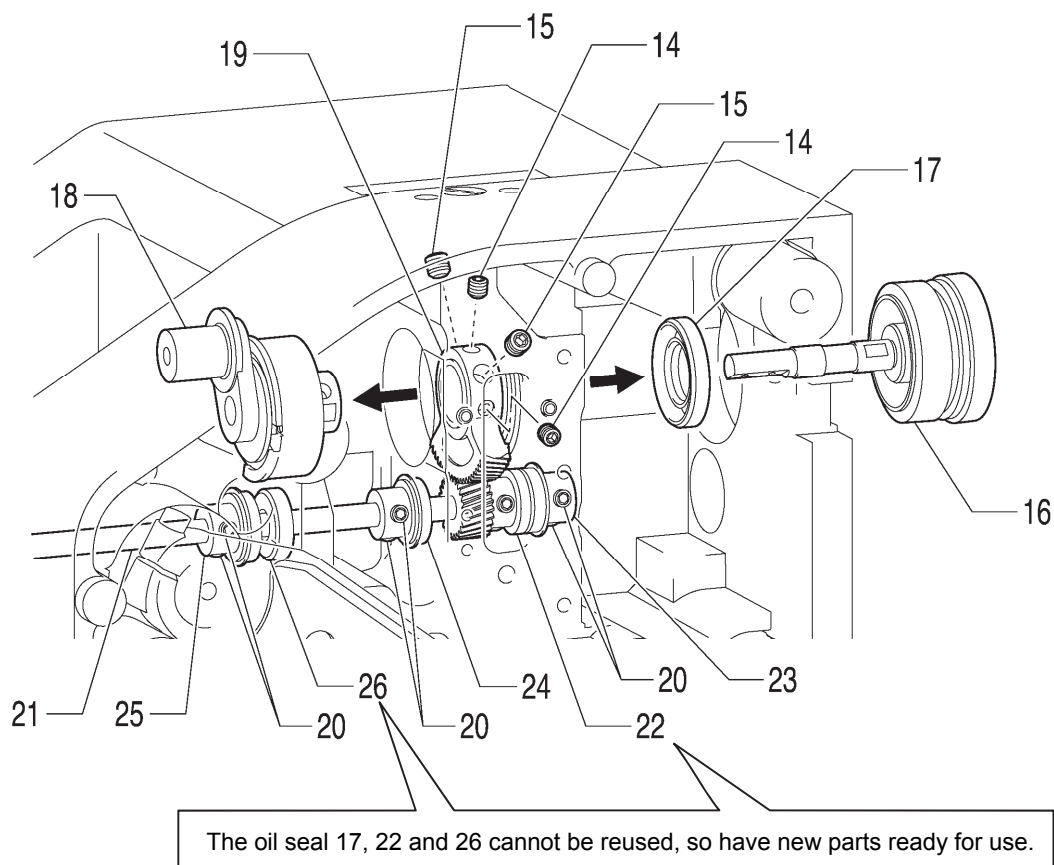
4008B

- | | |
|--|---|
| 7. Screws [10 pcs] | 22. Bobbin winder unit |
| 8. Crank cover assembly | 23. Set screws [2 pcs: Loosen] |
| 9. Bolts [2 pcs] | 24. Set screws [2 pcs: Loosen] |
| 10. Crank rod [Upper half] | 25. Set screws [2 pcs: Loosen] |
| 11. Needle bearing | 26. Set screws [2 pcs: Loosen] |
| 12. Crank rod [Lower half: Shift to the left and pull out] | 27. Bolts [4 pcs] |
| 13. Screw | 28. Motor assembly |
| 14. Tube holder | 29. Joint assembly |
| 15. Screws [2 pcs] | 30. Oil seal |
| 16. Bolts [2 pcs] | 31. Upper shaft [Tap out toward the rear] |
| 17. Crank lubrication assembly | 32. Ball bearing bush M assembly |
| 18. Screw | 33. Oil seal |
| 19. Set screw [Loosen] | 34. Pulley gear R |
| 20. Screws [2 pcs] | 35. Thread take-up crank assembly |
| 21. Plain washers [2 pcs] | |

7-7. Lower shaft mechanism



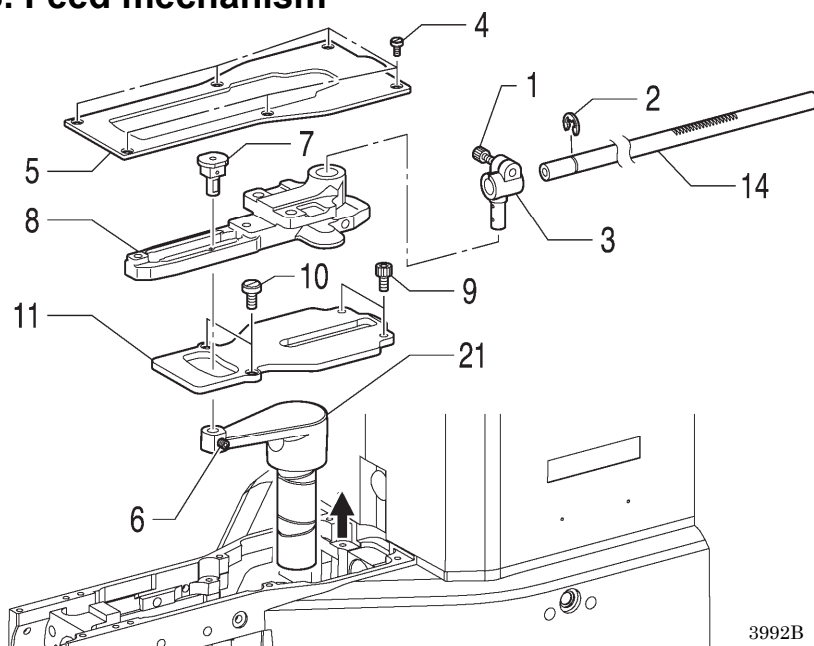
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3990B

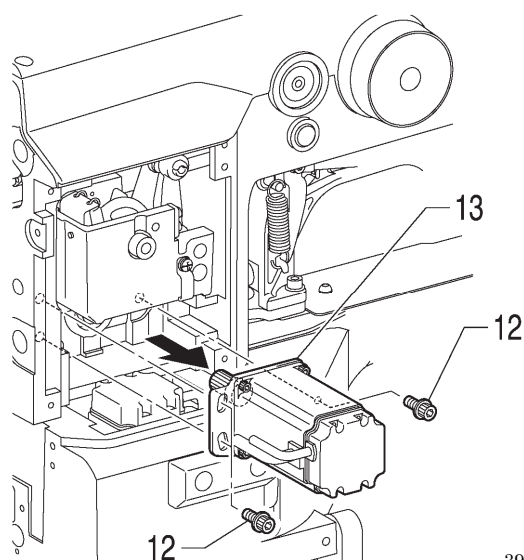
- | | |
|--|--|
| 14. Set screws [2 pcs] | 21. Lower shaft assembly |
| 15. Set screws [2 pcs] | 22. Oil seal |
| 16. Rock gear shaft R assembly [Pull backward] | 23. Bearing bush assembly [Tap out the oil seal 22 and the bearing bush together toward the rear.] |
| 17. Oil seal | 24. Bearing bush assembly |
| 18. Rock gear shaft F assembly [Pull forward] | 25. Bearing bush assembly |
| 19. Rock gear | 26. Oil seal |
| 20. Set screws [6 pcs: Loosen] | |

7-8. Feed mechanism

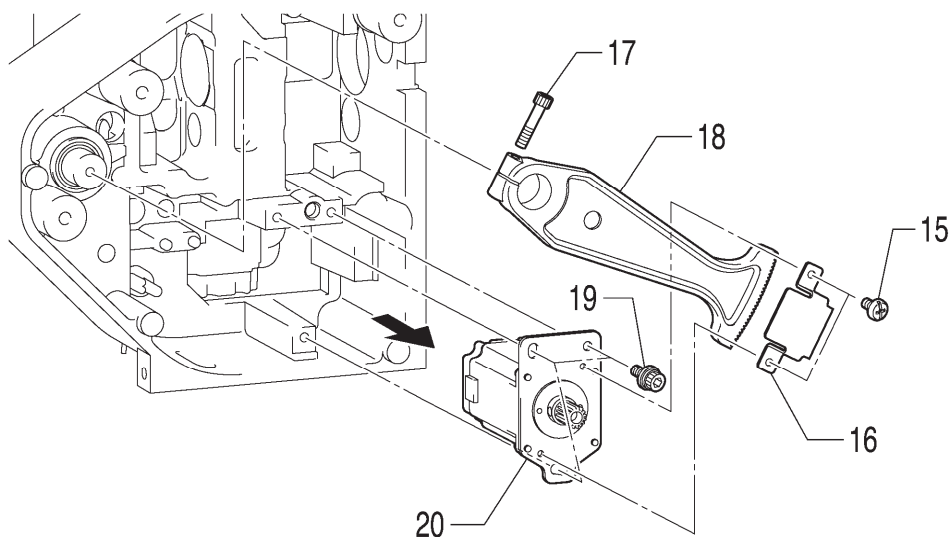


3992B

1. Bolt [Loosen]
2. Retaining ring E
3. Y-feed rack shaft clamp [Pull upward]
4. Screws [6 pcs]
5. Feed bar cover
6. Set screw [Loosen]
7. T-length regulator block assembly
8. Feed bar sub assembly
9. Bolts [2 pcs]
10. Screws [2 pcs]
11. Feed bar support plate
12. Socket bolts (S/P washer) [3 pcs]
13. Y-motor setting plate [Pulse motor Y assembly]
14. Y-feed rack shaft
15. Screws [2 pcs]
16. X-feed gear cover
17. Bolt [Loosen]
18. X-feed gear
19. Socket bolts (S/P washer) [3 pcs]
20. X-motor setting plate [Pulse motor X assembly]
21. X-feed lever [Pull upward]

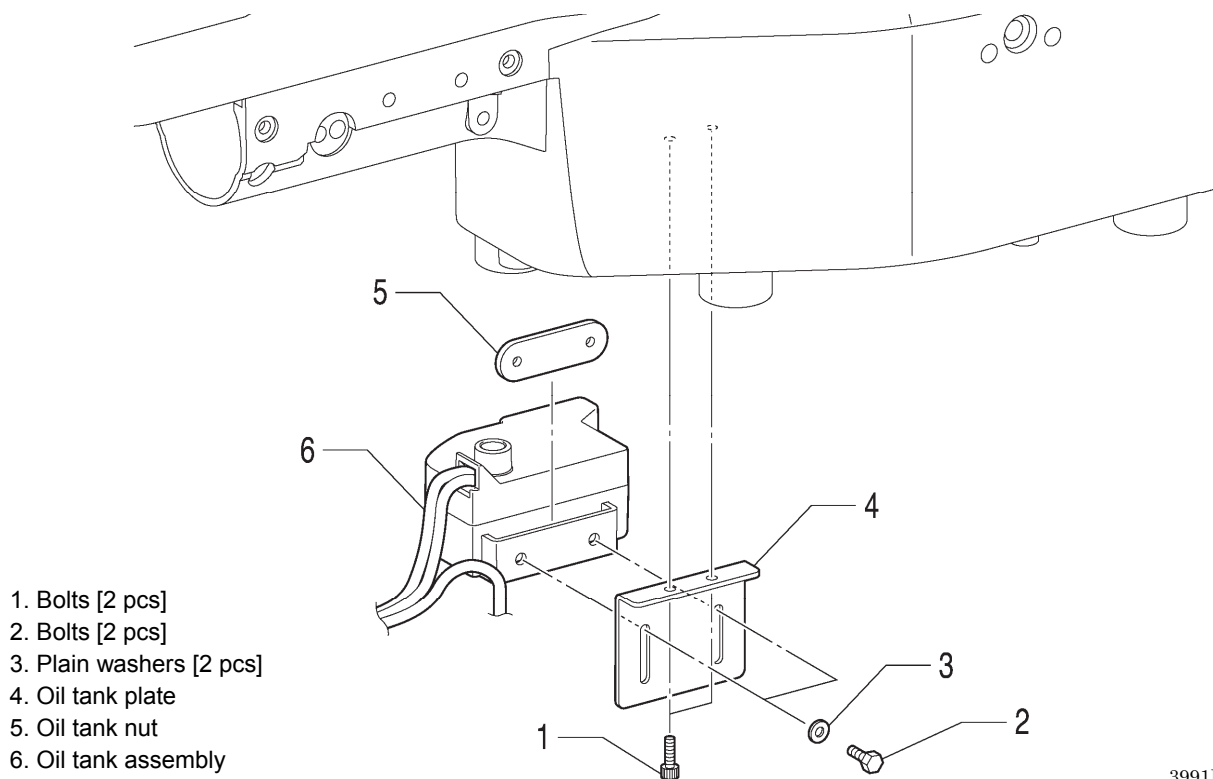


3993B

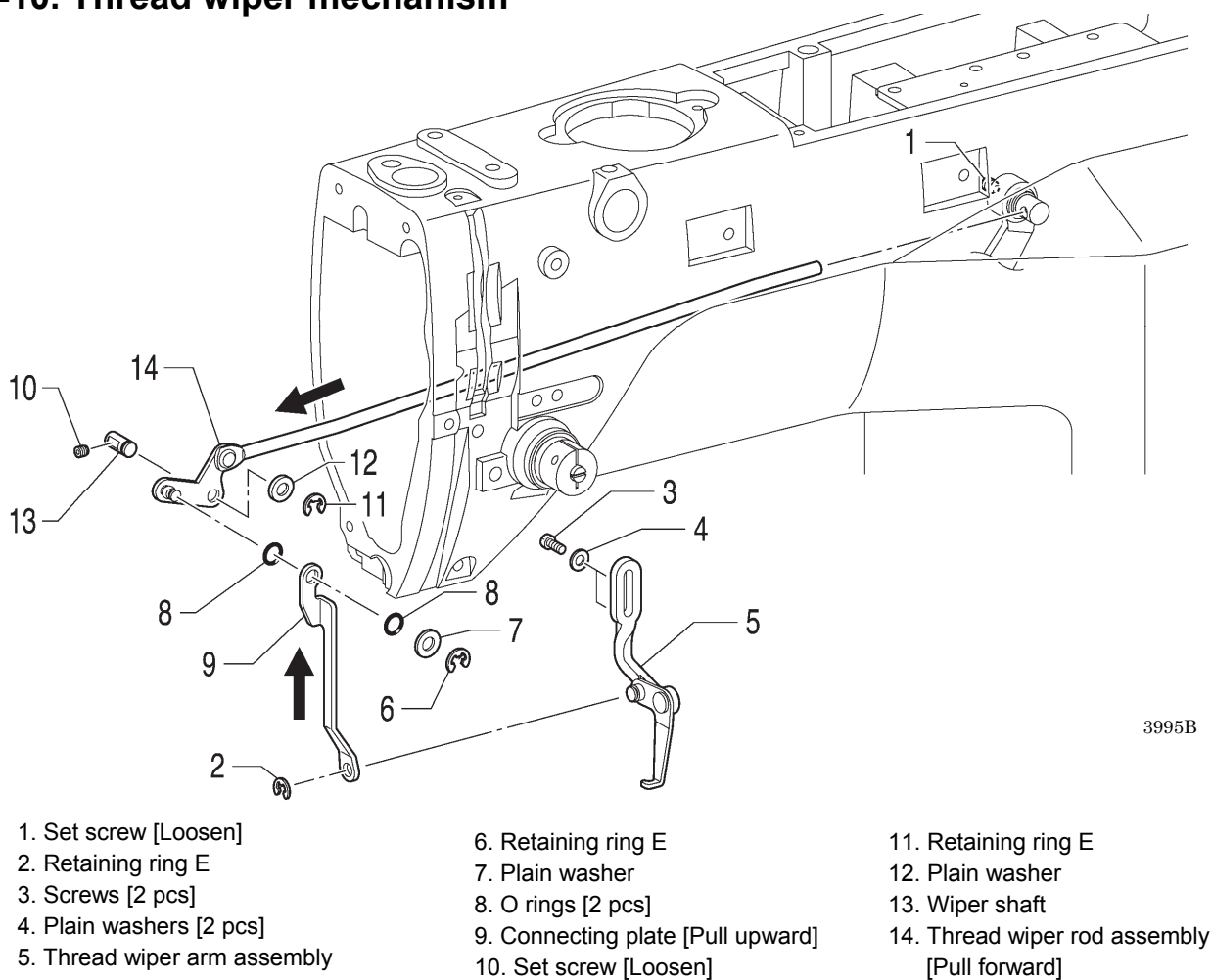


3994B

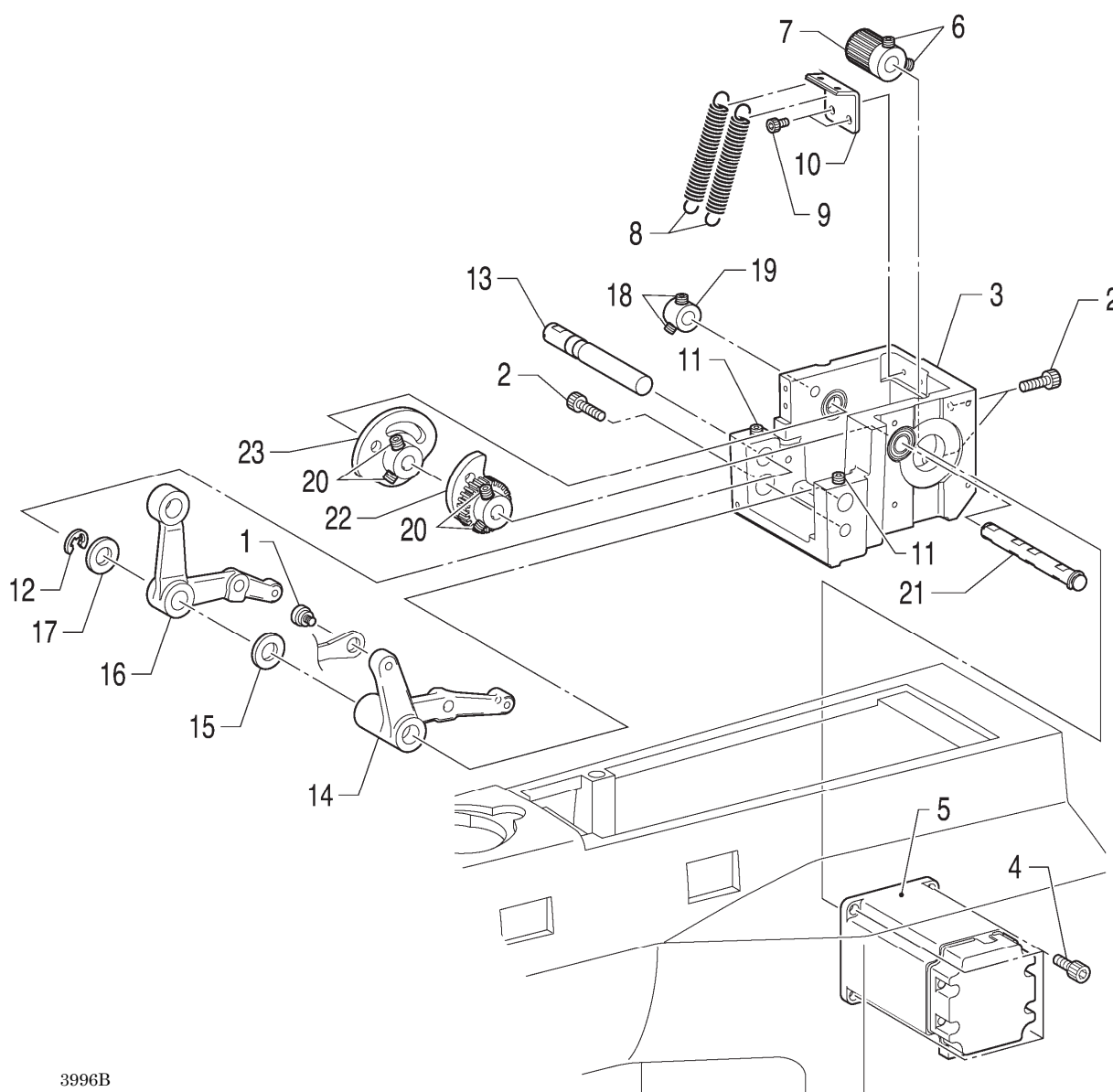
7-9. Lubrication



7-10. Thread wiper mechanism



7-11. Work clamp lifter mechanism

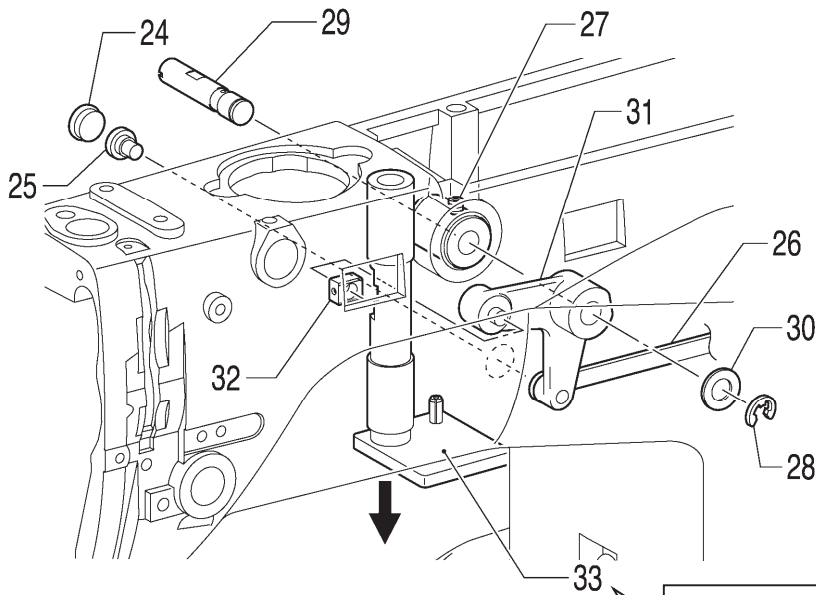


3996B

- | | | |
|-------------------------------|--------------------------------|--------------------------------|
| 1. Link shoulder screw | 9. Bolts [2 pcs] | 17. Plain washer |
| 2. Bolts [3 pcs] | 10. Spring hook | 18. Set screws [2 pcs: Loosen] |
| 3. Motor bracket assembly | 11. Set screws [2 pcs: Loosen] | 19. Set collar |
| 4. Bolts [4 pcs] | 12. Retaining ring E | 20. Set screws [4 pcs: Loosen] |
| 5. Pulse motor P assembly | 13. Work clamp cam lever shaft | 21. Work clamp cam shaft |
| 6. Set screws [2 pcs: Loosen] | 14. Work clamp driving lever | 22. Cam gear |
| 7. Gear 16 | 15. Plain washer | 23. Thread wiper cam |
| 8. Springs [2 pcs] | 16. Thread wiper driving lever | |

(Continued on next page.)

KE-430HX/KE-430HS



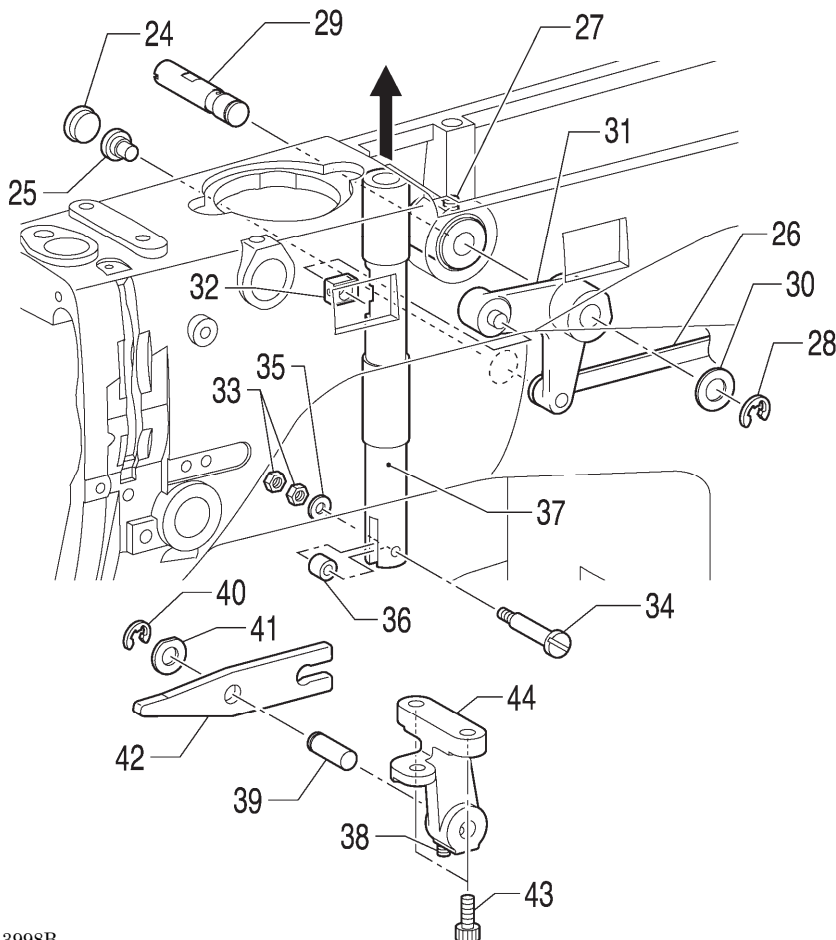
- 24. Rubber cap
- 25. Link shoulder screw
- 26. Link
- 27. Set screw [Loosen]
- 28. Retaining ring E
- 29. Lever shaft
- 30. Plain washer
- 31. Lifter lever assembly
- 32. Needle bar guide slide block
- 33. Work clamp lifter plate assembly

3997B

NOTE:

Hold the work clamp lifter plate assembly 33 when removing the lifter lever assembly 31, otherwise the work clamp lifter plate assembly 33 may fall down.

BE-438HX/BE-438HS



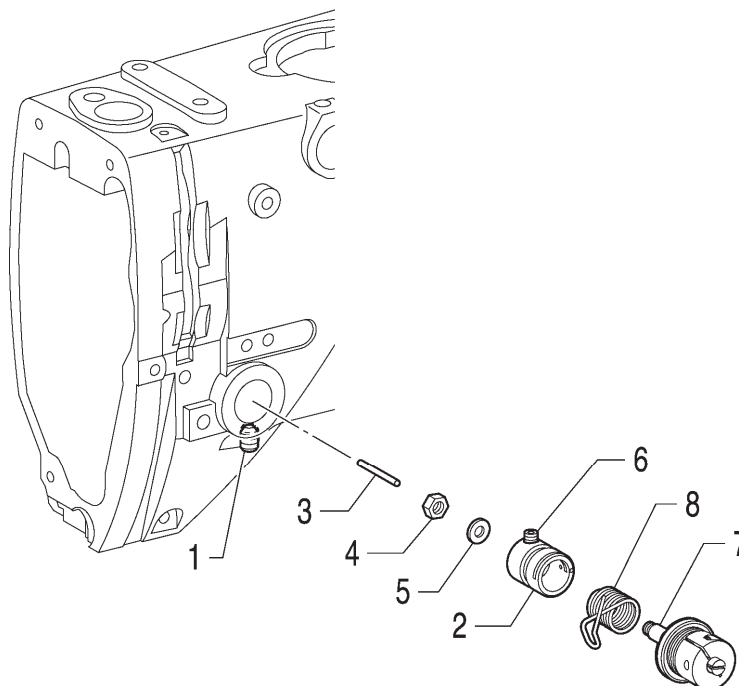
- 24. Rubber cap
- 25. Link shoulder screw
- 26. Link
- 27. Set screw [Loosen]
- 28. Retaining ring E
- 29. Lever shaft
- 30. Plain washer
- 31. Lifter lever assembly
- 32. Needle bar guide slide block
- 33. Nut [2 pcs]
- 34. Button clamp roller shaft
- 35. Plain washer
- 36. Roller
- 37. Button clamp rod [Pull upward]
- 38. Set screw [Loosen]
- 39. Button clamp lever shaft [Pull out]
- 40. Retaining ring E
- 41. Plain washer
- 42. Button clamp lever
- 43. Bolts [2 pcs]
- 44. Button clamp lever holder

3998B

7-12. Threading mechanism

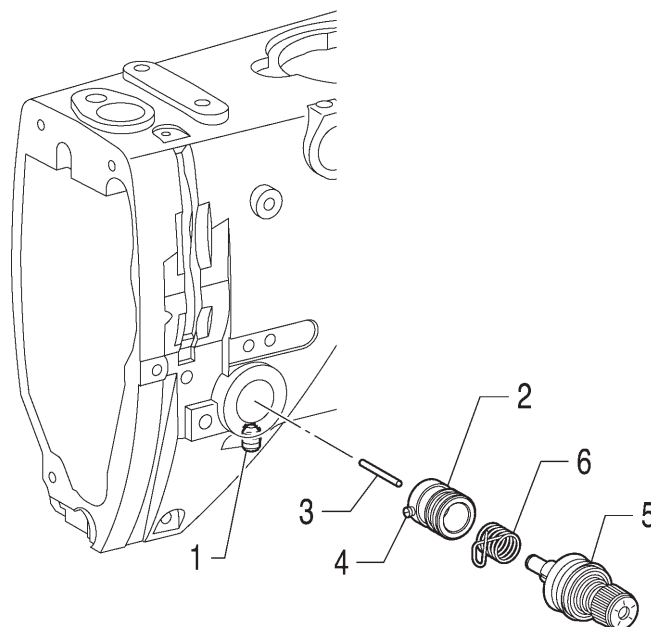
KE-430HX, BE-438HX

1. Set screw [Loosen]
2. Tension bracket [Pull out]
3. Tension release pin
4. Nut
5. Plain washer
6. Set screw [Loosen]
7. Tension stud assembly [Pull out]
8. Thread take-up spring



3999B

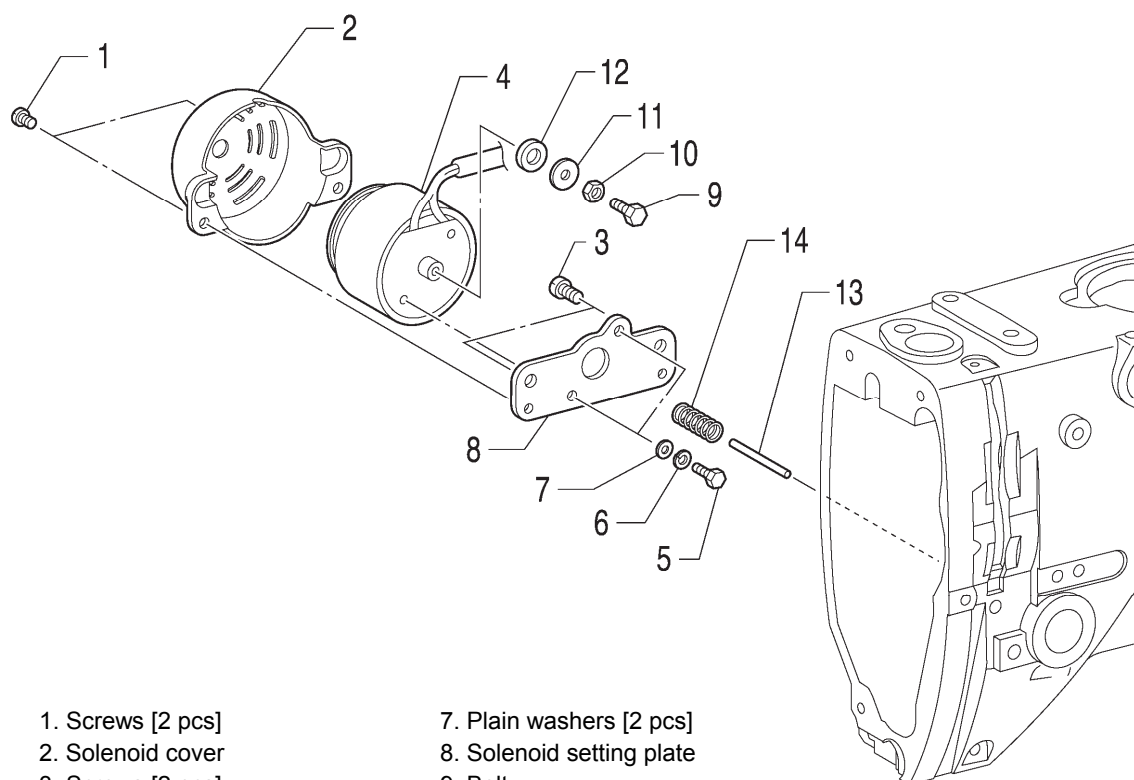
KE-430HS, BE-438HS



4000B

1. Set screw [Loosen]
2. Thread tension bracket [Pull out]
3. Tension release pin
4. Set screw [Loosen]
5. Tension stud assembly [Pull out]
6. Thread take-up spring

7-13. Tension release mechanism



1. Screws [2 pcs]

2. Solenoid cover

3. Screws [2 pcs]

4. Tension solenoid

(KE-430HX, BE-438HX)

Tension release solenoid

(KE-430HS, BE-438HS)

5. Bolts [2 pcs]

6. Spring washers [2 pcs]

7. Plain washers [2 pcs]

8. Solenoid setting plate

9. Bolt

10. Nut

11. Plain washer

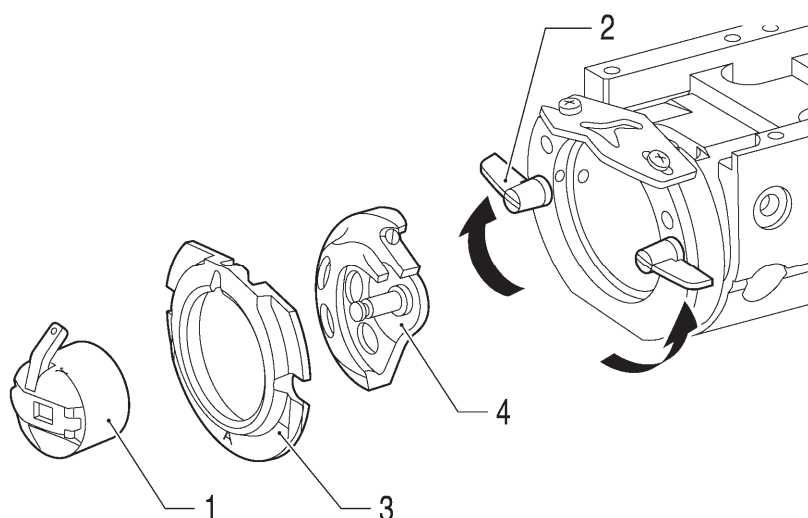
12. Solenoid cushion

13. Tension release bar

14. Tension release spring (KE-430HS, BE-438HS)

4001B

7-14. Hook mechanism



4004B

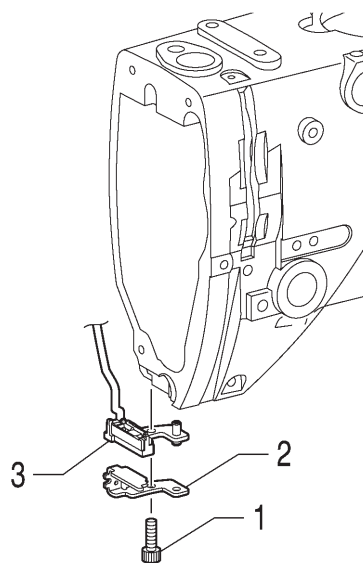
1. Bobbin case assembly

2. Shuttle race base setting claw [Open to right and left]

3. Shuttle race base

4. Shuttle hook

7-15. LED mechanism



- 1. Bolt
- 2. LED setting plate
- 3. LED assembly

4011B

8. ASSEMBLY

! CAUTION



Disassembly should only be carried out by a qualified technician.



Turn off the power switch before carrying out disassembly, otherwise the machine may operate if the foot switch is depressed by mistake, which may result in injury.



Be sure to wear protective goggles and gloves when handling the lubricating oil and grease, so that they do not get into your eyes or onto your skin, otherwise inflammation can result.

Furthermore, do not drink the oil or eat the grease under any circumstances, as they can cause vomiting and diarrhea.

Keep the oil out of the reach of children.



When replacing parts and installing optional accessories, be sure to use only genuine Brother parts.

Brother will not be held responsible for any accidents or problems resulting from the use of non-genuine parts.



If any safety devices have been removed, be absolutely sure to re-install them to their original positions and check that they operate correctly before using the machine.



To prevent accidents and problems, do not modify the machine yourself.

Brother will not be held responsible for any accidents or problems resulting from modifications made to the machine.

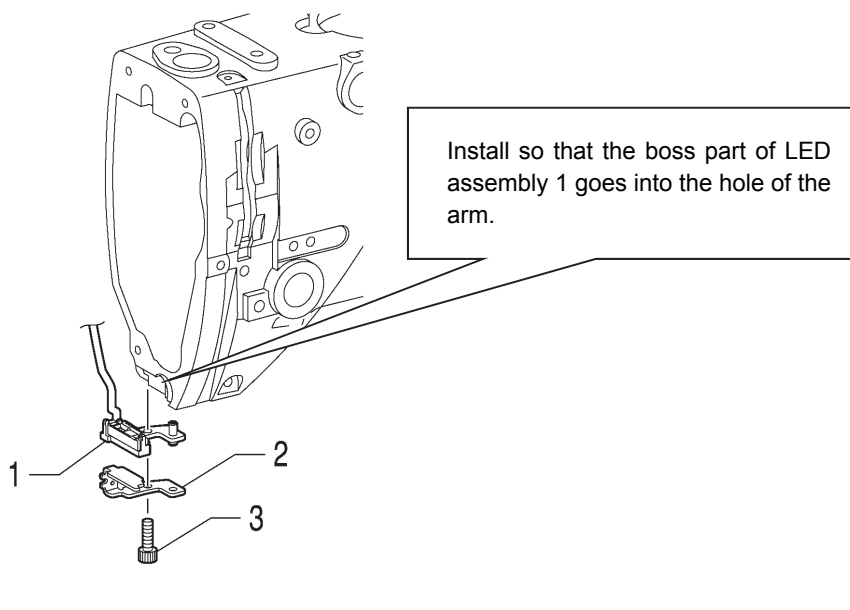
Assemble each part in order of the numbers.

* (number) indicates part names only. (The numbers do not indicate the order of assembly.)

* Apply grease to the required places when reassembling the parts and once every two years.

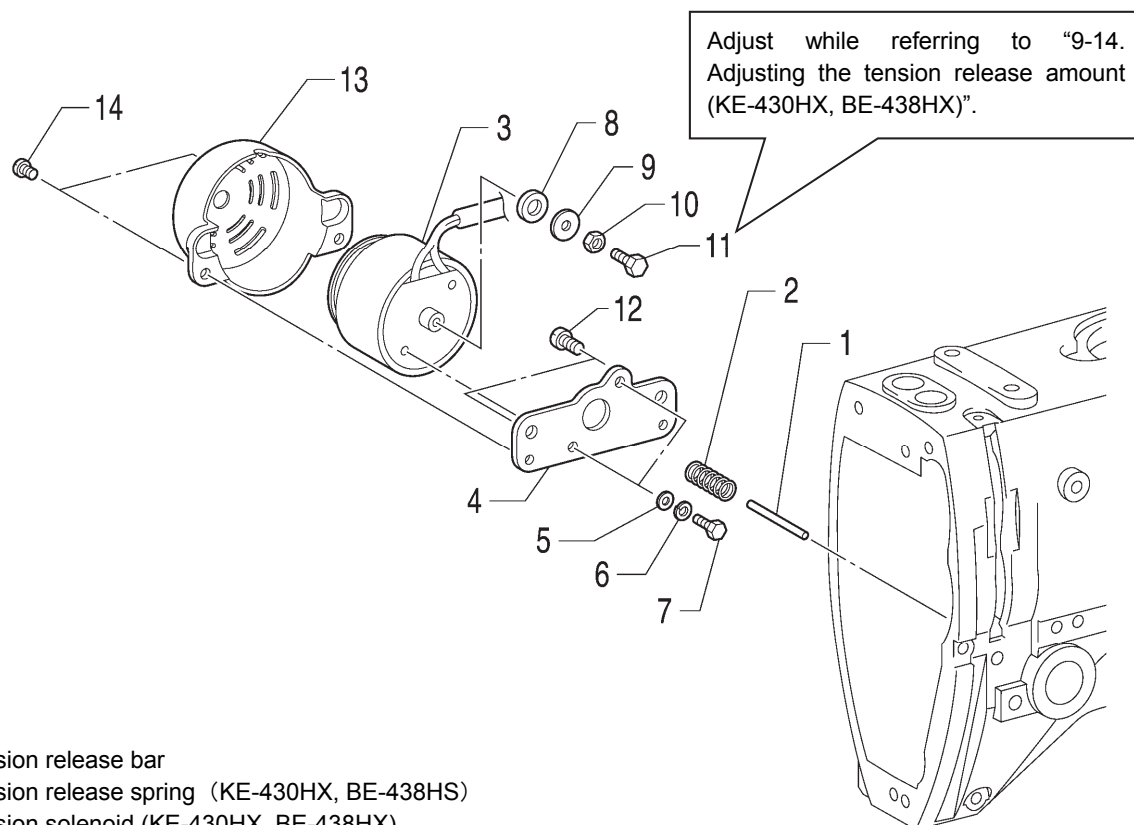
Unless otherwise specified, use Brother-specified "Grease unit (other than for Europe: SA8771-101, for Europe: SB6659-101)".

8-1. LED mechanism



4014B

8-2. Tension release mechanism

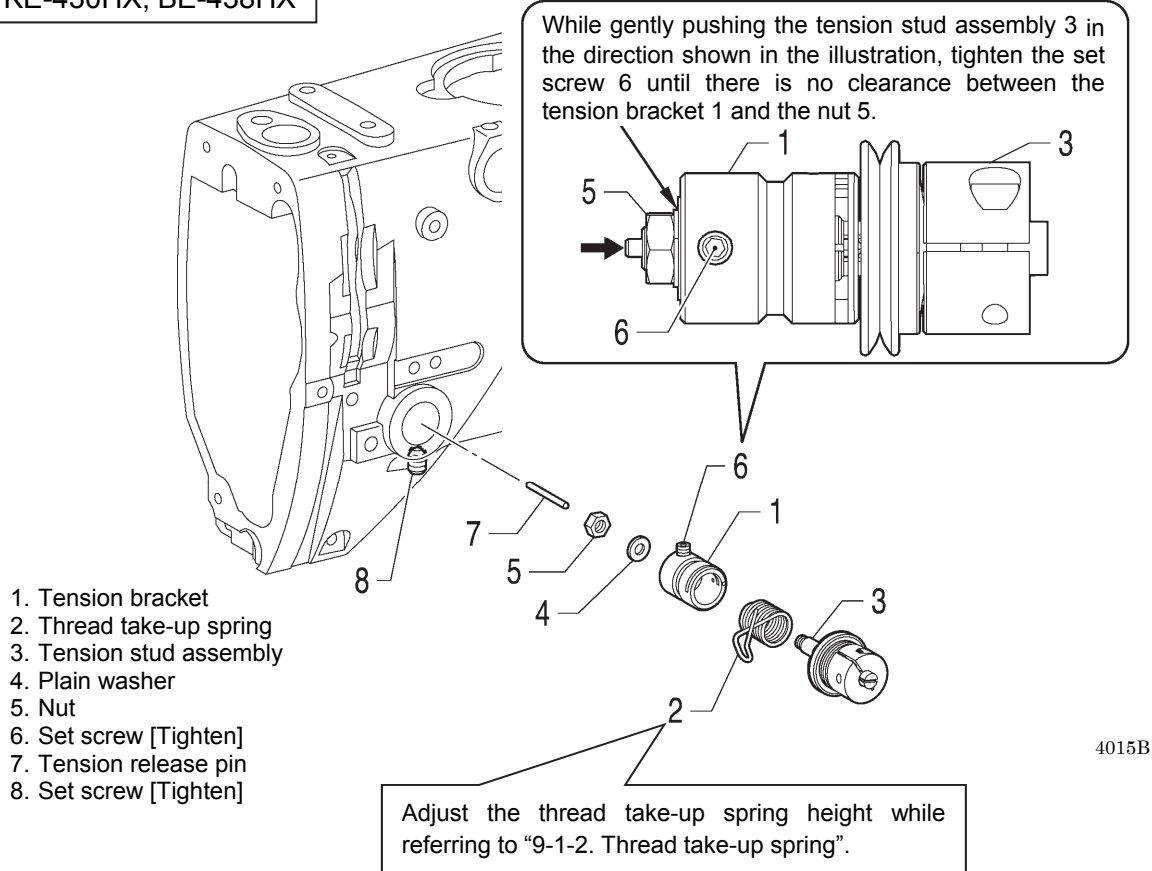


1. Tension release bar
2. Tension release spring (KE-430HX, BE-438HS)
3. Tension solenoid (KE-430HX, BE-438HX)
Tension release solenoid (KE-430HS, BE-438HS)
4. Solenoid setting plate
5. Plain washers [2 pcs]
6. Spring washers [2 pcs]
7. Bolts [2 pcs]
8. Solenoid cushion
9. Plain washer
10. Nut
11. Bolt
12. Screws [2 pcs]
13. Solenoid cover
14. Screws [2 pcs]

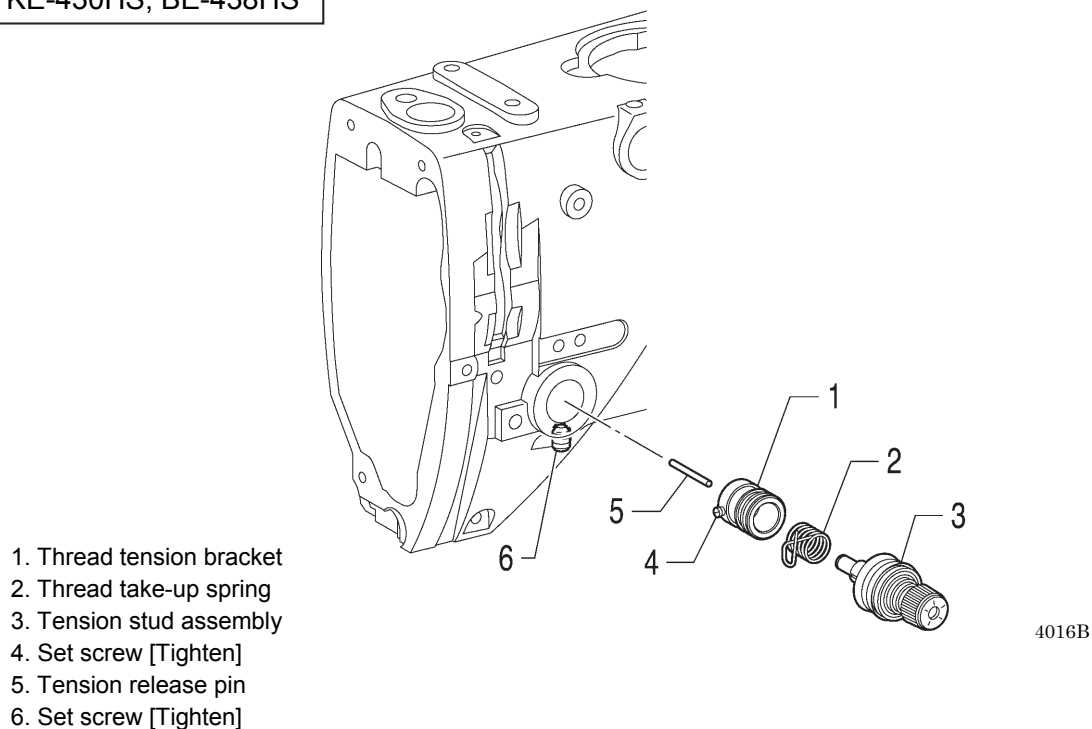
4013B

8-3. Threading mechanism

KE-430HX, BE-438HX

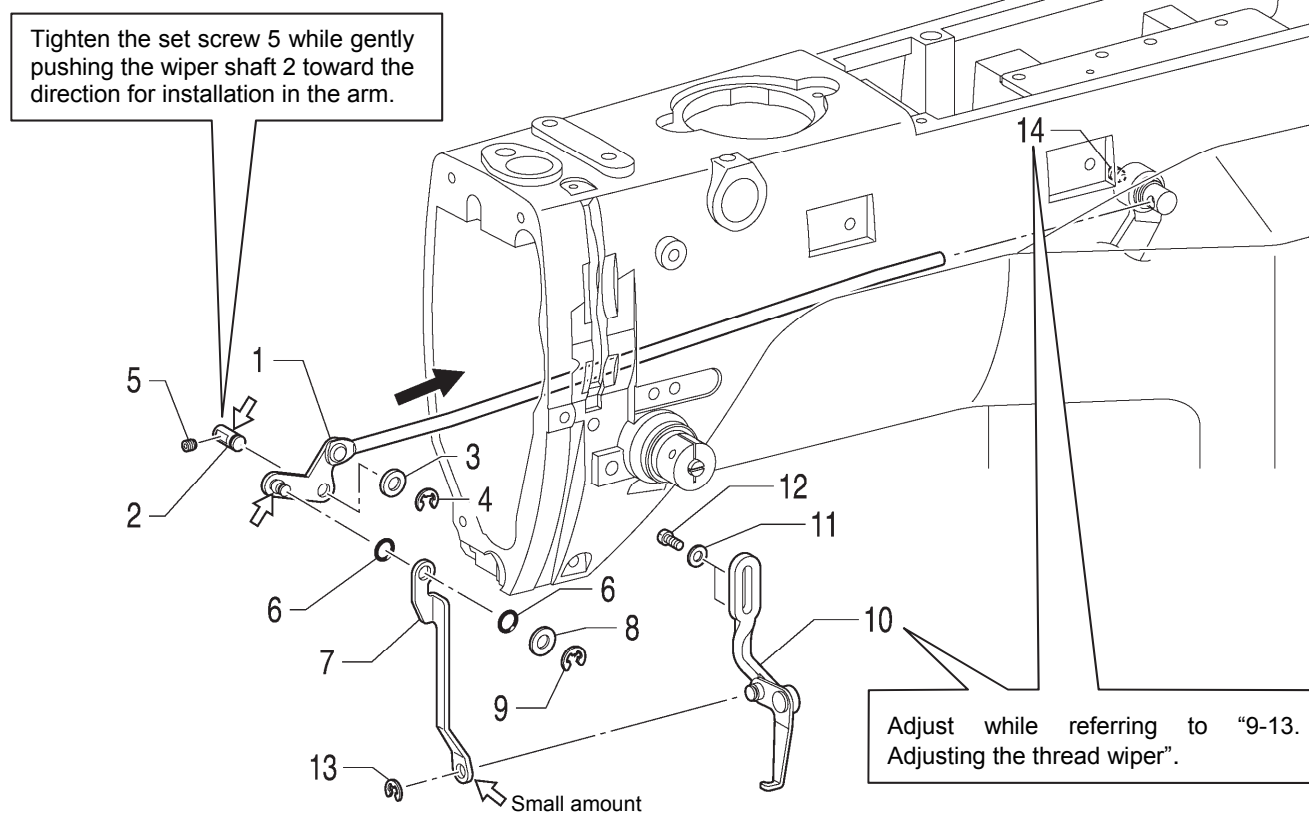


KE-430HS, BE-438HS



8-4. Thread wiper mechanism

Apply grease to the portions indicated by the white arrows.



4017B

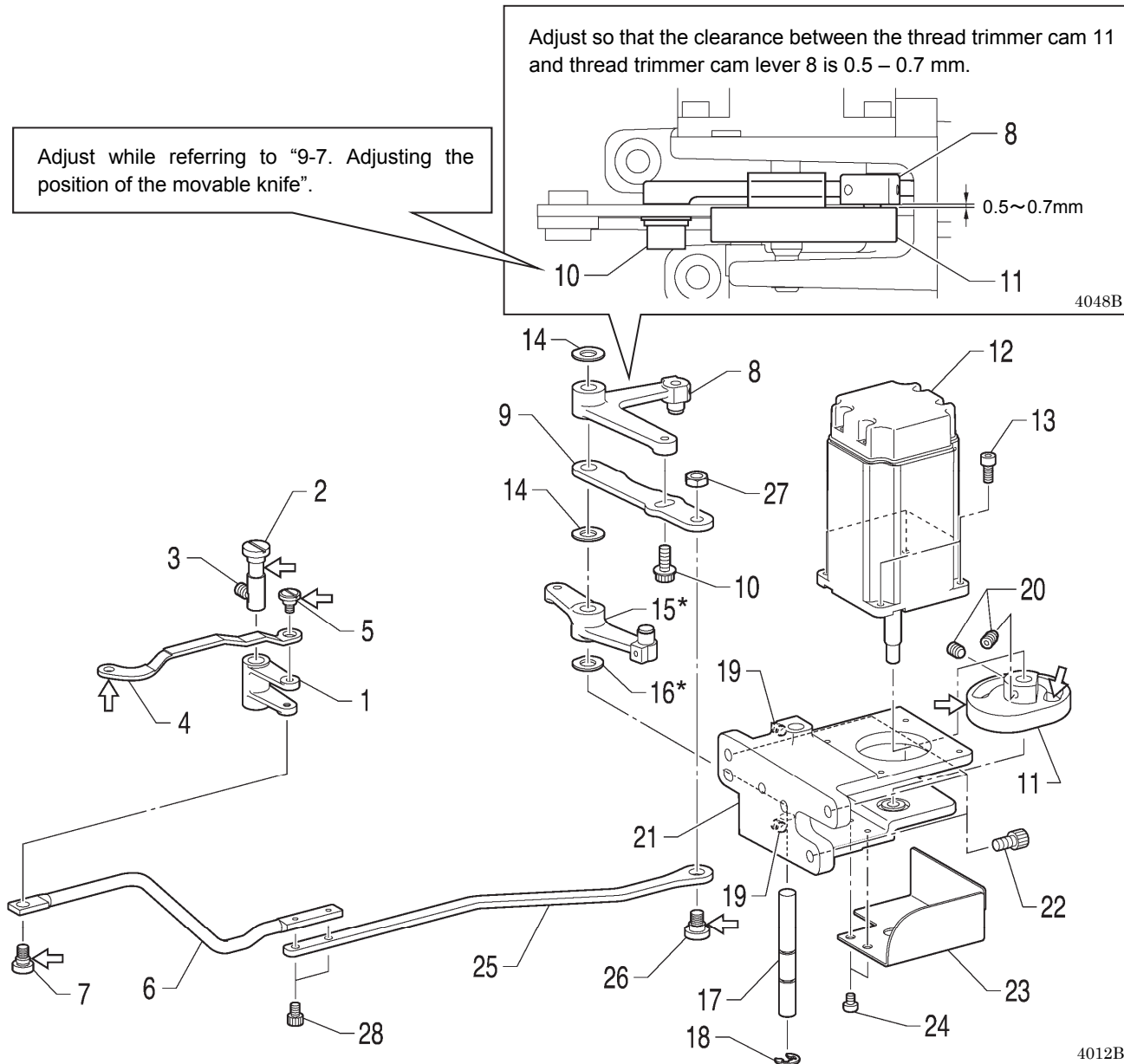
- 1. Thread wiper rod assembly [Insert from the front]
- 2. Wiper shaft
- 3. Plain washer
- 4. Retaining ring E
- 5. Set screw [Tighten]

- 6. O rings [2 pcs]
- 7. Connecting plate
- 8. Plain washer
- 9. Retaining ring E
- 10. Thread wiper arm assembly

- 11. Plain washers [2 pcs]
- 12. Screws [2 pcs]
- 13. Retaining ring E
- 14. Set screw [Tighten]

8-5. Thread trimmer mechanism (1)

Apply grease to the portions indicated by the white arrows.



1. Movable knife lever
2. Shaft M-knife lever
3. Set screw
4. M-knife connecting plate
5. Shoulder screw 6-1.8
6. Thread trimmer rod
7. Shoulder screw 6.5-3
8. Thread trimmer cam lever
9. Thread trimmer lever
10. Socket bolt (S/P washer)

11. Thread trimmer cam
12. Pulse motor Y assembly
13. Bolts [4 pcs]
14. Washers [2 pcs]
- * 15. Thread nipper lever
- * 16. Washer
17. Thread trimmer cam lever shaft
18. Retaining ring E
19. Set screws [2 pcs]
20. Set screws [2 pcs]

21. Motor bracket rear assembly
22. Bolts [3 pcs]
23. Motor bracket rear cover
24. Screws [2 pcs]
25. Thread trimmer link
26. Link shoulder screw
27. Nut
28. Bolts [2 pcs]

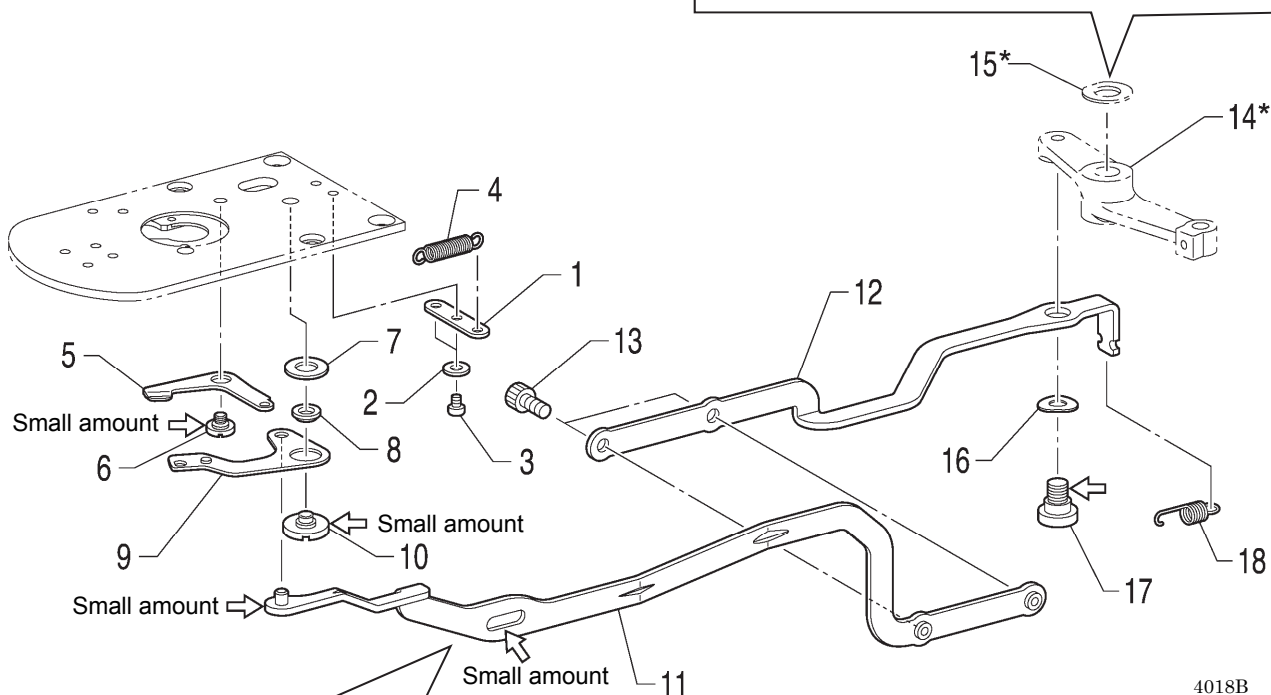
For the KE-430HX-03, -0K, -01, install the parts marked with * as the thread nipper parts.

8-6. Thread nipper mechanism (KE-430HX-03, -0K, -01 specifications only)

Apply grease to the portions indicated by the white arrows.

Adjust while referring to “9-21. Adjusting the thread nipper motor position”.

The TN-driving lever 14 and the washer 15 which are marked with * should be installed at the time of installation of the thread trimmer mechanism.



4018B

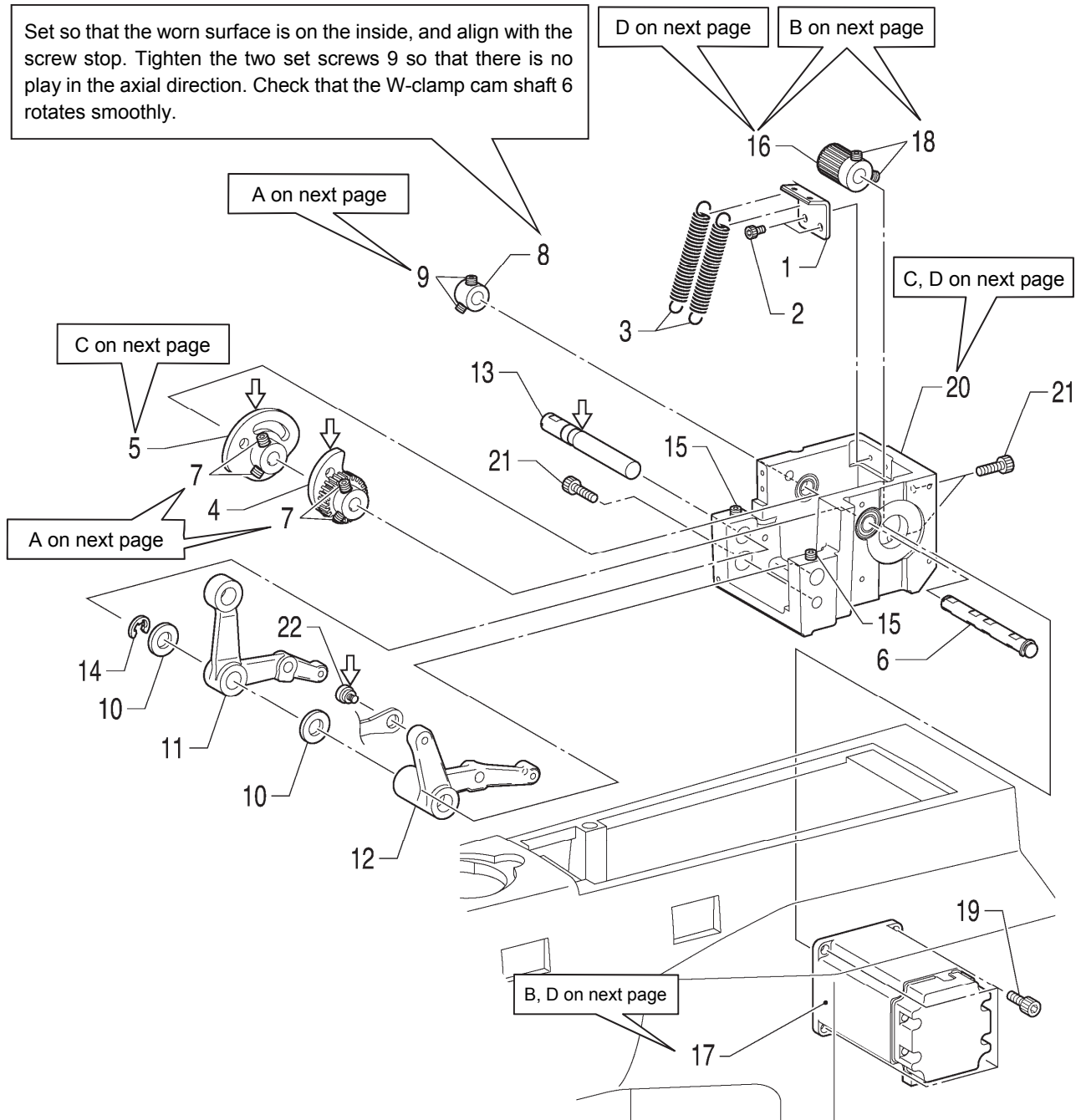
Install so that the pin of the arm goes into the open part of the slot in the TN-connecting plate F assembly 11.

- | | | |
|--------------------------|------------------------------------|-----------------------------|
| 1. Spring hook plate | 7. Thread nipper spacer | 13. Bolts [2 pcs] |
| 2. Plain washers [2 pcs] | 8. Thread nipper collar | *14. TN-driving lever |
| 3. Screws [2 pcs] | 9. Thread nipper D assembly | *15. Washer |
| 4. Spring extension | 10. Thread nipper shoulder screw | 16. Washer, wave spring |
| 5. Thread nipper U | 11. TN-connecting plate F assembly | 17. Link shoulder screw |
| 6. Shoulder screw, 6-0.9 | 12. TN-connecting plate R | 18. TN-driving lever spring |

8-7. Work clamp lifter mechanism

Apply grease to the portions indicated by the white arrows.

* Refer to the next page for the points to note during reassembly.



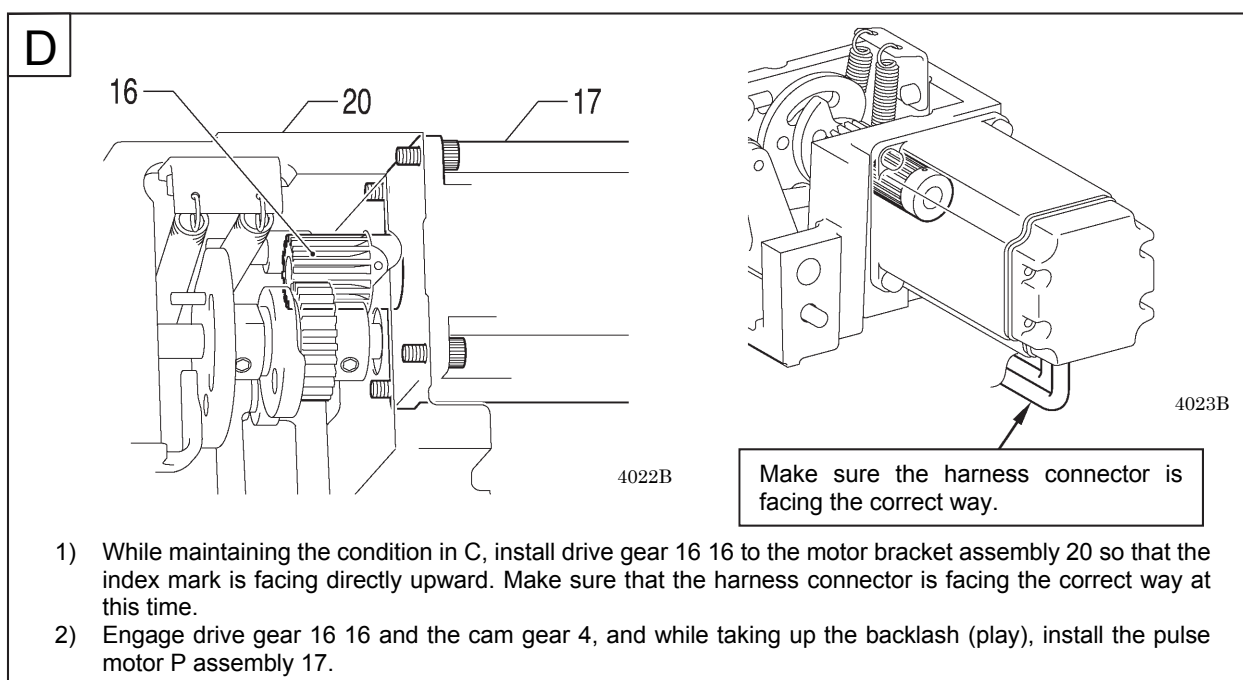
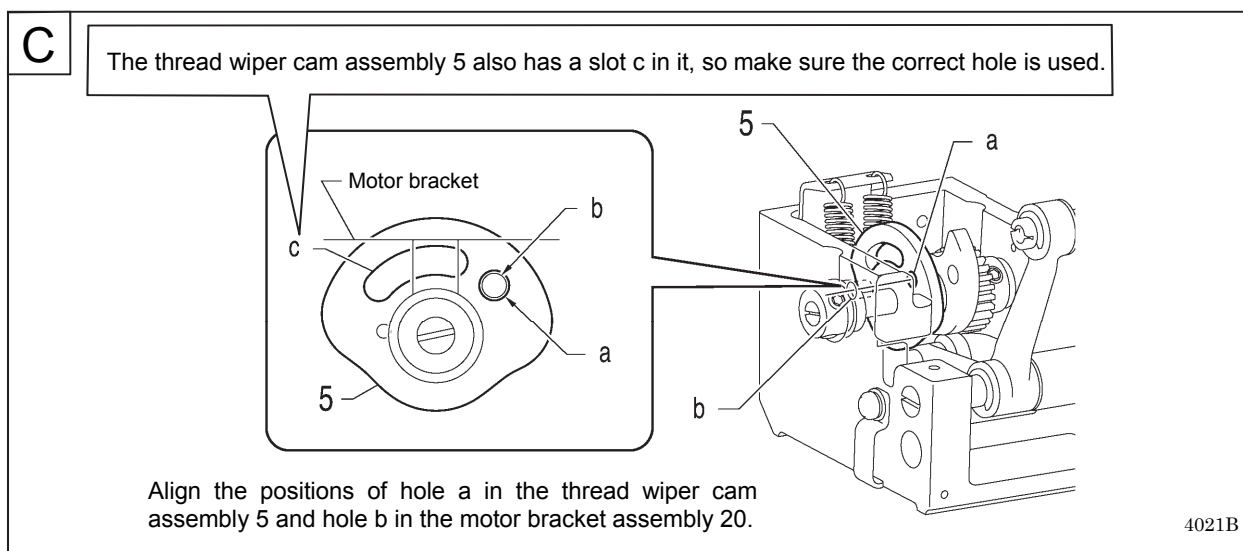
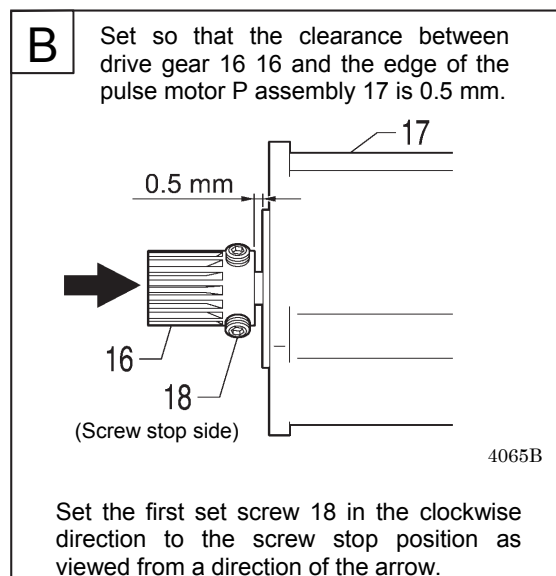
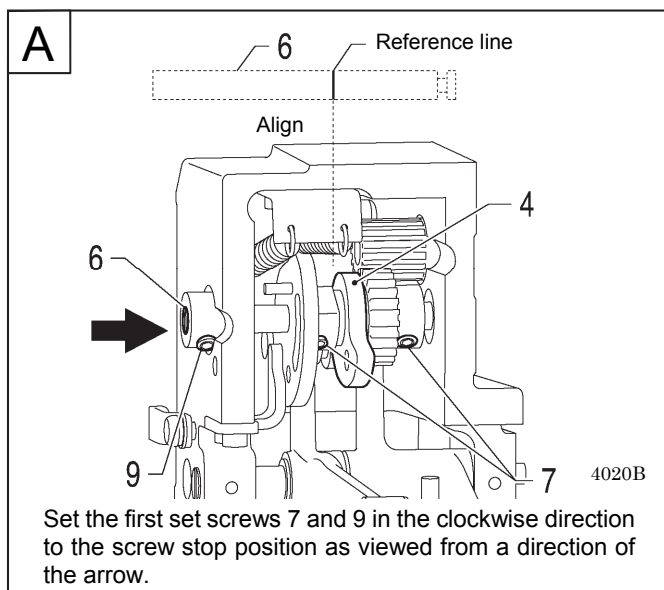
4019B

- 1. Spring hook
- 2. Bolts [2 pcs]
- 3. Springs [2 pcs]
- 4. Cam gear
- 5. Thread wiper cam
- 6. W-clamp cam shaft
- 7. Set screws [4 pcs: Tighten]
- 8. Set screw collar

- 9. Set screws [2 pcs: Tighten]
- 10. Plain washers [2 pcs]
- 11. Thread wiper driving lever
- 12. Drive lever w-clamp lifter
- 13. Lever shaft
- 14. Retaining ring E
- 15. Set screws [2 pcs: Tighten]
- 16. Gear 16

- 17. Pulse motor P assembly
- 18. Set screws [2 pcs: Tighten]
- 19. Bolts [4 pcs]
- 20. Motor bracket assembly
- 21. Bolts [3 pcs]
- 22. Link shoulder screw

(Continued on next page.)

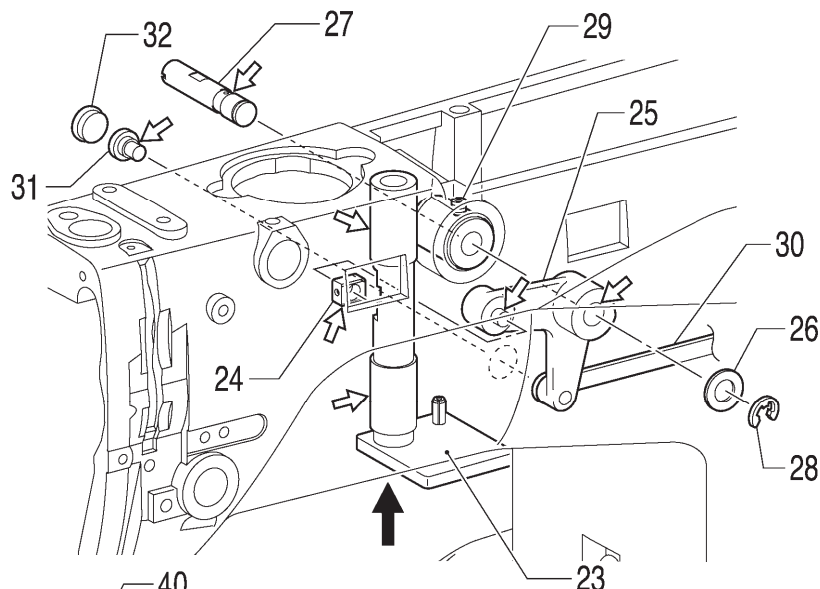


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8. ASSEMBLY

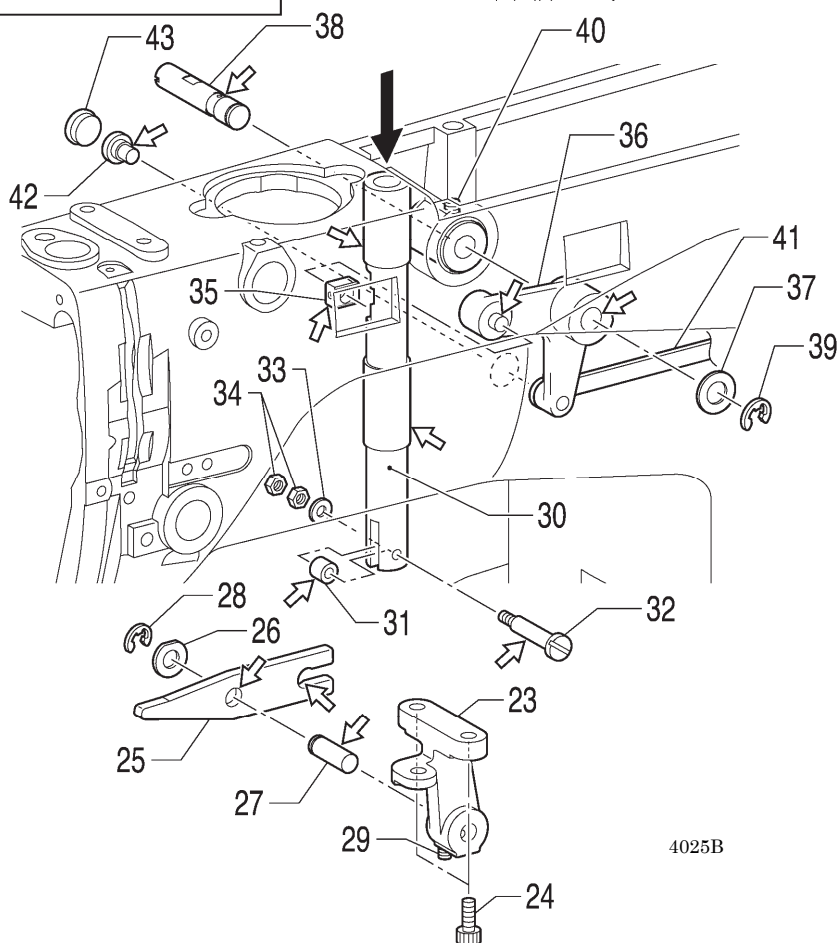
KE-430HX/KE-430HS

- 23. Work clamp lifter plate assembly
- 24. Needle bar guide slide block
- 25. Lifter lever assembly
- 26. Plain washer
- 27. Lever shaft
- 28. Retaining ring E
- 29. Set screw [Tighten]
- 30. Link
- 31. Link shoulder screw
- 32. Rubber cap



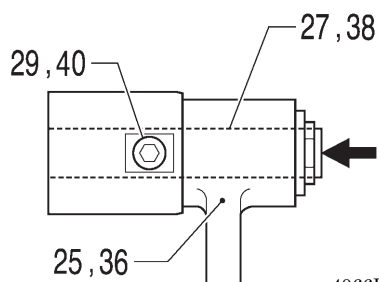
4024B

BE-438HX/BE-438HS



- 23. Button clamp lever holder
- 24. Bolts [2 pcs]
- 25. Button clamp lever
- 26. Plain washer
- 27. Button clamp lever shaft
- 28. Retaining ring E
- 29. Set screw [Tighten]
- 30. Button clamp rod
- 31. Roller
- 32. Button clamp roller shaft
- 33. Plain washer
- 34. Nuts [2 pcs]
- 35. Needle bar guide slide block
- 36. Lifter lever assembly
- 37. Plain washer
- 38. Lever shaft
- 39. Retaining ring E
- 40. Set screw [Tighten]
- 41. Link
- 42. Link shoulder screw
- 43. Rubber cap

4025B



4066B

Align the set screw 29 (430H) or 40 (438H) with the screw stop on the lever shaft 27 (430H) or 38 (438H), and then tighten the set screw 29 (430H) or 40 (438H) while gently pushing the lifter lever assembly 25 (430H) or 36 (438H) so that there is no play in it. Check that the lifter lever assembly 25 (430H) or 36 (438H) moves smoothly.

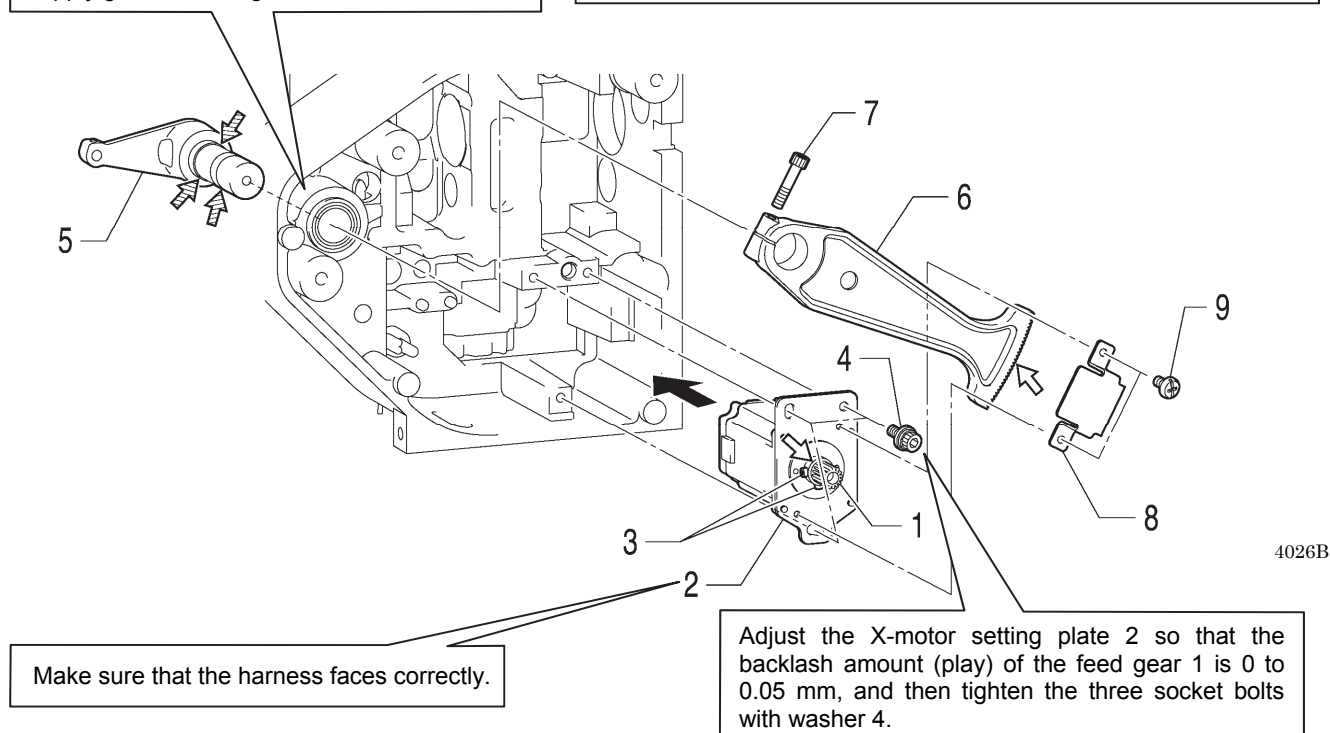
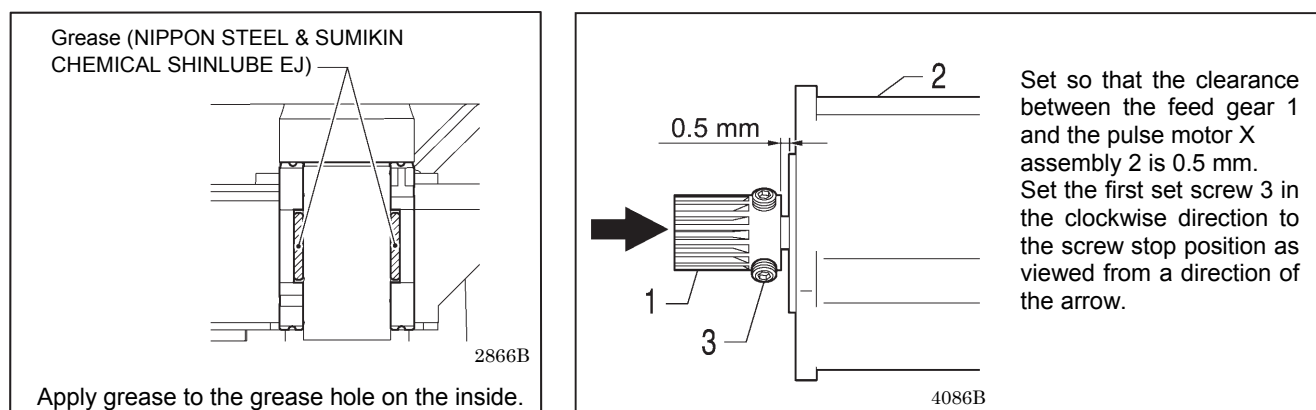
NOTE:

When the work clamp lifter plate assembly 23 (430H)/button clamp rod 30 (438H) is lowered, the needle bar guide slide block 24 (430H) or 35 (438H) moves away, so hold the work clamp lifter plate assembly 23 (430H)/button clamp rod 30 (438H) in place until the link shoulder screw 31 (430H) or 42 (438H) is tightened.

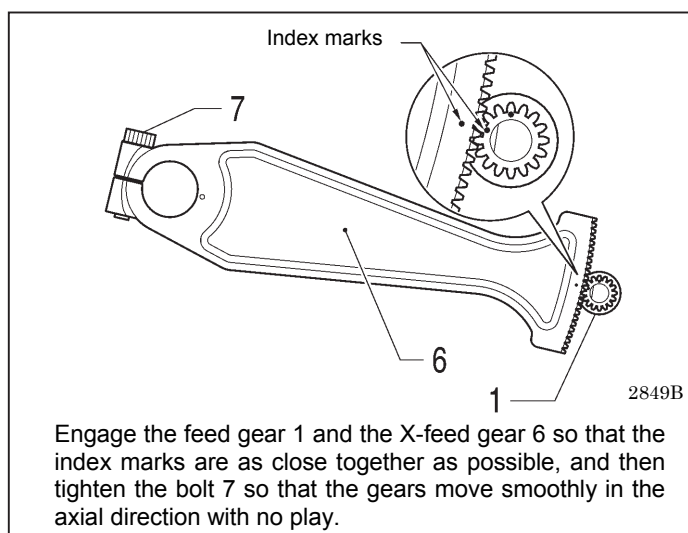
8-8. Feed mechanism

Apply grease to the portions indicated by the white arrows.

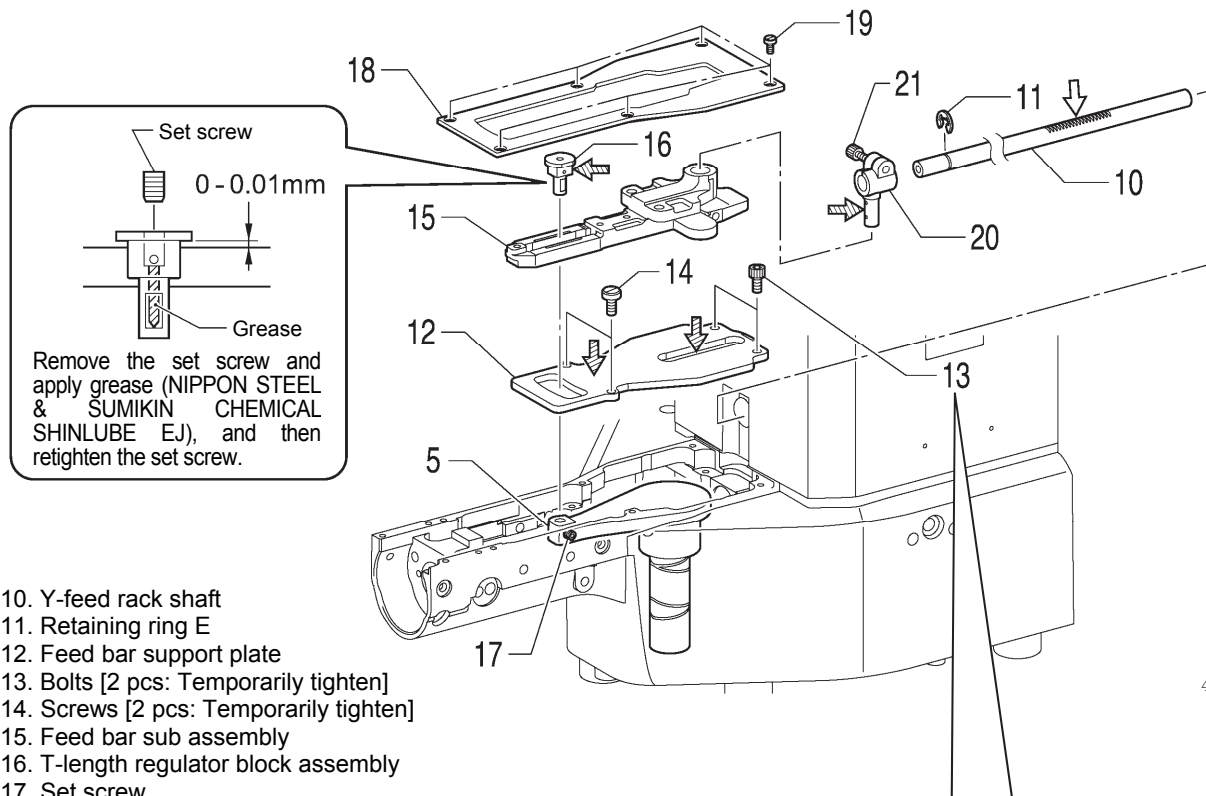
Apply grease (NIPPON STEEL & SUMIKIN CHEMICAL SHINLUBE EJ) to the portions indicated by the shaded arrows.



1. Feed gear
2. X-motor setting plate [Pulse motor X assembly]
3. Set screws [2 pcs: Tighten]
4. Socket bolts (S/P washer) [3 pcs]
5. X-feed lever
6. X-feed gear
7. Bolt
8. X-feed gear cover
9. Screws [2 pcs]

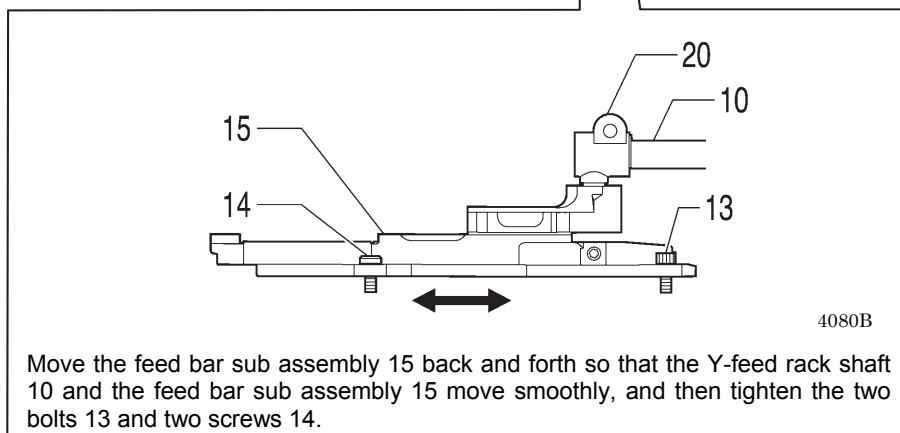


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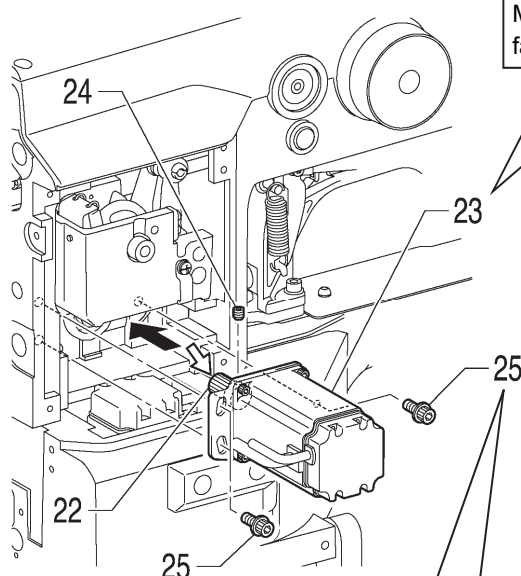
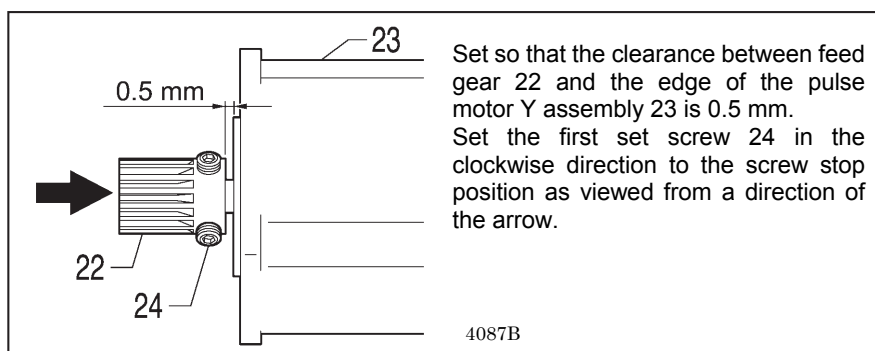


- 10. Y-feed rack shaft
- 11. Retaining ring E
- 12. Feed bar support plate
- 13. Bolts [2 pcs: Temporarily tighten]
- 14. Screws [2 pcs: Temporarily tighten]
- 15. Feed bar sub assembly
- 16. T-length regulator block assembly
- 17. Set screw
- 18. Feed bar cover
- 19. Screws [6 pcs]
- 20. Y-feed rack shaft clamp
- 21. Bolt

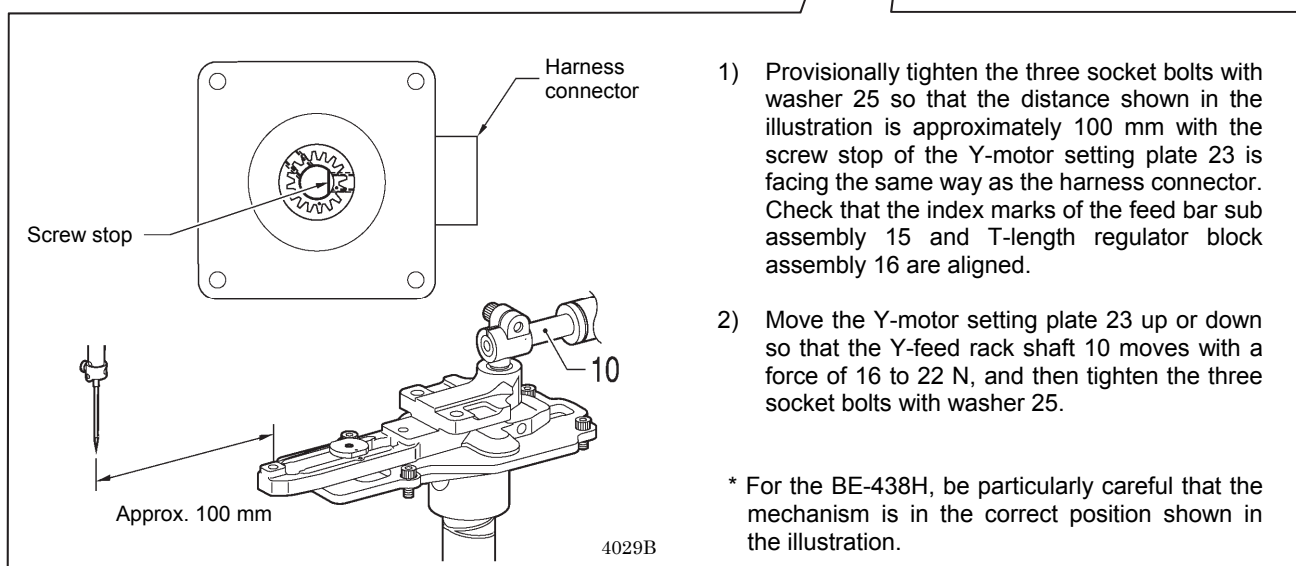
4027B



(Continued on next page.)



- 22. Feed gear
- 23. Y-motor setting plate (pulse motor Y assembly)
- 24. Set screws [2 pcs]
- 25. Socket bolts (S/P washer)
[3 pcs: Temporarily tighten]

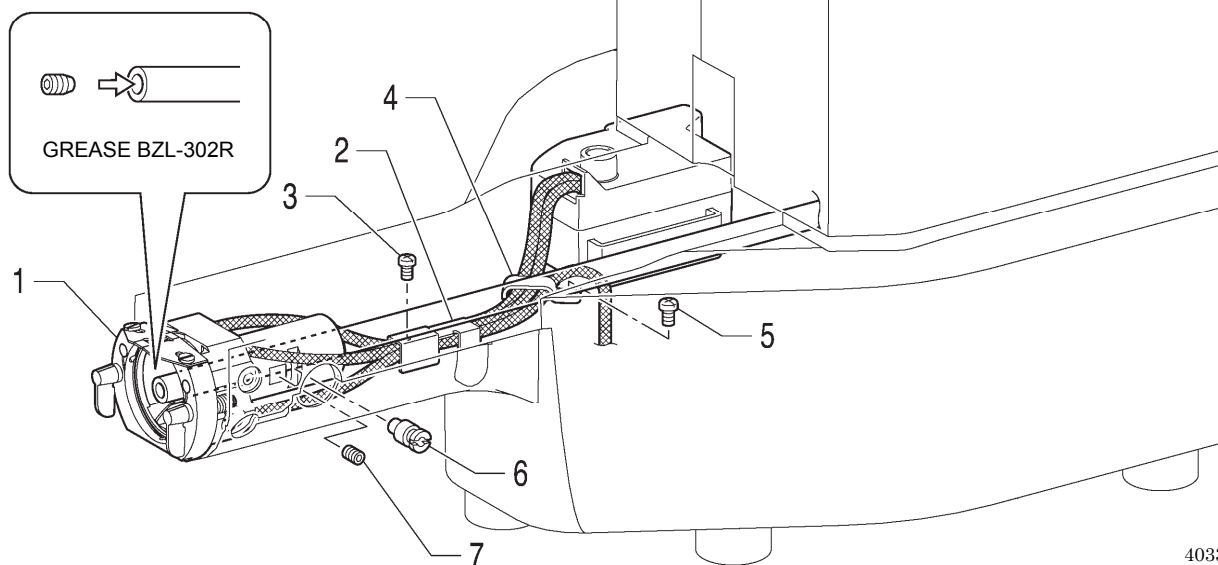
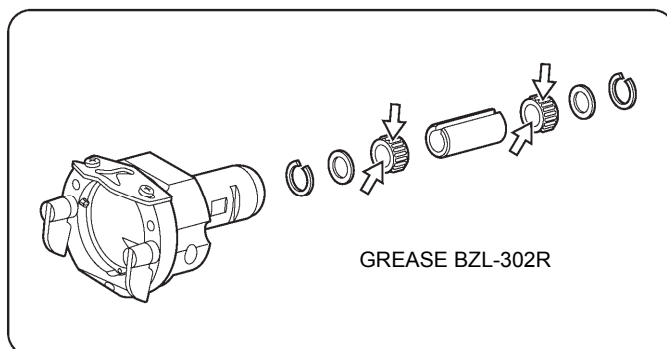


After installing, adjust while referring to "9-19. Adjusting the X- and Y-feed motor home position"

8-9. Lower shaft mechanism

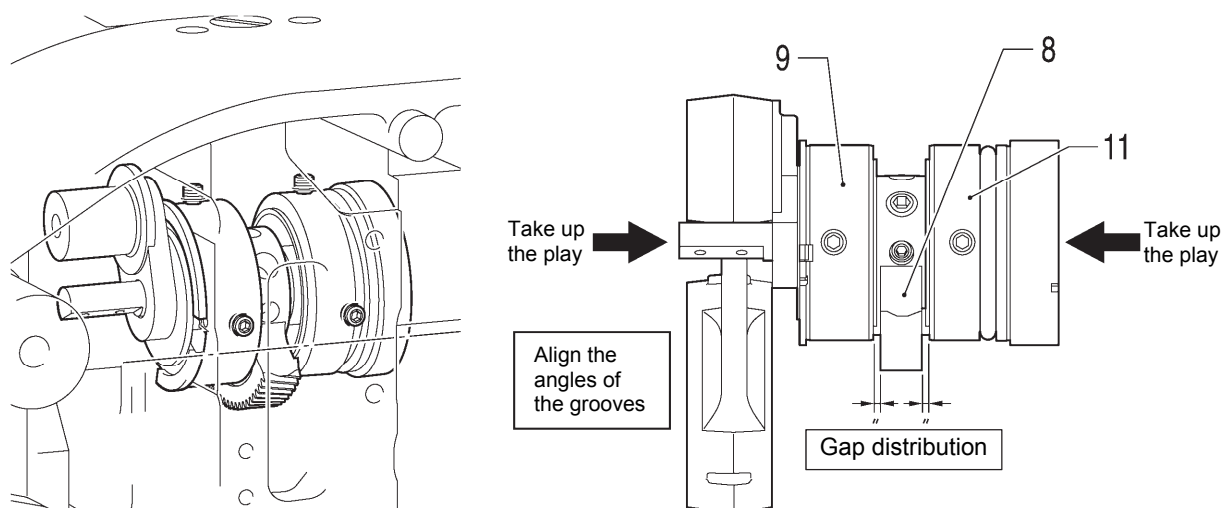
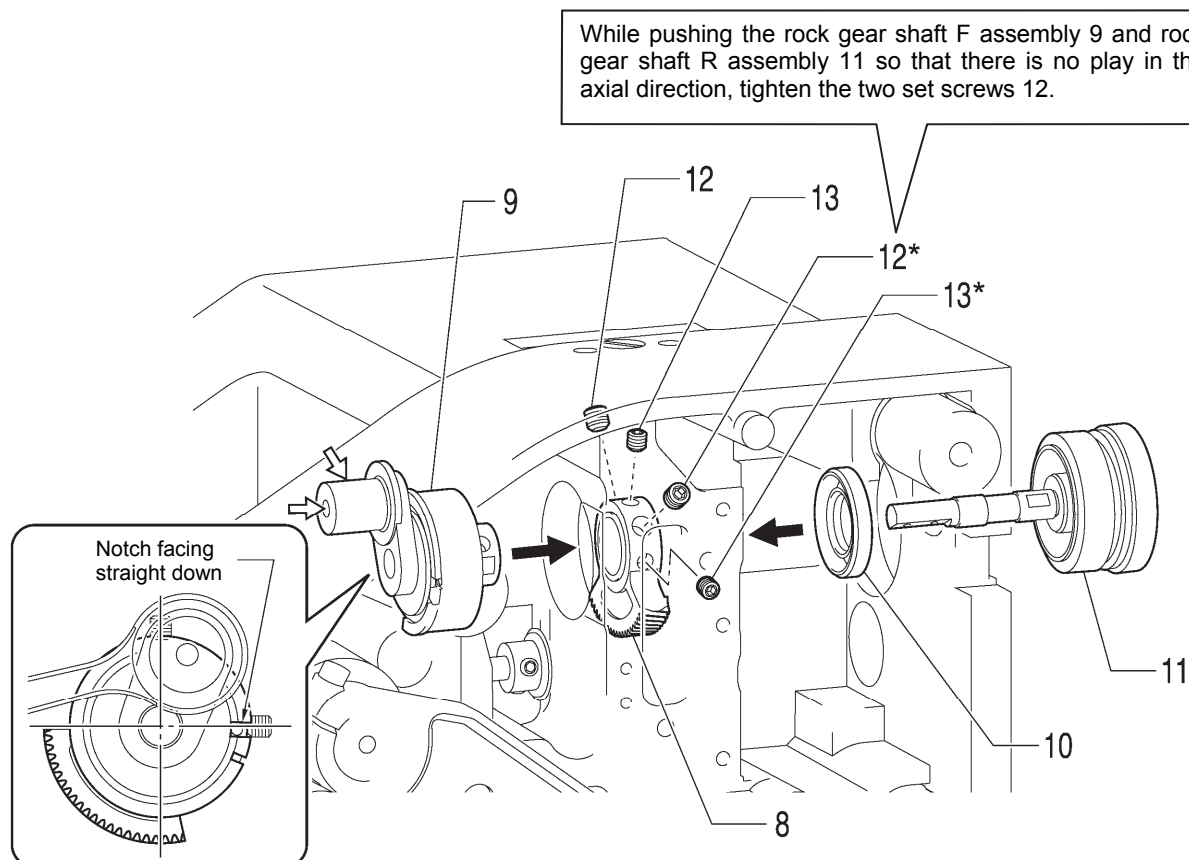
Apply grease to the portions indicated by the white arrows.

1. Shuttle race base assembly
2. Oil tube support
3. Screw
4. Cord holder U3
5. Screw
6. Adjusting stud
7. Set screw [Tighten]



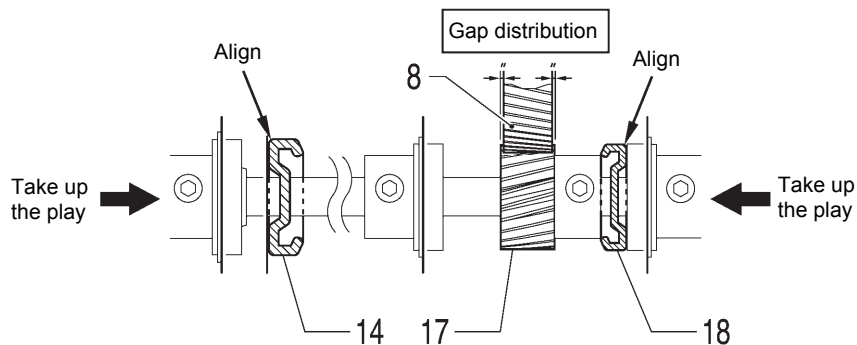
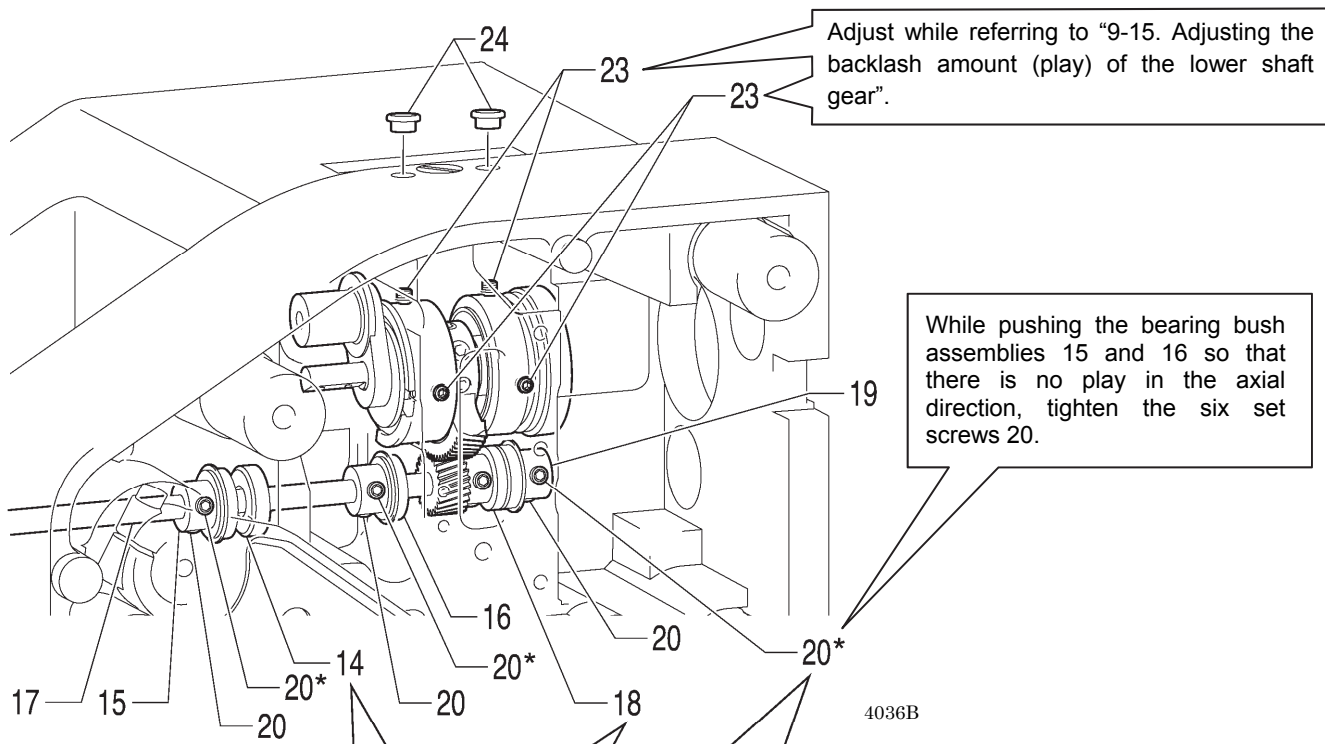
(Continued on next page.)

* Set screws marked with * in the illustration are on the screw stop side.

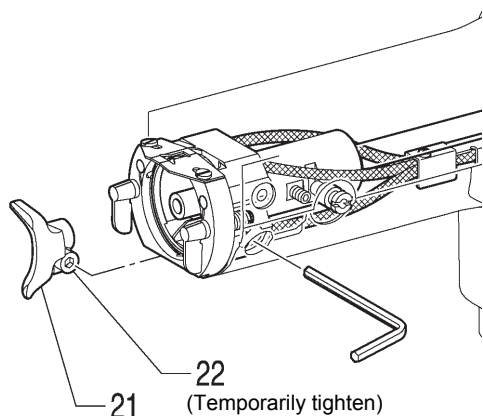


- 8. Rock gear
- 9. Rock gear shaft F assembly
- 10. Oil seal
- 11. Rock gear shaft R assembly
- 12. Set screws [2 pcs: Tighten]
- 13. Set screws [2 pcs: Tighten]

(Continued on next page.)



- 1) The oil seal 14, 18 cannot be reused, so have new parts ready for use.
- 2) Make sure that the installation directions for the oil seal 14, 18 are correct.
- 3) Apply grease (SEALUB S-14) to the outer circumferences of the oil seal 14, 18 when inserting the oil seal 14, 18 into the holes in the bed.



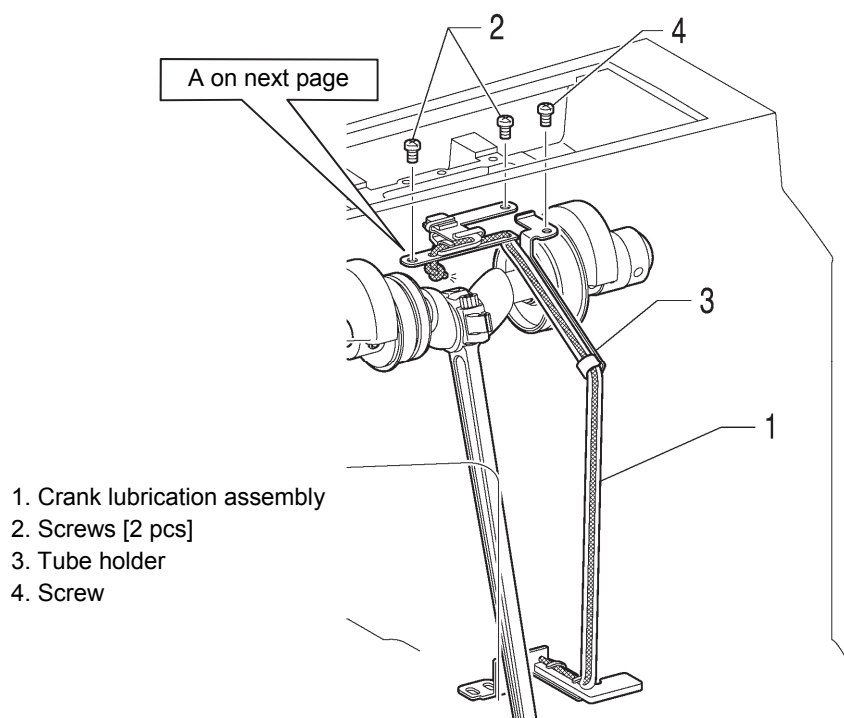
- 14. Oil seal
- 15. Bearing bush assembly
- 16. Bearing bush assembly
- 17. Lower shaft assembly
- 18. Oil seal
- 19. Bearing bush assembly
- 20. Set screws [6 pcs: Tighten]
- 21. Driver
- 22. Bolt [Temporarily tighten]
- 23. Set screws [4 pcs]
- 24. Rubber caps [2 pcs]

8-10. Upper shaft mechanism

Apply grease to the portions indicated by the white arrows.

* Set screws marked with * in the illustration are on the screw stop side.

* Refer to the next page for the points to note during reassembly.



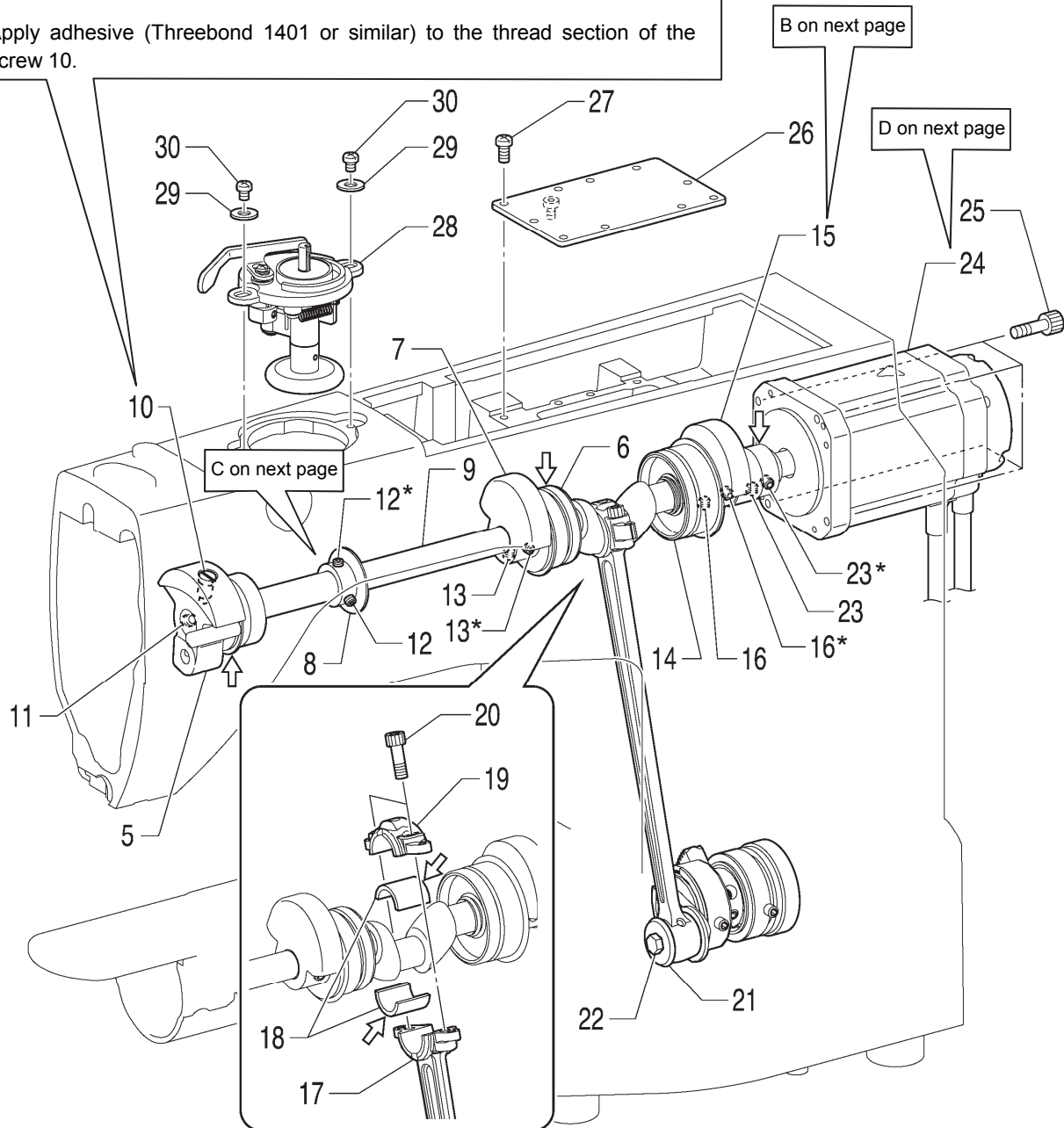
4030B

(Continued on next page.)

8. ASSEMBLY

- 1) Tighten the screw 10 of the thread take-up crank assembly 5 so that it is aligned with the hole in the upper shaft 9.
- 2) While pressing the thread take-up crank assembly 5 so that there is no play in it, tighten the two set screws 13.

Apply adhesive (Threebond 1401 or similar) to the thread section of the screw 10.



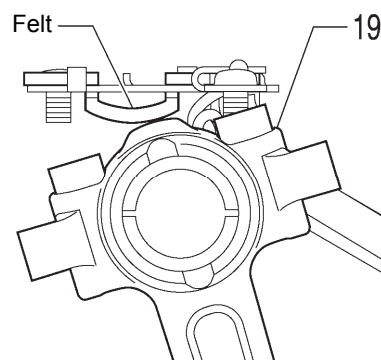
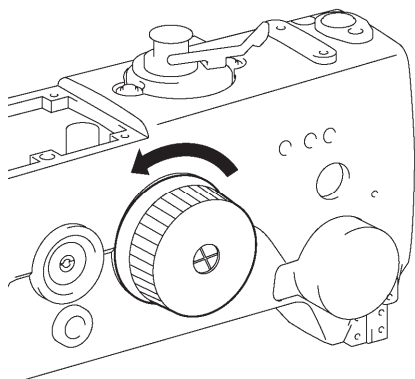
4047B

5. Thread take-up crank assembly
6. Oil seal
7. Ball bearing bush M assembly
8. Pulley gear R
9. Upper shaft
10. Screw
11. Set screw
12. Set screws [2 pcs]
13. Set screws [2 pcs]
14. Oil seal

15. Joint assembly
16. Set screws [2 pcs]
17. Crank rod [Lower half]
18. Needle bearing
19. Crank rod [Upper half]
20. Bolts [2 pcs]
21. Washer crank rod
22. Bolt

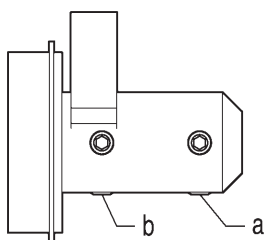
23. Set screws [2 pcs]
24. Motor assembly
25. Bolts [4 pcs]
26. Crank cover assembly
27. Screws [10 pcs]
28. Bobbin winder unit
29. Plain washers [2 pcs]
30. Screws [2 pcs]

(Continued on next page.)

A

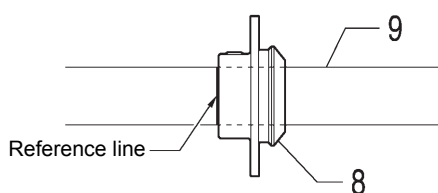
4064B

After installing, turn the pulley to check if the upper surface of the crank rod 19 touches the felt.

B

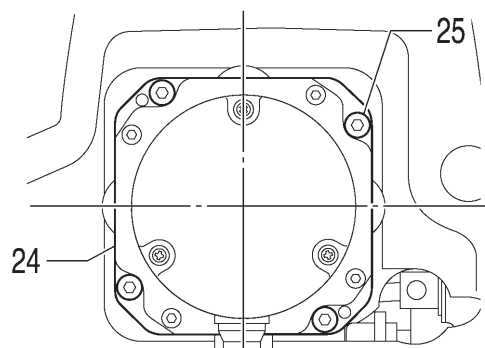
4085B

Tighten the set screw so that the screw stop a on the motor shaft is in the same phase as the screw stop b on the upper shaft 9.

C

4062B

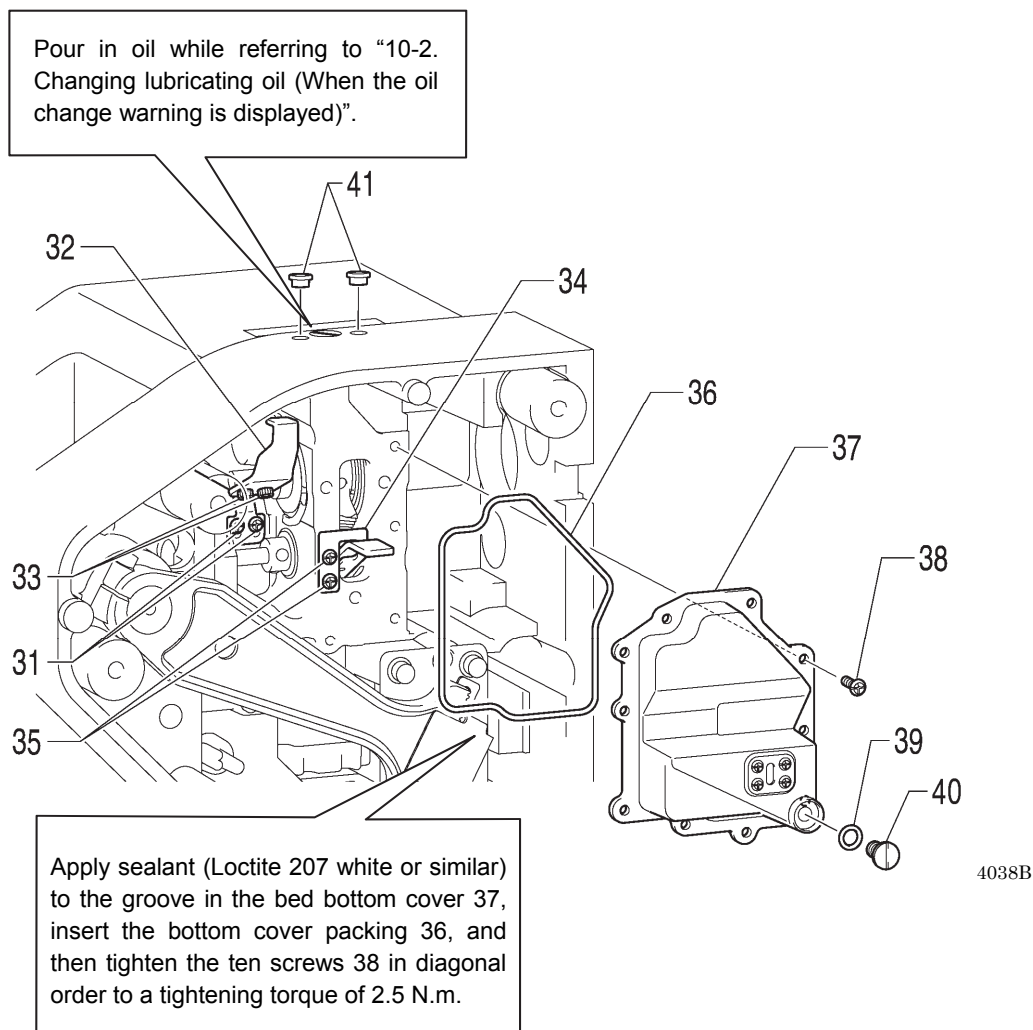
Install pulley gear R 8 so that it is aligned with the reference line on the upper shaft 9.

D

4072B

With the motor assembly 24 provisionally installed, check that the pulley turns smoothly, by turning the pulley in the direction of the arrow in A and then tighten the four bolts 25. If the pulley does not turn smoothly, loosen the screw a in B. Next turn the pulley to align the upper shaft 9 and motor shaft and then tighten the screw a in B again.

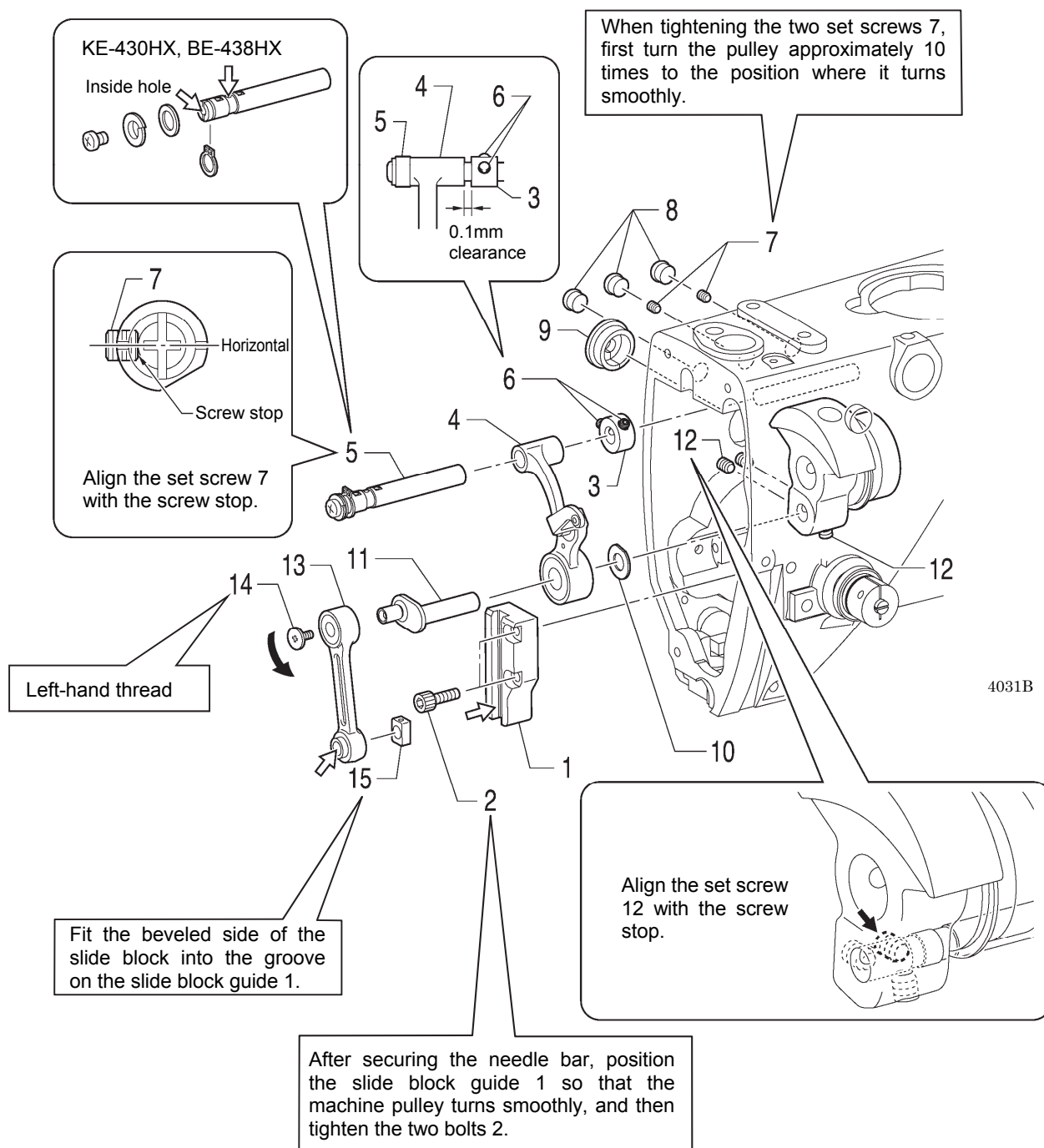
(Continued on next page.)



- | | |
|---------------------------------------|-------------------------|
| 31. Bolts [2 pcs] | 37. Bed bottom cover |
| 32. Splash plate | 38. Screws [10 pcs] |
| 33. Socket bolts (S/P washer) [2 pcs] | 39. O ring |
| 34. Lower shaft felt assembly | 40. Screw |
| 35. Screws [2 pcs] | 41. Rubber caps [2 pcs] |
| 36. Bottom cover packing | |

8-11. Needle bar mechanism

Apply grease to the portions indicated by the white arrows.

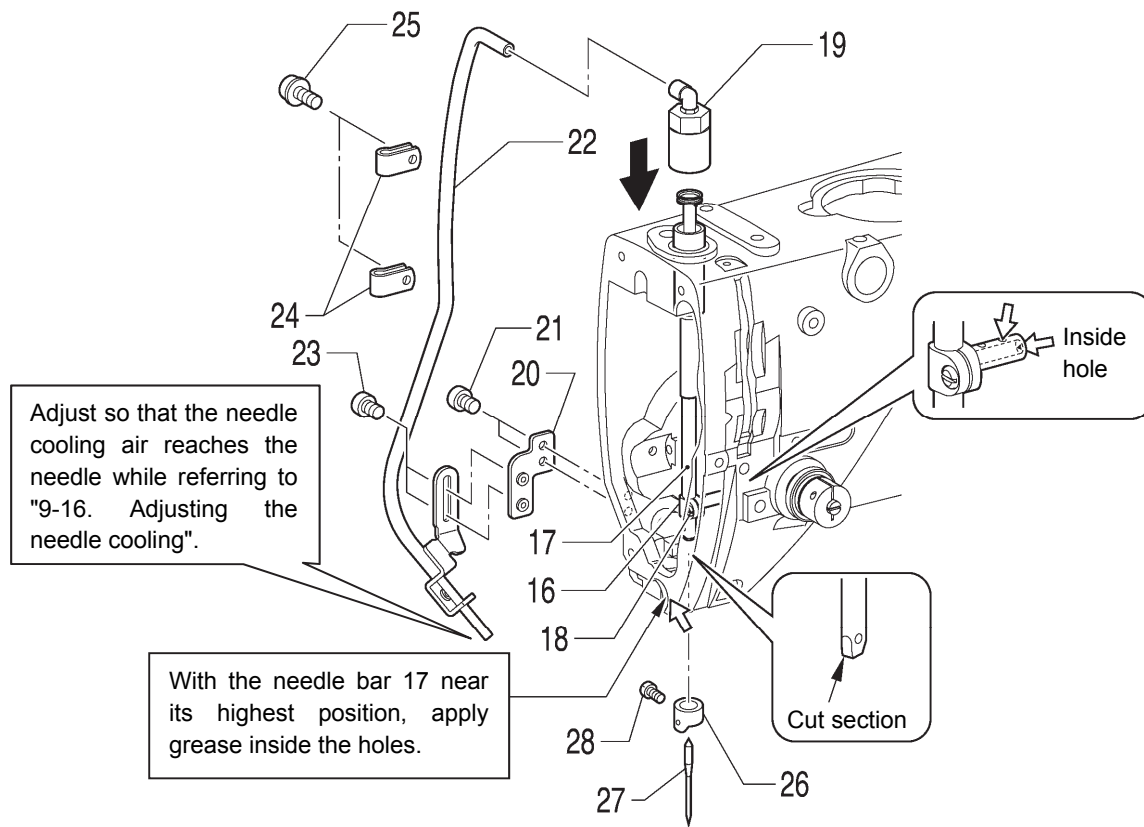


1. Slide block guide
2. Bolts [2 pcs: Temporarily tighten]
3. Set screw collar
4. Thread take-up lever assembly
5. Thread take-up support shaft assembly
6. Set screws [2 pcs: Tighten]
7. Set screws [2 pcs: Tighten]

8. Rubber caps [3 pcs]
9. Rubber cap
10. Washer
11. Needle bar crank
12. Set screws [3 pcs: Tighten]
13. Needle bar connecting rod assembly
14. Screw [Tighten]
15. Slide block

(Continued on next page.)

KE-430HX

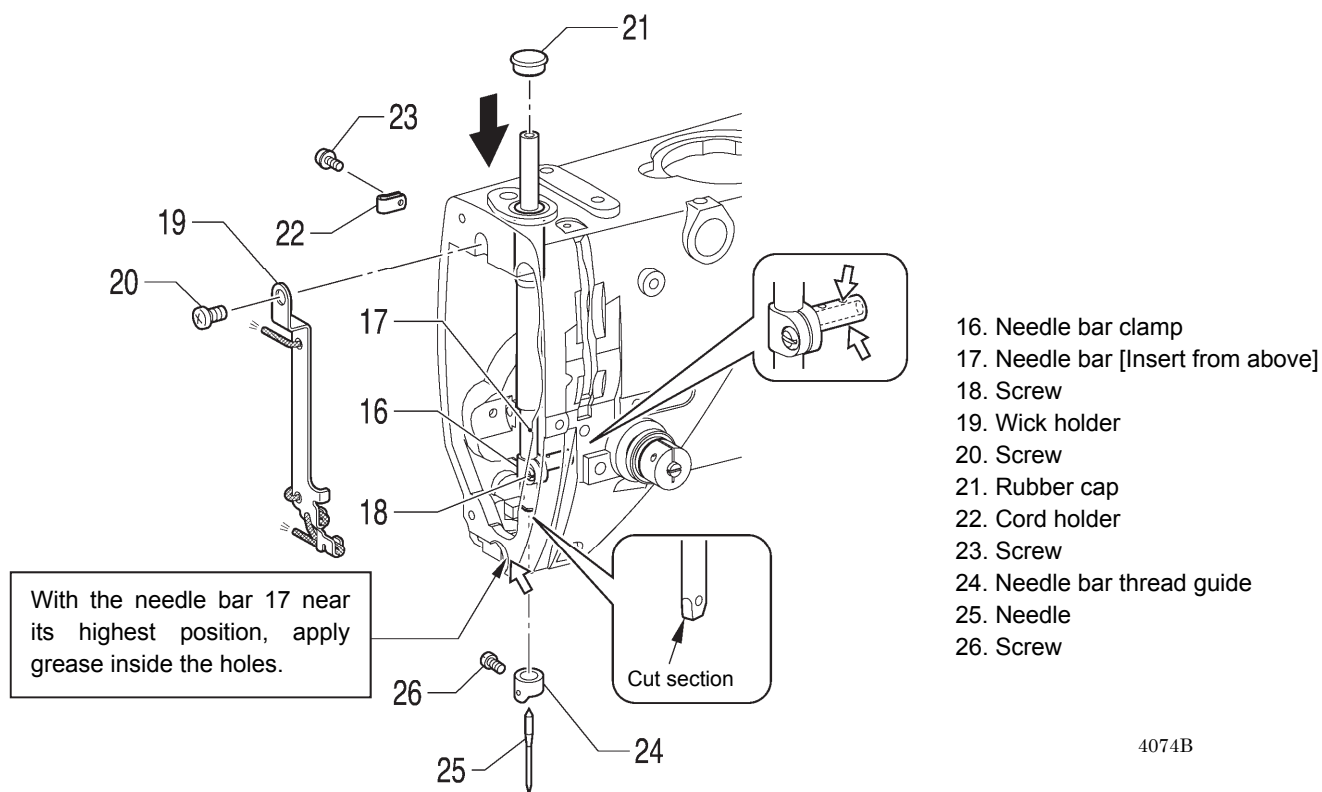


4032B

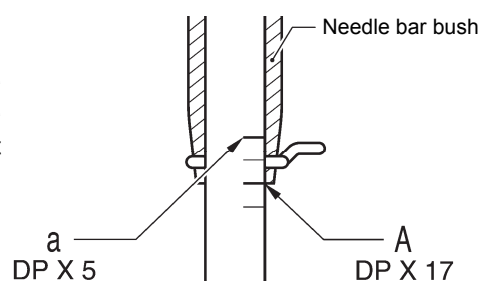
- | | | |
|------------------------------------|--------------------------|-----------------------------|
| 16. Needle bar clamp | 21. Screws [2 pcs] | 25. Screws [2 pcs] |
| 17. Needle bar [Insert from above] | 22. Air nozzle assembly | 26. Needle bar thread guide |
| 18. Screw | 23. Screws [2 pcs] | 27. Needle |
| 19. Air cylinder cap assembly | 24. Cord holders [2 pcs] | 28. Screw |
| 20. Middle plate | | |

(Continued on next page.)

KE-430HS, BE-438HX/BE-438HS

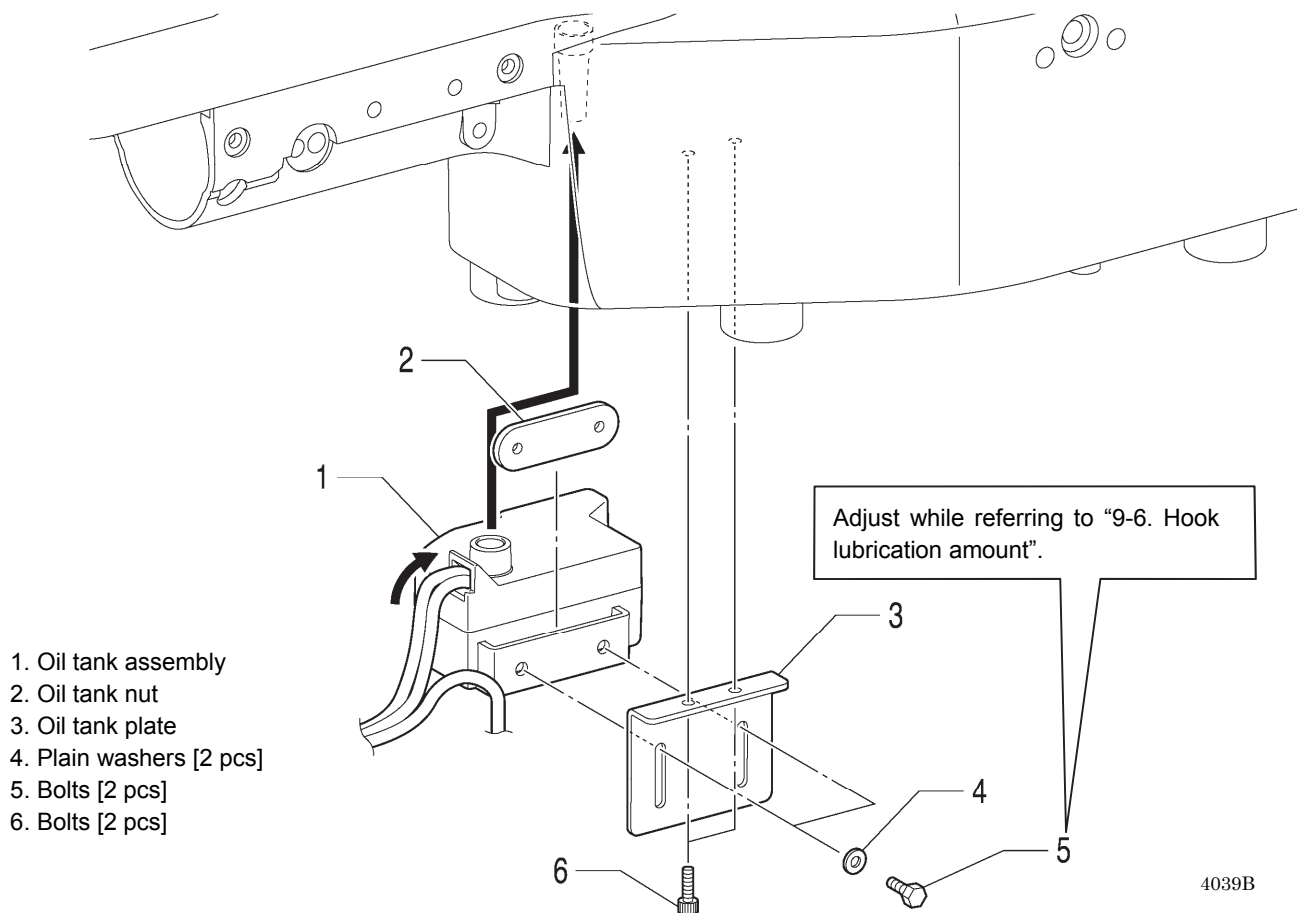


Align the needle bar 17 so that reference line A (the second lowest reference line on the needle bar 17) or reference line a (top reference line) is aligned with the lower edge of the needle bar bush when the machine pulley is turned to lower the needle bar 17 to its lowest position, and then set the cut section so that it is facing forward and tighten the screw 18.

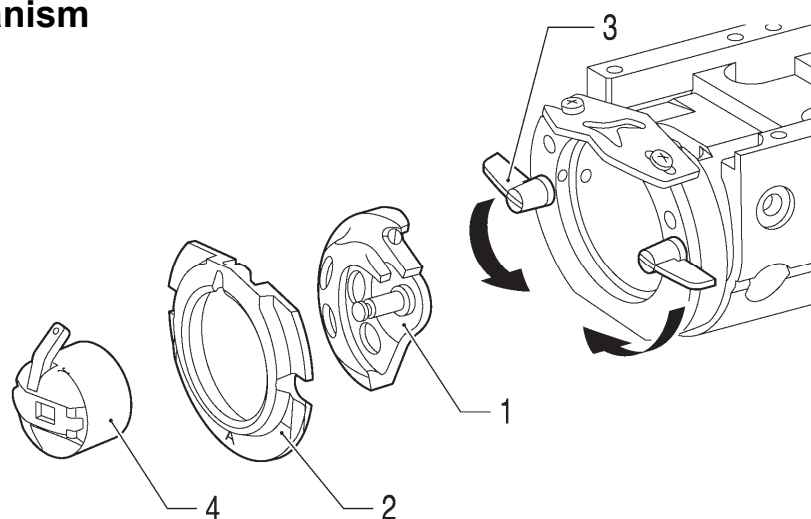


4647Q

8-12. Lubrication



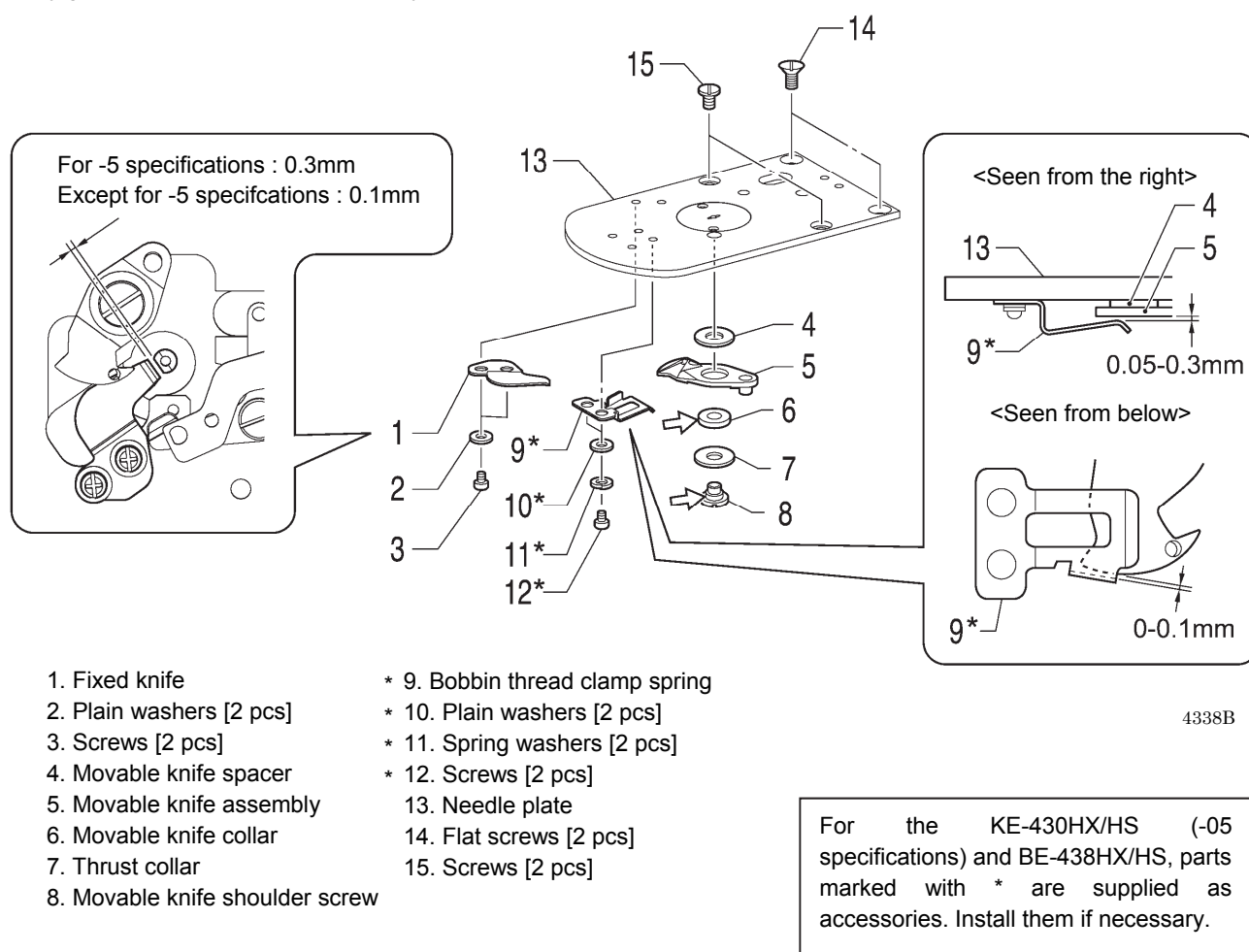
8-13. Hook mechanism



After installing the shuttle hook 1, carry out the adjustments in "9-3. Adjusting the needle bar lift amount and the driver needle guard" and "9-4. Adjusting the needle clearance".

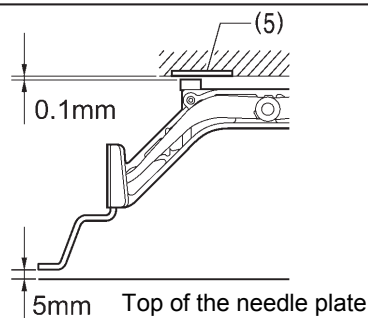
8-14. Thread trimmer mechanism (2)

Apply grease to the portions indicated by the white arrows.

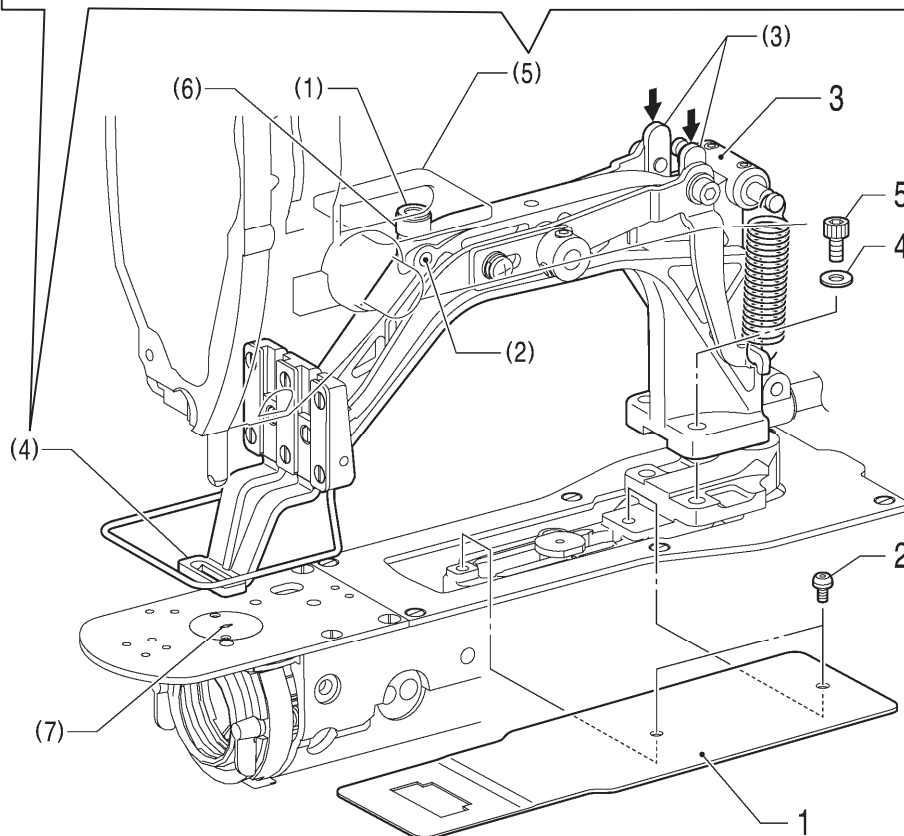


8-15. Work clamp arm mechanism (KE-430HX/HS)

Push the work clamp arm lever plate (3) in the direction of the arrow to raise the work clamp (4) by about 5 mm, set so that there is a clearance of 0.1 mm between the underside of the work clamp arm support plate (5) and the slider (1), and then secure the slider (1) of the work clamp support shaft (6) with the set screw (2).



2885B



1. Feed plate
2. Screws [2 pcs]
3. Work clamp arm assembly
4. Plain washers [2 pcs]
5. Bolts [2 pcs]

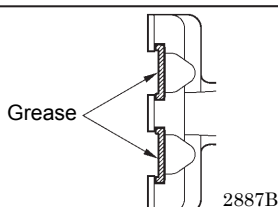
- (1) Slider
- (2) Set screw
- (3) Work clamp arm lever plate
- (4) Work clamp
- (5) Work clamp arm support plate
- (6) Support shaft
- (7) Needle hole

4042B

After installing, carry out test feeding and check that the needle hole (7) is not inside the work clamp (4) and the frame of the feed plate 1. If the needle hole (7) is inside the frame, adjust the position of the work clamp arm assembly 3 and the feed plate 1.

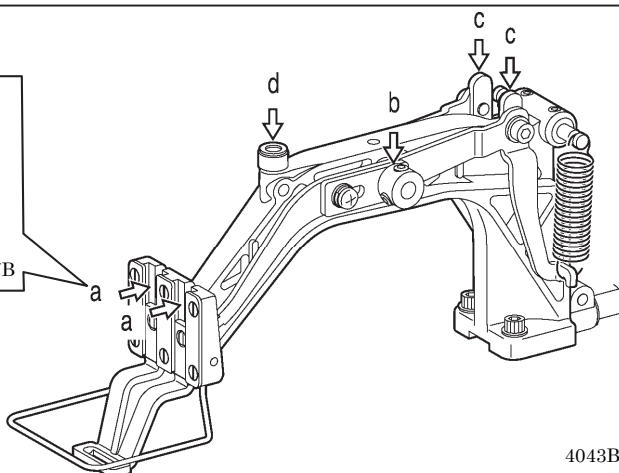
Apply grease to all sliding parts during assembly.

It is recommended that you use commercially-available JXTG Nippon Oil & Energy POWERNOC WB 2 as the grease to be applied to the sliding part of the work clamp.



2887B

- a. Sliding portion of the work clamp (Apply small amount of grease.)
- b. Sliding portion of the work clamp arm lever shaft and the work clamp arm levers R and L
- c. Round portion of the work clamp arm lever plate
- d. Slider of work clamp support shaft

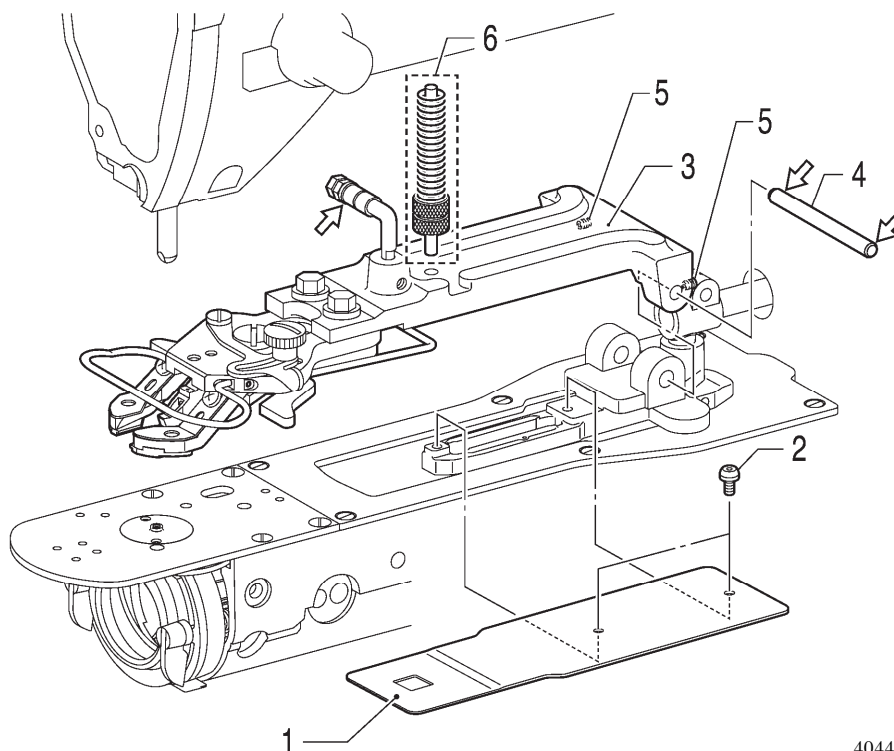


4043B

8-16. Work clamp arm mechanism (BE-438HX/HS)

Apply grease to the portions indicated by the white arrows.

1. Feed plate
2. Screws [2 pcs]
3. Button clamp holder
4. Button clamp holder shaft
5. Set screws [2 pcs]
6. Tension nuts [2 pcs]
Adjusting nut
Compression spring

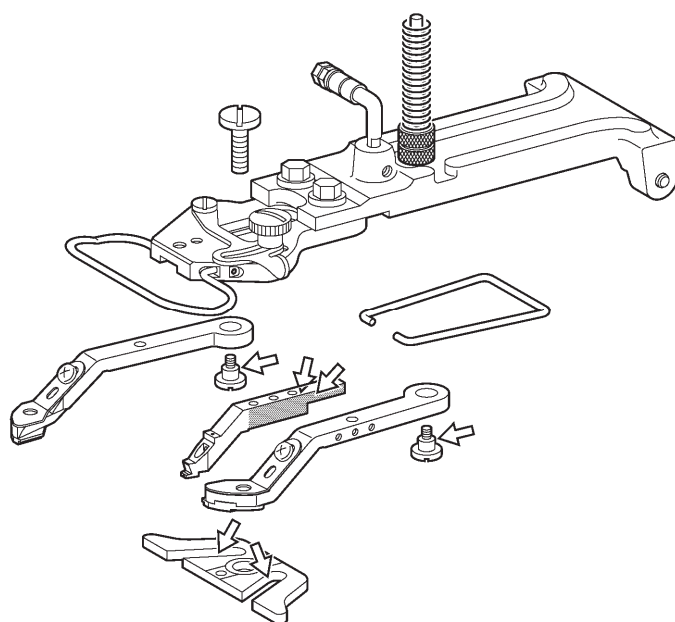


4044B

After installing, check that the needle passes into the button hole without touching the button. (Refer to “9-12. Adjusting the position of the button clamp (BE-438HX/HS)”.)

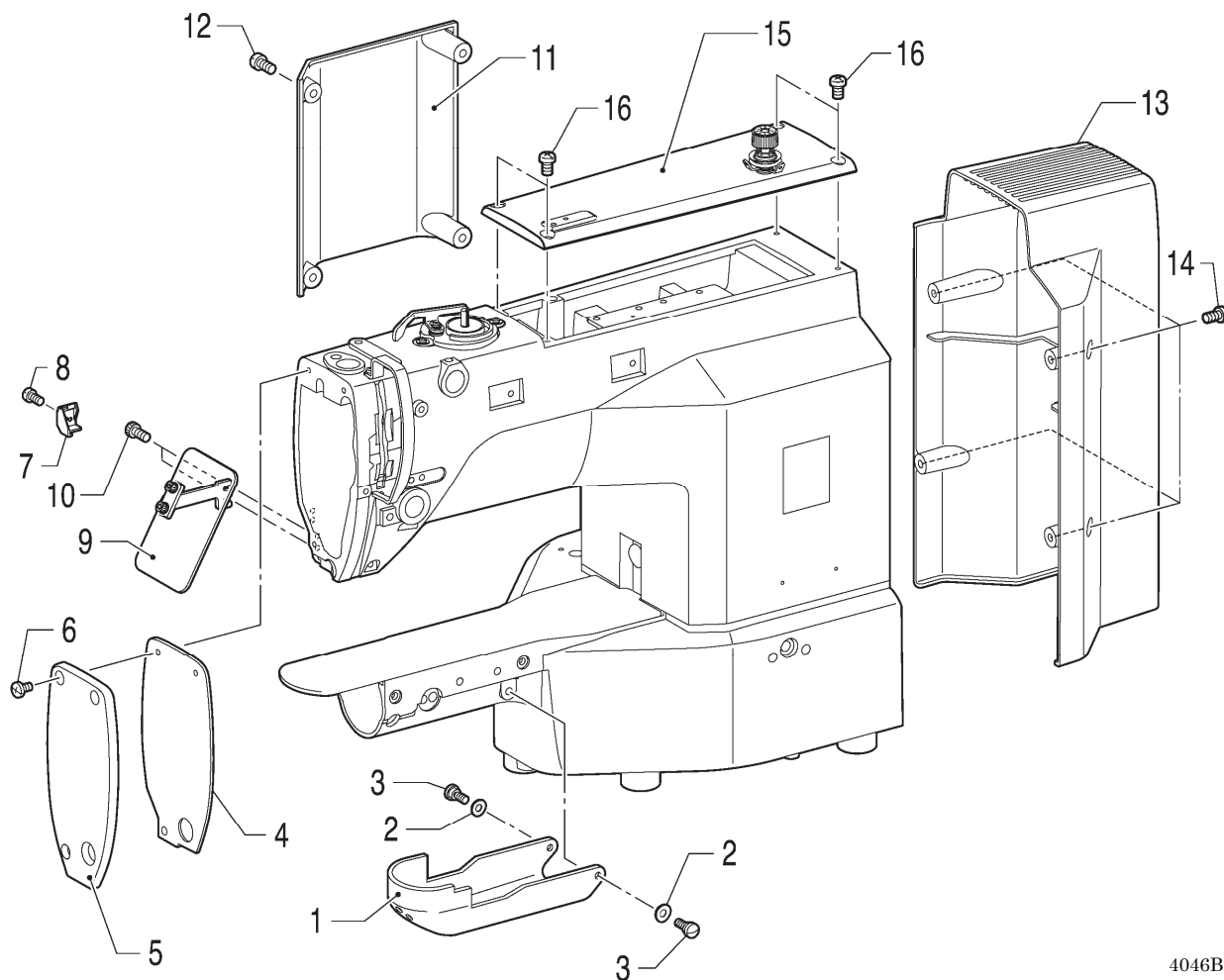
Apply grease to the portions indicated by the white arrows.

It is recommended that you use commercially-available <JXTG Nippon Oil & Energy POWERNOC WB 2>.



4045B

8-17. Covers



4046B

- | | |
|---------------------------------|-----------------------|
| 1. Shuttle race cover assembly | 9. Eye guard assembly |
| 2. Washers, wave spring [2 pcs] | 10. Bolts [2 pcs] |
| 3. Shoulder screws, 8-2 [2 pcs] | 11. Side cover |
| 4. Face plate packing | 12. Screws [4 pcs] |
| 5. Face plate | 13. Rear cover |
| 6. Screws [3 pcs] | 14. Screws [4 pcs] |
| 7. LED cover | 15. Top cover |
| 8. Screw | 16. Screws [4 pcs] |

9. ADJUSTMENT

CAUTION



Maintenance and inspection of the sewing machine should only be carried out by a qualified technician.



Ask your Brother dealer or a qualified electrician to carry out any maintenance and inspection of the electrical system.



If any safety devices have been removed, be absolutely sure to re-install them to their original positions and check that they operate correctly before using the machine.



Secure the table so that it will not move when tilting back the machine head. If the table moves, it may crush your feet or cause other injuries.



Use both hands to hold the machine head when tilting it back or returning it to its original position. If only one hand is used, the weight of the machine head may cause your hand to slip, and your hand may get caught.



Turn off the power switch and disconnect the power cord from the wall outlet at the following times, otherwise the machine may operate if the foot switch is depressed by mistake, which may result in injury.

- When carrying out inspection, adjustment and maintenance
- When replacing consumable parts such as the hook and knives



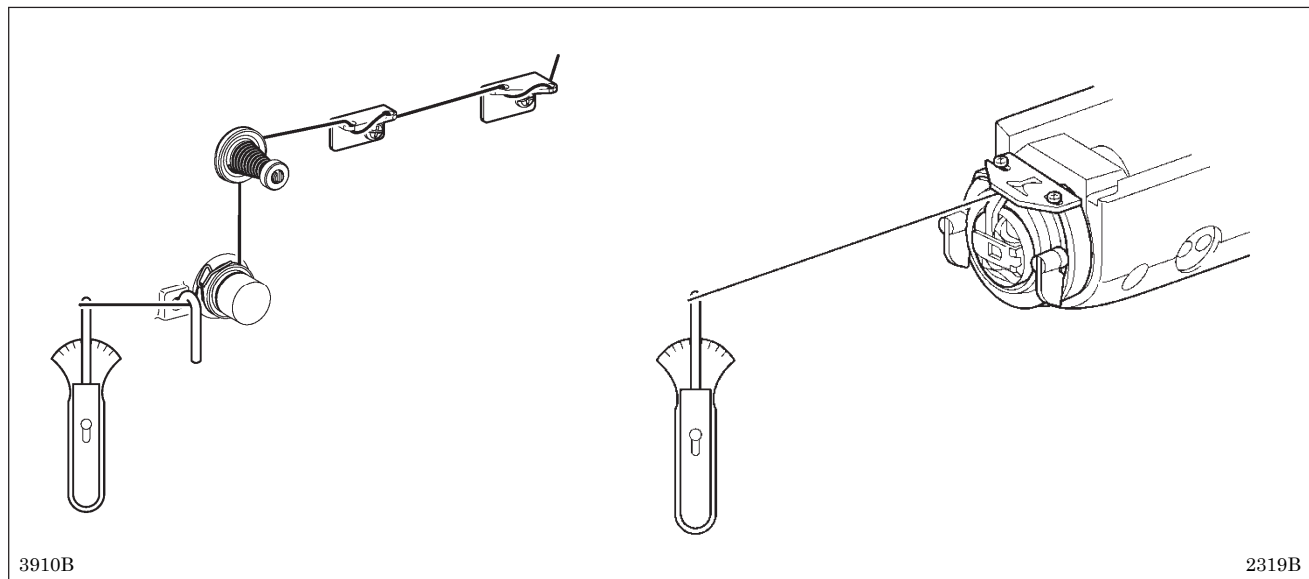
If the power switch needs to be left on when carrying out some adjustment, be extremely careful to observe all safety precautions.

9-1. Standard thread tension

Use	KE-430HX/HS				BE-438HX/HS
	Light-weight materials (-01)	Medium-weight materials (-03)	Knitted wear (-0K)	Heavy-weight materials (-05)	
Upper thread	#60 or equivalent	#50 or equivalent	#60 or equivalent	#30 or equivalent	#60 or equivalent
Lower thread	#60 or equivalent	#50 or equivalent	#80 or equivalent	#50 or equivalent	#60 or equivalent
Upper thread tension (N) [Tension value]	0.8 - 1.2 [80 - 120]* ¹			1.2 - 1.8 [70 - 130]* ¹	0.5 - 1.2 [50 - 150]* ¹
Lower thread tension (N)	0.2 - 0.3				0.2 - 0.3
Pre-tension (N)	0.05 - 0.3				0.1 - 0.4
Needle	DP x 5 #14	DP x 5 #14	DP x 5 #9	DP x 17NY #19	DP x 17NY #12

*1: This is the tension value when the pretension is 0.05 N.

9-1-1. Upper and lower thread tension



<Foot switch>

S001	00000	X %	100
001	042	Y %	100
00000		2700	
		L/H	1/14
		→)(←	75

<During test operation standby>

S001		X %	100
001	042	Y %	100
		2700	
		L/H	1/14
		→)(←	75

(1)

<Foot switch>

<2-pedal foot switch (option)>

<Operating the upper thread tension>

• When using a foot switch

In the sewing standby condition and during test operation standby, the thread tension will output a tension value (1) which has been set for the section while the foot switch is being depressed backward.

When the foot switch is returned to the neutral position, the thread tension will be released.

* If the work clamp/button clamp is lowered when the foot switch is returned to the neutral position, the thread tension output will be maintained.

• When using a 2-pedal foot switch (option)

In the sewing standby condition and during test operation standby, the thread tension will output a thread tension value (1) which has been set in the section No. 1 (sewing start section) appearing on the operation panel while the work clamp/button clamp is lowered (while the work clamp switch is being depressed).

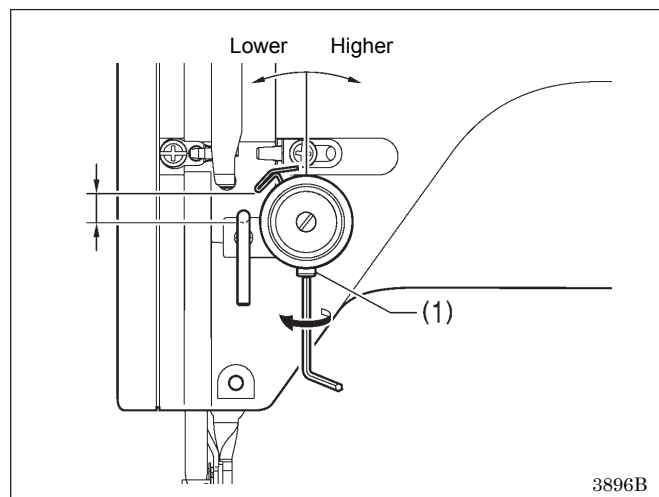
When the work clamp switch is released, the thread tension will be released.

* If the work clamp/button clamp is lowered when the work clamp switch is released, the thread tension output will be maintained.

* The above operations are enabled when memory switch No. 072 is set to "1" or "2".

9-1-2. Thread take-up spring

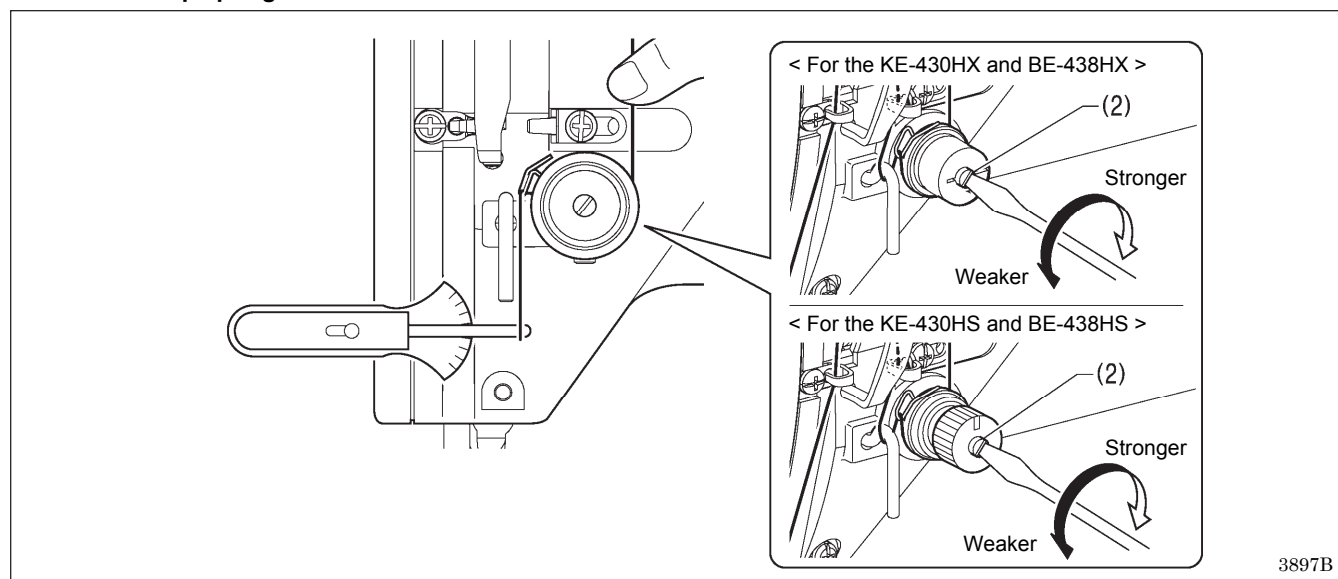
Use	KE-430HX/HS				BE-438HX/HS
	Light-weight materials (-01)	Medium-weight materials (-03)	Knitted wear (-0K)	Heavy-weight materials (-05)	
Thread take-up spring height (mm)	6 - 11				6 - 11
Thread take-up spring tension (N)	0.2 - 0.4			0.6 - 1.0	0.15 - 0.35



<Thread take-up spring height>

Loosen the set screw (1) and turn the tensioner body to adjust the thread take-up spring height.

<Thread take-up spring tension>

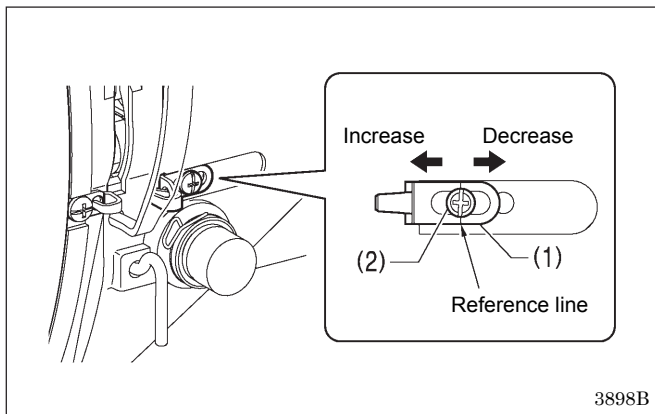


Adjust by turning the shaft (2) with a screwdriver.

NOTE:

When the spring height (stroke) is great or the spring tension is insufficient, it may cause the thread end length to vary after thread trimming.

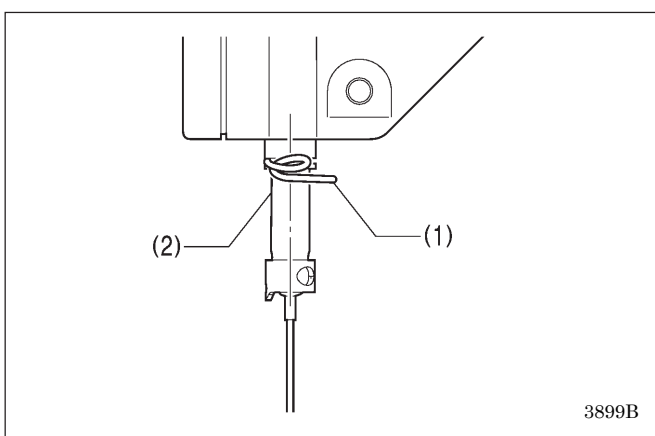
9-1-3. Arm thread guide C



The standard position of arm thread guide C (1) is when the screw (2) is aligned with the position of the reference line. To adjust the position, loosen the screw (2) and then move arm thread guide C (1).

- When sewing heavy-weight materials, move arm thread guide C (1) to the left. (The thread take-up amount will increase.)
- When sewing light-weight materials, move arm thread guide C (1) to the right. (The thread take-up amount will decrease.)

9-1-4. Needle bar bush thread guide D

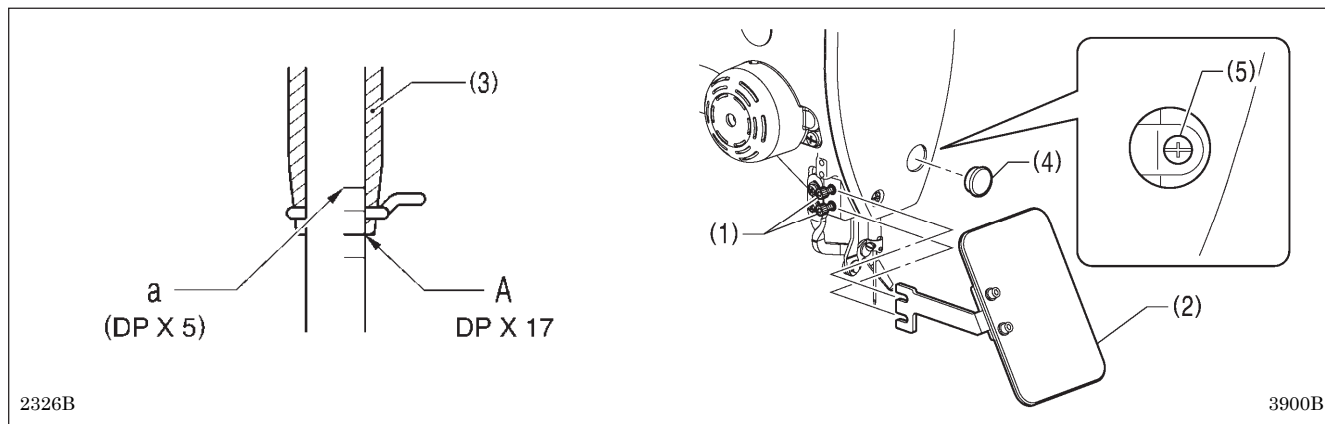


Check that needle bar bush thread guide D (1) is at the middle of the needle bar (2).

NOTE:

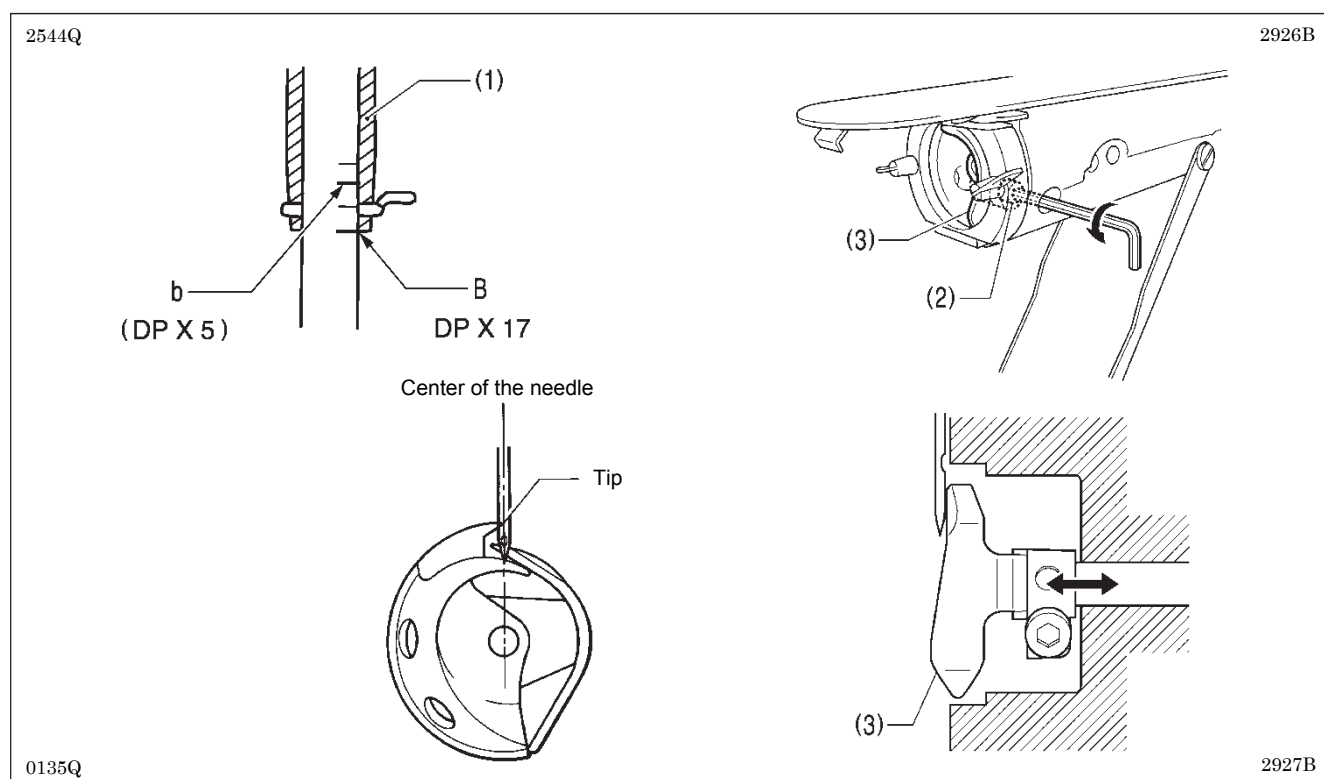
If needle bar bush thread guide D (1) is not in the correct position, the thread may pull out or skipped stitches may occur.

9-2. Adjusting the needle bar height



1. Loosen the two bolts (1), and then remove the eye guard assembly (2).
 2. Turn the pulley in the direction of the arrow to move the needle bar to the lowest position. Then remove the rubber cap (4), loosen the screw (5) and then move the needle bar up or down to adjust so that the second reference line from the bottom of the needle bar (reference line A) is aligned with the lower edge of the needle bar bush (3).
- * If using a DP x 5 needle, use the highest reference line (reference line a).

9-3. Adjusting the needle bar lift amount and the driver needle guard



1. Turn the pulley in the direction of the arrow to raise the needle bar from the lowest position until the lowest reference line on the needle bar (reference line B) is aligned with the lower edge of the needle bar bush (1).

* If using a DP x 5 needle, align with reference line b which is the second reference line from the top.

NOTE:

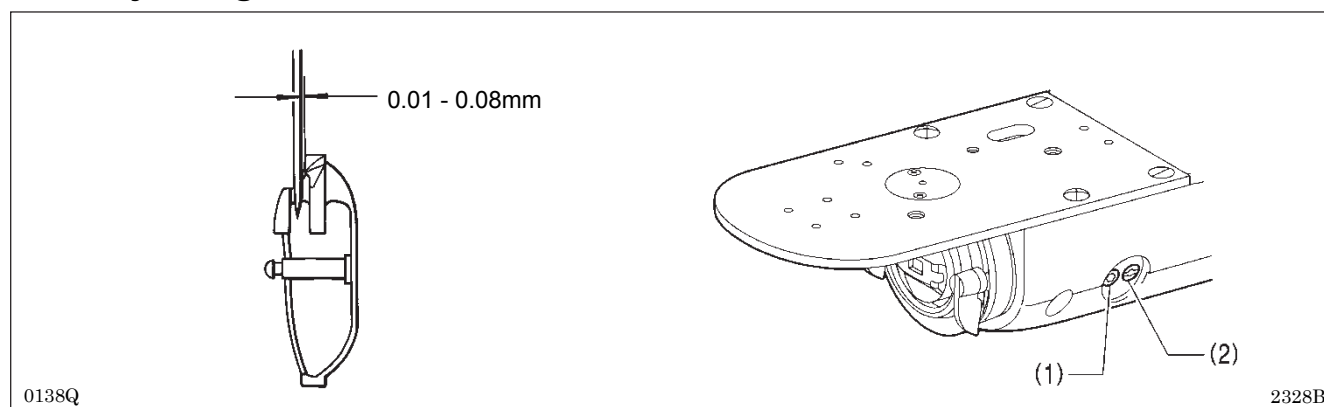
Check that the wiper and the needle do not touch when lowering the needle bar. If the wiper is in a position where it will touch the needle, move the wiper to the right by hand so that it does not touch the needle.

2. Loosen the bolt (2).
3. Move the driver (3) back and forth to adjust so that it is touching the needle when the tip of the shuttle hook is aligned with the middle of the needle, and then tighten the bolt (2).

NOTE:

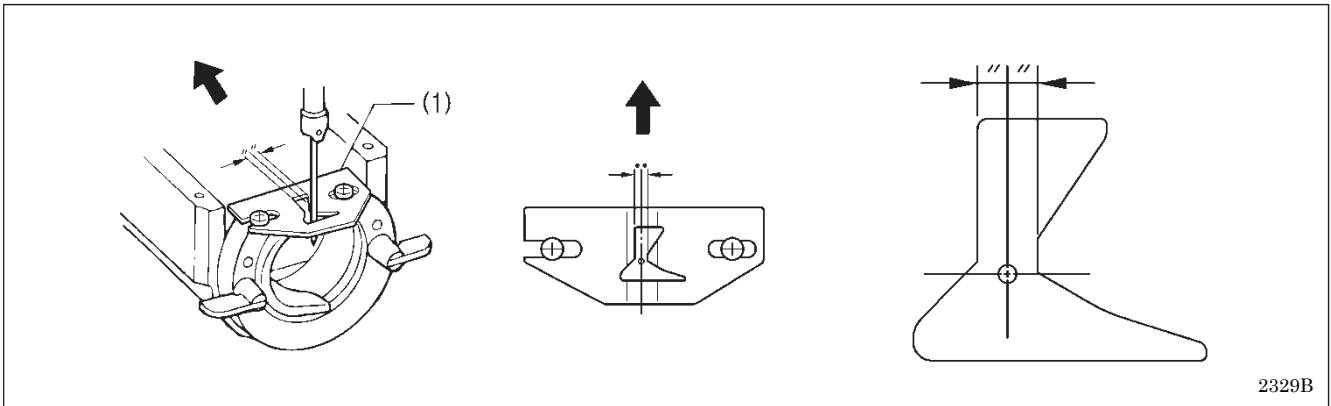
If the needle contact pressure from the driver (3) is too great, skipped stitches may occur. On the other hand, if the driver (3) is not touching the needle, the tip of the inner shuttle hook will obstruct the needle, resulting in an excessively high amount of friction.

9-4. Adjusting the needle clearance



Turn the pulley in the direction of the arrow to align the tip of the shuttle hook with the needle center line. Then loosen the set screw (1) and turn the adjusting stud (2) to adjust so that the clearance between the needle and the tip of the shuttle hook is 0.01 to 0.08 mm.

9-5. Adjusting the shuttle race thread guide

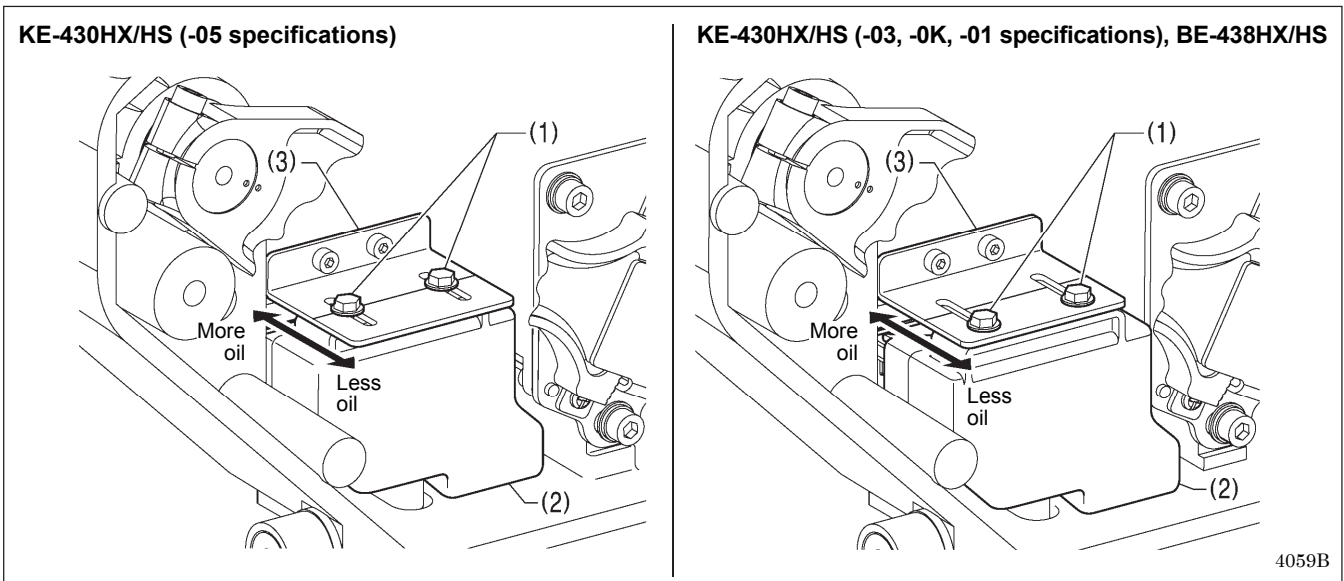


Install the shuttle race thread guide (1) by pushing it in the direction of the arrow so that the needle groove is aligned with the center of the needle plate hole.

NOTE:

If the shuttle race thread guide is in the wrong position, thread breakages, soiled thread or thread entanglements may occur. The position of the shuttle race thread guide is adjusted at the time of shipment from the factory. It should not be changed if at all possible.

9-6. Hook lubrication amount



1. Tilt back the machine head.
2. Loosen the two bolts (1), and then adjust by moving the oil tank assembly (2) up or down.

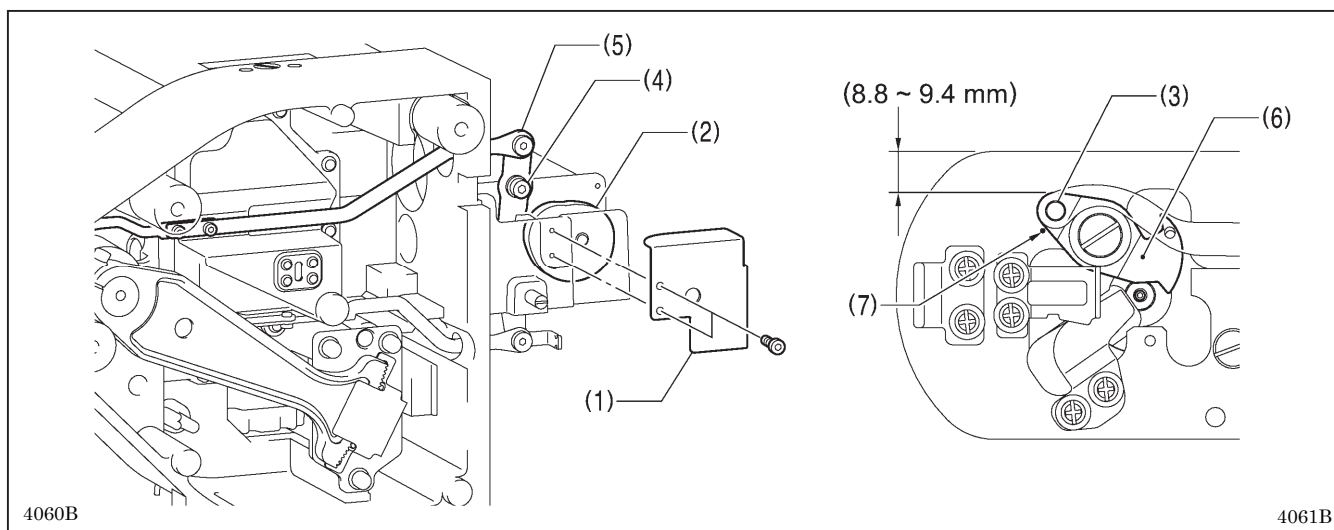
KE-430HX/HS (-05 specifications)

The standard position is when the centers of the two bolts (1) are aligned with the reference line on the oil tank plate (3).

KE-430HX/HS (-03, -0K, -01 specifications), BE-438HX/HS

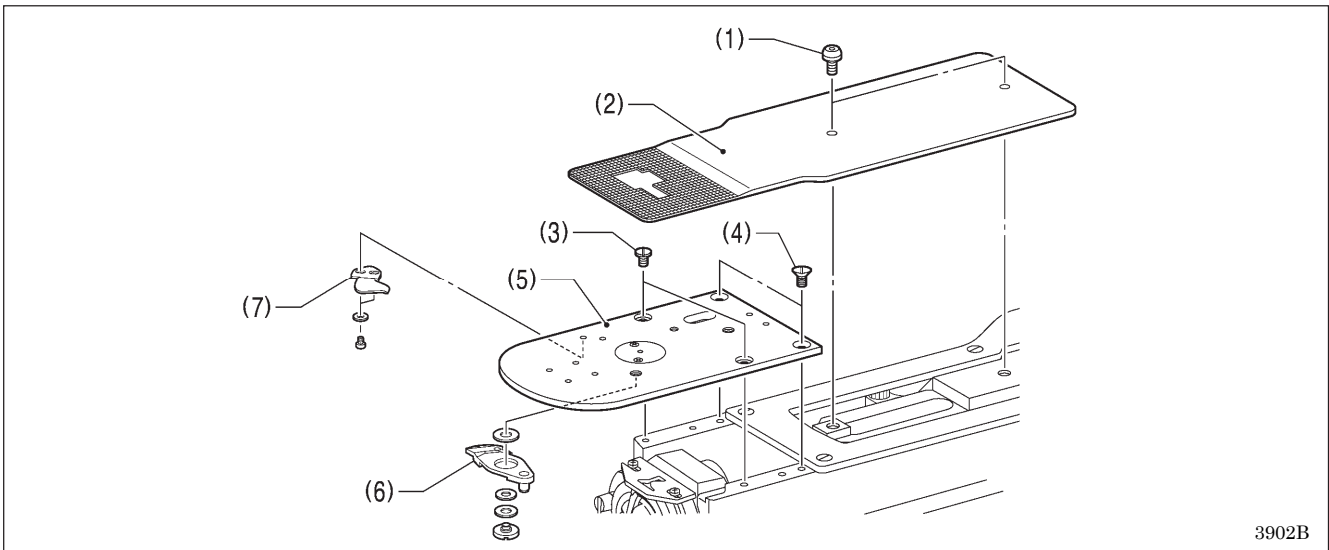
The standard position is when the oil tank assembly (2) is as far forward as possible as shown in the illustration.

9-7. Adjusting the position of the movable knife

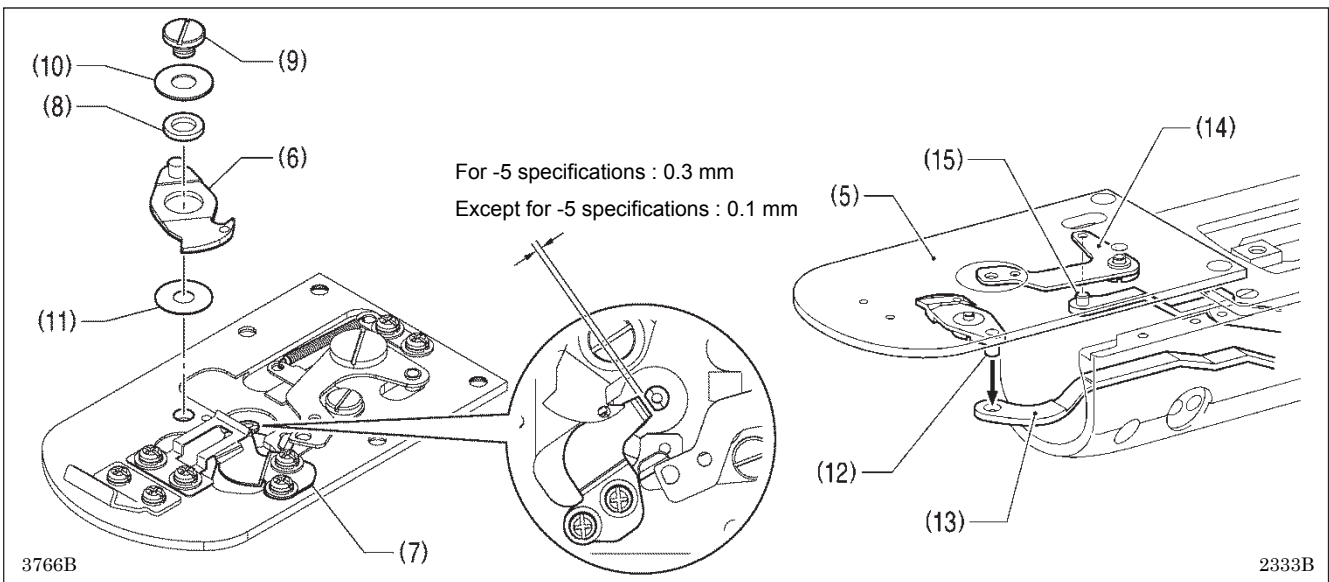


1. Remove the rear cover.
2. Tilt back the machine head.
3. Remove the motor bracket rear cover (1).
4. Turn the thread trimmer cam (2) by hand so that the movable knife pin (3) lowers to its lowest position.
5. Loosen the bolt (4).
6. Move the thread trimmer link (5) back and forth by turning the thread trimmer cam (2) by hand to adjust so that the ridge line on the movable knife (6) is touching the edge of the index mark (7), and then tighten the bolt (4) while keeping the parts in this position.
7. Install the motor bracket rear cover (1).
8. Return the machine head to its original position, and then install the rear cover.

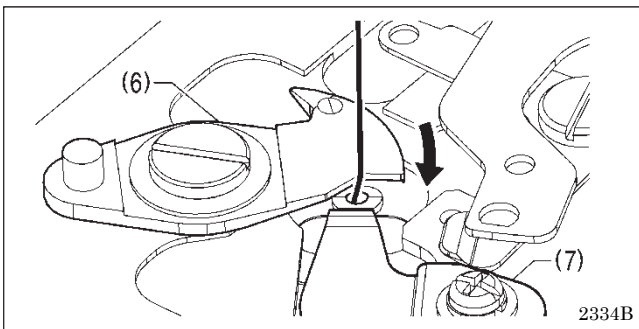
9-8. Replacing the movable knife and fixed knife



1. Open the shuttle race cover, remove the two screws (1), and then remove the feed plate (2).
2. Remove the two screws (3) and the two flat screws (4), and then remove the needle plate (5).
3. Remove the movable knife (6) and the fixed knife (7).



4. Install new fixed knife (7) in the position shown in the illustration.
5. Apply grease to the outside of the movable knife collar (8) and to the movable knife shoulder screw (9), and then install the new movable knife (6) together with the thrust washer (10) and movable knife spacer (11).
6. Check that the movable knife (6) and the fixed knife (7) cut the thread cleanly.
7. Apply grease to the pin (12) of the movable knife, and then insert it into the hole in the movable knife connecting plate (13), and then while inserting the pin (15) of TN-connecting plate assembly F into the hole in the thread nipper D assembly (14), install the needle plate (5).
8. Check that the center of the needle hole and the tip of the needle are aligned.



Select an accessory movable knife spacer (11) which allows the movable knife (6) to cross over the fixed knife (7) with a force that is as gentle as possible while still allowing the thread to be trimmed.

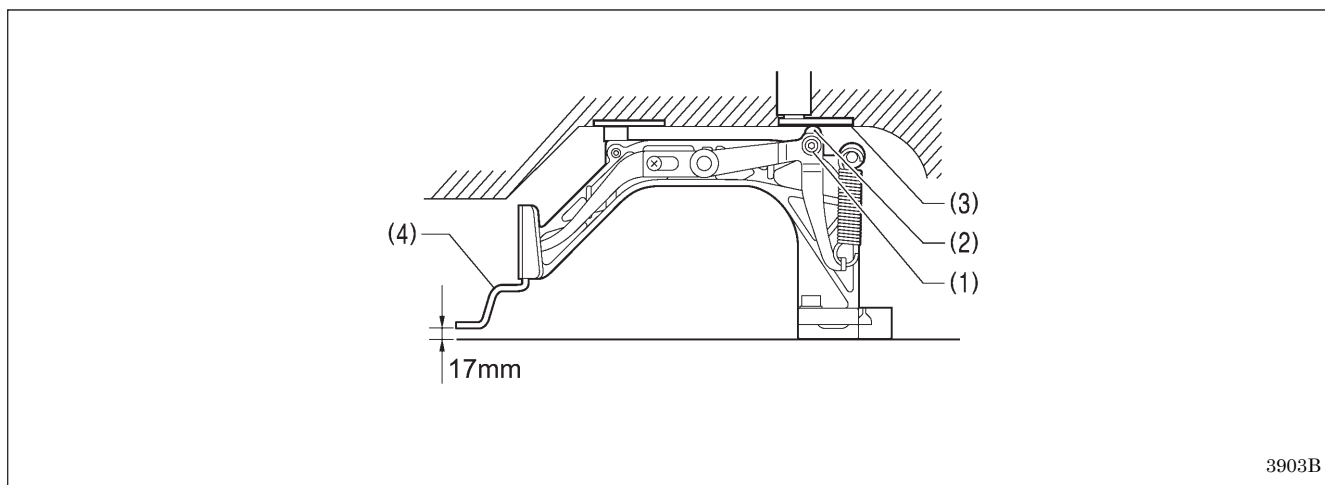
* Move the movable knife (6) as shown in the illustration and check that it trims the thread cleanly.

NOTE:

If the movable knife spacer (11) is too thick, it will not be possible to trim the thread.

9-9. Adjusting the work clamp lift amount (KE-430HX/HS)

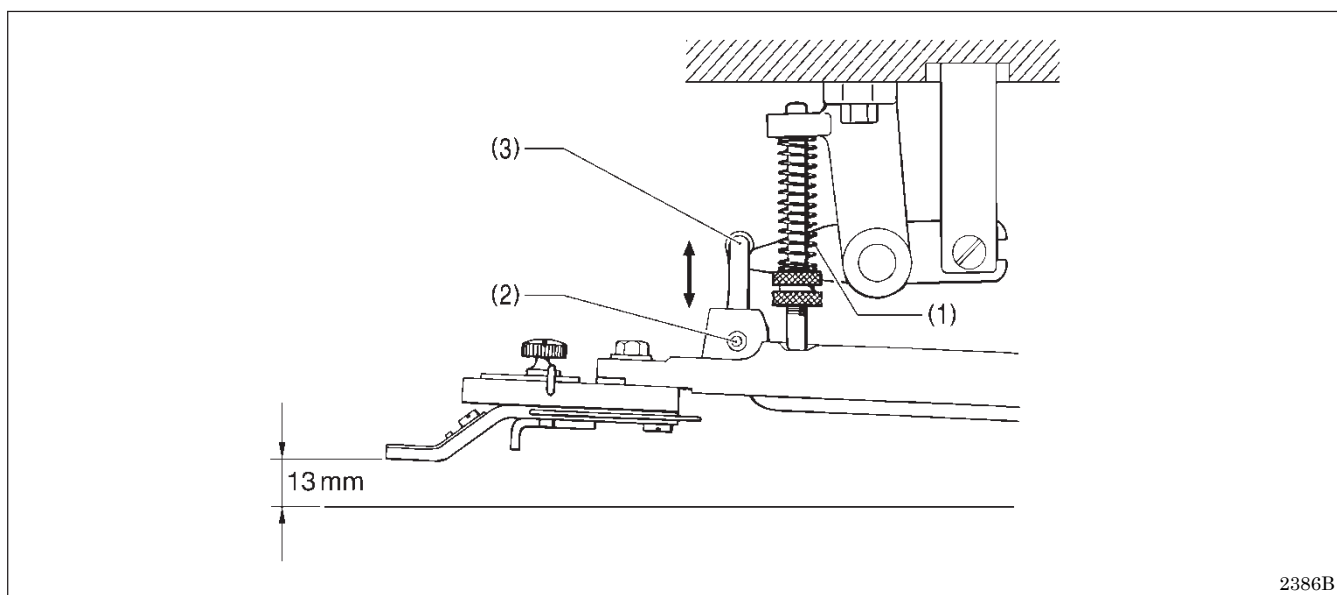
Adjust so that the actual maximum lift amount for the work clamp is 17 mm above the top of the needle plate when the maximum work clamp height has been set to 17 mm using the operation panel.



1. Loosen the bolt (1) and move the work clamp arm lever plate (2) up or down to adjust.
2. Apply grease to the bottom of the work clamp lifter plate assembly (3), to the top of the work clamp arm lever plate (2) and to the sliding part of the work clamp (4) (grease is already applied at the time of shipment), and check that the movement becomes easier.

9-10. Adjusting the button clamp lift amount (BE-438HX/HS)

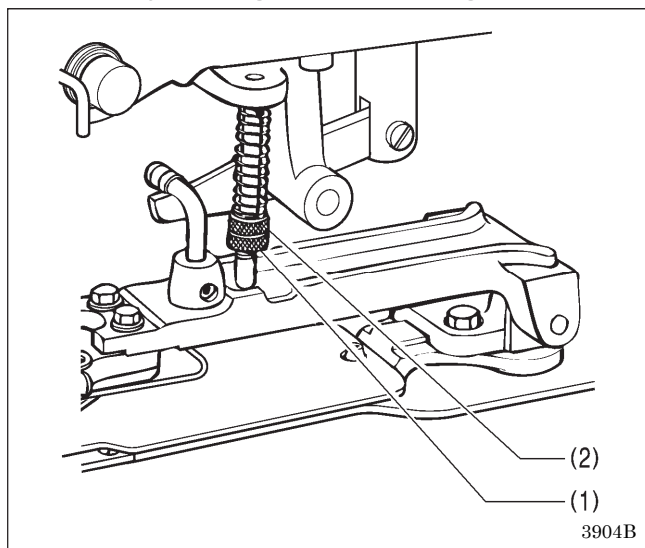
Adjust so that the actual maximum lift amount for the button clamp is 13 mm above the top of the needle plate when the maximum work clamp height has been set to 13 mm using the operation panel.



Remove the compression spring (1), loosen the screw (2) and adjust the button clamp holder hook assembly (3) by moving it up and down.

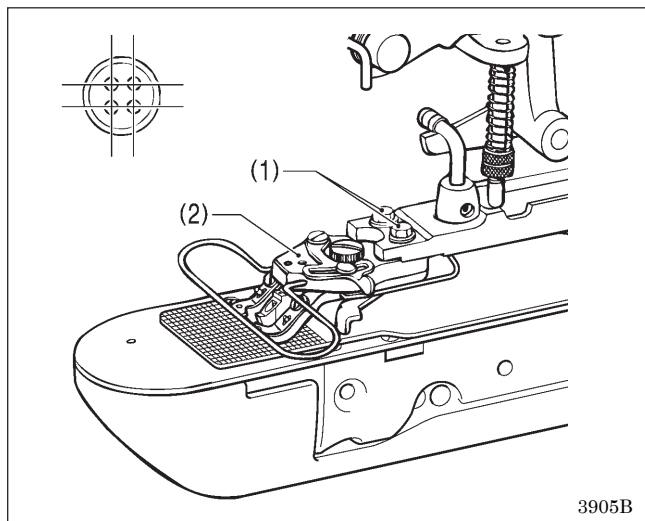
* If the button clamp lift amount is too large, the button clamp will not be raised.

9-11. Adjusting the holding pressure (BE-438HX/HS)



Loosen the nut (1) and turn the nut (2) to the point where it is just tight enough that the material will not slip out of place when it is slightly pulled (keeping pressure as slight as possible).

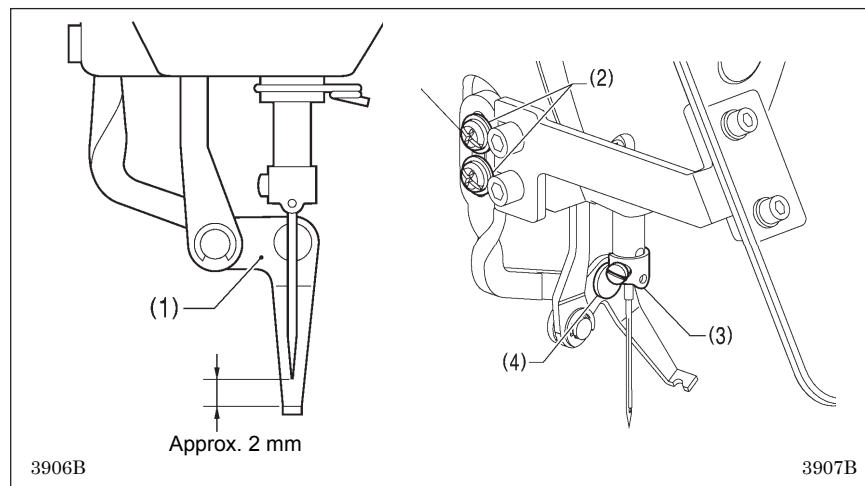
9-12. Adjusting the position of the button clamp (BE-438HX/HS)



1. Loosen the two bolts (1) and adjust the button clamp base (2) by moving it.
2. Carry out test feeding to check that the needle will go through the button hole with no contact.

9-13. Adjusting the thread wiper

<Thread wiper height adjustment>

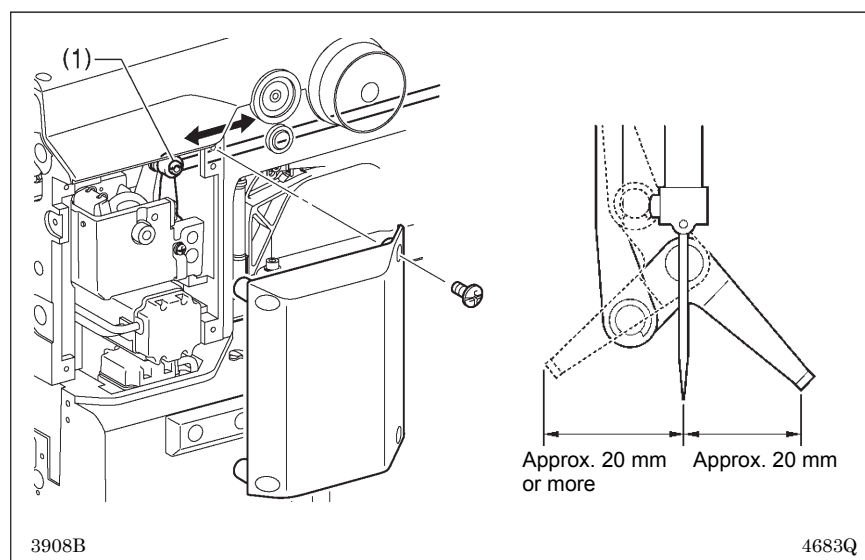


Loosen the two screws (2) and adjust the thread wiper (1) so that the clearance between the top of the thread wiper (1) and the tip of the needle is approximately 2 mm when the thread wiper (1) is aligned with the center of the needle at the needle bar stop position.

NOTE:

Check that the needle bar thread guide (3) and the thread wiper base shaft (4) are not touching at this time.

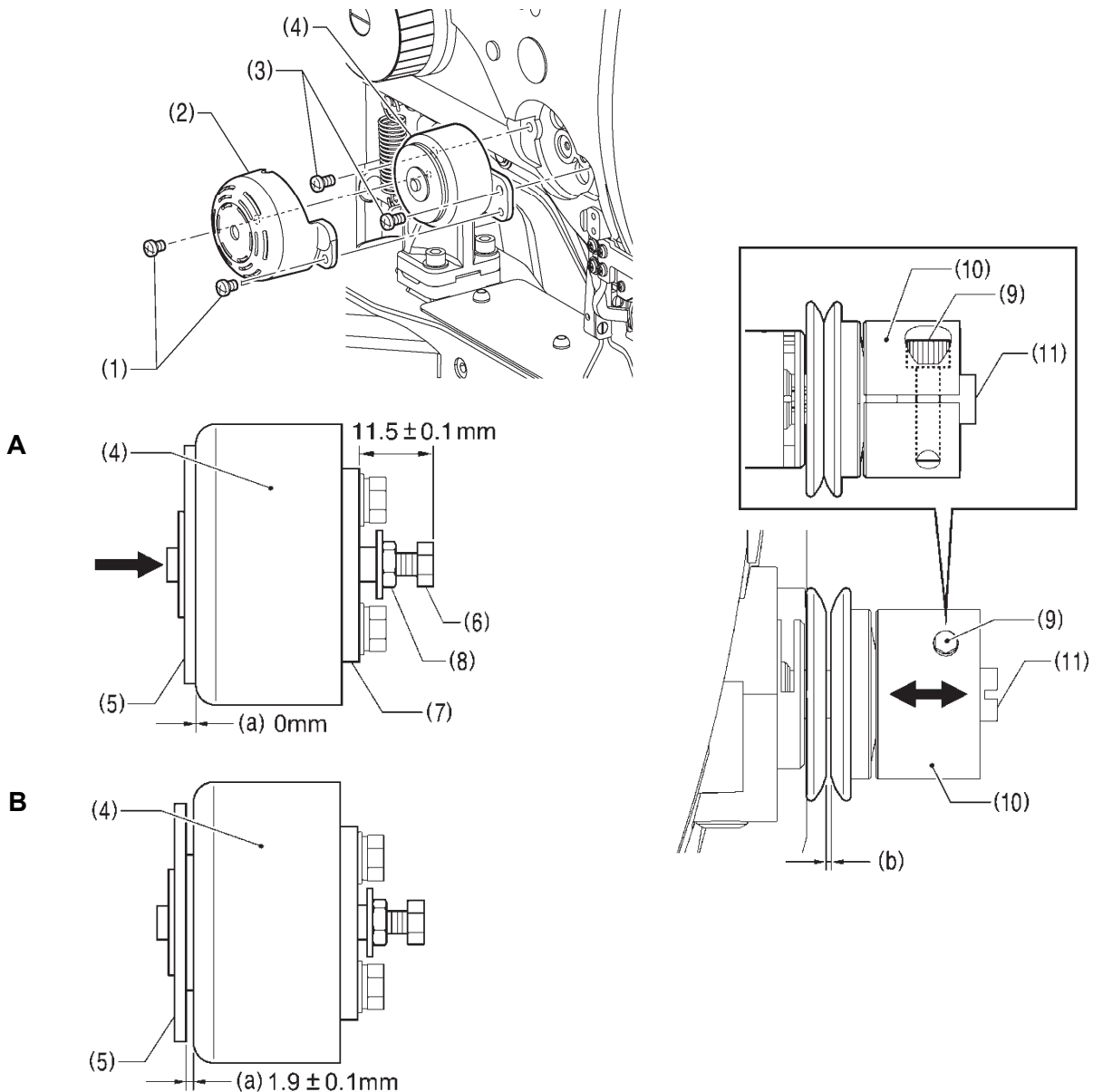
<Thread wiper racking width adjustment>



1. Remove the side cover.
2. Loosen the set screw (1) and adjust so that the distance from the thread wiper to the center of the needle is as shown in the illustration.

9-14. Adjusting the tension release amount (KE-430HX, BE-438HX)

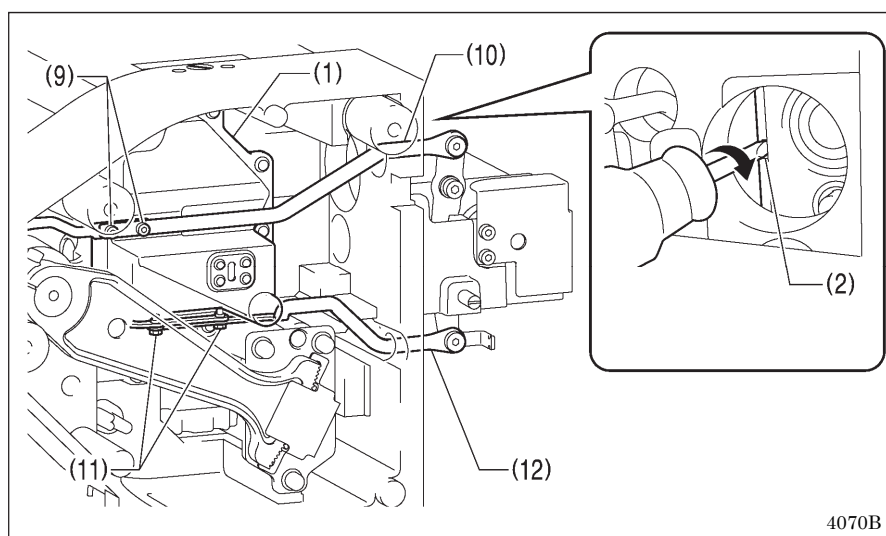
4069B

2738B
2928B

2739B

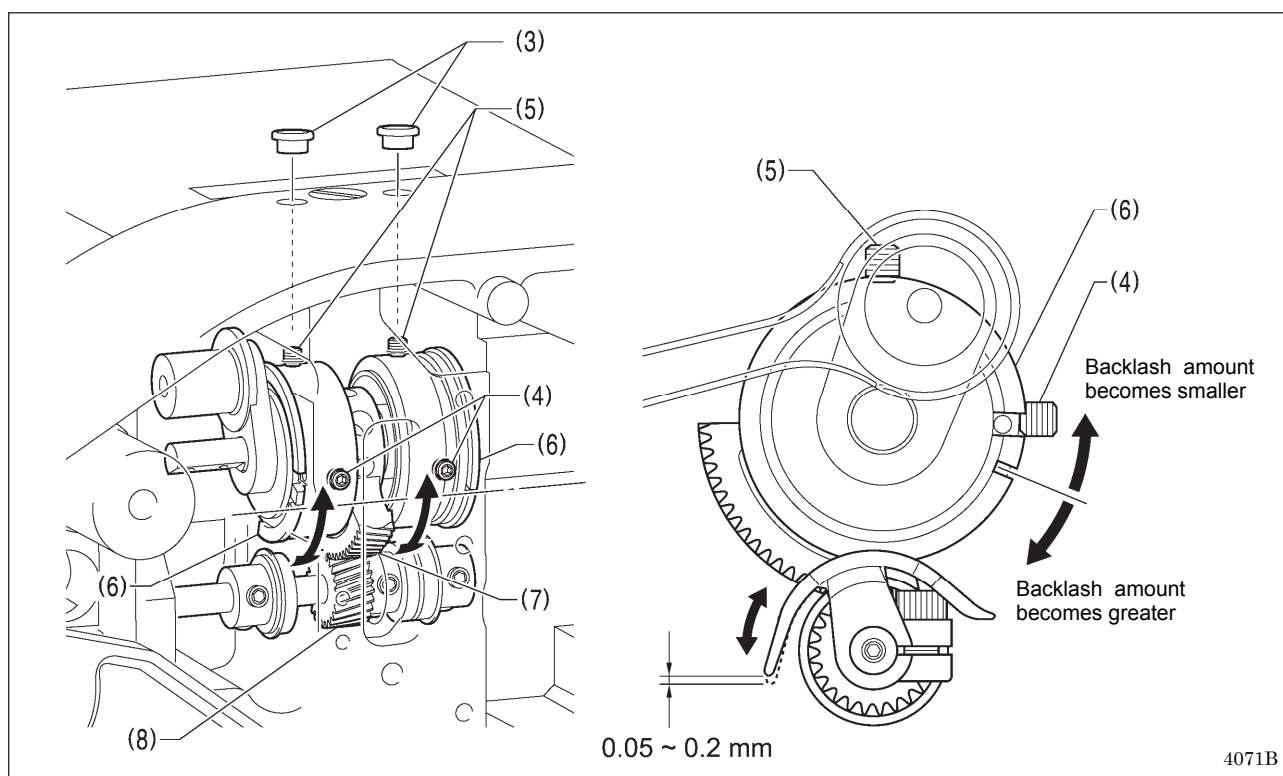
1. Remove the two screws (1), and then remove the solenoid cover (2).
2. Remove the two screws (3).
3. With the plunger (5) of the tension solenoid (4) pushed all the way in, loosen the nut (8) and turn the bolt (6) to adjust so that the end of the bolt (6) is 11.5 mm from the side of the solenoid setting plate (7). (Figure A)
4. With the tension solenoid (4) installed to the arm with the two screws (3), insert spacers or similar to operate the thread tension so that distance (a) is 1.9 ± 0.1 mm. (Figure B)
 - * Select a spacer that can maintain the distance (a) without becoming bent when the tension solenoid (4) operates.
 - * Refer to "9-1-1. Upper and lower thread tension" for details on operating the thread tension.
5. Loosen the bolt (9) and adjust the sideways position of the set collar (10) so that the opening amount (b) for the tension discs is 0 mm. After adjusting, retighten the bolt (9).
 - * Make sure that the movement range for the set collar (10) does not pass the edge of the shaft (11) at this time.
6. Release the thread tension.
 - Check that the opening amount (b) for the tension discs is 0.8 to 1.5 mm.
7. Install the solenoid cover (2) with the two screws (1).

9-15. Adjusting the backlash amount (play) of the lower shaft gear



1. Remove the rear cover.
2. Tilt back the machine head.
3. Open the shuttle race base cover.
4. Remove the shuttle hook, shuttle race base and bobbin case.
5. Remove the two bolts (9), and then remove the thread trimmer link (10).
6. (KE-430HX-03, -0K, -01 specifications only)
Remove the two bolts (11), and then remove the TN-connecting plate R (12).
7. Drain the oil in the bed bottom cover (1) while following the procedure in "10-2. Changing lubricating oil (When the oil change warning is displayed)".
8. Remove the ten screws.

9. Insert the tool (2) into the notch on the right side of the bed bottom cover (1), and then turn the tool (2) to remove the bed bottom cover from the bed. Be careful not to get any lubricating oil on the pulse motor X assembly at this time.



10. Remove the two rubber caps (3).
11. Loosen the two set screws (4) and the two set screws (5).
12. Use the tip of a screwdriver or similar to adjust the front and back rock gear collars (6) so that the backlash amount (play) is 0.05 to 0.2 mm.
 - * Alternately move the phase of the front and back rock gear collars (6) slightly to adjust so that the phase of the front and back rock gear collars (6) is about the same (within ± 10 degrees).
 - If the backlash amount (play) is small, it will result in wear in the rock gear (7) and the lower shaft gear (8).
 - If the backlash amount (play) is large, it will result in wear in the rock gear (7) and the lower shaft gear (8) and also result in increased noise during sewing.
13. First tighten the two set screws (4), and then tighten the two set screws (5).
14. Recheck that the backlash amount (play) which was adjusted in step 12 above is correct.
15. Install any parts which were removed from step 10 onward by carrying out the removal procedure in reverse.
 - * Refer to "10-2. Changing lubricating oil (When the oil change warning is displayed)" when adding the oil into the bed bottom cover.

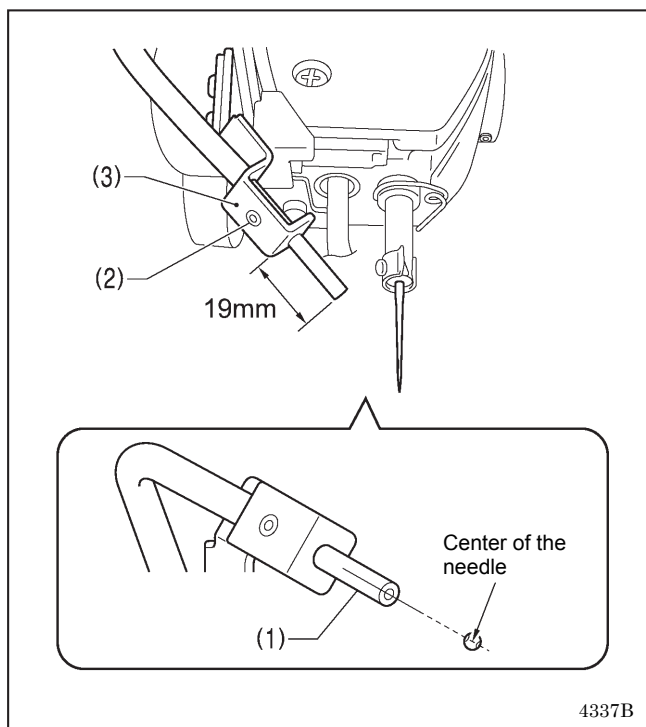
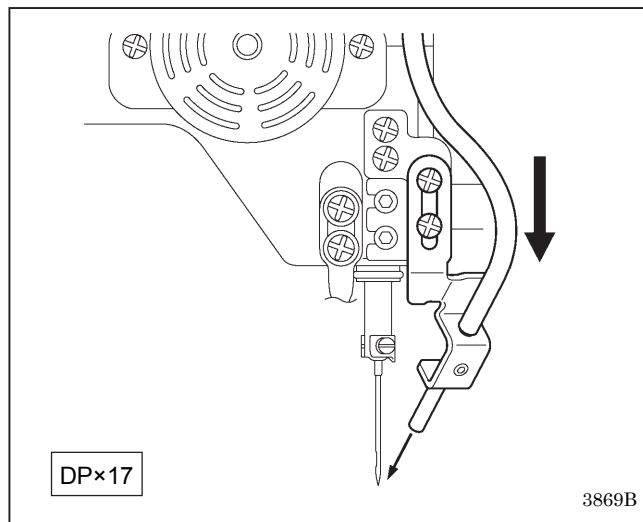
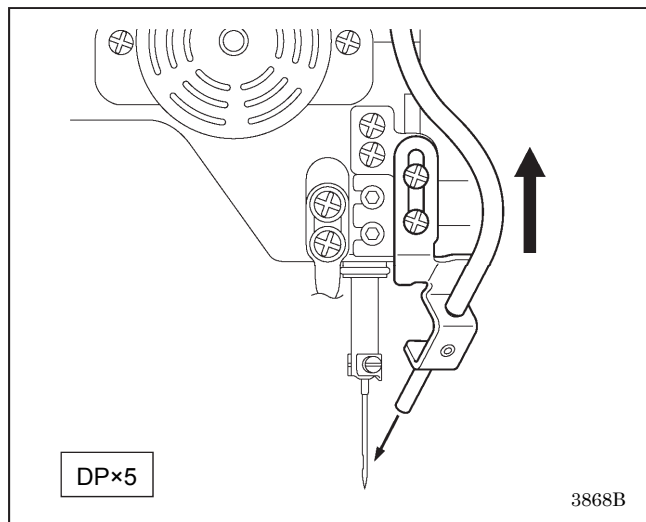
9-16. Adjusting the needle cooling

When securing the air nozzle assembly, change the height depending on the needle count.

* The height is adjusted at the time of shipment from the factory.

If the needle count is changed, adjust while referring to below.

- DPx5 (KE-430HX-3, -K, -1): Push the air nozzle assembly upward so that the open part of the slot is at the top.
- DPx17 (KE-430HX-5): Push the air nozzle assembly downward so that the open part of the slot is at the bottom.

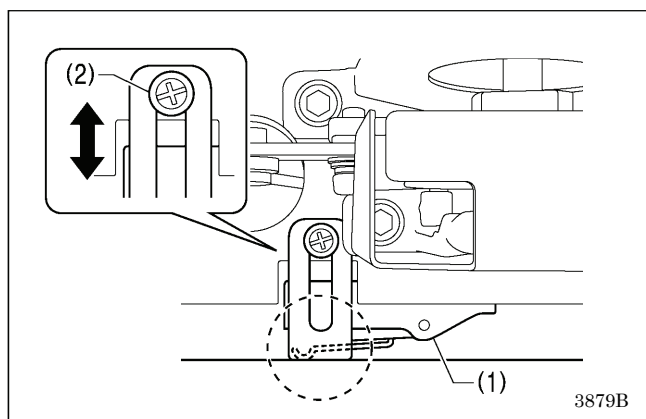


1. Turn on the power switch.
2. Depress the foot switch to the 2nd step.
The feed mechanism will move to the home position.
3. While sewing any program, depress the foot switch to the 2nd step to complete one cycle of operation.
4. Turn off the power switch.
5. Check that the center of the needle is aligned with the extension line from the air nozzle (1) at the needle bar stop position.
* Check that the extension line passes through the threading hole of the needle at this time.

NOTE:

- If the center of the needle is not aligned with the extension line from the air nozzle (1) at the needle bar stop position, the needle cooling may not function effectively.
- Do not loosen set screw (2). If accidentally loosened, tighten set screw (2) maintaining the distance of nozzle holder (3) and the tip of air nozzle (1) at 19 mm.

9-17. Checking the machine head switch



1. Turn on the power switch.
2. Check that no error numbers appear on the operation panel.

<If error “E050”, “E051” or “E055” is displayed>

If the machine head switch (1) is not turned on, error “E050”, “E051” or “E055” will occur.

Use the screw (2) to adjust the installation position of the machine head switch as shown in the illustration.

9-18. Adjusting the main shaft motor reference position

1	Remove the needle, the needle bar thread guide and the feed plate.
2	Adjustment menu - Main shaft motor reference position adjustment - Home position adjustment - Nipper adjustment While pressing the menu key, turn on the power switch. - If "Main shaft motor reference position adjustment" is selected in the adjustment menu screen, the display will switch to the main shaft motor reference position adjustment screen.
3	Main shaft motor reference position adjustment Turn the pulley to the setting 1 position OK Prepare a block (1) with a suitable thickness. Turn the pulley in the direction of the arrow (counterclockwise) so that the block (1) is clamped by the needle bar bush D (2) and the needle bar thread guide (3). At this time, press the OK key OK to set the current position as the setting 1 position. - If the setting 1 position is within the range, continue by adjusting the setting 2 position. - If the setting value is outside the range, adjust the main shaft again so that the machine pulley is at the setting 1 position. 4292B
4	Main shaft motor reference position adjustment Turn the pulley to the setting 2 position OK Remove the block (1), and then turn the pulley until the needle bar passes its highest position and starts to drop. Clamp the block (1) between the needle bar bush D (2) and the needle bar thread guide (3) once more. At this time, press OK key OK to set the current position as the setting 2 position. - If the setting 2 position is within the range, the message "Successfully stored the adjusted value." will be displayed, and then the display will return to the adjustment menu screen. - If the setting value is outside the range, adjust the main shaft again so that the machine pulley is at the setting 2 position. Main shaft motor reference position adjustment Successfully stored the adjusted value. OK
5	Turn off the power switch and install the needle, the needle bar thread guide and the feed plate.

9-19. Adjusting the X- and Y-feed motor home position

1	With the power switch turned off, turn the pulley in the direction of the arrow to set the needle to the needle up stop position (near the highest thread take-up position).	
2	Adjustment menu Main shaft motor reference position adjustment Home position adjustment Nipper adjustment	While pressing the menu key, turn on the power switch.  Select "Home position adjustment" in the adjustment menu screen, and the display will switch to the home position adjustment screen.
3	Home position adjustment X motor (1) 0 Y motor (2) 0 Work clamp motor 0 OK	After switching to the home position adjustment screen, touch the "X motor" setting value (1) or the "Y motor" setting value (2) to select it. When a setting value is touched, it is highlighted in black, and the value can then be changed.
4	Home position adjustment X motor 0 Y motor 0 Work clamp motor 0 OK	Touch the check key  (3). - Home position detection will be carried out based on the currently-displayed offset value. - When the reset key  (4) is pressed, the offset value which is selected at the time will be reset to "0", and the feed motor will then carry out home position detection.

(Continued to the next page.)

5

Home position

3950B

For example, if the offset value is 10

Home position adjustment			
X motor	(1)	10	+
Y motor	(2)	0	-
Work clamp motor	(5)	0	
			OK

Use the feed plate and press the or key to change the offset value so that the needle drops at the home position.

- The X-feed or Y-feed motor will operate according to the offset value.
- The offset value will flash after it has been changed.
- The offset value to be changed can be switched between the value for the X motor (1), Y motor (2) and work clamp motor (5) at any time simply by touching the value.

6

Home position adjustment			
X motor		10	+
Y motor		0	-
Work clamp motor		0	
(3)			OK

Touch the check key (3) to provisionally confirm the offset value.

- Home position detection for the feed motor will be carried out based on the currently-displayed offset value. Check the home position.

7

Repeat steps 5 and 6 to set the home position to the correct position.

8

Completion of setting

Home position adjustment			
X motor		10	+
Y motor		- 15	-
Work clamp motor		0	
			OK

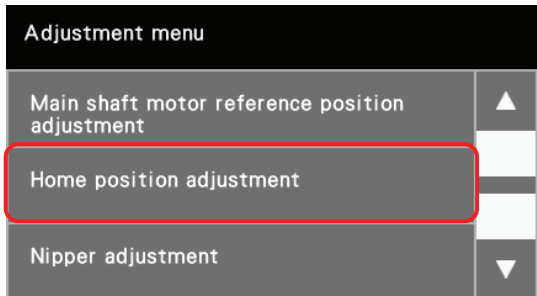
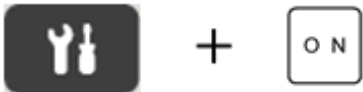
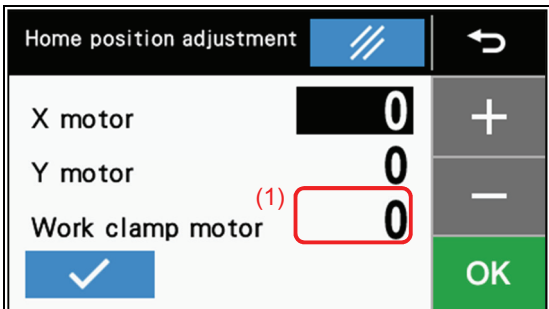
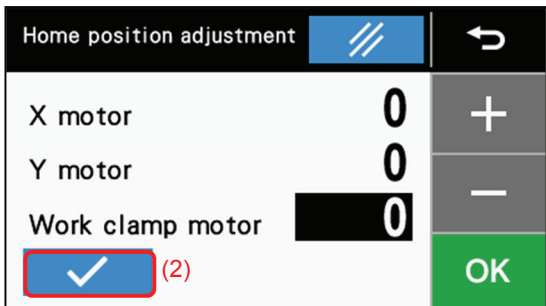
Touch the OK key to confirm the offset value.

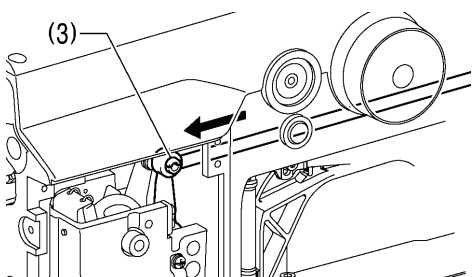
- The changed offset value will stop flashing and will be illuminated on the screen.
- * If the offset value has not been provisionally confirmed (if the check key was not touched), the invalid buzzer will sound.

When the setting has been completed, press the back key .

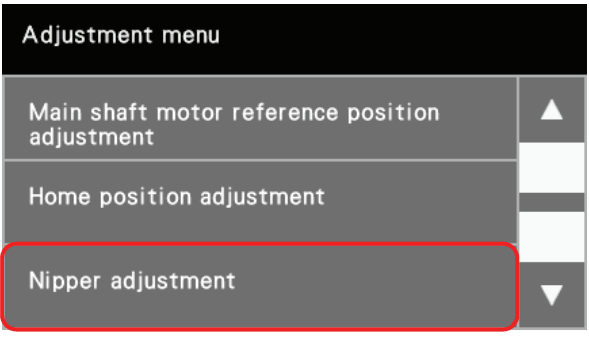
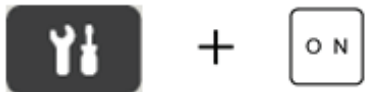
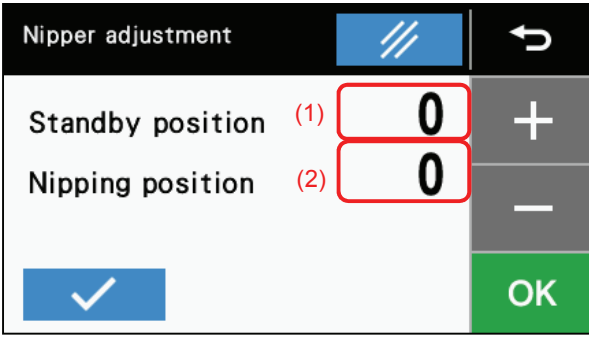
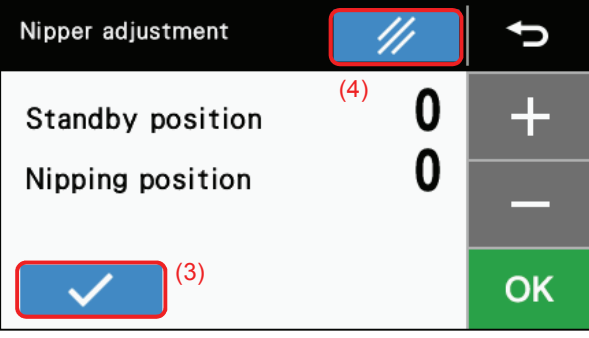
- The display will return to the adjustment menu screen.

9-20. Adjusting the work clamp motor home position

1	With the power switch turned off, turn the pulley in the direction of the arrow to set the needle to the needle up stop position (near the highest thread take-up position).	
2		While pressing the menu key, turn on the power switch.  Select "Home position adjustment" in the adjustment menu screen, and the display will switch to the home position adjustment screen.
3		After switching to the home position adjustment screen, touch the "Work clamp motor" setting value (1) to select it. When a setting value is touched, it is highlighted in black, and the value can then be changed.
4		Touch the check key  (2). - Home position detection for the work clamp motor will be carried out based on the currently-displayed offset value. * When the reset key  is pressed, the offset value which is selected at the time will be reset to "0", and the selected work clamp motor will then carry out home position detection.

5	 4009B Home position adjustment X motor0 + - Y motor0 + - Work clamp motor-2 + - ✓ OK For example, if the offset value is -2 Press the + key or the - key (4) until the thread wiper driving lever (3) is at the position where it starts moving to set the offset value. • The work clamp motor will operate in accordance with the offset value. • The offset value will flash after it has been changed.
6	Home position adjustment X motor0 + - Y motor0 + - Work clamp motor-1 + - ✓ (2) OK Touch the check key ✓ (2) to provisionally confirm the offset value. • Home position detection for the work clamp motor will be carried out based on the currently-displayed offset value. Check the home position.
7	Repeat steps 5 and 6 to set the home position to the correct position.
8	Home position adjustment X motor0 + - Y motor0 + - Work clamp motor-1 + - ✓ OK Touch the OK key OK to confirm the offset value. • The changed offset value will stop flashing and will be illuminated on the screen. * If the offset value has not been provisionally confirmed (if the check key was not touched), the invalid buzzer will sound. When the setting has been completed, press the back key . • The display will return to the adjustment menu screen.
9	Turn off the power switch.

9-21. Adjusting the thread nipper motor position

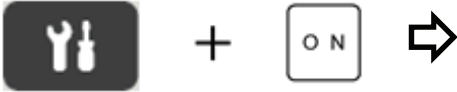
1	With the power switch turned off, turn the pulley in the direction of the arrow to set the needle to the needle up stop position (near the highest thread take-up position).	
2	Open the shuttle race base cover and remove the shuttle race base, shuttle hook and bobbin case assembly.	
3		While pressing the menu key, turn on the power switch.  If "Nipper adjustment" is selected in the adjustment menu screen, the display will switch to the nipper adjustment screen.
4		After switching to the nipper adjustment screen, touch the "Standby position" setting value (1) or the "Nipping position" setting value (2) to select it. When a setting value is touched, it is highlighted in black, and the value can then be changed.
5		Touch the check key (3). Home position detection will be carried out based on the currently-displayed offset value. * When the reset key (4) is pressed, the offset value which is selected at the time will be reset to "0", and the thread trimmer motor will then carry out home position detection.

6	3966B For example, if the standby position offset is "5" <Standby position> Press the + key or the - key (7) to change the offset value so that the hole (5) in the needle hole plate is near the center of the hole in the thread nipper D assembly (6). <Nipping position> Pass the thread through the hole (5) in the needle hole plate and the hole in the thread nipper D assembly (6), and then press the + key or the - key (7) to change the offset value. • The thread trimmer motor will operate in accordance with the offset value. • The offset value will flash after it has been changed.
7	Completion of setting Touch the OK key OK to confirm the offset value. When the setting has been completed, press the back key ↩. • The display will return to the adjustment menu screen.
8	Turn off the power switch, install the shuttle race base, shuttle hook and bobbin case assembly, and then close the shuttle race base cover.

9-22. Setting method for standard depression strokes (Foot switch)

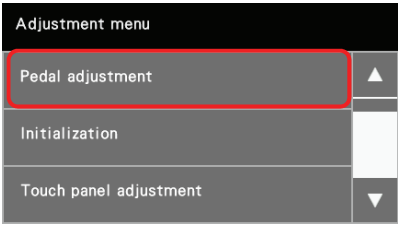
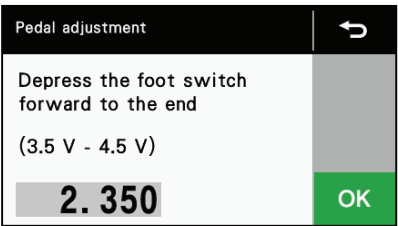
Use the following procedure to set the operating positions for the depression stroke.

- 1 The display will switch to the pedal adjustment screen.**

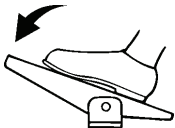


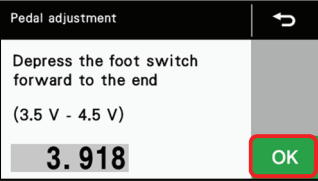
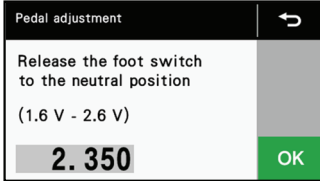
While pressing the menu key, turn on the power switch.

 - The panel display will switch to the adjustment menu screen.
- 2 The display will switch to the pedal adjustment screen.**

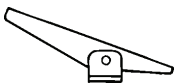
• Touch "Pedal adjustment" in the adjustment menu screen to switch to the pedal adjustment screen.
- 3 Memorizing the maximum forward position**

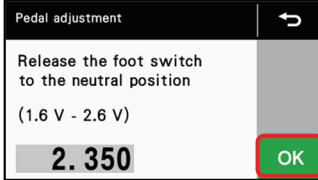
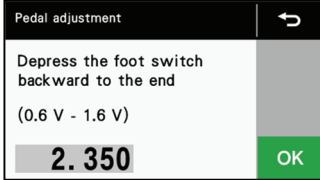


With the foot switch depressed all the way forward, wait until the voltage display value stabilizes and then press the OK key .

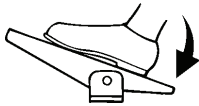
2827B
- 4 Memorizing the neutral position**

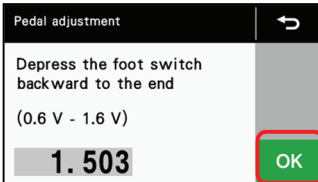
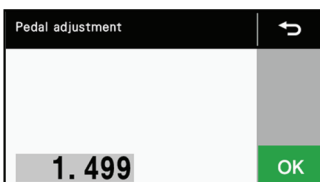


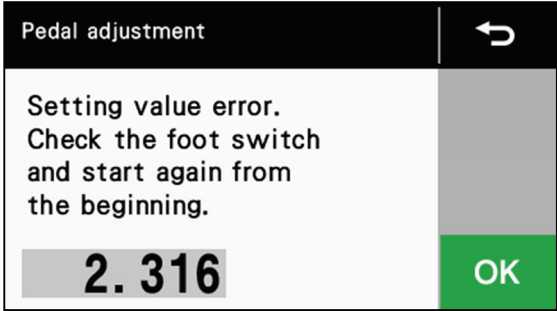
With your foot released from the foot switch, wait until the voltage display value stabilizes and then press the OK key .

2828B
- 5 Memorizing the maximum backward position**



With the foot switch depressed all the way backward, wait until the voltage display value stabilizes and then press the OK key .
- 6 Completion of setting**



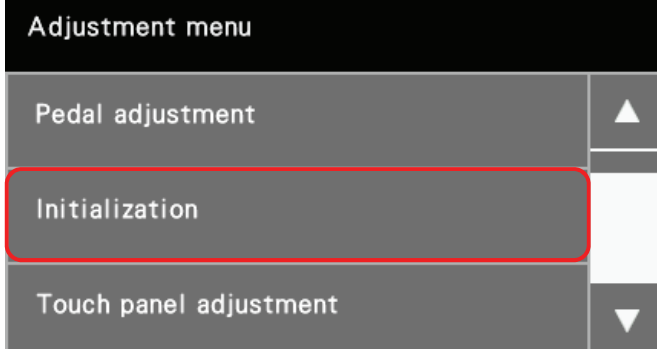
When the setting is complete, the display will return to the adjustment menu screen.

NOTE:
If the foot switch operation is not carried out correctly, the screen shown in the illustration at left will be displayed and the buzzer will sound twice. Repeat the procedure from recording the maximum pedal depression amount in step 3.

9-23. Data initialization method

If the sewing machine stops operating normally, the cause may be that an incorrect setting may have been made for the memory data by means of memory switches, for instance. In such cases, it may be possible to restore normal operation by following the steps given below to initialize the memory setting data.

1 Switch to initialization mode.



(While pressing the menu key, turn on the power switch.)

The adjustment menu screen will be displayed. Touch "Initialization" in the adjustment menu to switch to the initialization screen.

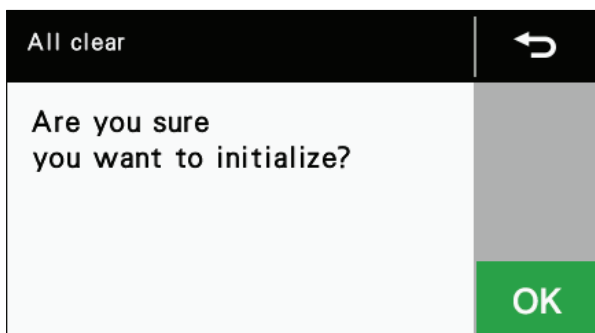
[Reference]

If the ▲ or ▼ key is touched at this time, you can select the items to be initialized as shown below.

* For details on initialization items other than "All clear", refer to the Service Manual.

	Items displayed on the screen	Initialization items
	All clear	All data in memory is cleared or initialized.
	Standard program initialization	The sewing condition settings are initialized.
	Additional program initialization	All additional program settings are initialized.
	User program initialization	All user program settings are initialized.
	Cycle program initialization	All cycle program settings are initialized.
	Memory switch initialization	All memory switches which can be initialized are initialized.

2

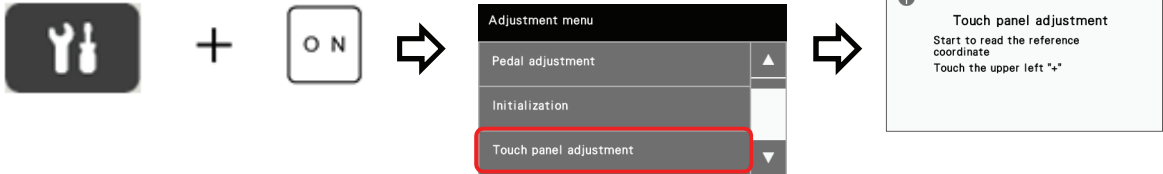
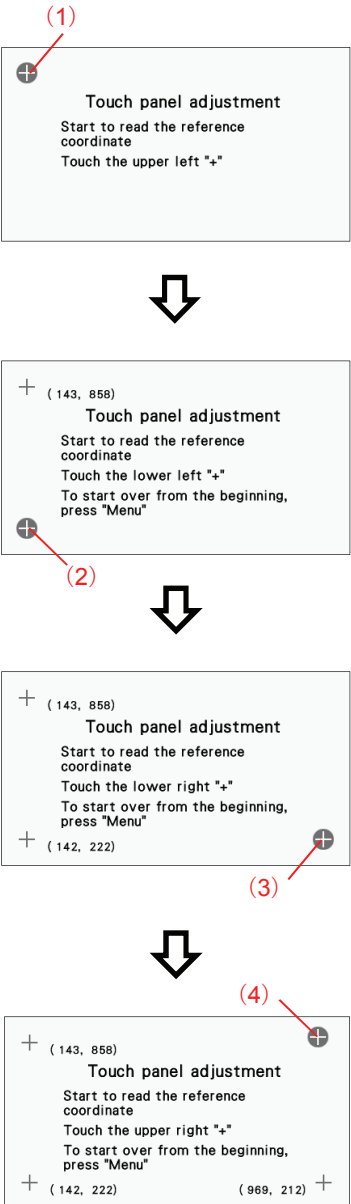


If you touch an item name in the list, the initialization check screen will be displayed.

- Touch the OK key  to start the initialization.
- When the screen display returns to the initialization screen, initialization is complete.

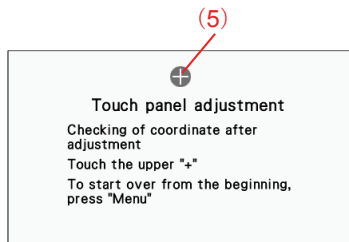
9-24. Adjusting the touch panel key position

If the touch key reading coordinates and the touch key (icon) display positions do not correspond, carry out the following adjustment.

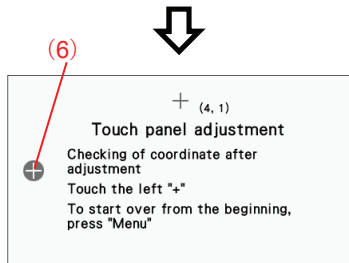
1	<Starting software version update mode> - While pressing the menu key, turn on the power to switch to the adjustment menu screen. - Touch "Touch panel adjustment" in the adjustment menu screen to switch to the touch panel adjustment screen. 
2	<Touch key adjustment mode (coordinate reading)> This procedure reads the coordinates at four places on the screen. Carry out the following operations using a touch pen or similar to press the positions accurately.  - Press the center of the "+" mark (1) in the top-left corner. - If the coordinates which are read are within the allowable range, the next screen is displayed. On the other hand, if the coordinates which are read are outside the allowable range, the buzzer will sound twice, so repeat the operation. - Press the center of the "+" mark (2) in the bottom-left corner. - If the coordinates which are read are within the allowable range, the next screen is displayed. On the other hand, if the coordinates which are read are outside the allowable range, the buzzer will sound twice, so repeat the operation. - Press the center of the "+" mark (3) in the bottom-right corner. - If the coordinates which are read are within the allowable range, the next screen is displayed. On the other hand, if the coordinates which are read are outside the allowable range, the buzzer will sound twice, so repeat the operation. - Press the center of the "+" mark (4) in the top-right corner. - If the coordinates which are read are within the allowable range, the coordinate checking screen is displayed. On the other hand, if the coordinates which are read are outside the allowable range, the buzzer will sound twice, so repeat the operation.

3 <Touch key adjustment mode (coordinate checking)>

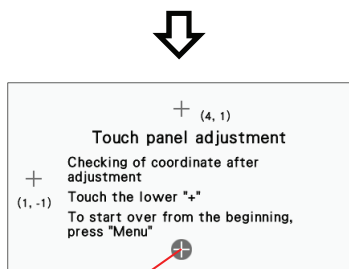
This procedure measures the error between the coordinates at the centers of the keys and the coordinates which are read at four places on the screen.



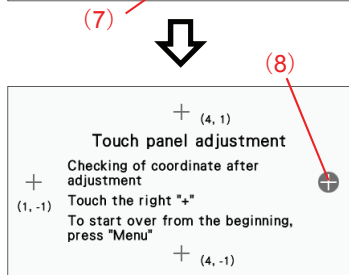
1. Press the center of the "+" mark (5) at the center-top.
2. If the error is within the allowable range (-5 to +5), the next screen is displayed.
On the other hand, if the error is outside the allowable range, the buzzer will sound twice, so repeat the operation.



3. Press the center of the "+" mark (6) at the center-left.
4. If the error is within the allowable range (-5 to +5), the next screen is displayed.
On the other hand, if the error is outside the allowable range, the buzzer will sound twice, so repeat the operation.

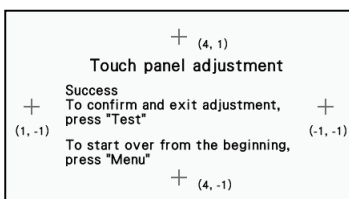


5. Press the center of the "+" mark (7) at the center-bottom.
6. If the error is within the allowable range (-5 to +5), the next screen is displayed.
On the other hand, if the error is outside the allowable range, the buzzer will sound twice, so repeat the operation.



7. Press the center of the "+" mark (8) at the center-right.
8. If the errors are within the allowable range (-5 to +5), the adjustment value confirmation checking screen is displayed.
On the other hand, if the error is outside the allowable range, the buzzer will sound twice, so repeat the operation.

4 <Adjustment value confirmation checking>



1. To confirm the adjustment values and exit, press the test key . On the other hand, if you would like to repeat the procedure from the beginning, press the menu key .
* The procedure returns to step 2.



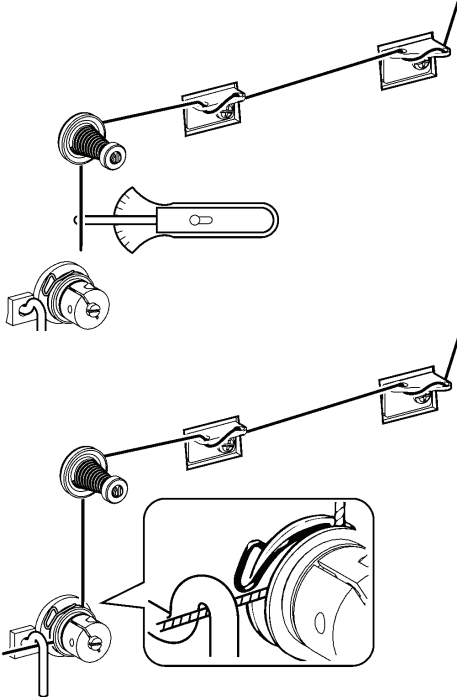
2. Once the adjustment values have been saved, turn off the power.

9-25. Correcting the digital tension output (KE-430HX, BE-438HX)

- The procedure for correcting the tension (output) from the digital tension involves carrying out the procedure of measuring the actual upper thread tension and entering the four measured values using the operation panel.
- The standard values for the four tensions for the digital tension vary for each measurement as shown in the table at right. By entering the actual measurement values each time, the digital tension output will be automatically corrected, and the tension that corresponds to the setting can then be obtained when sewing.

Standard tension value for digital tension (gf)
30
80
180
330

1



1. Adjust the pretension to 5 to 7 gf (0.05 to 0.07 N).
* At this time, use the TOYOBO polyester #50 (white) thread which is recommended by Brother.

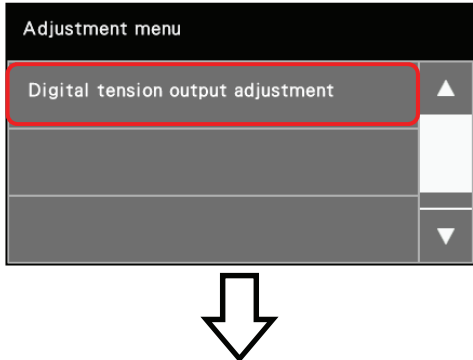
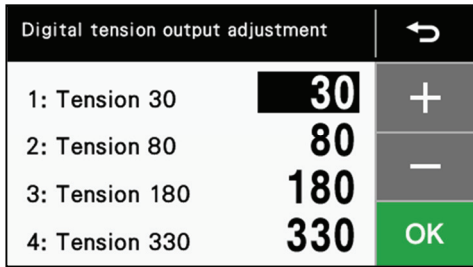
2. Thread the upper thread as shown in the illustration at left.

NOTE:
Do not thread the thread through the thread take-up spring.

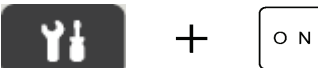
* If you do not observe the above note, the digital tension output correction may not be carried out correctly.

2816B 2817B

2

1. While pressing the menu key, turn on the power switch.



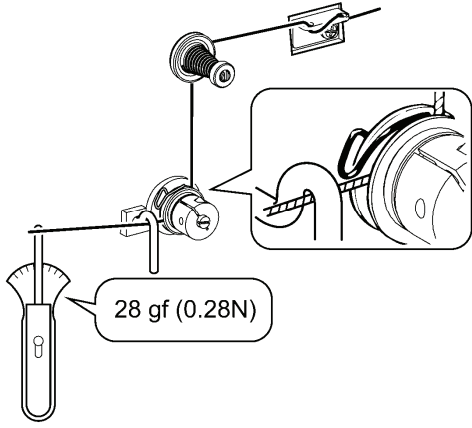
* Keep pressing the menu key  until the model name is displayed and the buzzer beeps once.

2. Touch "Digital tension output adjustment" in the adjustment menu screen to switch to the adjustment screen.

* This function can only be used with the KE-430HX and BE-438HX.

9. ADJUSTMENT

3



28 gf (0.28N)

2820B

Use a tension gauge to measure the upper thread tension as shown in the illustration at left.

Press the **+** or **-** key (1) to increase or decrease the value appearing in the menu display until it corresponds with the measurement value.

Digital tension output adjustment		↩
1: Tension 30	28	+
2: Tension 80	80	-
3: Tension 180	180	OK
4: Tension 330	330	

(1)

For example, if the upper thread tension is 28 gf

- * Round off any decimal places in the measurement value.
- * The units for the measurement values which are input to the operation panel are "gf". If the values are measured using units of "N", convert using a factor of 1 N ÷ 100 gf.
- * The value will illuminate, and it will change to flashing if it is increased or decreased.

4 Confirm the correction value.

Digital tension output adjustment		↩
1: Tension 30	28	+
2: Tension 80	80	-
3: Tension 180	180	OK
4: Tension 330	330	

1. Touch the OK key OK.

2. The selected value will be set as the correction value.

- The setting value will stop flashing and will illuminate.

NOTE:

If you need to set the range to a value which is higher than the value in the "Allowable setting range" column of the table, problems with operation of the digital tension may occur.

Carry out the steps in "9-14. Adjusting the tension release amount (KE-430HX, BE-438HX)", and then repeat the procedure from item 1.

5 Set the remaining three items for upper thread tension in the same was as in steps 3 and 4.

- You can change the value for an item by touching the setting value for the item you would like to change.

6 Completion of setting

Digital tension output adjustment		↩
1: Tension 30	28	+
2: Tension 80	85	-
3: Tension 180	172	OK
4: Tension 330	310	

You can return to the adjustment menu by pressing the back key ↩.

Standard value (gf)	Allowable setting range (gf)	Digital tension output adjustment		↩
30	[11 – 49]	1: Tension 30	30	+
80	[51 – 109]	2: Tension 80	80	
180	[121 – 239]	3: Tension 180	180	–
330	[241 – 491]	4: Tension 330	330	OK

10. MAINTENANCE

! CAUTION



Turn off the power switch before carrying out cleaning. The machine may operate if the foot switch is depressed by mistake, which may result in injury.



Be sure to wear protective goggles and gloves when handling the lubricating oil and grease, so that they do not get into your eyes or onto your skin, otherwise inflammation can result.

Furthermore, do not drink the oil or eat the grease under any circumstances, as they can cause vomiting and diarrhea. Keep the oil out of the reach of children.



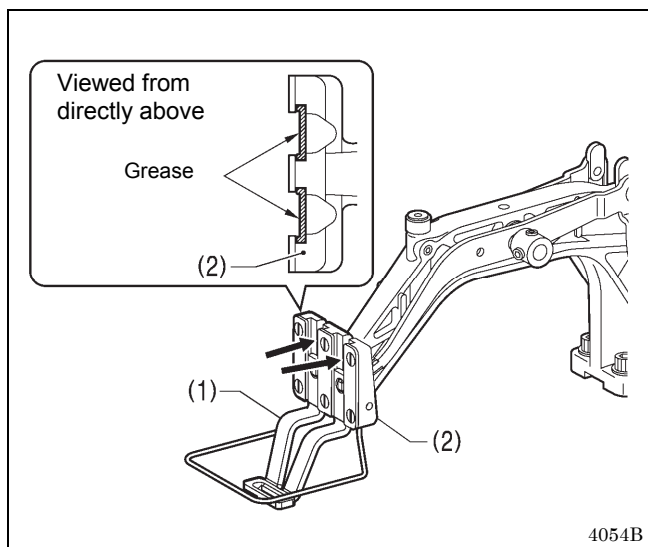
Secure the table so that it will not move when tilting back the machine head. If the table moves, it may crush your feet or cause other injuries.



Use both hands to hold the machine head when tilting it back or returning it to its original position.

If only one hand is used, the weight of the machine head may cause your hand to slip, and your hand may get caught.

10-1. Applying grease (Work clamp: KE-430HX/HS)



Periodically apply grease to the sliding parts of the work clamp (1) and the work clamp arm (2).

NOTE:

After replacing the work clamp (1), be sure to apply grease before using it.

- * It is recommended that you use commercially-available JXTG Nippon Oil & Energy POWERNOC WB 2 grease for the work clamp (1).

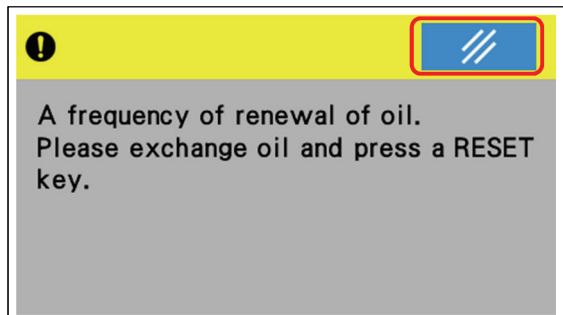
10-2. Changing lubricating oil (When the oil change warning is displayed)

If the oil change warning is displayed when the power switch is turned on, it is to notify you that it is time to change the oil.

(The sewing machine will not operate at this time, even if the foot switch is depressed.)

Change oil as required, while referring to the following for details.

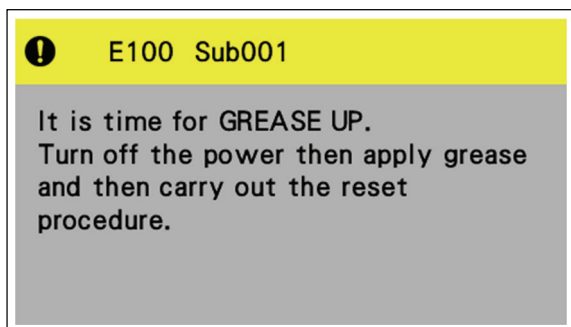
< To continue sewing temporarily without changing oil >



1. Press the reset key .
2. After you return to the home screen, you can depress the foot switch to resume sewing.

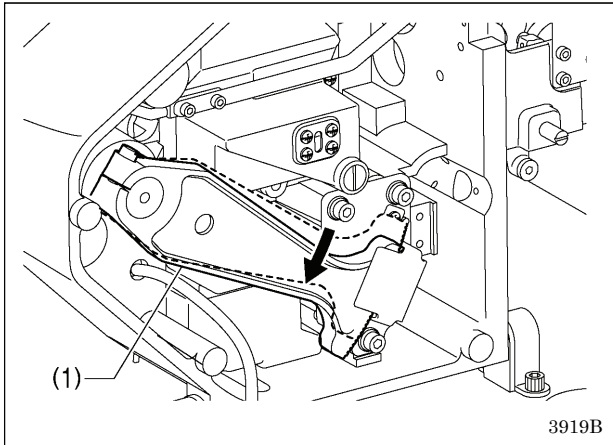
NOTE:

- The oil change warning will continue to be displayed each time the power is turned on until the oil is changed and the notification is reset by carrying out the procedure on page 164.
- If you continue to use the sewing machine after the oil change warning appears without changing the oil (or without carrying out the reset procedure), "E100" will appear after a certain period of time and the sewing machine will be forcibly prevented from operating for safety reasons.
If this happens, change the oil and then carry out the reset procedure.

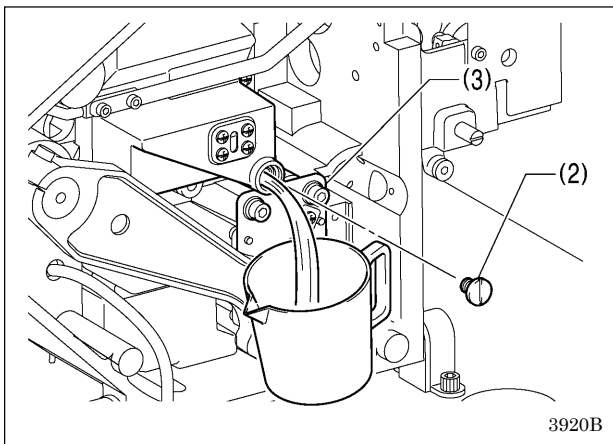


- * If you continue to use the sewing machine after carrying out the reset procedure but without changing the oil, problems with the sewing machine may result.

- Use only the lubricating oil <JXTG Nippon Oil & Energy SEWINGLUBE N 10; VG10> specified by Brother.



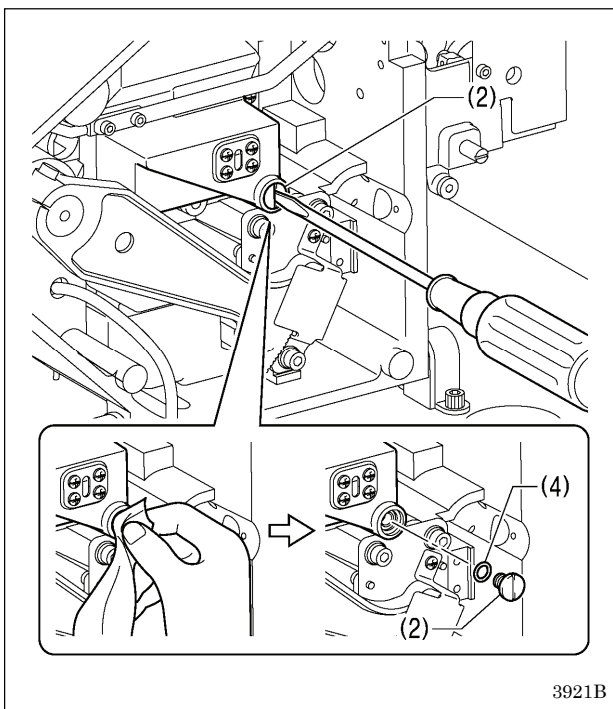
1. Turn off the power switch.
2. Tilt back the machine head.
3. Move the X-feed gear (1) in the direction of the arrow.



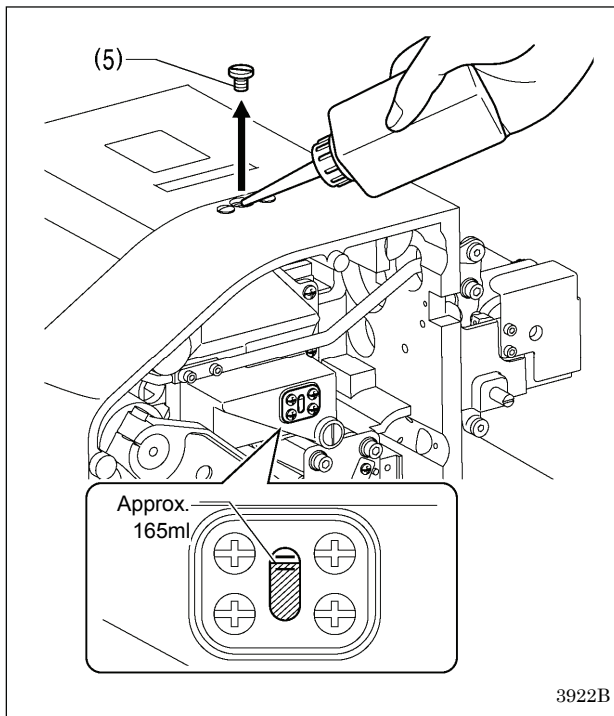
4. Have an oil pan ready, and then remove the screw (2).
- * When the screw is removed, the oil will gush out. Make sure the oil pan is facing the correct way.

NOTE:

- Be careful not to get any lubricating oil on the X-feed motor (3). Otherwise it may cause problems with the operation of the pulse motor.
- Do not deliberately tilt the sewing machine in order to fully drain the oil. Injury may result.



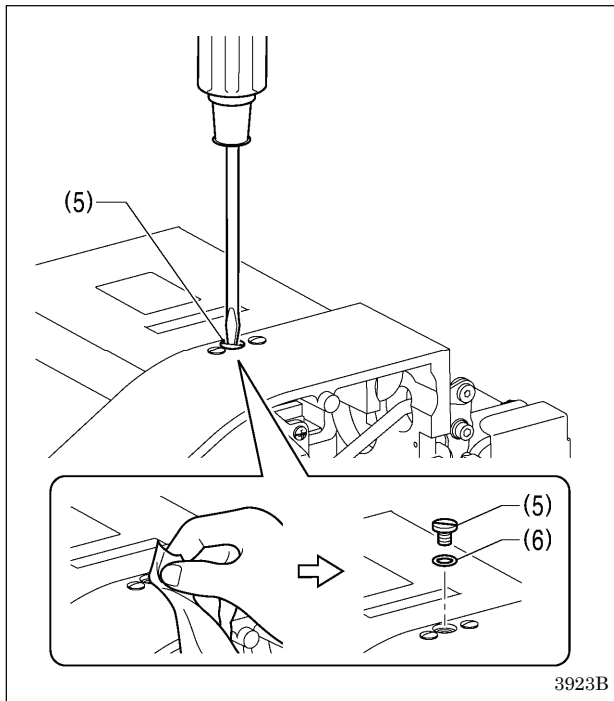
5. Install the screw (2).
- * Clean around the installation area before installing.
 - * When installing the screw (2), check that the O-ring (4) is fitted onto it.



6. Remove the screw (5).
7. Pour in oil through the hole where the screw (5) was installed.
- * Add approximately 165 ml of oil.

NOTE:

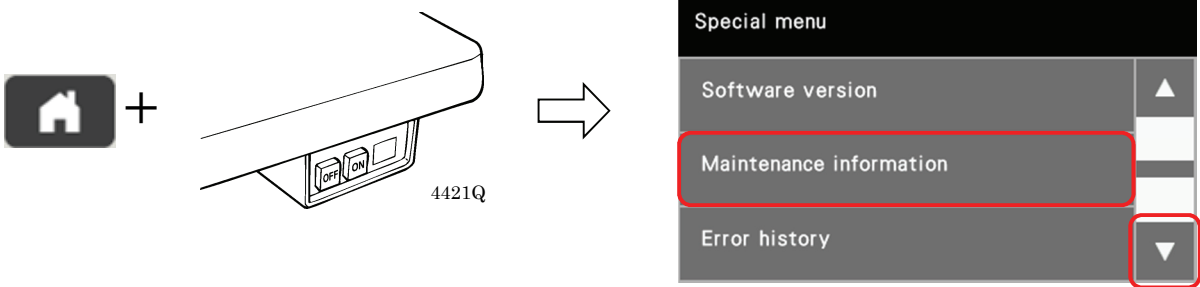
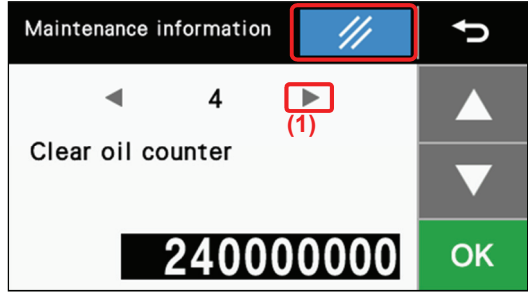
- Check that the oil level reaches in between the two reference lines in the oil gauge window. If the oil level is below the lower reference line, operating problems such as motor seizure may result.
- Do not allow the oil level to go past the top reference line. Oil leaks may occur during sewing machine operation.



8. Install the screw (5).
- * Clean around the installation area before installing.
- * When installing the screw (5), check that the O-ring (6) is fitted onto it.

<Oil change counter initialization method>

After the oil has actually been changed, carry out the procedure below to clear the cumulative number of stitches before the oil was changed.

1	 (While pressing the home key, turn on the power.) The special menu screen will be displayed. Press the  key to display "Maintenance information", and then touch "Maintenance information".
2	 Press the  key (1) to switch to clear oil counter mode. When the reset key  (2) is pressed, the oil counter will become zero and it will blink. When the OK key  is touched, the oil counter value will illuminate and the oil counter will be cleared. (This completes the clearing operation.)
3	 Press the back key  (3) to return to the special menu. Press the home key  to return to the home screen.

11. INSTALLING THE 2-PEDAL FOOT SWITCH (OPTION)

⚠ DANGER

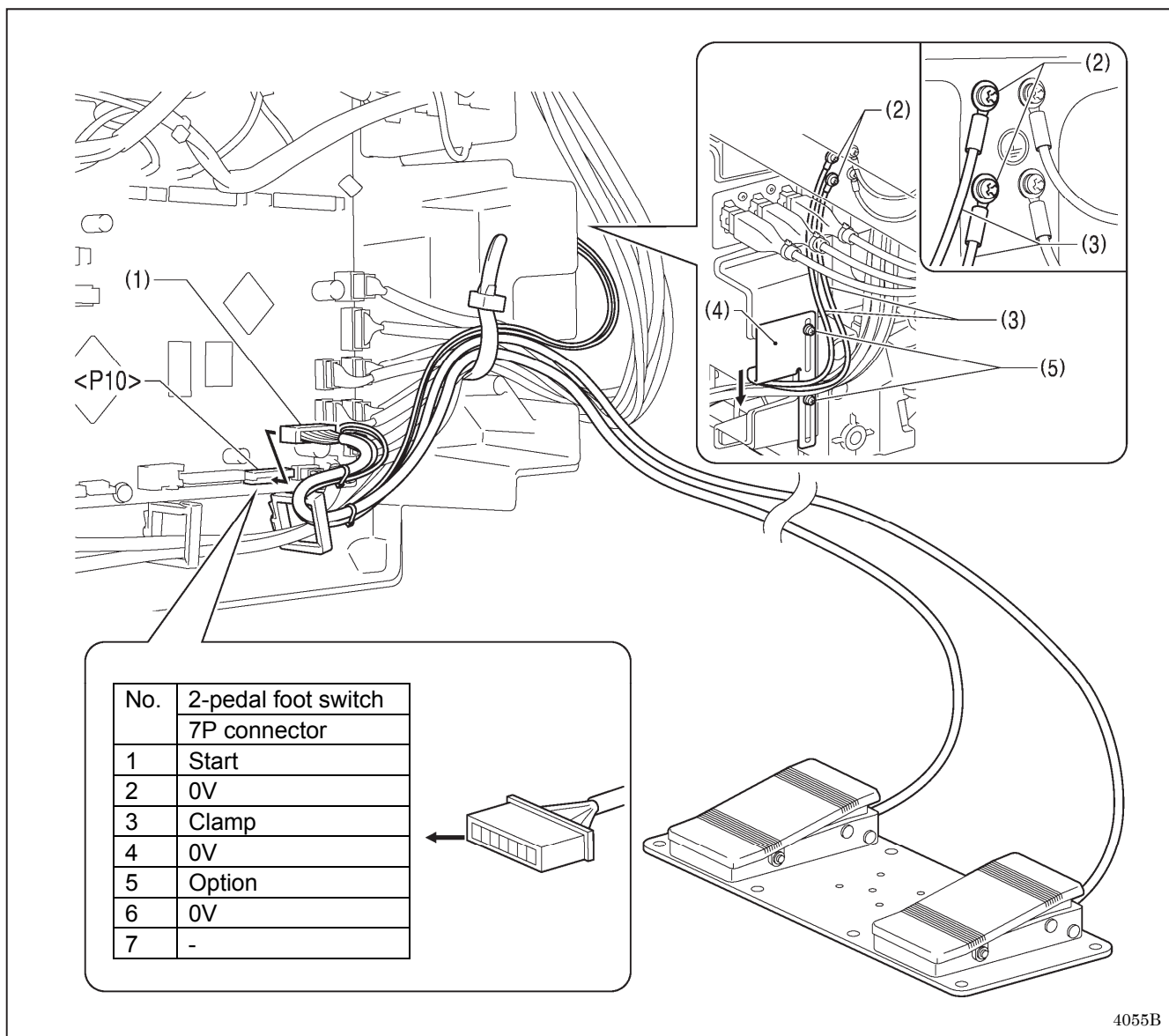


Wait at least 5 minutes after turning off the power switch and disconnecting the power cord from the wall outlet before opening the control box cover. Touching areas where high voltages are present will result in serious injury from electric shocks.



Be sure to connect the ground.

If the ground connection is not secure, you run a high risk of serious injury from electric shocks, and problems with correct operation will also occur.



1. Insert the connector (1) of the 2-pedal foot switch into connector <P10> on the main PCB.
2. Temporarily remove the two screws (2) from the ground terminal of the control box, and then connect the two ground wires (3) in the 2-pedal foot switch harness to the ground terminal and retighten the screws to the ground terminal.
3. Close the cord presser plate (4) in the direction of the arrow, and secure it by tightening the two screws (5).

NOTE:

- Make sure that the ground connections are secure in order to ensure safety.
- Close the cord presser plate (4) securely so that no foreign objects, insects or small animals can get inside the control box.

12. TABLE OF ERROR CODES

DANGER



Wait at least 5 minutes after turning off the power switch and disconnecting the power cord from the wall outlet before opening the control box cover. Touching areas where high voltages are present will result in serious injury from electric shocks.

If a malfunction should occur with the sewing machine, a buzzer will sound and an error code will appear in the touch panel. Follow the remedy procedure to eliminate the cause of the problem.

Switch-related errors

Code	Cause and remedy
E010	The stop switch was pressed. Turn off the stop switch to clear the error.
E011	The stop switch was pressed. Press the reset key to clear the error. You can press the change stitch count key [▼] to move the work clamp so that you can continue sewing.
E012	The stop switch was pressed. Press the reset key to clear the error, and then depress the foot switch to the 2nd step to carry out home position detection.
E015	The stop switch is being pressed when the power was turned on, or there is a problem with the connection of the stop switch. Turn off the power, and then check that connector P15 on the main P.C. board is properly connected.
E016	Problem with the stop switch connection when the power was turned on. Turn off the power, and then check that connector P15 on the main P.C. board is properly connected.
E020	The start switch was pressed without lowering the work clamp. Release the start switch to clear the error.
E025	The start switch (2nd foot switch) is being pressed, or there is a problem with the connection of the start switch (foot switch). Turn off the power and check the foot switch.
E035	The work clamp switch (1st foot switch) is being pressed, or there is a problem with the connection of the work clamp switch (foot switch). Turn off the power and check the foot switch.
E050	Machine head tilting was detected after the power was turned on. Turn off the power, and then return the machine head to its original position. Check that connector P9 on the main P.C. board is properly connected.
E051	Machine head tilting was detected while the sewing machine was operating. Turn off the power, and then check that connector P9 on the main P.C. board is properly connected.
E055	Machine head tilting was detected when the power was turned on. Turn off the power, and then return the machine head to its original position. Check that connector P9 on the main P.C. board is properly connected.
E065	The operation panel key is being pressed or there is a problem with the switch. Turn off the power, and then check.

12. TABLE OF ERROR CODES

Main shaft-related errors

Code	Cause and remedy
E100	It is time to change the oil. Turn off the power, and then carry out the reset procedure.
E110	Needle up stop position error. Turn the pulley and adjust the stop position.
E111	Upper shaft did not stop at the needle up stop position when the sewing machine stopped. Turn off the power, and then check that there are no problems with the thread trimming mechanism and the main shaft motor mechanism.
E121	Thread trimming was not completed. Turn off the power, and then check if the cutting edges of the fixed knife and movable knife are damaged or worn.
E130	Main shaft motor stopped due to a problem, or synchronizer is faulty. Turn off the power, and then turn the machine pulley to check if the machine has locked up. Check that connector P5 on the power supply motor P.C. board and the 4-pin main shaft motor connector are properly connected.
E131	Synchronizer is not connected correctly. Turn off the power, and then check that connector P5 on the motor P.C. board is properly connected.
E132	Problem detected with main shaft motor operation. Turn off the power, and then check that connector P4 and P5 on the motor P.C. board are properly connected.
E133	Main shaft motor stopping position is incorrect. Turn off the power, and then check that connector P5 on the motor P.C. board is properly connected.
E150	Main shaft motor is overheating, or temperature sensor is faulty. Turn off the power, and then check that connector P5 on the motor P.C. board is properly connected. (When sewing data with a small number of stitches (15 stitches or less) is sewn repeatedly (short cycle operation), the main shaft motor may overheat and the "E150" error code may be generated.)

Feed mechanism-related errors

Code	Cause and remedy
E200	X feed motor home position cannot be detected. Problem with X feed motor or poor X encoder connection. Turn off the power, and then check that connector P2 on the main P.C. board is properly connected.
E201	X feed motor stopped abnormally. Turn off the power and check that there is no problem with the X feed direction.
E204	X feed motor stopped abnormally during sewing. Turn off the power and check that there is no problem with the X feed direction.
E205	X-feed motor stopped due to a problem while moving to the sewing start position. Turn off the power and check that there is no problem with the X feed direction.
E206	X feed motor stopped abnormally during test feeding. Turn off the power and check that there is no problem with the X feed direction.
E210	Y feed motor home position cannot be detected. Problem with Y feed motor or poor Y encoder connection. Turn off the power, and then check that connector P3 on the main P.C. board is properly connected.
E211	Y feed motor stopped abnormally. Turn off the power and check that there is no problem with the Y feed direction.
E214	Y feed motor stopped abnormally during sewing. Turn off the power and check that there is no problem with the Y feed direction.
E215	Y-feed motor stopped due to a problem while moving to the sewing start position. Turn off the power and check that there is no problem with the Y feed direction.
E216	Y feed motor stopped abnormally during test feeding. Turn off the power and check that there is no problem with the Y feed direction.

Work clamp-related errors

Code	Cause and remedy
E300	Work clamp home position cannot be detected. Problem with work clamp motor or poor work clamp encoder connection. Turn off the power, and then check that connector P6 on the main P.C. board is properly connected.
E301	Work clamp / button clamp raised or lowered position cannot be detected. Turn off the power, and then check that there is no problem with the vertical work clamp / button clamp position.
E330	Work clamp motor has overrun in the lowering direction.
E331	Work clamp motor has overrun in the raising direction.

Communication and memory-related errors

Code	Cause and remedy
E400	Communication error was detected with the main P.C. board when the power was turned on. Turn off the power, and then check that connector P5 on the motor P.C. board is properly connected.
E401	Communication error was detected with the motor P.C. board when the power was turned on. Turn off the power, and then check that connector on the main P.C. board and connector P23 on the motor P.C. board are properly connected.
E410	Communication error was detected between main P.C. board and panel P.C. board. Turn off the power, and then turn it back on again.
E411	Communication error was detected between main P.C. board and motor P.C. board. Turn off the power, and then turn it back on again.
E420	No USB memory is inserted. Press the reset key to clear the error.
E421	Data contents are incorrect and cannot be used, or no data. Press the reset key to clear the error. Check that data for this pattern number is present in the USB memory.
E422	Error occurred while reading from USB memory. Press the reset key to clear the error. Check the data in the USB memory.
E424	Insufficient space in USB memory. Press the reset key to clear the error. Use different USB memory.
E425	Error occurred while reading from USB memory. Press the reset key to clear the error. Use the specified type of USB memory. Check if the USB memory is write-protected and if it has enough free space.
E428	A pattern registered in a program has been erased. Press the reset key to clear the error. Re-register the program, or add the pattern.
E430	Flash memory of the main P.C. board is faulty. Turn off the power and change the main P.C. board.
E440	Data cannot be backed up to the main P.C. board. Turn off the power and change the main P.C. board.
E450	Model selection has not been loaded from the machine head memory. Turn off the power and check that connector P5 on the power supply motor P.C. board is properly connected.
E452	Machine head memory is not connected. Turn off the power, and then check that connector P4 on the main P.C. board is properly connected.
E474	Internal memory is full. Press the reset key to clear the error. Clear any unnecessary additional patterns and then secure a free space.

12. TABLE OF ERROR CODES

Data editing-related errors

Code	Cause and remedy
E500	The enlargement ratio setting caused the sewing data to extend outside the sewing area. Press the reset key to clear the error. Set the enlargement ratio again.
E502	The enlargement ratio caused the stitch length to exceed the maximum length of 12.7 mm. Press the reset key to clear the error. Set the enlargement ratio again.
E510	Error in pattern data. Press the reset key to clear the error. Re-read the pattern data from the USB memory, or recreate the pattern data.
E511	No end code has been input into pattern data. Press the reset key to clear the error. Recreate the pattern data containing the end code, or change the pattern number being read.
E512	Number of stitches exceeds allowed maximum. Press the reset key to clear the error. Change the number of the pattern being read.
E581	Memory switch file cannot be read correctly. Copy source model and copy destination model are incorrect. (Maybe the 430HX/HS is attempting to read data for the 438HX/HS.) Read data for the same sewing machine model.
E582	Memory switch versions do not match. Read data for the same version.
E583	Parameter versions do not match. Read data for the same version.

Device-related errors

Code	Cause and remedy
E600	Upper thread breakage has occurred. Thread the upper thread and then press the reset key to clear the error. Re-sewing is then possible.
E690	Thread trimmer pulse motor home position cannot be detected. Problem with thread trimmer pulse motor or poor thread nipper encoder connection. Turn off the power, and then clean underneath the needle plate to remove any dust and thread scraps. Check that connector on the main P.C. board is properly connected.
E692	Thread trimmer pulse motor stopped abnormally. Turn off the power, and then check.

P.C. board-related errors

Code	Cause and remedy
E700	Abnormal rise in power supply voltage. Turn off the power and check the input voltage.
E701	Abnormal rise in main shaft motor drive voltage. Turn off the power, and then check the voltage.
E705	Abnormal drop in power supply voltage. Turn off the power and check the input voltage.
E710	Abnormal current was detected in main shaft motor. Turn off the power, and then check if there are any problems with the sewing machine.
E711	Abnormal current was detected in pulse motor. Turn off the power, and then check if there are any problems with each motor.

Version updating-related errors

Code	Cause and remedy
E870	No panel control program is present. Write the panel control program.
E880	Version updating requests could not be received when power was turned on.
E881	Communication error occurred during version updating. Turn off the power, and then check.
E882	No USB memory is inserted. Turn off the power, and then check.
E883	No control program is present in the USB memory. Turn off the power, and then check.
E884	There is a problem with the control program. Write the correct file into the USB memory.
E885	Communication error within the USB memory occurred during main version updating. Turn off the power, and then check the connection of the USB memory.
E886	Fault is found in the version up file during main version updating. Turn off the power, and then repeat the version update procedure.
E887	Failed to save the version up file during main version updating. Turn off the power, and then repeat the version update procedure.

Others errors

Code	Cause and remedy
E990	There is an error with an optional part. Turn off the power, and then check if there are any problems with the optional parts.

If an error code that is not listed above appears or if carrying out the specified remedy does not solve the problem, contact the place of purchase.

13. TROUBLESHOOTING

- Please check the following points before calling for repairs or service.
- If the following remedies do not fix the problem, turn off the power switch and consult a qualified technician or the place of purchase.

DANGER



Wait at least 5 minutes after turning off the power switch and disconnecting the power cord from the wall outlet before opening the the control box cover. Touching areas where high voltages are present will result in serious injury from electric shocks.

CAUTION



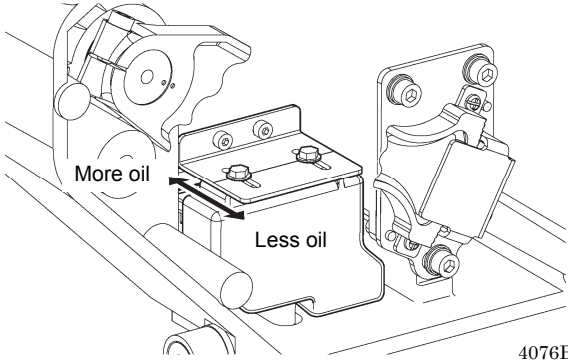
Turn off the power switch and disconnect the power cord before carrying out troubleshooting, otherwise the machine may operate if the foot switch is depressed by mistake, which may result in injury.

13-1. Upper thread breakage / lower thread breakage

Items with a “*” in the “Page” column should only be handled by a qualified technician.

Cause	Check	Remedy	Page
Threading	Upper thread threading	Thread the upper thread correctly.	Instruction manual
	Lower thread threading	Thread the lower thread correctly.	Instruction manual
Thread path	Flaw or abrasion on the thread path	Polish with buff or replace the part. Especially, pay attention to finishing around needle plate hole.	*
Upper thread amount	Arm thread guide C position	Adjust the arm thread guide C position.	135
Upper thread	Upper thread tension	Adjust the upper thread tension appropriately.	132, Instruction manual
Thread take-up spring	Thread take-up spring tension and height	Loosen the thread take-up spring tension or lower the height to such a degree that does not cause double hooking.	134
Needle	Needle facing	Attach the needle so that the long groove on the needle is at the front.	Instruction manual
	Needle installation height	Insert the end of the needle shank so that it touches the top edge of the needle hole of the needle bar.	—
	Bent needle	Replace needle.	—
	Blunt needle tip or burr	Replace needle.	—
	Needle and thread	Replace needle that fits the thread.	132

(Continued on next page)

Cause	Check	Remedy	Page
Hook	Needle bar height and needle bar lift amount	Adjust the needle bar height. Adjust the needle and hook timing.	135*
	Gap between needle and the tip of the shuttle hook	Adjust the gap between the needle and the tip of the shuttle hook to 0.01 - 0.08 mm. (Adjust to as large as possible without causing skipped stitches to occur.) * Move the needle bar back and forth with your finger, and check that the tip of the shuttle hook does not strike the needle.	136*
	Hook lubrication	Too small quantity of lubrication oil will cause thread breakage. Adjust the quantity of lubrication oil. 	137*
	Thread twisted around hook	Remove any twisted thread from the inner and shuttle hook.	*
	Flaw on the tip of the shuttle hook and shuttle hook edge	Polish with buff or replace the part. * Use the KE-430H/BE-438H bobbin case.	*
Lower thread	Lower thread tension	Adjust the lower thread tension appropriately.	132, Instruction manual
Bobbin case	Damage outside of bobbin case and bent bobbin holder spring, etc.	Buff it to correct the damage, or replace the part. * Use the KE-430H/BE-438H bobbin case.	*
Heat (of needle or thread)	Install needle cooler	When using synthetic thread, use the needle cooler (silicone oil).	Instruction manual
	Air amount for needle cooling	Remove any dust and thread scraps inside the air cylinder cover.	Instruction manual *
	Air direction for needle cooling	Adjust the height or lateral position of the air nozzle for needle cooling to point to the needle.	145*

13-2. Skipped stitches

Items with a “*” in the “Page” column should only be handled by a qualified technician.

Cause	Check	Remedy	Page
Thread tension	Upper thread tension	Adjust the upper thread tension appropriately.	132, Instruction manual
Thread take-up spring	Thread take-up spring tension and height	Loosen the thread take-up spring tension or lower the height to such a degree that does not cause double hooking.	134
Needle	Needle facing	Attach the needle so that the long groove on the needle is at the front.	Instruction manual
	Needle installation height	Insert the end of the needle shank so that it touches the top edge of the needle hole of the needle bar.	—
	Bent needle	Replace needle.	—
	Blunt needle tip or burr	Replace needle.	—
	Needle count	Use a needle with a count that matches the thread and material or one with a smaller size.	132
Needle bar play	Vertical and longitudinal needle bar play	Reduce the needle bar play, or replace the parts.	*
Hook	Blunt shuttle hook tip	Polish with buff or replace the part. * Use the KE-430H/BE-438H hook.	*
	Needle bar lift amount	Adjust the needle bar lift amount. Adjust the needle and hook timing	135*
	Driver needle guard	Adjust the driver so that the needle contact pressure is not too great.	136*
	Gap between needle and the tip of the shuttle hook	Adjust the gap between the needle and the tip of the shuttle hook to 0.01 - 0.08 mm. (Adjust to as large as possible without causing skipped stitches to occur.) * Move the needle bar back and forth with your finger, and check that the tip of the shuttle hook does not strike the needle.	136*
Needle plate	Relation between material and needle plate	Use a needle plate that matches the material.	—
Feed plate	Relation between material and feed plate	Use a feed plate that matches the material.	—
Work clamp	Relation between material and work clamp	Use a work clamp that matches the material.	—

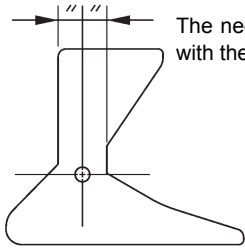
13-3. Uneven seams (1) Incorrect thread tightness

Items with a “*” in the “Page” column should only be handled by a qualified technician.

Cause	Check	Remedy	Page
Upper thread	Upper thread tension	Adjust the upper thread tension appropriately.	132, Instruction manual
Thread take-up spring	Thread take-up spring tension	• If the upper thread trailing length varies considerably, increase the thread take-up spring tension. • If the upper thread trailing length is short, decrease the thread take-up spring tension. • If the thread take-up spring operation is unstable, increase the thread take-up tension, and also increase the stroke.	134*
Lower thread	Lower thread tension	Adjust the lower thread tension appropriately.	132, Instruction manual

13-4. Uneven seams (2) Thread jamming

Items with a "*" in the "Page" column should only be handled by a qualified technician.

Cause	Check	Remedy	Page
Thread take-up spring	Thread take-up spring tension	- If the upper thread trailing length varies considerably, increase the thread take-up spring tension. - If the upper thread trailing length is short, decrease the thread take-up spring tension. - If the thread take-up spring operation is unstable, increase the thread take-up tension, and also increase the stroke.	134*
Shuttle race thread guide	Shuttle race thread guide position	Adjust the shuttle race thread guide position.  4077B	137*
Hook	Needle bar lift amount	Adjust the needle bar lift amount. Adjust the needle and hook timing.	136*

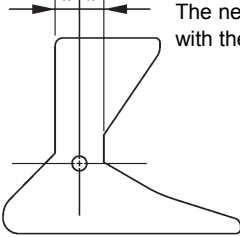
13-5. Uneven seams (3) Upper thread mis-winding

Items with a "*" in the "Page" column should only be handled by a qualified technician.

Cause	Check	Remedy	Page
Upper thread	Upper thread tension	Decrease upper thread tension to such a degree that it does not influence stitch quality.	Instruction manual
	Upper thread trailing length is long or uneven.	Turn the sub-tension nut to adjust the tension. * When using the thread nipper device, adjust the upper thread trailing length to 35 - 40 mm.	Instruction manual
	Tension at the beginning of sewing	Change the value of memory switch No. 551 to change the upper thread tension releasing stitch count at the sewing start. * Although the upper thread mis-winding will decrease, the upper thread pulling out will increase if the upper thread tension is not released at the sewing start.	21
Thread take-up spring	Thread take-up spring tension	- If the upper thread trailing length varies considerably, increase the thread take-up spring tension. - If the upper thread trailing length is short, decrease the thread take-up spring tension. - If the thread take-up spring operation is unstable, increase the thread take-up tension, and also increase the stroke.	134*
Needle	Button position (BE-438HX/HS)	Check that the needle is not touching the button.	141

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13. TROUBLESHOOTING

Cause	Check	Remedy	Page
Shuttle race thread guide	Position of the shuttle race thread guide	Adjust the shuttle race thread guide position.  The needle groove is aligned with the center of the needle. 4077B	137*
Thread nipper	Stitch count	Disable memory switch No. 500 setting: thread nipper device operation.	21
Bobbin thread clamp spring	Bobbin thread clamp spring	Install the bobbin thread clamp spring.	—

13-6. Uneven seams (4) All stitches

Items with a "*" in the "Page" column should only be handled by a qualified technician.

Cause	Check	Remedy	Page
Threading	Upper thread threading	Thread the upper thread correctly.	—
	Lower thread threading	Thread the lower thread correctly.	—
Thread path	Flaw or abrasion on the thread path	Polish with buff or replace the part. Especially, pay attention to finishing around needle plate hole.	*
Thread chips or dust	Shuttle race base rim Hook edge Around needle plate hole	Remove thread chips or dust.	*
Upper thread	Upper thread tension	Increase upper thread tension while checking stitches.	Instruction manual
	Needle and thread	Replace needle that fits the thread.	132
Needle	Installing a needle	Insert the end of the needle shank so that it touches the top edge of the needle hole of the needle bar.	—
	Needle count	Replace needle with thicker one.	—
Lower thread	Lower thread tension	Adjust the lower thread tension appropriately.	132, Instruction manual
Bobbin case	Damaged outside of bobbin case and bent bobbin holder spring, etc.	Polish with buff or replace the part. * Use the KE-430H/BE-438H bobbin case.	*
Needle plate	Relation between material and needle plate	Use a needle plate that matches the material.	—
Feed plate	Relation between material and feed plate	Use a feed plate that matches the material.	—
Work clamp	Relation between material and work clamp	Use a work clamp that matches the material.	—

13-7. Upper thread pulling out

Items with a “*” in the “Page” column should only be handled by a qualified technician.

Cause	Check	Remedy	Page
Upper thread	Upper thread tension	Decrease upper thread tension to such a degree that it does not influence stitch quality.	Instruction manual
	Upper thread trailing length is short.	Enable memory switch No. 500 setting: thread nipper device operation.	21
	Upper thread trailing length is short or uneven.	Turn the sub-tension nut to adjust the tension. * When using the thread nipper device, adjust the upper thread trailing length to 35 - 40 mm.	Instruction manual
	Tension at the beginning of sewing	Change the value of memory switch No. 551 to change the upper thread tension releasing stitch count at the sewing start. * Although the upper thread pulling out will decrease, the upper thread mis-winding will increase if the upper thread tension is releasing at the sewing start.	21
Thread take-up spring	Thread take-up spring tension	Heighten take-up spring tension.	134*
Hook	Needle bar height and needle bar lift amount	Adjust the needle bar height. Adjust the needle and hook timing.	135*
	Shuttle race base rim Hook edge Around needle plate hole	Remove thread chips or dust.	*
Lower thread	Lower thread tension is low.	Heighten lower thread tension.	Instruction manual
	Bobbin thread clamp spring	Install the bobbin thread clamp spring.	—
Sewing start speed	Slow start	Change the slow start pattern.	52
	Sewing speed for the 1st stitch	Reduce the sewing speed for the 1st stitch. (Approximately 600 - 1000 sti/min)	52
Skipped stitches	Skipped stitches at the sewing start	Refer to “13-2. Skipped stitches”.	173

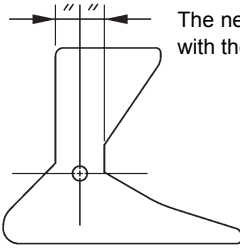
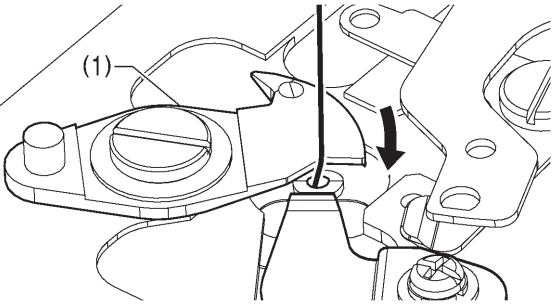
13-8. If the upper thread trailing length is unstable

Items with a “*” in the “Page” column should only be handled by a qualified technician.

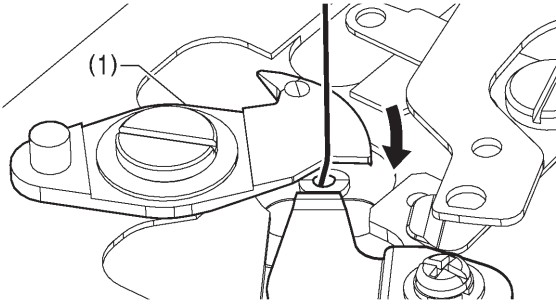
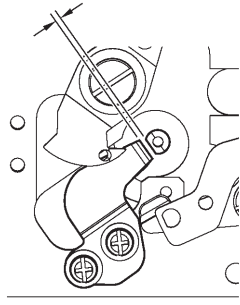
Cause	Check	Remedy	Page
Thread take-up spring	Thread take-up spring tension	- If the upper thread trailing length varies considerably, increase the thread take-up spring tension. - If the upper thread trailing length is short, decrease the thread take-up spring tension. - If the thread take-up spring operation is unstable, increase the thread take-up tension, and also increase the stroke.	134*

13-9. Incorrect upper thread trimming

Items with a "*" in the "Page" column should only be handled by a qualified technician.

Cause	Check	Remedy	Page
Sub-tension	Sub-tension is weak.	Turn the sub-tension nut to adjust the tension.	Instruction manual
Needle	Needle bar lift amount	Confirm the needle lift amount.	136*
Shuttle race thread guide	Shuttle race thread guide position	Adjust the shuttle race thread guide position.  The needle groove is aligned with the center of the needle. 4077B	137*
Lower thread	Lower thread tension is low.	Adjust to heighten the lower thread tension.	132, Instruction manual
Movable knife	Cutting quality of movable knife	Remove the needle plate, thread the needle hole plate, and then move the movable knife (1) as shown in the illustration and check that it trims the thread cleanly.  4078B If it does not trim the thread cleanly: • Adjust the position of the movable knife. • Replace the movable knife spacer with a thinner one. • Replace the movable knife assembly. * Use the KE-430H/BE-438H movable knife assembly.	139*

(Continued on next page)

Cause	Check	Remedy	Page
Fixed knife	Cutting quality of fixed knife	Remove the needle plate, thread the needle hole plate, and then move the movable knife (1) as shown in the illustration and check that it trims the thread cleanly.  4078B If it does not trim the thread cleanly: • Adjust the position of the fixed knife. (Refer to the next section.) • Sharpen the fixed knife. • Replace the fixed knife assembly. * Use the KE-430H/BE-438H fixed knife assembly.	139*
	Installation position of the fixed knife	Check that the fixed knife is installed in the position shown in the illustration. For -5 specifications : 0.3 mm Except for -5 specifications : 0.1 mm  4079B	128*
Skipped stitches	Skipped stitches at the final stitch	Refer to "13-2. Skipped stitches".	173

13-10. Needle breakage

Items with a “*” in the “Page” column should only be handled by a qualified technician.

Cause	Check	Remedy	Page
Needle	Needle facing	Attach the needle so that the long groove on the needle is at the front.	Instruction manual
	Needle installation height	Insert the end of the needle shank so that it touches the top edge of the needle hole of the needle bar.	—
	Bent needle	Replace needle.	—
	Blunt needle tip or burr	Replace needle.	—
	Needle and thread	Replace with a needle suitable for the thread.	132
	Button position (BE-438HX/HS)	Check that the needle is not touching the button.	141
Hook	Needle bar height and needle bar lift amount	Adjust the needle bar height. Adjust the needle and hook timing.	135*
	The clearance between the needle and the tip of the hook	Adjust so that the clearance between the needle and the tip of the shuttle hook is 0.01 - 0.08 mm. (Make as large as possible to the extent the skipped stitches will not occur.) * Check that the needle and the tip of the shuttle hook do not touch while moving the needle bar back and forth with your finger.	136*
Needle plate	Installation position of the needle plate	Adjust the position of the needle plate back and forth so that the needle is aligned with the center of the needle hole.	*
Thread wiper	Thread wiper height	- Adjust the height of the thread wiper so that the work clamp does not press the thread. - Adjust so that the needle and thread wiper do not interfere with each other.	142*

13-11. Entanglement of upper thread with the thread nipper (KE-430HX)

Items with a “*” in the “Page” column should only be handled by a qualified technician.

Cause	Check	Remedy	Page
Sub-tension	Upper thread at the sewing start is long.	Turn the tension nut to adjust the upper thread trailing length to 35 - 40 mm.	Instruction manual

13-12. Feed mechanism does not move and go out of step

Items with a “*” in the “Page” column should only be handled by a qualified technician.

Cause	Check	Remedy	Page
Feed motor	Cord connections	Check if there are any problems with the connection and contact of the feed motor connector on the side face of the control box.	*
Feed motor home position	Incorrect feed home position	Adjust the home positions of the X-feed and Y-feed pulse motors respectively.	148*
Home position sensor of feed motor (When using the optional expansion area kit)	Position of the home position sensor of feed motor	Adjust the position of the home position sensor of the feed motor.	*

13-13. Button breakage (BE-438HX/HS)

Items with a "*" in the "Page" column should only be handled by a qualified technician.

Cause	Check	Remedy	Page
Needle	Button position (BE-438HX/HS)	Check that the needle is not touching the button.	141

13-14. Work clamp height is unsuitable (KE-430HX/HS)

Items with a "*" in the "Page" column should only be handled by a qualified technician.

Cause	Check	Remedy	Page
Work clamp	Grease for work clamp	Grease the sliding part of the work clamp.	129*
Work clamp arm lever plate	Clearance between work clamp arm lever plate and work clamp lifter plate	• Apply grease to the upper surface of the work clamp arm lever plate and to the underside of the work clamp lifter plate. • Adjust the work clamp lift amount.	129* 140*
Thread wiper	Contact between thread wiper and work clamp	Check that the thread wiper does not touch the work clamp, and then adjust the thread wiper to an appropriate position.	142*
Work clamp pulse motor home position	Incorrect work clamp home position	Adjust the home position of the work clamp pulse motor.	150*

13-15. Button clamp height is unsuitable (BE-438HX/HS)

Items with a "*" in the "Page" column should only be handled by a qualified technician.

Cause	Check	Remedy	Page
Button clamp holder hook	Button clamp holder hook position	Adjust the button clamp lift amount.	140*
Thread wiper	Contact between thread wiper and work clamp	Check that the thread wiper does not touch the work clamp, and then adjust the thread wiper to an appropriate position.	142*
Work clamp pulse motor home position	Incorrect work clamp home position	Adjust the home position of the work clamp pulse motor.	150*

13-16. Thread wiper does not operate correctly

Items with a "*" in the "Page" column should only be handled by a qualified technician.

Cause	Check	Remedy	Page
Thread wiper	Thread wiper height	• Adjust the thread wiper height so that it does not interfere with the needle. • Adjust the operating distance of the thread wiper.	142*

13-17. Lower thread winds to one side, lower thread winding amount is incorrect

Items with a "*" in the "Page" column should only be handled by a qualified technician.

Cause	Check	Remedy	Page
Bobbin winder tension	Bobbin winder tension assembly height	Adjust the bobbin winder tension height.	*
Bobbin presser	Bobbin presser position	Adjust the bobbin presser height.	*

13-18. Operation panel display freeze and operation is not possible

Items with a "*" in the "Page" column should only be handled by a qualified technician.

Cause	Check	Remedy	Page
Poor connection inside control box	P.C. board cord connections	• Check if there are any problems with the connection and contact of main P.C. board connector. • Check if there are any problems with the connection and contact of the power supply motor P.C. board connector and the operation panel P.C. board connector.	*

13-19. Sewing machine does not operate when the power is turned on

Items with a "*" in the "Page" column should only be handled by a qualified technician.

Cause	Check	Remedy	Page
Machine head switch	Machine head switch height	• Check if there are any problems with the connection and contact of the machine head switch connector. • Adjust the position of the switching plate.	146*

14. ELECTRIC MECHANISM



DANGER



Wait at least 5 minutes after turning off the power switch and disconnecting the power cord from the wall outlet before opening the control box cover. Touching areas where high voltages are present will result in serious injury from electric shocks.

14-1. Precautions at the time of adjustment

Pay attention to the following when opening the control box for maintenance.

Electric shock

Some large capacitors may have a high voltage remaining in them for up to 5 minutes after the power is turned off. To prevent electric shock, wait at least 5 minutes after the power is turned off before doing the following:

- Opening and closing the control box
- Replacing fuses
- Separating and joining connectors
- Measuring resistance
- Doing anything with a possibility of touching something inside the control box

Some adjustments require measuring the voltage while the power is turned on with the control box kept open.

In such a case, be careful not to touch any place other than that for the measurement. In addition, always keep in mind that a high voltage remains for about 5 minutes after power is turned off.

Injury

When separating or rejoining connectors, and measuring something, be careful not to cut your fingers on metal parts such as heat sinks and covers.

14-2. Components inside and outside the control box and in the operation panel

Main PCB

Secure to the surface opposite the operator. This PCB drives the pulse motor and solenoids.

Power supply motor PCB

Secure to the surface on the operator side. This PCB drives the main shaft motor. There are two fuses on this PCB.

Conversion transformer box (100V, 110V, 380V, 400V AC specifications only)

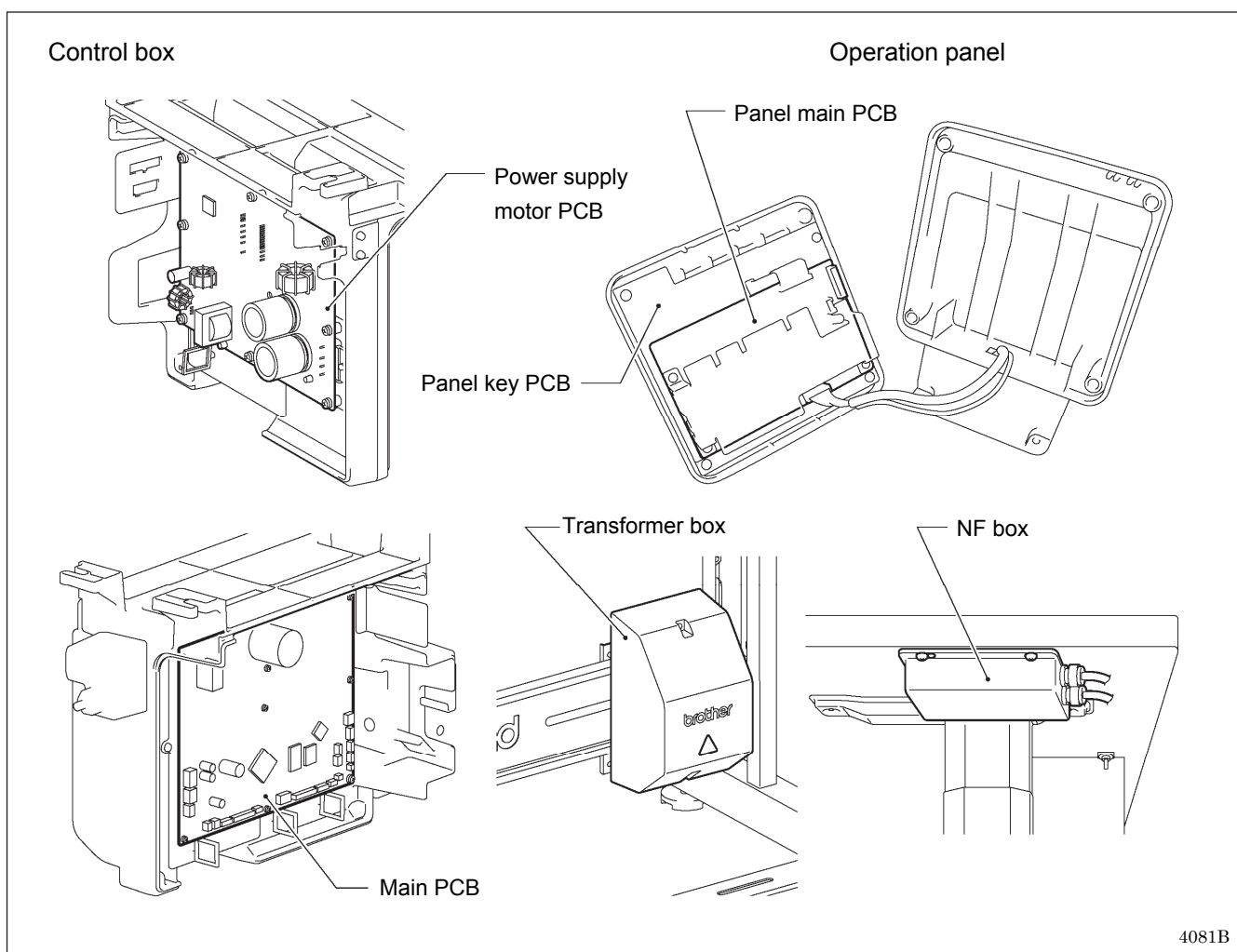
Steps-down the power supply voltage and generates the voltages that are required for the control box.

NF box (For Europe)

Eliminates electrical noise that is transmitted along the power supply line.

Panel PCB

Secured inside the operation panel. This PCB controls indications of the machine status and the input operation.



4081B

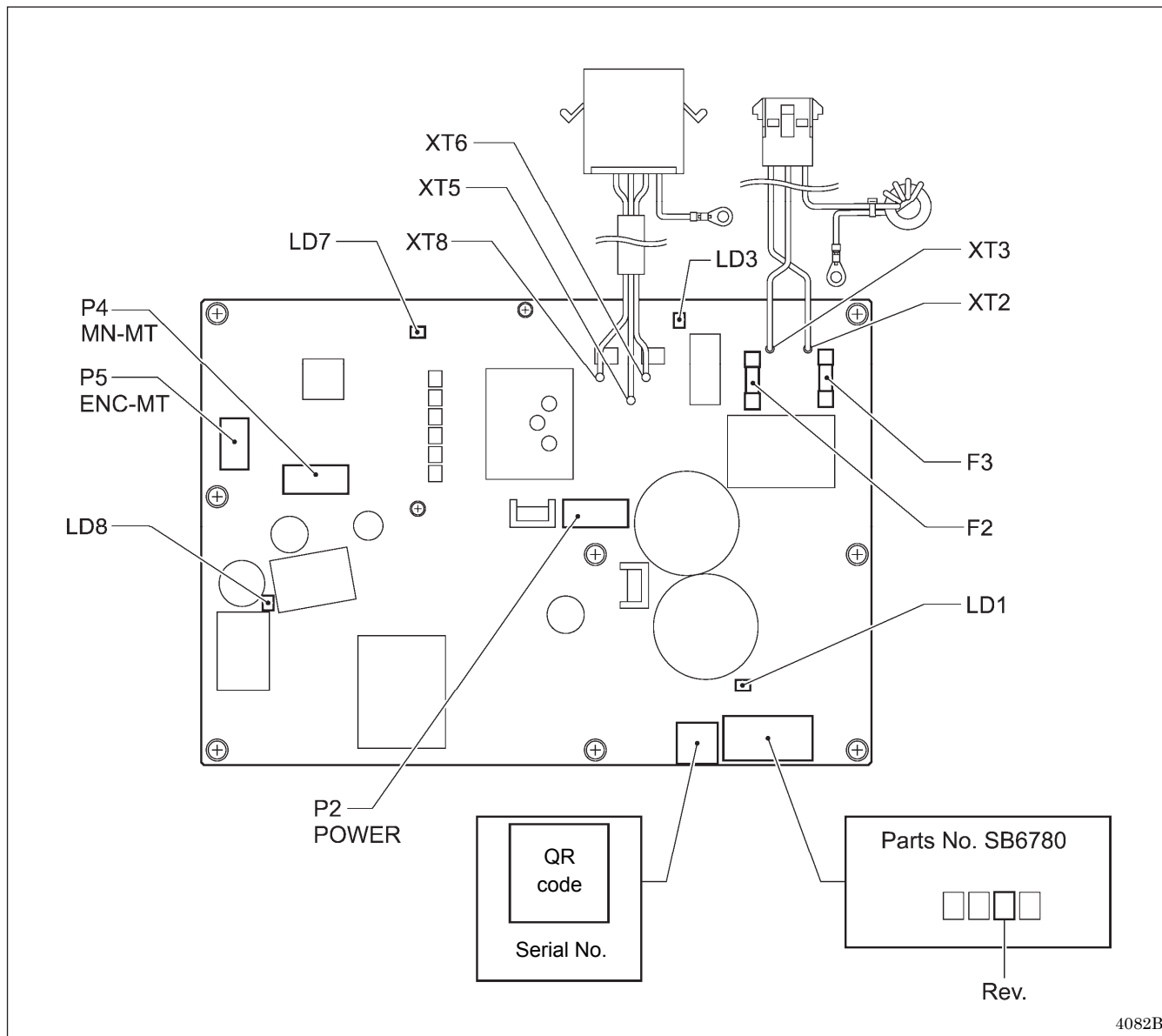
14-3. Fuse explanation

When replacing a fuse, be sure to use the specified ones listed below.

If a component on a PCB is damaged, the fuses may blow again soon even after they have been replaced.

No.	Part name	Part code	When a fuse has blown
F2 F3	GLASS FUSE 10A-50T100H	SB3275-001	The power supply LED is not illuminated, and nothing operates.

Power supply motor PCB



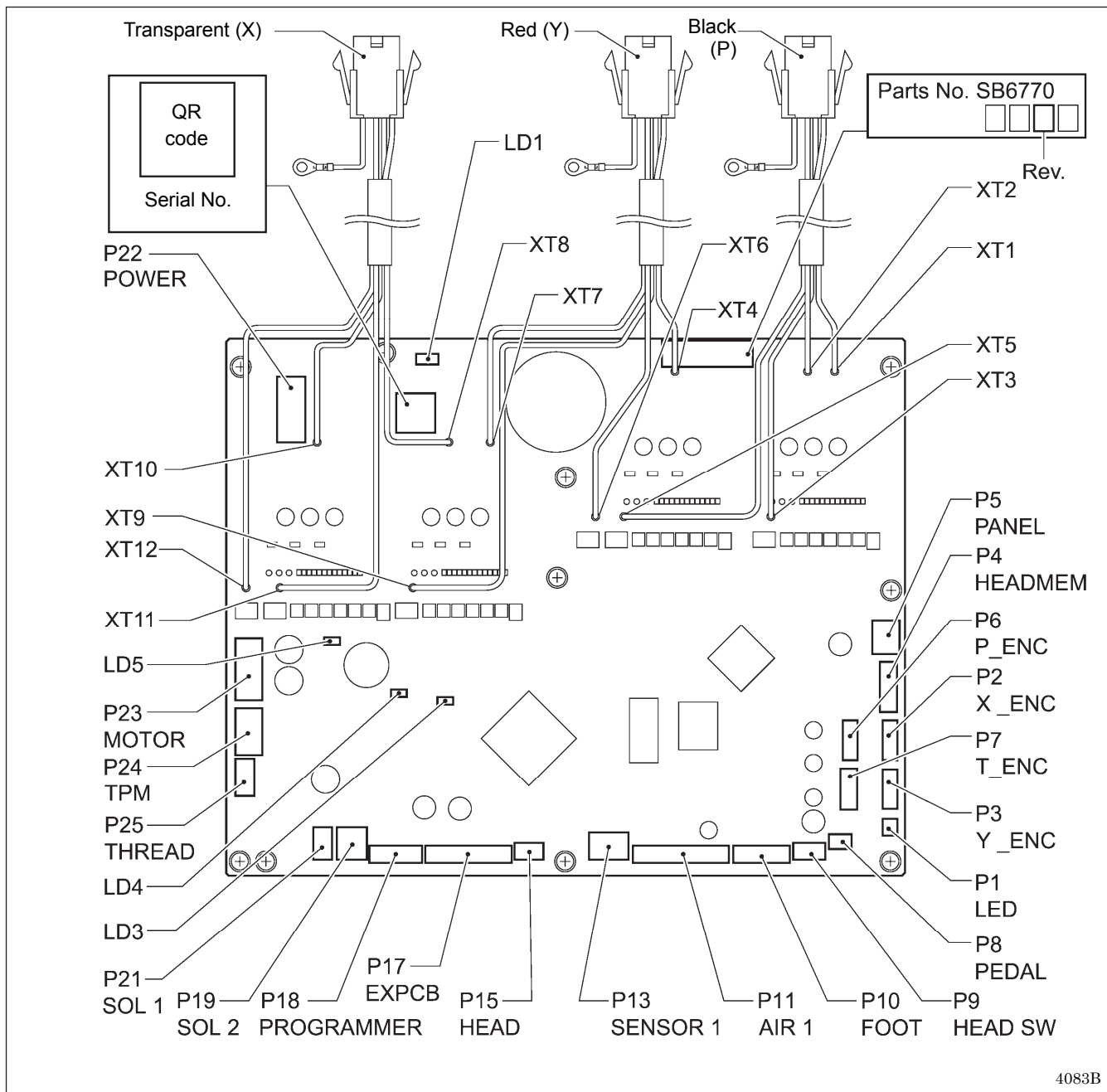
4082B

14-4. Connectors

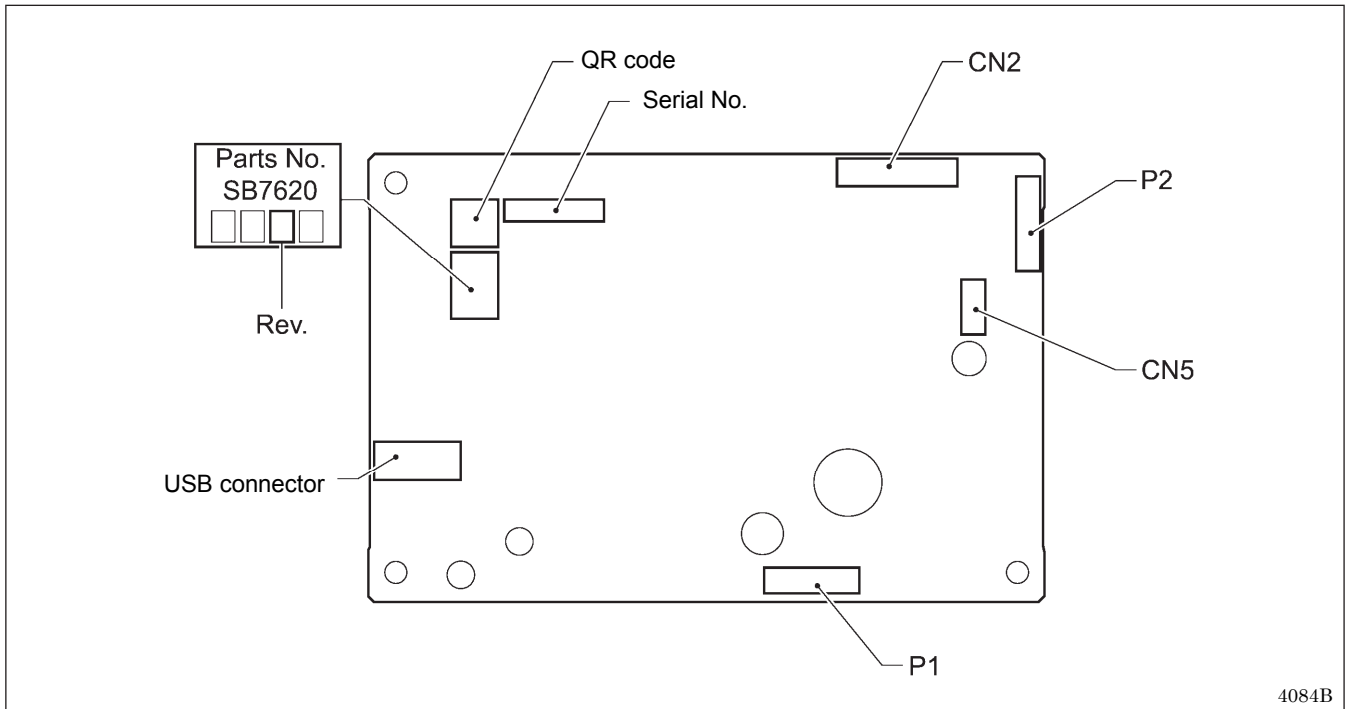
Most of the machine trouble is due to connector problems including improper connection or sufficient contact. Therefore, be sure to check if each connector is correctly inserted and that there is no contact failure between pins and wires before starting troubleshooting procedures.

14-4-1. Connector positions

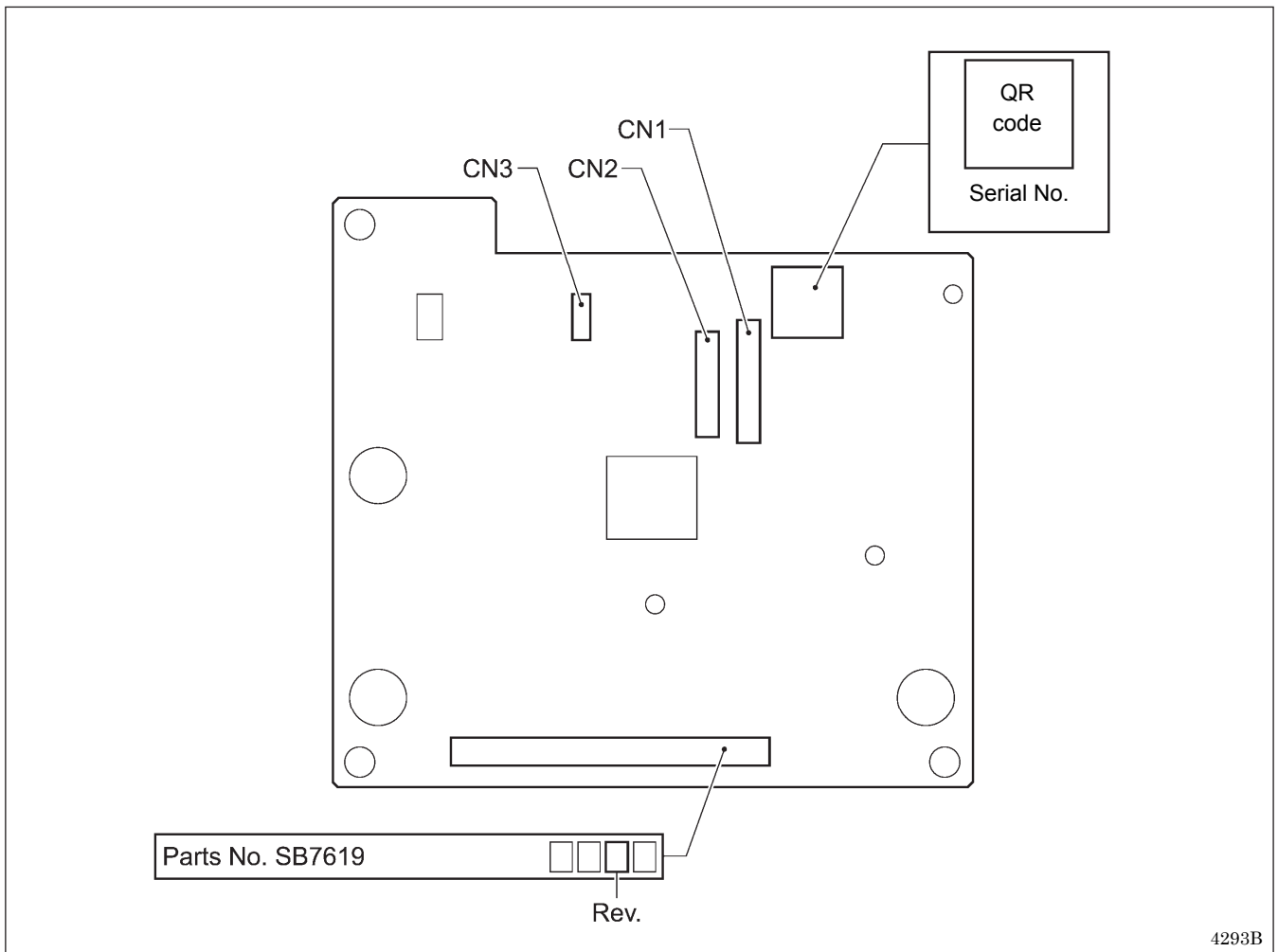
Main PCB



Panel main PCB



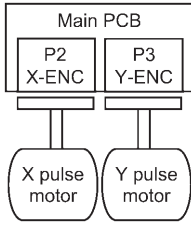
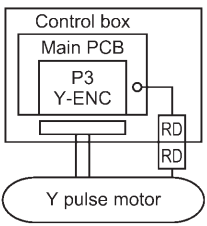
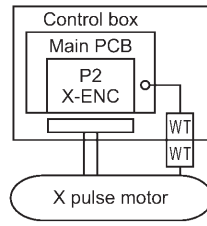
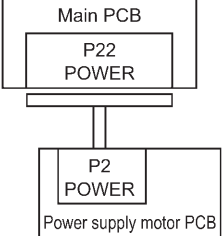
Panel key PCB



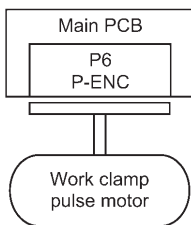
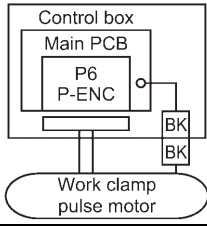
14-4-2. Contact failure

The connectors functions are divided into five categories. Some connectors may belong to more than one group. Be sure to investigate another category if a problem is not found in one category.

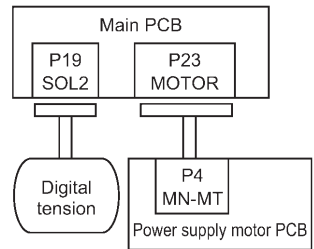
Feed mechanism

Problem	Connector No. and position
- The work clamp/button clamp operates for a moment but the home position is not detected correctly. - Error E200 or E210 is displayed.	 4297B
- Problem with work clamp/button clamp operation in the forward-back (Y) direction. - Error E210, E211, E214, E215 or E216 is displayed.	 4298B
- Problem with work clamp/button clamp operation in the sideways (X) direction. - Error E200, E201, E204, E205 or E206 is displayed.	 4299B
- The feed motor does not operate for home position detection immediately after the power is turned on. - Error E200 or E210 is displayed.	 4300B

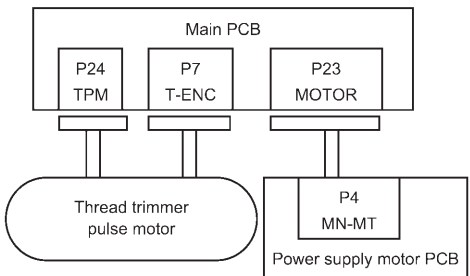
Work clamp mechanism

Problem	Connector No. and position
- The work clamp pulse motor rotates but the home position is not detected correctly. - Error E300 is displayed.	 4301B
- The work clamp pulse motor does not rotate correctly. - Error E300, E301, E330 or E331 is displayed.	 4302B

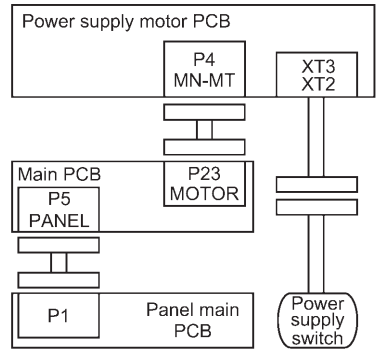
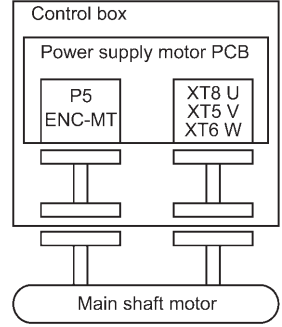
Tension release mechanism

Problem	Connector No. and position
Digital tension does not operate. (There is no error display.)	 4303B

Thread trimmer mechanism

Problem	Connector No. and position
Error E690 or E692 is displayed.	 4304B

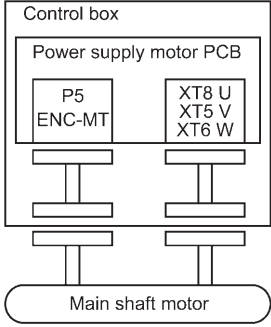
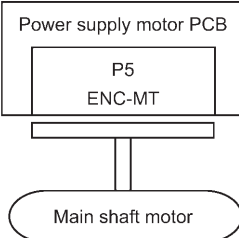
Sewing operation

Problem	Connector No. and position
- The power supply LED is not illuminated, and nothing operates. - Machine operation is unstable.	 4305B
- The main shaft motor does not operate correctly. - Error E130 is displayed.	 4306B

Others

Problem	Connector No. and position
- Panel indication is strange. - The power supply LED is not illuminated. - Any operation panel key does not have any effect. - The operation panel is completely dark.	4307B
The sewing machine does not operate when the foot switch is depressed. (No error display.)	4308B
Error E055 is displayed after the power is turned on.	4309B
Error E452 is displayed after the power is turned on.	4310B
The machine does not start up with the “brother” display appearing.	4307B

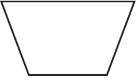
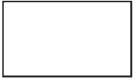
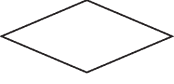
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Problem	Connector No. and position
• The sewing machine does not operate when the foot switch is depressed. • Error E130 is displayed.	 4306B
• Error E131 is displayed.	 4311B

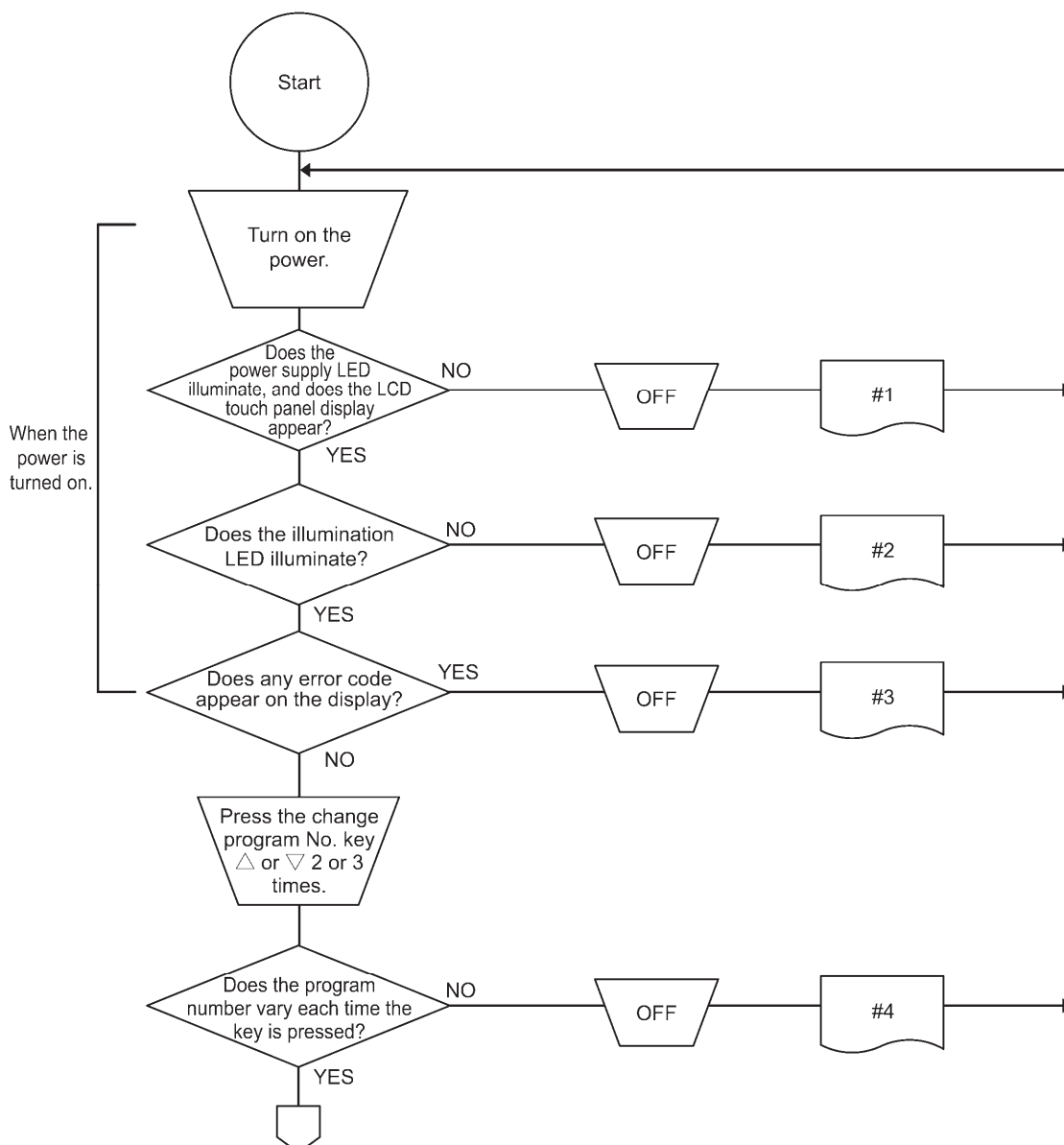
14-5. Troubleshooting

14-5-1. Troubleshooting flowchart

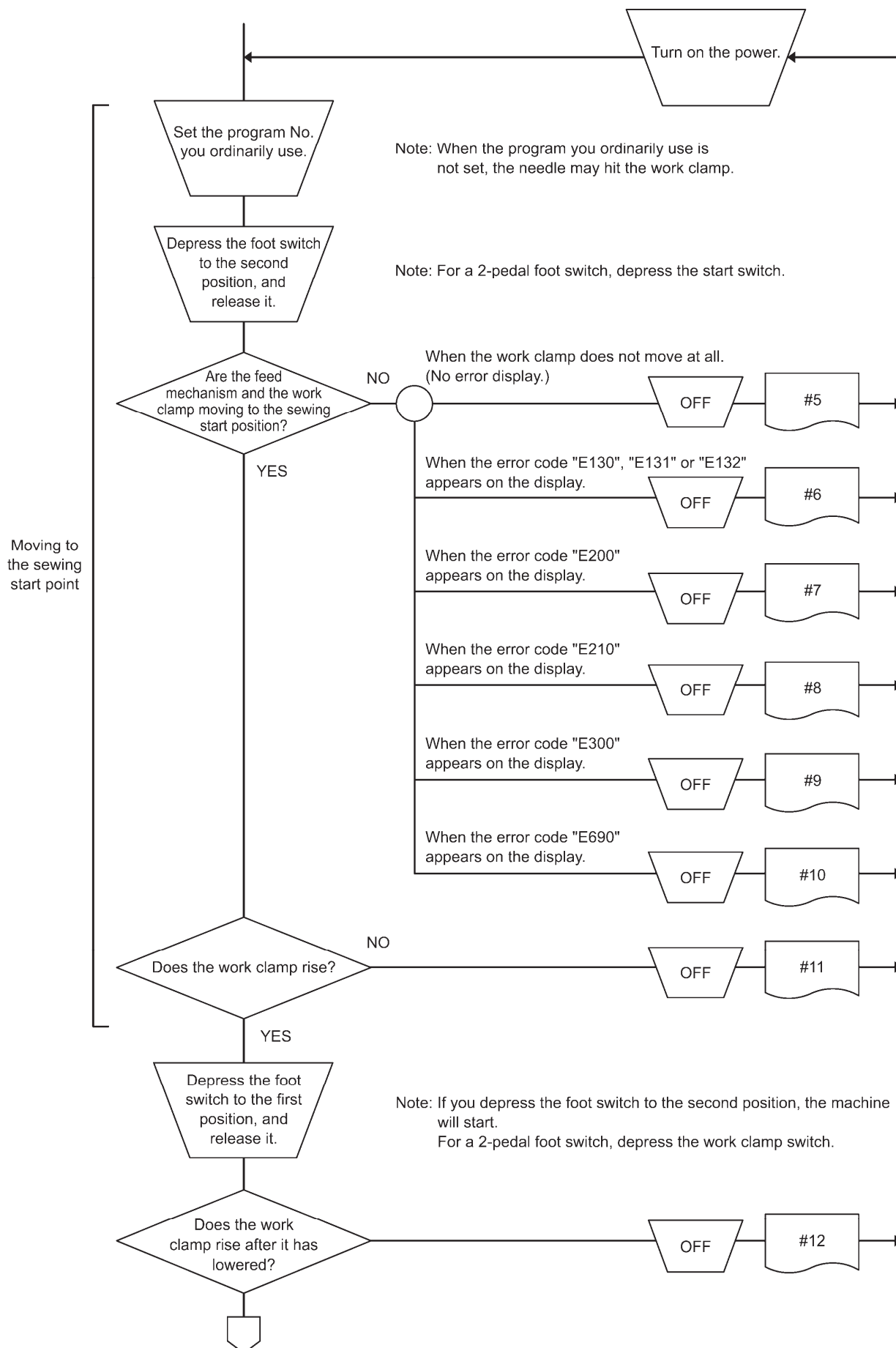
Symbols and their meanings

	Switch operation		Set-up operation or condition
	Yes-or-no decision-making process		The procedure to follow continues on the next page
	The error status number in the first column of the table of "14-5-2. Problem solution and measures"		Turning-off of the power switch

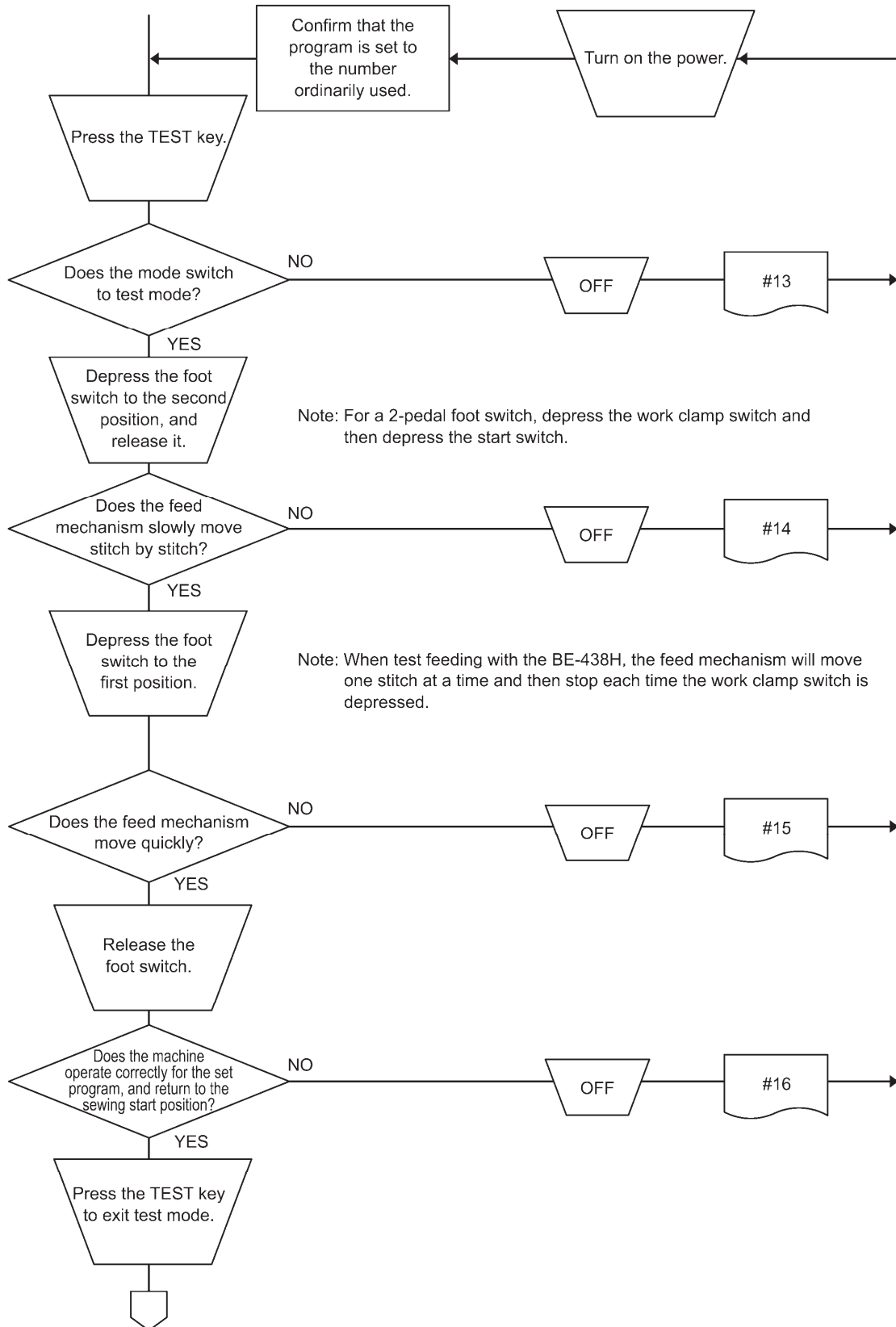
2707B

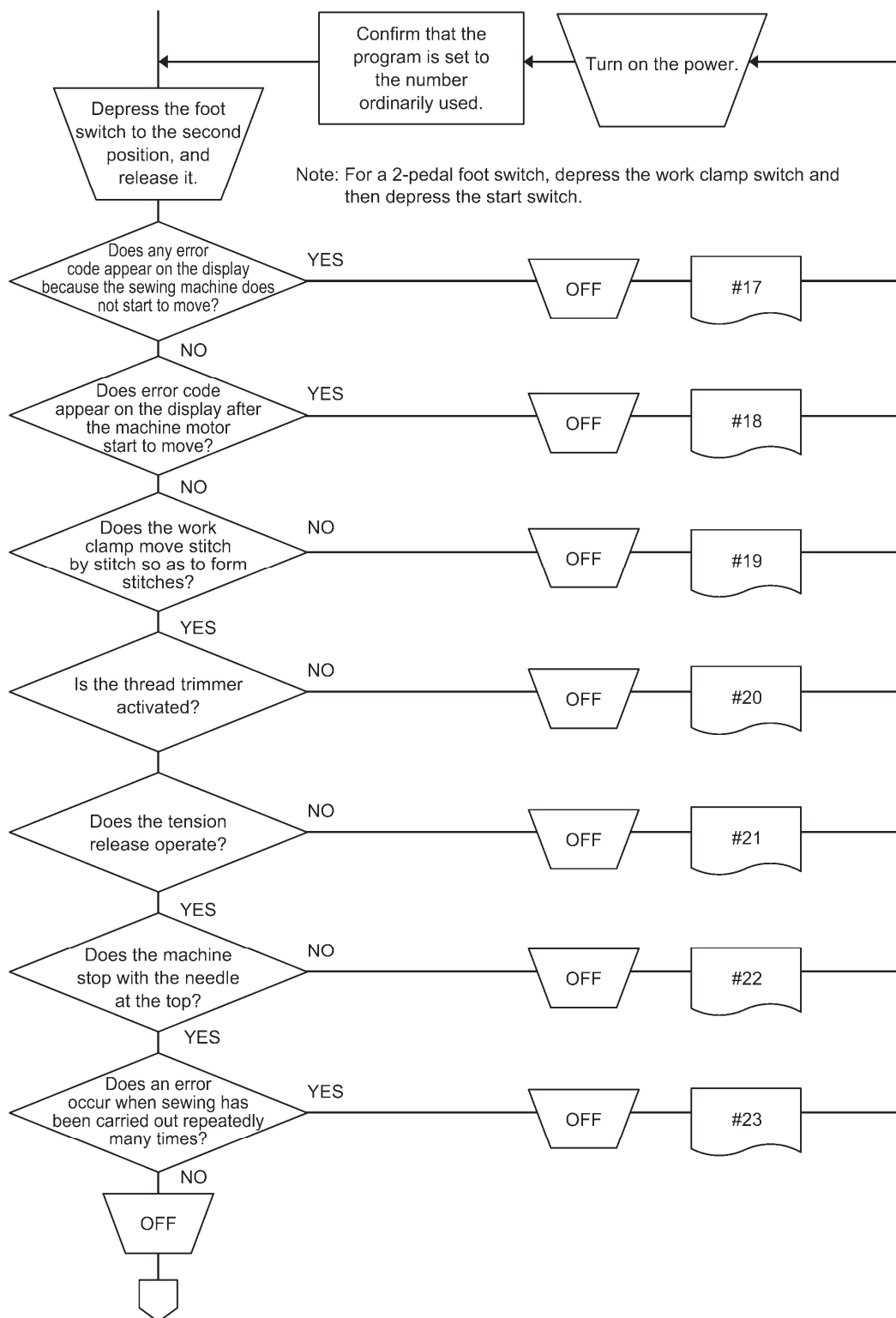


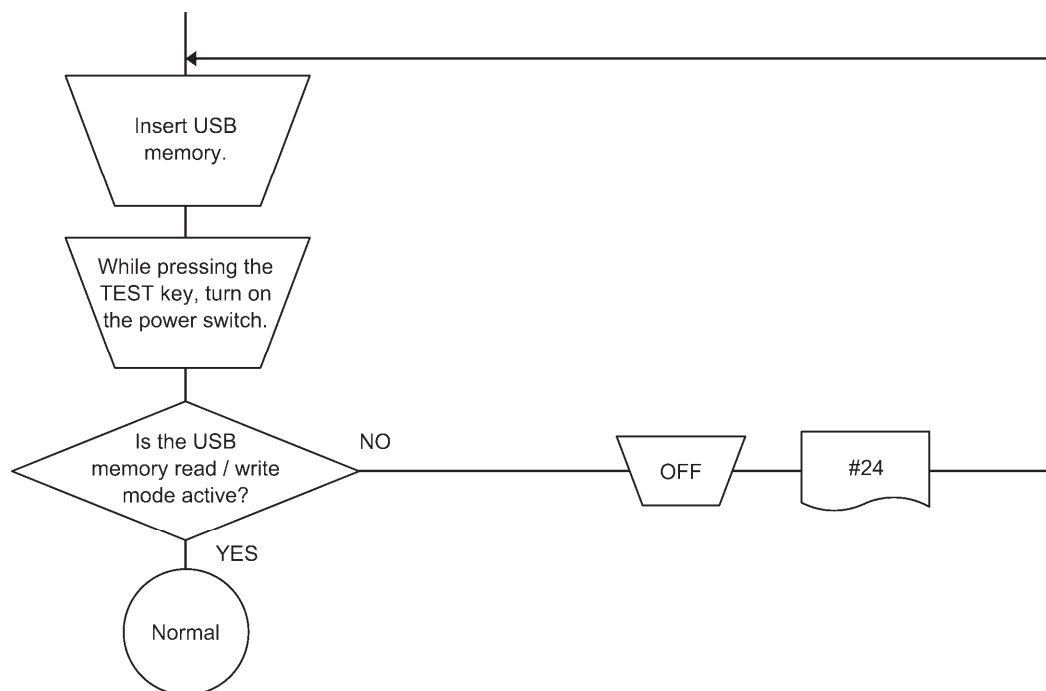
4312B



4313B







4316B

14-5-2. Problem solution and measures

DANGER



Wait at least 5 minutes after turning off the power switch and disconnecting the power cord from the wall outlet before opening the control box cover. Touching areas where high voltages are present will result in serious injury from electric shocks.

1. Pay attention to the following when opening the control box for maintenance.

Electrical shock

Some large capacitors may have a high voltage remaining in them for up to 5 minutes after the power is turned off. To prevent electrical shock, wait at least 5 minutes after the power is turned off before doing the following:

- Opening and closing the control box
- Replacing fuses
- Separating and joining connectors
- Measuring resistance
- Doing anything with a possibility of touching something inside the control box

Some adjustments require measuring the voltage while the power is turned on with the control box kept open.

In such a case, be careful not to touch any place other than that for the measurement. In addition, always keep in mind that a high voltage remains for about 5 minutes after power is turned off.

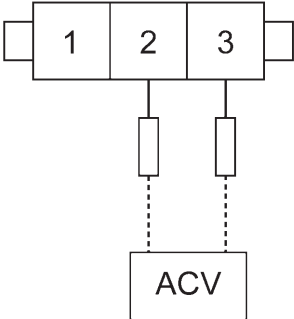
Injury

When separating or rejoining connectors, and measuring something, be careful not to cut your fingers on metal parts such as heat sinks and covers.

2. When replacing a fuse, be sure to use a new one of the same quality and capacity as the old one.

Before adjustment

1. While the power is turned off, check each connector is securely plugged in by referring to "14-4. Connectors".
2. Find the error status number in the troubleshooting flowchart.
3. From the applicable part of the flowchart, take the reference number to find the correspondingly numbered details of the problem in the following table.

Error status #1 The power supply LED does not light when the power is turned on.		
Probable causes	Check/ repair/ adjust	Parts to be replaced
1. (Check the LEDs)	Turn on the power switch and check that the various LEDs illuminate in the following order. a. <Check 300V> LD1 (red) on power supply motor PCB does not illuminate. →#1-2, #1-3, #1-4 b. <Check 24V> LD8 (green) on power supply motor PCB does not illuminate. →#1-5, #1-6 c. <Check 5V> LD3 (green) on power supply motor PCB does not illuminate. →#1-5, #1-6 d. <Check panel connection> Power supply LED on panel does not illuminate. →#1-5, #1-6, #1-7	
2. Malfunction of power cord (Check 300V)  4175B	Disconnect the power supply connector which links the power cord for the power switch and the power supply motor PCB, turn on the power switch, and then measure the voltage between terminals 1-2 of the cord-side connector. OK if the voltage is the same as mains voltage. In addition, check that the power supply voltage at the wall outlet is within the range of the specification voltage +/- 10%. After inspecting, turn off the power switch and connect the power supply connector.	
3. Blown fuse (Check 300V)	Remove fuses F2 and F3 from the power supply motor PCB and check the continuity. OK if there is continuity.	GLASSFUSE 10A-50T100H (10A-250V)
4. Malfunction of power supply motor PCB 1	Disconnect connector P2 (POWER) from the power supply motor PCB, and then turn on the power switch and check LD1 (red) on the power supply motor PCB. OK if illuminated. After checking, turn off the power switch, wait for five minutes or more, turn the power back on, and then with P2 inserted into the power supply motor PCB, check the voltage between terminals 1-3 of P2. OK if 254 to 358 V DC.	Power supply motor PCB assembly
5. Malfunction of power supply motor PCB 2	Disconnect connector P2 (POWER), connector P4 (MN-MT) and connector P5 (ENC-MT) from the power supply motor PCB, and then turn on the power switch and check LD3 (green) and LD8 (green) on the power supply motor PCB. OK if illuminated.	Power supply motor PCB assembly

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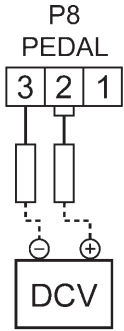
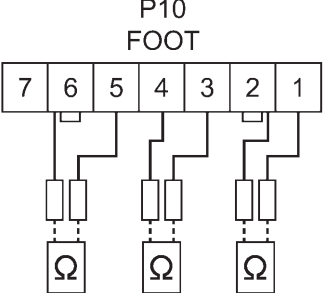
Error status #1 The power supply LED does not light when the power is turned on.		
Probable causes	Check/ repair/ adjust	Parts to be replaced
6. Malfunction of main PCB	Disconnect all connectors other than P22 (POWER) and P23 (MOTOR) from the main PCB, and then turn on the power switch and check LD3 (green) and LD8 (green) on the power supply motor PCB. OK if illuminated. After checking, turn off the power switch, wait for five minutes or more, and then insert all connectors.	Main PCB assembly
7. Malfunction of operation panel (Check 24V)	Check that connector P5 (PANEL) of the main PCB is inserted.	Operation panel assembly
8. Malfunction of LED assembly	Disconnect connector P1 (LED) from the main PCB, and then turn on the power switch and check LD8 (green) on the power supply motor PCB. OK if illuminated.	LED assembly

Error status #2 Illumination LED does not light.		
Probable causes	Check/ repair/ adjust	Parts to be replaced
1. Malfunction of main PCB	Disconnect connector P1 (LED) from the main PCB, and then turn on the power switch. Set the brightness level of illumination LED to "10" by means of panel operation, and measure the voltage between terminals 1-2 of P1 (LED) of main PCB. OK if approximately 24 V DC.	Main PCB assembly
2. Malfunction of LED assembly	Check if there is a broken wire somewhere in the harness which is inserted into connector P1 (LED) of main PCB.	LED assembly

14. ELECTRIC MECHANISM

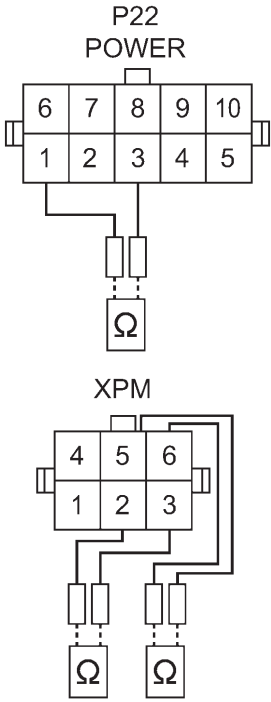
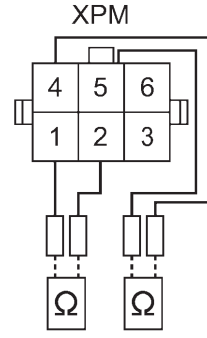
Error status #3 Error code appears on the display when the power is turned on.		
Probable causes	Check/ repair/ adjust	Parts to be replaced
1. If "E025" or "E035" is displayed, the foot switch is still depressed.	a. Check if the foot switch is still depressed. b. Check if there is a short-circuit in the harness. c. Check if connector P8 (PEDAL) is connected to the main PCB. d. Reset the depression stroke for the foot switch while referring to "9-22. Setting method for standard depression strokes (Foot switch)". * For a 2-pedal foot switch, the connector for the main PCB will be P10 (FOOT).	Treadle unit 2-pedal foot switch assembly
2. If "E055" is displayed, there is a malfunction of the machine head switch.	a. Check if the machine head switch is off. b. Check if there is a broken wire in the harness. c. Check if connector P9 (HEAD-SW) is connected to the main PCB.	Switch assembly machine head
3. If "E065" is displayed, one of the keys on the operation panel is still depressed.	a. Check that there is no incorrect sensitivity when the surface of the panel sheet and the keys are pressed. b. Check that connector P5 (PANEL) is connected to the main PCB, and that connector P1 is connected to the panel main PCB. c. Check that CN2 is inserted into the panel main PCB.	Operation panel assembly
4. If "E131" is displayed, there is a poor connection of the synchronizer.	Check that connector P5 (ENC-MT) is connected to the power supply motor PCB, and that connector P23 (MOTOR) is connected to the main PCB.	
5. If "E400" or "E401" is displayed, there is a connection fault between the main PCB and the power supply motor PCB or between the main PCB and the panel main PCB.	a. Check LD3 (green) on the main PCB. OK if illuminated. b. Check LD3 (green) on the power supply motor PCB. OK if illuminated. c. Check that connector P23 (MOTOR) is connected to the main PCB, and that connector P4 (MN-MT) is connected to the power supply motor PCB. d. Check that connector P5 (PANEL) is connected to the main PCB, and that connector P1 is connected to the panel main PCB. e. Check if there is a broken wire in the harness.	Main PCB assembly Power supply motor PCB assembly Operation panel assembly
6. If "E450" or "E452" is displayed, the machine head memory cannot be recognized.	a. Check if connector P4 (HEAD-MEM) is connected to the main PCB. b. Check if there is a broken wire in the harness.	
7. If "E700" is displayed, the power supply voltage is abnormally high.	a. Check that the power supply voltage at the mains is at the specification voltage +/-10%. b. See #1-2.	
8. If "E705" is displayed, the power supply voltage is abnormally low.	a. Check that the power supply voltage at the mains is at the specification voltage +/- 10%. b. See #1-2.	

Error status #4 No keys on the operation panel are not activated.		
Probable causes	Check/ repair/ adjust	Parts to be replaced
1. Malfunction of panel main PCB or panel key PCB	a. Check that connector P5 (PANEL) is connected to the main PCB, and that connector P1 is connected to the panel main PCB. b. Check that CN2 of the panel main PCB and CN2 of the panel key PCB are inserted.	Operation panel assembly
2. Main software not installed correctly	Carry out the restore operation by referring to "4-10. Updating the software version".	

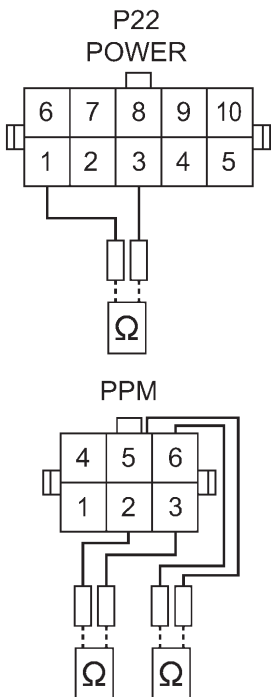
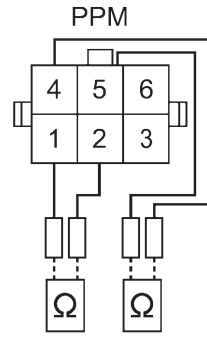
Error status #5 Work clamp does not return to the home position even after the foot switch is depressed.		
Probable causes	Check/ repair/ adjust	Parts to be replaced
1. Incorrect depression position adjustment	Adjust the position while referring to "9-22. Setting method for standard depression strokes (Foot switch)".	
2. Malfunction of treadle unit  4176B	With the main PCB connector P8 (PEDAL) connected, turn on the power and measure the voltage between terminals 2-3 of the cord connector (No. 2 +, No. 3 -). OK if the voltage is approximately 1.6-2.6 V at neutral, approximately 3.5-4.5 V when the foot switch is depressed as far as it will go, and approximately 0.6-1.6 V when it is depressed backward.	Treadle unit
3. Malfunction of foot switch and cord (option)  4177B	(Check that the power is turned off.) Disconnect the foot switch from the main PCB, and measure the voltage between terminals 1-2 and 3-4 of the foot switch connector. (Check between terminals 5-6 if using a 2-pedal foot switch.) OK if the resistance is normally ∞ ohms but 0 ohms when depressed.	Foot switch

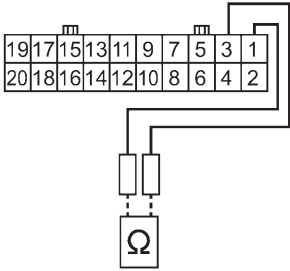
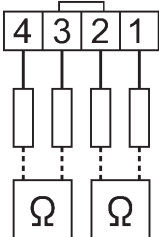
14. ELECTRIC MECHANISM

Error status #6 Error code appears on the display when the foot switch is depressed.		
Probable causes	Check/ repair/ adjust	Parts to be replaced
1. If "E130", "E131" or "E132" is displayed, there is a poor connection of the synchronizer.	Refer to inspection #3-4.	
2. If "E130", "E131" or "E132" is displayed, there is a poor connection of the machine motor.	Check the connection of the relay connector between the machine motor and the power supply motor PCB. * Relay connector: connected to P5 (ENC-MT) and XT8, XT5 and XT6.	
3. If "E130", "E131" or "E132" is displayed, there is a blown fuse.	a. Remove fuses F2 and F3 from the power supply motor PCB and check the continuity. OK if there is continuity. b. If fuse F2 or F3 is blown, check the resistance values between each terminal of the relay connector between the machine motor and the power supply motor PCB and between each terminal of the fuses. OK if ∞ ohms. c. If the fuses blow again after they are replaced, replace the power supply motor PCB.	GLASS FUSE 10A-50T100H (10A-250V) Power supply motor PCB assembly
4. If "E130", "E131" or "E132" is displayed, there is a malfunction of the power supply motor PCB.	a. Check the encoder input while referring to "4-4. Checking input". b. If step a. (above) is OK, there is a malfunction of the power supply motor PCB.	Motor 430 assembly Power supply motor PCB assembly

Error status #7 Error code appears on the display when the foot switch is depressed.		
Probable causes	Check/ repair/ adjust	Parts to be replaced
1. If the X feed does not move and "E200" is displayed, there is a malfunction of the power PCB.	Disconnect connector P2 (POWER) from the power supply motor PCB, and then turn on the power switch and check LD1 (red) on the power supply motor PCB. OK if illuminated. After checking, turn off the power switch, wait for five minutes or more, and then insert connector P2 (POWER) into the power supply motor PCB.	Power supply motor PCB assembly
2. If the X feed does not move and "E200" is displayed, there is a malfunction of the main PCB.  P22 POWER XPM 4178B 4179B	a. With LD1 (red) on the main PCB not lit, disconnect connector P22 (POWER) from the main PCB, and check the continuity between terminals 1-3 of P22. OK if ∞ ohms. b. Disconnect the relay connector (translucent and connected to XT12/XT10/XT11/XT8) for connecting the pulse motor to the main PCB, and check the continuity between terminals 2-3 and 5-6 of the connector. OK if ∞ ohms. c. Insert P22 and the translucent relay connector for connecting the pulse motor connection to the main PCB, turn on the power switch and then check LD1 (red) on the power supply motor PCB. OK if illuminated. If LD1 is illuminated during the check in step 1 in #7 but it does not illuminate when the translucent relay connector for connecting the pulse motor and P22 are connected, there is a malfunction of the main PCB. d. Check LD1 (red) on the main PCB. OK if illuminated.	Main PCB assembly
3. If the X feed moves slightly and "E200" is displayed, there is a malfunction of the encoder.	a. Check that connector P2 (X-ENC) is inserted into the main PCB and that the color matches. b. Check the encoder input while referring to "4-4. Checking input".	Pulse motor X assembly Main PCB assembly
4. If the X feed does not move and "E200" is displayed, there is a malfunction of the pulse motor and cord.  XPM 4180B	a. Disconnect the relay connector (translucent and connected to XT12/XT10/XT11/XT8) for connecting the pulse motor to the main PCB, and measure the resistance between terminals 2-3 and 5-6 of the cord connector. OK if 1.89-2.3 ohms (2.1 ohms +/- 10%). b. If step a. (above) is OK, there is a malfunction of the main PCB.	Pulse motor X assembly Main PCB assembly

Error status #8 Error code appears on the display when the foot switch is depressed.		
Probable causes	Check/ repair/ adjust	Parts to be replaced
1. If the Y feed does not move and "E210" is displayed, there is a malfunction of the power supply motor PCB.	Disconnect connector P2 (POWER) from the power supply motor PCB, and then turn on the power switch and check LD1 (red) on the power supply motor PCB. OK if illuminated. After checking, turn off the power switch, wait for five minutes or more, and then insert connector P2 (POWER) into the power supply motor PCB.	Power supply motor PCB assembly
2. If the Y feed does not move and "E210" is displayed, there is a malfunction of the main PCB. P22 POWER YPM 4178B 4181B	a. With LD1 (red) on the main PCB unlit, disconnect connector P22 (POWER) from the main PCB, and check the continuity between terminals 1-3 of P22. OK if ∞ ohms. b. Disconnect the relay connector (red and connected to XT9/XT7/XT6/XT4) for connecting the pulse motor to the main PCB, and check the continuity between terminals 2-3 and 5-6 of the connector. OK if ∞ ohms. c. Insert P22 and the red relay connector for connecting the pulse motor to the main PCB, turn on the power switch and then check LD1 (red) on the power supply motor PCB. OK if illuminated. If LD1 is illuminated during the check in step 1 in #8 but it does not illuminate when P22 is connected, there is a malfunction of the main PCB. d. Check LD1 (red) on the main PCB. OK if illuminated.	Main PCB assembly
3. If the Y feed moves slightly and "E210" is displayed, there is a malfunction of the encoder.	a. Check that connector P3 (Y-ENC) is inserted into the main PCB and that the color matches. b. Check the encoder input while referring to "4-4. Checking input".	Pulse motor Y assembly Main PCB assembly
4. If the Y feed does not move and "E210" is displayed, there is a malfunction of the pulse motor and cord. YPM 4182B	a. Disconnect the relay connector (red and connected to XT9/XT7/XT6/XT4) for connecting the pulse motor to the main PCB, and measure the resistance between terminals 2-3 and 5-6 of the cord connector. OK if 1.89-2.3 ohms (2.1 ohms +/- 10%). b. If step a. (above) is OK, there is a malfunction of the main PCB.	Pulse motor Y assembly Main PCB assembly

Error status #9 Error code appears on the display when the foot switch is depressed.		
Probable causes	Check/ repair/ adjust	Parts to be replaced
1. If the work clamp pulse motor does not operate and "E300" is displayed, there is a malfunction of the power supply motor PCB.	Disconnect connector P2 (POWER) from the power supply motor PCB, and then turn on the power switch and check LD1 (red) on the power supply motor PCB. OK if illuminated. After checking, turn off the power switch, wait for five minutes or more, and then insert connector P2 (POWER) into the power supply motor PCB.	Power supply motor PCB assembly
2. If the work clamp pulse motor does not move and "E300" is displayed, there is a malfunction of the main PCB.  4178B 4183B	a. With LD1 (red) on the main PCB unlit, disconnect connector P22 (POWER) from the main PCB, and check the continuity between terminals 1-3 of P22. OK if ∞ ohms. b. Disconnect the relay connector (black and connected to XT5/XT2/XT3/XT1) for connecting the pulse motor to the main PCB, and check the continuity between terminals 2-3 and 5-6 of the connector. OK if ∞ ohms. c. Insert P22 and the black relay connector for connecting the pulse motor connection to the main PCB, turn on the power switch and then check LD1 (red) on the power supply motor PCB. OK if illuminated. If LD1 is illuminated during the check in step 1 in #9 but it does not illuminate when the black relay connector for connecting the pulse motor and P22 are connected, there is a malfunction of the main PCB. d. Check LD1 (red) on the main PCB. OK if illuminated.	Main PCB assembly
3. If the work clamp pulse motor moves slightly and "E300" is displayed, there is a malfunction of the encoder.	a. Check that connector P6 (P-ENC) is inserted into the main PCB and that the color matches. b. Check the encoder input while referring to "4-4. Checking input".	Pulse motor P assembly Main PCB assembly
4. If the work clamp pulse motor does not move and "E300" is displayed, there is a malfunction of the pulse motor and cord.  4184B	a. Disconnect the relay connector (black and connected to XT5/XT2/XT3/XT1) for connecting the pulse motor to the main PCB, and measure the resistance between terminals 2-3 and 5-6 of the cord connector. OK if 1.89-2.3 ohms (2.1 ohms +/- 10%). b. If step a. (above) is OK, there is a malfunction of the main PCB.	Pulse motor P assembly Main PCB assembly

Error status #10 Error code appears on the display when the foot switch is depressed.		
Probable causes	Check/ repair/ adjust	Parts to be replaced
1. If "E690" is displayed during home position detection, there is a malfunction of the thread nipper.	a. Check the position of the thread nipper driving lever. b. Remove any thread scraps from behind the needle plate.	
2. If "E690" is displayed during home position detection, the thread nipper home position adjustment is incorrect.	Adjust the home position while referring to "9-21. Adjusting the thread nipper motor position".	
3. If the thread trimmer pulse motor does not operate and "E690" is displayed, there is a malfunction of the power supply motor PCB.	Disconnect connector P4 (MN-MT) from the power supply motor PCB, and then turn on the power switch and check LD8 (green) on the power supply motor PCB. OK if illuminated. After checking, turn off the power switch, wait for five minutes or more, and then insert connector P4 (MN-MT) into the power supply motor PCB.	Power supply motor PCB assembly
4. If the thread trimmer pulse motor does not move and "E690" is displayed, there is a malfunction of the main PCB. P23 MOTOR  P24 TPM  4295B 4185B	a. Disconnect connector P23 (MOTOR) from the main PCB, and check the continuity between terminals 1-3 of P23. OK if ∞ ohms. b. Disconnect connector P24 (TPM) from the main PCB, and check the continuity between terminals 1-2 and 3-4 of P24. OK if ∞ ohms. c. Insert P23 and P24 into the main PCB, turn on the power switch and then check LD5 (green) on the main PCB. OK if illuminated.	Main PCB assembly
5. If the thread trimmer pulse motor moves slightly and "E690" is displayed, there is a malfunction of the encoder.	a. Check that connector P7 (T-ENC) is inserted into the main PCB and that the color matches. b. Check the encoder input while referring to "4-4. Checking input".	Pulse motor T assembly Main PCB assembly

(Continued on next page.)

Error status #10 Error code appears on the display when the foot switch is depressed.

Probable causes	Check/ repair/ adjust	Parts to be replaced
6. If the thread trimmer pulse motor does not move and "E690" is displayed, there is a malfunction of the pulse motor and cord. P24 TPM 4 3 2 1 Ω Ω 4185B	a. Disconnect the main PCB connector P24 (TPM) and measure the resistance between terminals 1-2 and 3-4 of the cord connector. OK if 1.53-1.87 ohms (1.7 ohms +/- 10%). b. If step a. (above) is OK, there is a malfunction of the main PCB.	Pulse motor T assembly Main PCB assembly

Error status #11 Work clamp does not rise.

Probable causes	Check/ repair/ adjust	Parts to be replaced
1. Incorrect work clamp home position adjustment	a. Adjust the home position while referring to "9-20. Adjusting the work clamp motor home position". b. Check if the work clamp/button clamp is touching anything.	
2. Incorrect mechanism adjustment	Check if the work clamp arm moves smoothly.	Work clamp arm assembly

Error status #12 Work clamp does not lower.

Probable causes	Check/ repair/ adjust	Parts to be replaced
1. Incorrect work clamp home position adjustment	a. Adjust the home position while referring to "9-20. Adjusting the work clamp motor home position". b. Check if the work clamp/button clamp is touching anything.	
2. Incorrect mechanism adjustment	Check if the work clamp arm moves smoothly.	Work clamp arm assembly

Error status #13 The mode does not switch to test mode even when the TEST key is pressed.

Probable causes	Check/ repair/ adjust	Parts to be replaced
Malfunction of panel main PCB or panel key PCB	a. Check that connector P5 (PANEL) on the main PCB and connector P1 on the panel main PCB are inserted. b. Check that CN2 on the panel main PCB and CN2 on the panel key PCB are inserted.	Operation panel assembly

14. ELECTRIC MECHANISM

Error status #14 The feed mechanism does not slowly move stitch by stitch during test feeding.		
Probable causes	Check/ repair/ adjust	Parts to be replaced
1. Incorrect memory switch setting	Set memory switch No. 200 to OFF. (The default setting is ON for the BE-438H.)	
2. Malfunction of foot switch and cord	See #5.	

Error status #15 Quick feeding cannot be performed during test feeding.		
Probable causes	Check/ repair/ adjust	Parts to be replaced
Malfunction of foot switch and cord	See #5.	

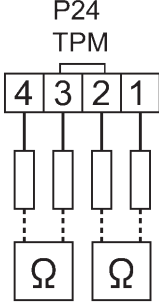
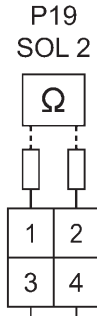
Error status #16 The machine does not operate correctly for the set program during test feeding.		
Probable causes	Check/ repair/ adjust	Parts to be replaced
Malfunction of foot switch and cord	See #5.	

Error status #17 The machine does not operate during sewing, and the error code [E130], [E131] or [E132] appears on the display.		
Probable causes	Check/ repair/ adjust	Parts to be replaced
Malfunction of machine motor, fuses or power supply motor PCB.	Refer to steps 1 to 4 in #6.	

Error status #18 Error code appears on the display after the machine operates.		
Probable causes	Check/ repair/ adjust	Parts to be replaced
1. If "E111", "E130", "E131", "E132" or "E133" is displayed after the sewing machine operates, there is a poor connection of the machine motor encoder.	Check that connector P4 (MN-MT) of the power supply motor PCB and connector P23 (MOTOR) of the main PCB are inserted.	
2. If "E111", "E130", "E131", "E132" or "E133" is displayed after the sewing machine operates, there is a poor connection of the machine motor.	Check the connection of the relay connector between the machine motor and the power supply motor PCB.	
3. If "E111", "E130", "E131", "E132" or "E133" is displayed after the sewing machine operates, interference is causing operating errors.	Check that the ground wire is securely connected and that the sewing machine is not close to any equipment that generates strong electrical interference.	
4. If "E111", "E130", "E131", "E132" or "E133" is displayed after the sewing machine operates, there is a malfunction of the power supply motor PCB.	Refer to steps 1 to 4 in #6.	Power supply motor PCB assembly
5. If "E111", "E130", "E132" or "E133" is displayed after the sewing machine operates, there is a malfunction of the power supply motor PCB or of the machine motor.	If an error occurs after inspection steps 1 to 4 above have been carried out, there is a malfunction of the machine motor.	Motor 430 assembly
6. If "E121" is displayed after the sewing machine operates, there is a malfunction of a component.	Check if the blades of the fixed knife and movable knife are damaged or worn.	
7. If "E121" is displayed after the sewing machine operates, the home position is incorrectly adjusted.	Adjust the home position while referring to "9-18. Adjusting the main shaft motor reference position".	
8. If "E121" is displayed after the sewing machine operates, there is a problem with the operation of the thread trimmer pulse motor.	Refer to steps 3 to 6 in #10.	Power supply motor PCB assembly Main PCB assembly Pulse motor T assembly

14. ELECTRIC MECHANISM

Error status #19 The machine cannot produce correct stitches.		
Probable causes	Check/ repair/ adjust	Parts to be replaced
1. Malfunction of machine motor encoder	Refer to step 1 in #18.	
2. If uneven seams are being sewn, there is a malfunction of the pulse motor or the mechanism is incorrectly adjusted.	a. Refer to step 4 in #7 and step 4 in #8 and step 4 in #9. b. If the uneven seams are due to insufficient work clamp pressure, adjust while referring to "9-9. Adjusting the work clamp lift amount (KE-430HX/HS)" or "9-10. Adjusting the button clamp lift amount (BE-438HX/HS)". c. If there is play in the feed mechanism, adjust the feed mechanism.	

Error status #20 The thread trimmer does not operate.		
Probable causes	Check/ repair/ adjust	Parts to be replaced
1. Incorrect memory switch setting	Set memory switch No. 164 to OFF. * If it is set to ON, thread trimming will not be carried out.	
2. Incorrect mechanism adjustment	Adjust the thread trimmer mechanism while referring to "9-21. Adjusting the thread nipper motor position". * If it seems that the thread trimming mechanism is not operating properly, carry out steps 3 to 6 below.	
3. Poor connector connection	Check that main PCB connector P24 (TPM) and P7 (T-ENC) are inserted.	
4. Malfunction of thread trimmer pulse motor  4185B	Disconnect main PCB connector P24 (TPM) and measure the resistance between terminals 1-2 and 3-4 of the cord connector. OK if 2-3 ohms.	Thread trimmer pulse motor assembly
5. Malfunction of tension solenoid  4188B	Disconnect main PCB connector P19 (SOL2) and measure the resistance between terminals 1-2 of the cord connector. OK if 4.99-5.51 ohms (5.25 ohms +/- 10%).	Tension solenoid

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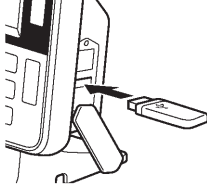
Error status #20 The thread trimmer does not operate.		
Probable causes	Check/ repair/ adjust	Parts to be replaced
6. Malfunction of main PCB. P24 TPM 4189B	a. Turn on the power switch and check LD5 (green) on the main PCB. OK if illuminated. b. Turn off the power switch, and with connector P24 (TPM) disconnected from the main PCB, check the continuity between terminals 1-2 and 3-4 of connector P24 on the main PCB. OK if ∞ ohms. c. With connector P24 (TPM) inserted into the main PCB, turn on the power switch, carry out sewing and measure the voltage between terminals 1-2 and 3-4 of connector P24. OK if voltage is output at sewing end thread trimming.	Main PCB assembly

Error status #21 The tension release does not operate.		
Probable causes	Check/ repair/ adjust	Parts to be replaced
1. Malfunction of main PCB P19 SOL 2 4296B	a. With connector P19 (SOL2) disconnected from the main PCB, check the continuity between terminals 1-2 of connector P19 on the main PCB. OK if ∞ ohms. b. With main PCB connector P19 (SOL2) connected, turn on the power and carry out sewing, and measure the voltage between terminals 1-2 of connector P19 (SOL2). OK if there is voltage output momentarily after sewing stops.	Main PCB assembly

Error status #22 The machine does not stop at the needle up stop position. ("UP" is frequently displayed after sewing)		
Probable causes	Check/ repair/ adjust	Parts to be replaced
1. Incorrect adjustment	Adjust while referring to "9-18. Adjusting the main shaft motor reference position".	
2. Problem with machine motor operation	Refer to steps 1 to 5 in #18.	Power supply motor PCB assembly Motor 430 assembly

14. ELECTRIC MECHANISM

Error status #23 Error code is displayed after the sewing machine operates (sewing program is sewn several times).		
Probable causes	Check/ repair/ adjust	Parts to be replaced
1. If "E150" is displayed after the sewing machine operates, the motor is abnormally overheating.	a. Turn off the power and let the motor stand for 30 minutes or more. b. Turn the power back on, and OK if operation is normal. c. If the area around the motor is not hot, carry out steps 1 to 5 in #18. * Avoid repeated sewing of sewing data that contains 15 stitches or less.	Power supply motor PCB assembly Motor 430 assembly

Error status #24 USB memory read/write mode cannot be activated.		
Probable causes	Check/ repair/ adjust	Parts to be replaced
1. If "E420" is displayed, the USB memory is incorrectly inserted.  3951B	a. Check the direction of insertion of the USB memory. b. Check the insertion of the USB memory.	
2. If "E420" is displayed, the USB memory is incorrectly formatted.	Check the format of the USB memory.	
3. Malfunction of USB memory	Use a PC to check if the contents of the USB memory can be read.	
4. Malfunction of operation panel	See #13.	Operation panel assembly



SERVICE MANUAL

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